

The Annotated Rudin

Principles of Mathematical Analysis, Third Edition

Walter Rudin

CHAPTERS 1–4

The Real and Complex Number Systems · Basic Topology · Numerical
Sequences and Series · Continuity

Rudin's text appears here complete and unaltered — his sentences, his paragraph breaks, his numbering. Nothing has been paraphrased away. The annotation is strictly additive: numbered markers in his prose key to notes in the outer margin, and longer commentary sits in ruled boxes between his paragraphs, where it can be skipped without breaking his argument.

How to read this book

Everything set in the main column below is Rudin, verbatim. You can read it straight through and never look at an annotation. The annotation layer sits in two places, both outside his text.

Margin notes. A small superscript number in Rudin's text — like this¹ — keys to a numbered note in the outer margin. Since the book prints in black and white, the *tier* of a note is carried by how its number is set:

¹ *The marker is the only mark added to his page, and it is the same for every tier so his line spacing is undisturbed. The number out here tells you what kind of note it is.*

7

Plain number — unpacking. What the sentence is actually saying, where a formula came from, an omitted algebraic step.

7

Outlined number — going deeper. Context beyond the course: what breaks in general, which later theorem this sets up. Skippable.

7

Reversed number — warning. A specific belief students form here that is false, or a place where Rudin's wording will mislead you.

Interleaved boxes. When a note runs longer than a margin will hold, it becomes a box between Rudin's paragraphs, framed by the same three tiers: a **solid** left rule for unpacking, a **dashed** left rule for going deeper, and rules **top and bottom** with a black tab for a warning.

Every figure is an addition. Rudin's Chapter 1 contains no diagrams at all. Any picture you see here is annotation, drawn to expose geometry that his algebra states without illustrating; each carries a grey italic caption.

Where the text is corrected. The main column follows a retyped edition whose own foreword concedes that fresh typos were probably introduced. Where one is evident, the reading is restored from Rudin's third edition and a warning-tier note records the change — silent emendation would make “verbatim” meaningless.

Structural commentary. Two further boxes carry no mathematics at all. They explain the book rather than its contents — why a chapter exists, why Rudin ordered it as he did, why a section that looks like a detour is not one.

HEAVY FRAME, BLACK TITLE BAR

Opens and closes every chapter: what it is for, how it is organized and why in that order, which parts are load-bearing. Where Rudin has explained his own choices — usually in the Preface — these boxes quote him rather than guess.

THIN FRAME, INLINE HEADING | Opens a section: why this material sits here, and what to hold on to.

Books cited in the notes. Where an annotation says another treatment does something differently, it means one of these. They are named because seeing the same theorem set up two ways is often what makes Rudin’s choice legible.

- | | |
|-------------------------------|--|
| <i>Tao, Analysis I–II</i> | Terence Tao, 3rd ed. Builds $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}$ from the Peano axioms and constructs $\mathbb{R}$ by Cauchy sequences rather than cuts. Volume II covers metric spaces and uniform convergence — Rudin’s Chapters 2, 4 and 7. |
| <i>Yu Pin, Tsinghua notes</i> | 数学分析讲义, Tsinghua / Yau Mathematical Sciences Center. Axiomatizes $\mathbb{R}$ by field, order, archimedean and <i>nested interval</i> axioms, then derives the supremum principle. |
| <i>ECNU text</i> | 数学分析 (Mathematical Analysis), 5th ed., 2019, East China Normal University. Standard Chinese two-volume text. Its §7.1 collects six equivalent formulations of completeness and proves them interderivable. |
| <i>USTC notes</i> | 中国科学技术大学少年班数学分析讲义 (程艺 et al., 2005). Teaches the main thread of calculus first and defers real-number theory to a later volume; lists Rudin among its three principal references. |

Chapter 1

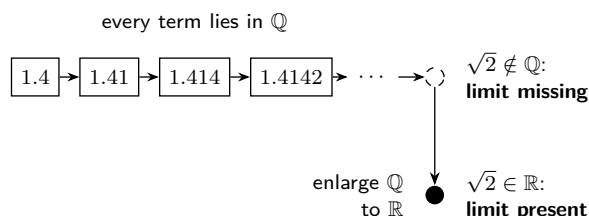
THE REAL AND COMPLEX NUMBER SYSTEMS

WHY THIS CHAPTER EXISTS

Consider the sequence

$$1, \quad 1.4, \quad 1.41, \quad 1.414, \quad 1.4142, \quad \dots$$

Every number in this sequence is rational. The sequence approaches $\sqrt{2}$, but $\sqrt{2}$ is not rational. Therefore the sequence has no limit that belongs to $\mathbb{Q}$.



The real numbers fill this gap by including $\sqrt{2}$. Rudin secures the property that rules out such gaps before using it throughout analysis — he states it as the *least-upper-bound property* (1.10), and it is what the word *complete* will name in 3.12.

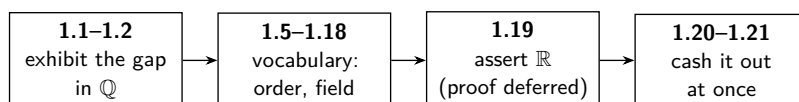
Everything asserting that something *exists* depends on it: that a bounded monotone sequence converges (3.14), that a continuous function on $[a, b]$ attains its bounds (4.16) and takes every intermediate value (4.23), that a continuous function is integrable (6.8). Each fails, or loses its meaning, over $\mathbb{Q}$ — $x^2 - 2$ changes sign on $[1, 2] \cap \mathbb{Q}$ without ever vanishing. Purely algebraic results survive; the existence theorems do not.

HOW IT IS ORGANIZED, AND WHY IN THAT ORDER

Rudin's own account. The obvious order would be to construct $\mathbb{R}$ from $\mathbb{Q}$ first and then do analysis on it. He declines, and says why in the Preface: beginning with the construction is “pedagogically unsound (though logically correct),” because “most students simply fail to appreciate the need for doing this.” Accordingly $\mathbb{R}$ “is introduced as an ordered field with the least-upper-bound

property,” and Dedekind’s construction is moved to the appendix, “where it may be studied and enjoyed whenever the time seems ripe.”

Reading that off the chapter, the shape is one designed to make you feel the need before paying the cost:



Four consequences are worth knowing in advance.

The construction is missing on purpose. Theorem 1.19 is stated and not proved; nothing in Chapters 2–11 depends on having read the appendix.

The abstraction comes before the object. Ordered sets (1.5–1.11) and fields (1.12–1.18) are defined *in general* before $\mathbb{R}$ is named, which lets Rudin characterise $\mathbb{R}$ by properties rather than by construction. He gives the second half of the reason himself at 1.13(c): proving the elementary consequences once, for an arbitrary field, means “we will not need to do it again for the real numbers and for the complex numbers.”

Two of the sections are equipment. The complex field (1.24–1.35) and Euclidean space (1.36–1.38) prove nothing about $\mathbb{R}$; they supply objects Chapters 2–11 need. The one result among them with real downstream weight is the Schwarz inequality, 1.35, which yields the metric on $\mathbb{R}^k$ and reappears for integrals in Exercise 6.10 (Hölder’s inequality, whose case $p = q = 2$ is there named Schwarz) and for square-integrable functions in 11.35.

Where the difficulty lies — a reader’s judgement, not Rudin’s. Not in 1.19, which is asserted, nor in the field axioms, which are routine. It is in 1.21 and in the appendix, the two places where a real number is produced out of nothing but a supremum. Follow those two arguments and you have the chapter.

HOW THE PROOFS IN THIS CHAPTER ARE FOUND

A proof you can only verify is a proof you cannot reproduce. Every proof in this chapter is built from one of seven discovery patterns, catalogued here once; the box at each proof works the details, and the same patterns carry most of Chapters 2 and 3.

1. Unwind the definition. When the objects are abstract, the axioms are the entire arsenal. Ask which axiom can *produce or*

remove the symbol standing between you and the goal, and the proof assembles itself (1.14, 1.15, 1.16, 1.18).

2. Hunt an invariant. To prove something impossible, find a quantity the two sides of the supposed equation cannot share — parity in 1.1, an integer strictly between 0 and 1 in 3.32, strict decrease in Tennenbaum’s dissection.

3. Transport the problem to a supremum. Completeness is the chapter’s only existence tool, so a proof that must *produce* a number has one move: build a set whose supremum is the number wanted — L in 1.11, $\{nx\}$ in 1.20(a), E in 1.21 — then interrogate the supremum with the two clauses of 1.8: step back to find members, step past to find bounds.

4. Work backwards for the constants. Every mysterious choice (h in 1.21, the coefficients B, C in 1.35) is found by writing down the target inequality and solving for the hypothesis, then presenting the steps in reverse.

5. Prove identities through uniqueness. To show two expressions are equal, show both answer a question that has exactly one answer (Corollary to 1.21, 1.33(c)).

6. Square away the absolute values. Statements about $|\cdot|$ become inner-product algebra after squaring; expand, bound the cross term, take roots at the end (1.33(e), 1.35, 1.37(e)).

7. Recognise pure verification. Some proofs (1.25–1.29, 1.31, 1.37 (a)–(c), most of the appendix) contain no idea to find — the idea lives in the *definition* being verified, and the proof is bookkeeping. Knowing that nothing is hidden is itself the skill: do not search a verification for insight that was spent earlier, at the definition.

8. Transport structure along f or f^{-1} . Added by Chapter 4, whose theorems mostly carry Chapter 2’s structures across a continuous map; listed here so the index stays complete. Nothing in this chapter uses it.

2 Rudin is choosing a starting line, and he defends the choice in the Preface: beginning with a construction of $\mathbb{R}$ is “pedagogically unsound (though logically correct)” because “most students simply fail to appreciate the need for doing this.” So the construction is exiled to the appendix, and $\mathbb{R}$ will arrive in 1.19 as an assertion. This sentence marks everything he is willing to take for granted.

INTRODUCTION

A satisfactory discussion of the main concepts of analysis (such as convergence, continuity, differentiation, and integration) must be based on an accurately defined number concept. We shall not, however, enter into any discussion of the axioms that govern the arithmetic of the integers, but assume familiarity with the rational numbers (i.e., the numbers of the form m/n where m and n are integers and $n \neq 0$).²

The rational number system is inadequate for many purposes, both as a field and as an ordered set. (These terms will be defined in Def-

initions 1.6 and 1.12.) For instance, there is no rational p such that $p^2 = 2$. (We shall prove this presently.) This leads to the introduction of so-called “irrational numbers” which are often written as infinite decimal expansions and are considered to be “approximated” by the corresponding finite decimals. Thus the sequence

$$1, 1.4, 1.41, 1.414, 1.4142, \dots$$

“tends to $\sqrt{2}$.”³ But unless the irrational number $\sqrt{2}$ has been clearly defined, the question must arise: Just what is it that this sequence “tends to”?

This sort of question can be answered as soon as the so-called “real number system” is constructed.

1.1 Example. We now show that the equation

$$p^2 = 2 \tag{1.1}$$

is not satisfied by any rational p . If there were such a p , we could write $p = m/n$ where m and n are integers that are not both even.⁴ Let us assume this is done. Then (1.1) implies

$$m^2 = 2n^2, \tag{1.2}$$

This shows that m^2 is even. Hence m is even (if m were odd, m^2 would be odd), and so m^2 is divisible by 4. It follows that the right side of (1.2) is divisible by 4, so that n^2 is even, which implies that n is even.

The assumption that (1.1) holds thus leads to the conclusion that both m and n are even, contrary to our choice of m and n . Hence (1.1) is impossible for rational p .

HOW THE PARITY ARGUMENT IS FOUND

Faced with $m^2 = 2n^2$, the search is for an *invariant*: some property that the equation forces on both sides but that the two sides cannot share. The right side is visibly even, so ask what evenness of m^2 says about m — and the cheapest arithmetic invariant, parity, turns out to propagate: m even makes m^2 divisible by 4, pushing evenness onto n^2 and then n .

Two design choices become visible once you look for them. The normalisation “not both even” exists so that the invariant has something to contradict — without it, “both even” is not absurd, merely reducible. And the argument is really about the exponent of 2: each side of $m^2 = 2n^2$ contains the prime 2 an even number of times on the left and an odd number on the right, which is the version that generalises (Exercise 1.2 runs it with the prime 3 on $m^2 = 12n^2$). Parity is that count, read modulo 2.

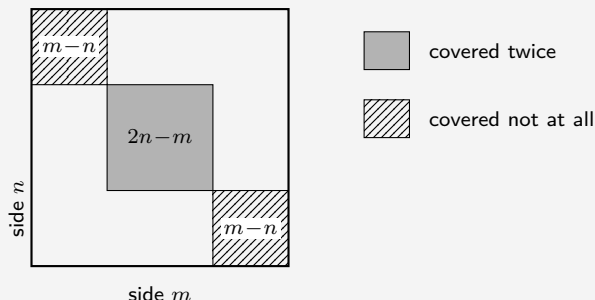
3 The scare quotes are doing real work. “Tends to” is undefined until Definition 3.1, and the thing it tends to is undefined until Theorem 1.19. Rudin is pointing at a circularity he is about to break, not relying on one.

4 Why we may assume this: put p in lowest terms. If m and n were both even we could cancel a factor of 2 and repeat, and the process must stop because $|m|$ strictly decreases. That appeal to “must stop” is well-ordering, so the step is not quite free. It can be avoided: Tao (Analysis I, 4.4.4) drops the assumption and instead turns each solution $p^2 = 2q^2$ into a smaller one, contradicting the principle of infinite descent. Same content, with the well-ordering moved into the open.

An alternative proof, done entirely by cutting up a square

Rudin argues by parity. There is a second proof, due to Stanley Tennenbaum, which uses no arithmetic beyond areas — and it makes the descent visible rather than hidden in the phrase “not both even.”

Suppose $m^2 = 2n^2$ with m, n positive integers. Then $n < m < 2n$, since $m^2 = 2n^2$ lies strictly between n^2 and $4n^2$. Draw a square of side m and lay a square of side n into two opposite corners. Because $m < 2n$ they overlap; because $m > n$ two corners are left bare.



Two squares of side n are dropped into opposite corners of a square of side m .

The two n -squares have total area $2n^2$, which by hypothesis is exactly m^2 , the area of the big square. So whatever is counted twice must have the same area as whatever is missed:

$$(2n - m)^2 = 2(m - n)^2.$$

That is the same equation again, in strictly smaller positive integers — $0 < 2n - m < m$ and $0 < m - n < n$. Repeating produces an infinite strictly decreasing sequence of positive integers, which is impossible. Hence no such m, n exist.

The two proofs use the same principle. Rudin’s “not both even” silently invokes well-ordering to stop a descent; here the descent is performed in the open, and each step is a picture.

We now examine this situation a little more closely. Let A be the set of all positive rationals p such that $p^2 < 2$ and let B consist of all positive rationals p such that $p^2 > 2$. We shall show that A contains no largest number and B contains no smallest.⁵

More explicitly, for every p in A we can find a rational q in A such that $p < q$, and for every p in B we can find a rational q in B such that $q < p$.

To do this, we associate with each rational $p > 0$ the number

$$q = p - \frac{p^2 - 2}{p + 2} = \frac{2p + 2}{p + 2}. \quad (1.3)$$

⁵ This, and not the irrationality of $\sqrt{2}$, is the real content of §1.1. Irrationality says $\sqrt{2} \notin \mathbb{Q}$. This says $\mathbb{Q}$ has a gap where $\sqrt{2}$ should be — a structural defect, and the one Definition 1.10 will name.

Then

$$q^2 - 2 = \frac{2(p^2 - 2)}{(p + 2)^2}. \quad (1.4)$$

If p is in A , then $p^2 - 2 < 0$, (1.3) shows that $q > p$ and (1.4) shows that $q^2 < 2$. Thus q is in A .

If p is in B , then $p^2 - 2 > 0$, (1.3) shows that $0 < q < p$, and (1.4) shows that $q^2 > 2$. Thus q is in B .

WHERE (1.3) COMES FROM

Rudin produces this formula with no derivation, and it is the first place in the book where a reader stalls. It is not Newton's method. Ask instead: which map $p \mapsto q$ has $\sqrt{2}$ as a *fixed point*? Try the simplest shape, a ratio of linear terms $q = (ap + b)/(cp + d)$. Setting $q = p$ gives $p(cp + d) = ap + b$, and we want that to collapse to $p^2 = 2$. The smallest-integer solution is $q = (2p + 2)/(p + 2)$: it gives $p^2 + 2p = 2p + 2$, that is, $p^2 = 2$.

Why it moves p toward $\sqrt{2}$ without crossing it becomes visible once you factor the error:

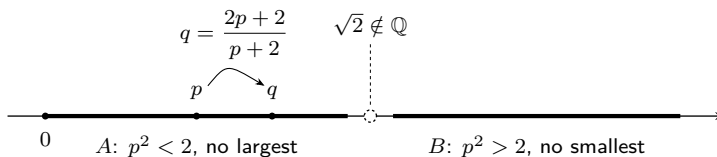
$$q - \sqrt{2} = \frac{2 - \sqrt{2}}{p + 2} (p - \sqrt{2}).$$

For $p > 0$ the multiplier lies strictly between 0 and $(2 - \sqrt{2})/2 \approx 0.29$. That it is *positive* keeps q on the same side of $\sqrt{2}$ as p — Rudin's " q is in A " and " q is in B ." That it is *less than one* makes the error shrink, so q is strictly better than p and no member of A can be largest.

Equation (1.4) is worth checking rather than accepting:

$$(2p + 2)^2 - 2(p + 2)^2 = (4p^2 + 8p + 4) - (2p^2 + 8p + 8) = 2p^2 - 4.$$

There is a second way to say what this establishes, worth knowing because it is the form other books use. Tao (*Analysis I*, 4.4.5) proves: for every rational $\varepsilon > 0$ there is a rational $x \geq 0$ with $x^2 < 2 < (x + \varepsilon)^2$. Rudin's version is about the sets A and B having no extreme member; Tao's is about how narrowly they can be straddled. They say the same thing — the gap is real, and it is narrower than any rational you care to name.



Every $p \in A$ is beaten by some $q \in A$, and every $p \in B$ beats some $q \in B$. The two sets crowd a point that is not there.

The estimate behind the contraction principle

That factorisation says $p \mapsto (2p + 2)/(p + 2)$ shrinks the distance to $\sqrt{2}$ by a factor of at most 0.29 at every step. It is the same estimate that Theorem 9.23, the contraction principle, later abstracts — though 9.23 itself does not apply here, since it needs a *complete* space and the positive rationals are exactly what is not complete. That is the point: the limit $\sqrt{2}$ does not exist in $\mathbb{Q}$, which is why Rudin must argue by hand that no member of A is largest rather than simply passing to a fixed point.

Iterating from $p_0 = 1$ gives

$$1, \frac{4}{3}, \frac{7}{5}, \frac{24}{17}, \frac{41}{29}, \frac{140}{99}, \dots$$

Every term stays below $\sqrt{2}$, as the positive multiplier requires. Some are convergents of $\sqrt{2} = [1; 2, 2, 2, \dots]$ — $\frac{7}{5}$ and $\frac{41}{29}$ are — but not all: the convergents run $1, \frac{3}{2}, \frac{7}{5}, \frac{17}{12}, \frac{41}{29}, \dots$, and this iteration visits every second one.

Newton's method on $f(p) = p^2 - 2$ would give $p \mapsto (p^2 + 2)/(2p)$ instead: faster, but from any starting point it lands *above* $\sqrt{2}$, so it would prove “ B has no smallest” and say nothing about A . Rudin's map does both halves at once.

6 The property Rudin describes here — between any two elements lies a third — is not the property Definition 1.10 will demand, and this sentence is the first warning that the two are independent. $\mathbb{Q}$ has the first and lacks the second. $\mathbb{Z}$ has the second (every nonempty bounded-above set of integers has a largest element) and lacks the first. So no amount of “no visible gaps between neighbours” delivers completeness.

7 Rudin's $\subset$ permits equality — it is what most modern texts write $\subseteq$. He writes “proper subset” when he means strict containment. Read every $\subset$ in this book as $\subseteq$, or you will misread hypotheses for the next 300 pages.

1.2 Remark. The purpose of the above discussion has been to show that the rational number system has certain gaps, in spite of the fact that between any two rationals there is another: if $r < s$, then $r < (r + s)/2 < s$.⁶ The real number system fills these gaps. This is the principal reason for the fundamental role which it plays in analysis.

In order to elucidate its structure, as well as that of the complex numbers, we start with a brief discussion of the general concepts of *ordered set* and *field*.

Here is some of the standard set-theoretic terminology that will be used throughout this book.

1.3 Definitions. If A is any set (whose elements may be numbers or any other objects), we write $x \in A$ to indicate that x is a member (or an element) of A . If x is not a member of A , we write $x \notin A$.

The set which contains no element will be called the *empty set*. If a set has at least one element, it is called *nonempty*.

If A and B are sets, and if every element of A is an element of B , we say that A is a *subset* of B , and we write $A \subset B$, or $B \supset A$.⁷ If, in addition, there is an element of B which is not in A , then A is said to be a *proper subset* of B . Note that $A \subset A$ for every set A .

If $A \subset B$ and $B \subset A$, we write $A = B$. Otherwise, $A \neq B$.

1.4 Definition. Throughout Chap. 1, the set of all rational numbers will be denoted by $\mathbb{Q}$.

ORDERED SETS

WHY THIS SECTION IS HERE | Nothing in this section mentions $\mathbb{R}$, and on a first reading it feels like an unmotivated detour into abstraction. It is not. Rudin has just shown that $\mathbb{Q}$ fails in a specific way, and he now needs language precise enough to *name* that failure — because the name will become the defining property of $\mathbb{R}$ in 1.19. Definitions 1.5 to 1.7 are only scaffolding; 1.8 and 1.10 are the payload. If you read one thing here, read Definition 1.10, and notice that it is exactly what Example 1.9(a) showed $\mathbb{Q}$ does not have.

1.5 Definition. Let S be a set. An *order* on S is a relation, denoted by $<$, with the following two properties:

- (i) If $x \in S$ and $y \in S$, then one and only one of the statements

$$x < y, \quad x = y, \quad y < x$$

is true.⁸

- (ii) If $x, y, z \in S$, if $x < y$ and $y < z$, then $x < z$.

The statement “ $x < y$ ” may be read as “ x is less than y ” or “ x is smaller than y ” or “ x precedes y ”.

It is often convenient to write $y > x$ in place of $x < y$.

The notation $x \leq y$ indicates that $x < y$ or $x = y$, without specifying which of these two is to hold. In other words, $x \leq y$ is the negation of $x > y$.⁹

1.6 Definition. An *ordered set* is a set S in which an order is defined.

For example, $\mathbb{Q}$ is an ordered set if $r < s$ is defined to mean that $s - r$ is a positive rational number.

1.7 Definition. Suppose S is an ordered set, and $E \subset S$. If there exists a $\beta \in S$ such that $x \leq \beta$ for every $x \in E$, we say that E is *bounded above*, and call β an *upper bound* of E .

Lower bounds are defined the same way (with $\geq$ in place of $\leq$).

1.8 Definition. Suppose S is an ordered set, $E \subset S$, and E is bounded above. Suppose there exists an $\alpha \in S$ with the following properties:

- (i) α is an upper bound of E .
(ii) If $\gamma < \alpha$ then γ is not an upper bound of E .¹⁰

Then α is called the *least upper bound* of E [that there is at most one such α is clear from (ii)] or the *supremum* of E , and we write

$$\alpha = \sup E.$$

The *greatest lower bound*, or *infimum*, of a set E which is bounded

8 *Trichotomy.* “One and only one” packs in two separate demands: at least one holds, so the order is total; and at most one holds, so $<$ is irreflexive and antisymmetric. Drop totality and you get a partial order, where suprema can fail to exist for reasons having nothing to do with gaps.

9 That last sentence is a small theorem, not a definition: “ $x < y$ or $x = y$ ” is equivalent to “not $x > y$ ” precisely because of trichotomy in (i). In a partial order the two come apart, which is why $\leq$ is normally taken as the primitive notion there.

10 This is the clause you will actually use, and it is stated contrapositively. The working form: for every $\gamma < \alpha$ there is an $x \in E$ with $x > \gamma$. In $\mathbb{R}$, equivalently: for every $\varepsilon > 0$ there is $x \in E$ with $x > \alpha - \varepsilon$. Reach for that sentence every time a proof opens “let $\alpha = \sup E$.”

below is defined in the same manner: The statement

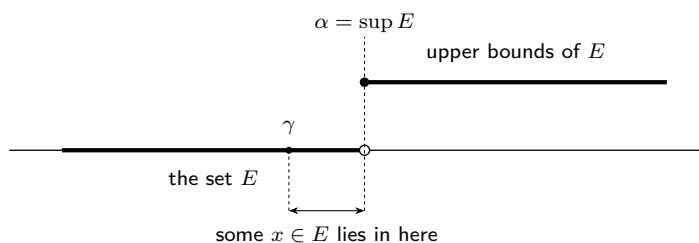
$$\alpha = \inf E$$

means that α is a lower bound of E and that no β with $\beta > \alpha$ is a lower bound of E .

WHY THE LEAST UPPER BOUND AND NOT THE LARGEST ELEMENT

Because the largest element usually does not exist, and the supremum is the repair. Take $E_1 = \{r \in \mathbb{Q} : r < 0\}$. It has no largest element: for any $r < 0$ there is $r/2$ with $r < r/2 < 0$. But the set of its upper bounds — everything ≥ 0 — does have a smallest one, namely 0.

The manoeuvre is to move the question from E , where the answer is often “none,” to the set of upper bounds of E , where it is often “yes.” Rudin’s bracketed aside is the payoff: uniqueness follows immediately from (ii), so speaking of “the” supremum is legitimate.



α is the seam. Everything to its right bounds E ; nothing to its left does.

The picture also shows why clause (ii) is the useful one. Stepping back from α by any amount lands you at a γ that is *not* an upper bound, which is to say some element of E sticks out past it. That is the sentence you will use.

1.9 Examples.

- (a) Consider the sets A and B of Example 1.1 as subsets of the ordered set $\mathbb{Q}$. The set A is bounded above. In fact, the upper bounds of A are exactly the members of B .¹¹ Since B contains no smallest member, A has no least upper bound in $\mathbb{Q}$.

Similarly, B is bounded below: the set of all lower bounds of B consists of A and of all $r \in \mathbb{Q}$ with $r \leq 0$. Since A has no largest member, B has no greatest lower bound in $\mathbb{Q}$.

- (b) If $\alpha = \sup E$ exists, then α may or may not be a member of E . For instance, let E_1 be the set of all $r \in \mathbb{Q}$ with $r < 0$. Let E_2 be the set of all $r \in \mathbb{Q}$ with $r \leq 0$. Then

$$\sup E_1 = \sup E_2 = 0,$$

$$\text{and } 0 \notin E_1, 0 \in E_2.¹²$$

11 Not exactly, and Rudin is being loose. Every $\beta \in B$ bounds A , and no member of A does. But one must also rule out a positive rational with $\beta^2 = 2$, and that is impossible only by Example 1.1. The claim holds, but it leans on the irrationality result proved a page earlier.

12 Both suprema exist in $\mathbb{Q}$. Part (a) fails not because $\mathbb{Q}$ is too small in general but because of where that particular gap sits. Whether $\sup E$ lies in E is a separate question from whether $\sup E$ exists, and running the two together is the most common first-week error.

- (c) Let E consist of all numbers $1/n$, where $n = 1, 2, 3, \dots$. Then $\sup E = 1$, which is in E , and $\inf E = 0$, which is not in E .

1.10 Definition. An ordered set S is said to have the *least-upper-bound property* if the following is true:

If $E \subset S$, E is not empty, and E is bounded above, then $\sup E$ exists in S .¹³

Example 1.9(a) shows that $\mathbb{Q}$ does not have the least-upper-bound property.

Rudin's choice of axiom is one of several

Completeness can be installed in an ordered field in several equivalent ways, and which one a book takes as primitive shapes everything after it.

Rudin takes the least-upper-bound property as the defining property (1.10), asserts a field with it (1.19), and *derives* the archimedean property from it (1.20a).

Yu Pin (Tsinghua) axiomatises $\mathbb{R}$ by field, order, archimedean, and the *nested interval* axiom — every decreasing sequence of closed bounded intervals has a common point. He observes that $\mathbb{Q}$ satisfies the first three and fails only the fourth, then derives the supremum principle from it. Archimedes must be assumed separately, because nested intervals alone does not imply it; he proves the converse too, so over an archimedean ordered field the two are equivalent.

Tao (*Analysis I*) assumes nothing, builds $\mathbb{R}$ from $\mathbb{Q}$ as equivalence classes of Cauchy sequences, and *proves* the least-upper-bound property as a theorem.

The ECNU text lays out the whole menu. Its §7.1 collects six statements — the supremum principle, monotone convergence, nested intervals, Heine–Borel, Bolzano–Weierstrass, and the Cauchy criterion — and proves the cycle $1 \Rightarrow 2 \Rightarrow \dots \Rightarrow 6 \Rightarrow 1$, so that any one of them may serve as the axiom and the other five follow.

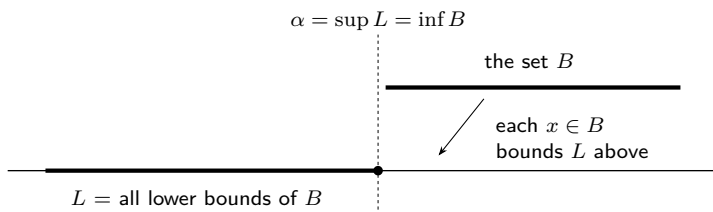
That last point is worth carrying into Chapter 2. Rudin does not present these as six faces of one property; he proves them separately and far apart — 2.41 and 2.42 for compactness and limit points in $\mathbb{R}^k$, 3.11 for Cauchy sequences, 3.14 for monotone ones. Each is a theorem earned from 1.19. Knowing they are interderivable tells you that those later proofs are not accumulating new strength about $\mathbb{R}$: they are unpacking, in different vocabularies, the single property assumed here.

13 Both hypotheses are load-bearing, and dropping either breaks the definition rather than merely weakening it. In $\mathbb{R}$: every real is an upper bound of $\emptyset$, so $\emptyset$ has no least one; and $\mathbb{Z}$ is unbounded above, so it has none either. Any proof that produces a supremum owes you a check of both.

We shall now show that there is a close relation between greatest lower bounds and least upper bounds, and that every ordered set with the least-upper-bound property also has the greatest-lower-bound property.

THE PICTURE BEHIND THEOREM 1.11

We are handed a hypothesis about suprema and asked to produce an infimum, so the only move available is to find a set whose *supremum* is the infimum we want. The set of lower bounds L is that set — it sits entirely below B , every element of B bounds it from above, and the two meet at a single point.



The two sets share one boundary point, and the theorem says that point is both $\sup L$ and $\inf B$.

1.11 Theorem. Suppose S is an ordered set with the least-upper-bound property, $B \subset S$, B is not empty, and B is bounded below. Let L be the set of all lower bounds of B . Then

$$\alpha = \sup L$$

exists in S and $\alpha = \inf B$.

In particular, $\inf B$ exists in S .

Proof. Since B is bounded below, L is not empty. Since L consists of exactly those $y \in S$ which satisfy the inequality $y \leq x$ for every $x \in B$, we see that every $x \in B$ is an upper bound of L .¹⁴ Thus L is bounded above. Our hypothesis about S implies therefore that L has a supremum in S ; call it α .

If $\gamma < \alpha$ then (see Definition 1.8) γ is not an upper bound of L , hence $\gamma \notin B$.¹⁵ It follows that $\alpha \leq x$ for every $x \in B$. Thus $\alpha \in L$.

If $\alpha < \beta$ then $\beta \notin L$, since α is an upper bound of L .

We have shown that $\alpha \in L$ but $\beta \notin L$ if $\beta > \alpha$. In other words, α is a lower bound of B , but β is not if $\beta > \alpha$. This means that $\alpha = \inf B$. $\square$

HOW THE PROOF FLOWS

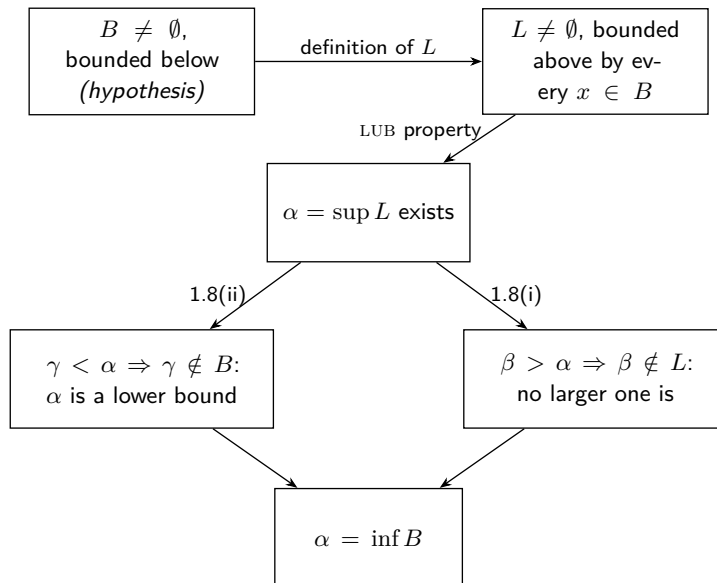
- 1. Choose the method.** The hypothesis speaks of suprema, the goal of an infimum, and the least-upper-bound property is the only existence tool in the chapter. So the method is forced — pattern 3 of the strategy index: *find a set whose supremum is the number you want*, then take its sup. The candidate is L , the set of all lower bounds of B ; the picture before the theorem shows why it is right: L ends exactly where B begins.
- 2. Buy the supremum.** The LUB property sells only to sets that are nonempty and bounded above, so both tickets must be pro-

14 This single sentence is the whole idea. We are handed a hypothesis about suprema and asked for an infimum, so we must find a set whose supremum is the infimum we want. L is that set: B presses down on it from above, and the two meet at one point.

15 Compressed to the point of opacity. Unpacked: γ failing to be an upper bound of L produces some $y \in L$ with $y > \gamma$. Since y is a lower bound of B , every $x \in B$ satisfies $x \geq y > \gamma$. So γ itself cannot belong to B .

duced. $L \neq \emptyset$ is the hypothesis “ B is bounded below” restated. L bounded above is the proof’s one genuine observation: every $x \in B$ bounds L , by the very definition of “lower bound.” Payment accepted: $\alpha = \sup L$ exists.

3. **Show α is a lower bound** (first clause of the goal). Rudin argues by testing every $\gamma < \alpha$: clause (ii) of 1.8 says γ is not an upper bound of L , so some $y \in L$ exceeds γ ; since y underlies all of B , so does γ , strictly — hence $\gamma \notin B$. Nothing in B lies below α , which is precisely $\alpha \in L$.
4. **Show nothing larger works** (second clause). Any $\beta > \alpha$ exceeds the supremum of L , hence $\beta \notin L$: it fails to be a lower bound. This step uses clause (i) of 1.8 — α is an upper bound of L .
5. **Read off the conclusion.** Steps 3 and 4 are exactly the two clauses of Definition 1.8, mirrored for the infimum, so $\alpha = \inf B$. Note the audit: clause (ii) of the supremum was spent in step 3, clause (i) in step 4 — each exactly once, the usual sign that L was the right set to build (compare the same remark at 1.21).



The proof as a flow diagram: hypotheses qualify L , the least-upper-bound property mints α , and the two clauses of Definition 1.8 each certify one half of “ $\alpha = \inf B$.”

Why this is not a triviality

The argument runs equally well in reverse, so for ordered sets the least-upper-bound property and the greatest-lower-bound property are *equivalent*. That is why Rudin never states a separate greatest-lower-bound axiom for $\mathbb{R}$: Theorem 1.19 gives him one and Theorem 1.11 hands him the other.

Every later appearance of $\inf$ rests on this — lower Riemann–Stieltjes sums in 6.2, $\liminf$ in 3.16, the distance from a point to a set in the exercises. It is a two-paragraph theorem that the rest of the book quietly leans on.

FIELDS

WHY THIS SECTION IS HERE | This is the second half of the vocabulary, and the less interesting half. An ordered set gave Rudin the language for *gaps*; a field gives him the language for *arithmetic*. $\mathbb{R}$ will be defined in 1.19 as an object having both, plus one extra property, so both notions must be on the table first.

Rudin tells you outright how to read 1.14–1.18: they are “some of the familiar properties of $\mathbb{Q}$ ” rederived from the axioms so that “once we do this, we will not need to do it again for the real numbers and for the complex numbers.” That is the entire point. Nothing here will surprise you; the value is that every one of these facts is now available for $\mathbb{R}$, for $\mathbb{C}$, and for any other field, purchased once. Skim the proofs, note Definition 1.17, and move on — but see the warning after 1.18, because the ordering axioms are where students later import assumptions that are not there.

1.12 Definition. A *field* is a set F with two operations, called *addition* and *multiplication*, which satisfy the following so-called “field axioms” (A), (M), and (D):

(A) Axioms for addition

- (A1) If $x \in F$ and $y \in F$, then their sum $x + y$ is in F .
- (A2) Addition is commutative: $x + y = y + x$ for all $x, y \in F$.
- (A3) Addition is associative: $(x + y) + z = x + (y + z)$ for all $x, y, z \in F$.
- (A4) F contains an element 0 such that $0 + x = x$ for every $x \in F$.
- (A5) To every $x \in F$ corresponds an element $-x \in F$ such that

$$x + (-x) = 0.$$

(M) Axioms for multiplication

- (M1) If $x \in F$ and $y \in F$, then their product xy is in F .¹⁶
- (M2) Multiplication is commutative: $xy = yx$ for all $x, y \in F$.
- (M3) Multiplication is associative: $(xy)z = x(yz)$ for all $x, y, z \in F$.
- (M4) F contains an element $1 \neq 0$ such that $1x = x$ for every $x \in F$.¹⁷

16 Source correction. The retyped edition this book follows drops “and $y \in F$ ” here, and drops the opening “If” from (A1); its own foreword warns that new typos were likely introduced. Both clauses are restored above from Rudin’s third edition. Without $y \in F$ the axiom does not say what it must — that F is closed under multiplication.

17 The clause $1 \neq 0$ is not pedantry. Without it the one-element set $\{0\}$ satisfies every other axiom, and then Proposition 1.16(a) would make every element equal to 0. Excluding it is what keeps “field” from being vacuous.

(M5) If $x \in F$ and $x \neq 0$ then there exists an element $1/x \in F$ such that

$$x \cdot (1/x) = 1.$$

(D) The distributive law

$$x(y + z) = xy + xz$$

holds for all $x, y, z \in F$.¹⁸

1.13 Remarks.

(a) One usually writes (in any field)

$$\begin{array}{ccccccc} x - y, & \frac{x}{y}, & x + y + z, & xyz, & & & \\ x^2, & x^3, & 2x, & 3x, \dots & & & \end{array}$$

in place of

$$\begin{array}{ccccccc} x + (-y), & x \cdot \left(\frac{1}{y}\right), & (x + y) + z, & (xy)z, & & & \\ xx, & xxx, & x + x, & x + x + x, \dots & & & \end{array}$$

- (b) The field axioms clearly hold in $\mathbb{Q}$, the set of all rational numbers, if addition and multiplication have their customary meaning. Thus $\mathbb{Q}$ is a field.
- (c) Although it is not our purpose to study fields (or any other algebraic structures) in detail, it is worthwhile to prove that some of the familiar properties of $\mathbb{Q}$ are consequences of the field axioms; once we do this, we will not need to do it again for the real numbers and for the complex numbers.

1.14 Proposition. *The axioms for addition imply the following statements.*

- (a) *If $x + y = x + z$ then $y = z$.*
- (b) *If $x + y = x$ then $y = 0$.*
- (c) *If $x + y = 0$ then $y = -x$.*
- (d) *$-(-x) = x$.*

Statement (a) is a cancellation law. Note that (b) asserts the uniqueness of the element whose existence is assumed in (A4), and that (c) does the same for (A5).

Proof. If $x + y = x + z$, the axioms (A) give

$$\begin{aligned} y &= 0 + y = (-x + x) + y = -x + (x + y) \\ &= -x + (x + z) = (-x + x) + z = 0 + z = z. \end{aligned}$$

This proves (a). Take $z = 0$ in (a) to obtain (b). Take $z = -x$ in (a) to obtain (c). Since $-x + x = 0$, (c) (with $-x$ in place of x) gives (d). $\square$

[18] (D) is the only axiom linking the two operations, and every proof below that mixes $+$ and $\cdot$ must route through it. Watch 1.16(a): the fact that $0x = 0$ — which looks like arithmetic so basic it needs no proof — is derivable only because (D) exists.

THE PATTERN OF ALL FOUR PROOFS

Every step in that display is a single named axiom, and it is worth naming them once: $0 + y = y$ is (A4); $0 = -x + x$ is (A5) with (A2); the regrouping is (A3). Nothing else is used — in particular, no multiplication and no order.

The economy of (b), (c), (d) is the point. Once cancellation (a) is available, the rest are not new arguments but *substitutions* into it. Rudin's "take $z = 0$ " and "take $z = -x$ " are complete proofs, and the habit of proving one general statement and specialising it repeatedly is one he will use constantly.

How is the chain for (a) *found*, rather than checked? Work from the goal. You must delete x from $x + y$; the only axiom that annihilates anything is (A5), so $-x$ must enter; the only license to bring it next to x is regrouping, (A3); and once $-x + x$ becomes 0, (A4) removes it. Each step is forced — the axioms offer exactly one tool per obstacle, so the proof writes itself left to right once you know what must disappear. This is pattern 1 of the strategy index: with abstract objects, unwind the definitions, because there is nothing else to use.

1.15 Proposition. *The axioms for multiplication imply the following statements.*

- (a) *If $x \neq 0$ and $xy = xz$ then $y = z$.*
- (b) *If $x \neq 0$ and $xy = x$ then $y = 1$.*
- (c) *If $x \neq 0$ and $xy = 1$ then $y = 1/x$.*
- (d) *If $x \neq 0$ then $1/(1/x) = x$.*

The proof is so similar to that of Proposition 1.14 that we omit it.¹⁹

1.16 Proposition. *The field axioms imply the following statements, for any $x, y, z \in F$.*

- (a) $0x = 0$.
- (b) *If $x \neq 0$ and $y \neq 0$ then $xy \neq 0$.*
- (c) $(-x)y = -(xy) = x(-y)$.
- (d) $(-x)(-y) = xy$.

Proof. $0x + 0x = (0 + 0)x = 0x$. Hence Proposition 1.14(b) implies that $0x = 0$, and (a) holds.²⁰ Next, assume $x \neq 0$, $y \neq 0$, but $xy = 0$. Then

$$1 = \left(\frac{1}{y}\right) \left(\frac{1}{x}\right) xy = \left(\frac{1}{y}\right) \left(\frac{1}{x}\right) 0 = 0,$$

a contradiction. Thus (b) holds.

The first equality in (c) comes from

$$(-x)y + xy = (-x + x)y = 0y = 0,$$

¹⁹ This is Exercise 1.3. Copy the proof of 1.14 with $+$ $\rightarrow \cdot$, $0 \rightarrow 1$, $-x \rightarrow 1/x$, and carry the hypothesis $x \neq 0$ along; the axioms (M1)–(M5) match (A1)–(A5) exactly except for that restriction. The restriction is unavoidable: (M5) grants inverses only to nonzero elements, which is precisely why the multiplicative statements need a hypothesis the additive ones did not.

²⁰ The trick in one line: manufacture an equation of the form $u + u = u$, then quote 1.14(b) to conclude $u = 0$. How is it found? Ask what the axioms know about 0: only (A4), $0 = 0 + 0$. So substitute that and distribute — the sole fact about 0 meets the sole axiom linking $+$ to $\cdot$, and the proof is the only sentence one can write. First place (D) does any work.

combined with Proposition 1.14(c); the other half of (c) is proved in the same way. Finally,

$$(-x)(-y) = -[x(-y)] = -[-(xy)] = xy$$

by (c) and Proposition 1.14(d). $\square$

What (b) actually says

Part (b) is the statement that a field has *no zero divisors*, and unlike the rest of this proposition it is not a formality. There are perfectly good rings where it fails: in the integers mod 6, $2 \cdot 3 = 0$ with neither factor zero. What rules that out here is (M5) — the existence of inverses — and the proof shows exactly how: multiply by $1/x$ and $1/y$ and the supposed $xy = 0$ collapses 1 to 0, contradicting $1 \neq 0$ in (M4).

This is why $\mathbb{Z}/6\mathbb{Z}$ is not a field while $\mathbb{Z}/5\mathbb{Z}$ is, and it is quietly required every time a later argument divides by a nonzero quantity.

1.17 Definition. An *ordered field* is a field F which is also an ordered set, such that

- (i) $x + y < x + z$ if $x, y, z \in F$ and $y < z$,
- (ii) $xy > 0$ if $x \in F$, $y \in F$, $x > 0$, and $y > 0$.

If $x > 0$, we call x *positive*; if $x < 0$, x is *negative*.²¹

For example, $\mathbb{Q}$ is an ordered field.

All the familiar rules for working with inequalities apply in every ordered field: Multiplication by positive [negative] quantities preserves [reverses] inequalities, no square is negative, etc. The following proposition lists some of these.

1.18 Proposition. *The following statements are true in every ordered field.*

- (a) *If $x > 0$ then $-x < 0$, and vice versa.*
- (b) *If $x > 0$ and $y < z$ then $xy < xz$.*
- (c) *If $x < 0$ and $y < z$ then $xy > xz$.*
- (d) *If $x \neq 0$ then $x^2 > 0$. In particular, $1 > 0$.*
- (e) *If $0 < x < y$ then $0 < 1/y < 1/x$.*

Proof.

- (a) If $x > 0$ then $0 = -x + x > -x + 0$, so that $-x < 0$. If $x < 0$ then $0 = -x + x < -x + 0$, so that $-x > 0$. This proves (a).
- (b) Since $z > y$, we have $z - y > y - y = 0$, hence $x(z - y) > 0$, and therefore

$$xz = x(z - y) + xy > 0 + xy = xy.$$

²¹ Only two axioms, and they are the minimum needed to tie the order to the arithmetic: translation preserves order, and positives are closed under multiplication. Everything else you believe about inequalities — 1.18(a)–(e), and the whole of Chapter 1's estimating — is derived from these two. Rudin is not listing convenient facts; he is exhibiting a complete basis.

(c) By (a), (b), and Proposition 1.16(c),

$$-[x(z - y)] = (-x)(z - y) > 0,$$

so that $x(z - y) < 0$, hence $xz < xy$.

(d) If $x > 0$, part (ii) of Definition 1.17 gives $x^2 > 0$. If $x < 0$, then $-x > 0$, hence $(-x)^2 > 0$. But $x^2 = (-x)^2$ by Proposition 1.16(d). Since $1 = 1^2$, $1 > 0$.

(e) If $y > 0$ and $v \leq 0$, then $yv \leq 0$. But $y \cdot (1/y) = 1 > 0$. Hence $1/y > 0$. Likewise, $1/x > 0$. If we multiply both sides of the inequality $x < y$ by the positive quantity $(1/x)(1/y)$, we obtain $1/y < 1/x$.

□

THE ONE MOVE USED IN EVERY PART

Each proof converts an inequality into a statement about something being positive, applies an axiom, and converts back. Concretely: $y < z$ becomes $z - y > 0$ by (i) of 1.17, then (ii) applies, then (i) restores the inequality.

That is the whole technique. It is worth watching in (b), where it is cleanest: $z > y \Rightarrow z - y > 0 \Rightarrow x(z - y) > 0 \Rightarrow xz > xy$. Part (c) is (b) applied to $-x$, and (e) is (b) applied with the positive multiplier $(1/x)(1/y)$. Only (d) does something different, splitting into cases on the sign of x .

PART (D) IS THE SOURCE OF MOST OF CHAPTER 1'S LEVERAGE

$x^2 > 0$ for $x \neq 0$ looks too obvious to be worth stating. It is the reason $\mathbb{C}$ cannot be ordered (Exercise 1.8: $-1 = i^2$ would have to be both positive and negative), the reason $a^2 + b^2 > 0$ in the proof of (M5) for $\mathbb{C}$ at 1.25, and the reason $|z| = (z\bar{z})^{1/2}$ makes sense at 1.32.

The corollary $1 > 0$ deserves separate notice. It is not an axiom, and in a field with no order there is no sense in which 1 exceeds 0. Here it is *derived*, from $1 = 1^2$ — and once you have it, (i) of 1.17 gives $0 < 1 < 2 < 3 < \dots$, so every ordered field contains a copy of $\mathbb{N}$ in the right order. That fact is used without comment in the proof of 1.20.

THE REAL FIELD

THE HINGE OF THE CHAPTER | Everything so far has been preparation, and everything after is consequence. This section contains the one theorem the chapter was written to state.

Watch what Rudin does with it. He states 1.19, declines to prove it, and then immediately proves 1.20 and 1.21 — results he could have extracted from the construction “with very little extra effort,” as he says. He derives them from the least-upper-bound property

instead, on purpose: he wants you to see the property doing work before you see where it came from. Read 1.20 and 1.21 less as facts about $\mathbb{R}$ than as two worked demonstrations of how a supremum is used.

We now state the existence theorem which is the core of this chapter.

1.19 Theorem. *There exists an ordered field $\mathbb{R}$ which has the least-upper-bound property.*

Moreover, $\mathbb{R}$ contains $\mathbb{Q}$ as a subfield.

The second statement means that $\mathbb{Q} \subset \mathbb{R}$ and that the operations of addition and multiplication in $\mathbb{R}$, when applied to members of $\mathbb{Q}$, coincide with the usual operations on rational numbers; also, the positive rational numbers are positive elements of $\mathbb{R}$.²² The members of $\mathbb{R}$ are called *real numbers*.

The proof of Theorem 1.19 is rather long and a bit tedious and is therefore presented in an Appendix to Chap. 1. The proof actually constructs $\mathbb{R}$ from $\mathbb{Q}$.²³

The next theorem could be extracted from this construction with very little extra effort. However, we prefer to derive it from Theorem 1.19 since this provides a good illustration of what one can do with the least-upper-bound property.

1.20 Theorem.

- (a) *If $x \in \mathbb{R}$, $y \in \mathbb{R}$, and $x > 0$, then there is a positive integer n such that*

$$nx > y.$$

- (b) *If $x \in \mathbb{R}$, $y \in \mathbb{R}$, and $x < y$, then there exists a $p \in \mathbb{Q}$ such that $x < p < y$.*

Part (a) is usually referred to as the *archimedean property* of $\mathbb{R}$. Part (b) may be stated by saying that $\mathbb{Q}$ is *dense* in $\mathbb{R}$: Between any two real numbers there is a rational one.²⁴

Proof. (a) Let A be the set of all nx , where n runs through the positive integers. If (a) were false, then y would be an upper bound of A . But then A has a least upper bound in $\mathbb{R}$. Put $\alpha = \sup A$. Since $x > 0$, $\alpha - x < \alpha$, and $\alpha - x$ is not an upper bound of A . Hence $\alpha - x < mx$ for some positive integer m . But then $\alpha < (m+1)x \in A$, which is impossible, since α is an upper bound of A .

(b) Since $x < y$, we have $y - x > 0$, and (a) furnishes a positive integer n such that

$$n(y - x) > 1.$$

Apply (a) again, to obtain positive integers m_1 and m_2 such that $m_1 > nx$, $m_2 > -nx$. Then

$$-m_2 < nx < m_1.$$

22 “Subfield” is a stronger claim than “subset.” It says the copy of $\mathbb{Q}$ sitting inside $\mathbb{R}$ has the arithmetic and the order it had before — so nothing you already know about rationals is invalidated by the enlargement.

23 Worth pausing on. Everything in the next ten chapters is a consequence of this one theorem plus the field and order axioms. If you accept 1.19, you never need the appendix; if you doubt it, the appendix settles it by exhibiting $\mathbb{R}$ as the set of Dedekind cuts of $\mathbb{Q}$. Either way the rest of the book is unaffected, which is why Rudin can safely postpone it.

24 Part (a) is a theorem, not an axiom, and it is not automatic. There are ordered fields in which it fails — fields containing an element larger than every integer. Such fields cannot have the least-upper-bound property, and (a) is precisely the proof of that.

Hence there is an integer m (with $-m_2 \leq m \leq m_1$) such that

$$m - 1 \leq nx < m.$$

If we combine these inequalities, we obtain

$$nx < m \leq 1 + nx < ny.$$

Since $n > 0$, it follows that

$$x < \frac{m}{n} < y.$$

This proves (b), with $p = m/n$. □

PART (A): GRAB THE ONLY HANDLE

How would one find this proof? Negate the conclusion: $nx \leq y$ for every n . That is a *boundedness* statement, and the only machinery 1.19 offers is that bounded sets have suprema — so the proof is forced to form $A = \{nx\}$ and take $\alpha = \sup A$. There is nothing else to try.

The attack on α is then the standard one from 1.8(ii): step back by a positive amount — here x itself, the natural unit — to find some $mx > \alpha - x$; step forward one term to $(m+1)x > \alpha$, and the supremum is beaten by a member of its own set. Keep the pattern: *when the hypothesis is “bounded,” the proof is “take the sup and derive a contradiction one step past it.”* It reappears at 1.21, 2.28, 2.38 and 3.14, and it is pattern 3 of the chapter’s strategy index.

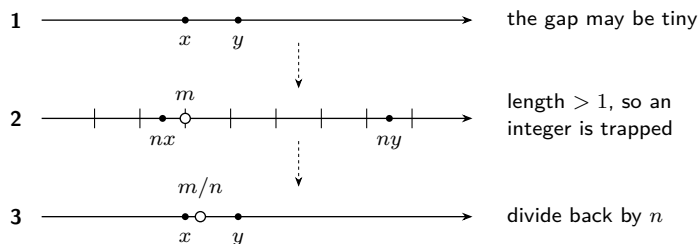
THE THREE MOVES IN PART (B)

Rudin gives the computation and not the plan. The plan is this: to fit a rational strictly between x and y , first *stretch* the gap until it is wider than 1, then *catch* an integer inside the stretched gap, then *shrink* back.

Stretch. Part (a) with the roles $x \rightsquigarrow y - x$ and $y \rightsquigarrow 1$ gives an n with $n(y - x) > 1$. The interval (nx, ny) now has length greater than 1.

Catch. An interval of length greater than 1 must contain an integer. Rudin proves this rather than asserting it, and that is what m_1, m_2 are for: they trap nx between two integers, so the set of integers exceeding nx is nonempty and bounded below, and therefore has a least element m . That gives $m - 1 \leq nx < m$. The existence of a *least* such integer is well-ordering again — the same principle used silently in 1.1. Rudin never states it as an axiom, but this step and the archimedean property both consume it.

Shrink. From $m \leq 1 + nx$ and $1 < n(y - x)$ we get $m < nx + n(y - x) = ny$, and dividing by $n > 0$ puts m/n strictly between x and y .



Stretch until an integer must fall inside, catch it, then shrink back. The integer m becomes the rational m/n between x and y .

We shall now prove the existence of n th roots of positive reals. This proof will show how the difficulty pointed out in the Introduction (irrationality of $\sqrt{2}$) can be handled in $\mathbb{R}$.

1.21 Theorem. For every real $x > 0$ and every integer $n > 0$ there is one and only one positive real y such that $y^n = x$.

This number y is written $\sqrt[n]{x}$ or $x^{1/n}$.

Proof. That there is at most one such y is clear, since $0 < y_1 < y_2$ implies $y_1^n < y_2^n$.

Let E be the set consisting of all positive real numbers t such that $t^n < x$. If $t = x/(1+x)$ then $0 < t < 1$. Hence $t^n \leq t < x$. Thus $t \in E$, and E is not empty.²⁵

If $t > 1 + x$ then $t^n \geq t > x$, so that $t \notin E$. Thus $1 + x$ is an upper bound of E .

Hence Theorem 1.19 implies the existence of

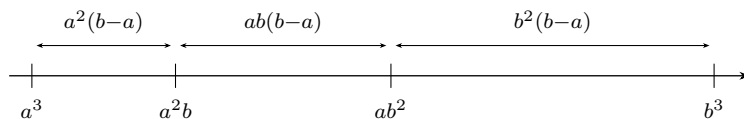
$$y = \sup E.$$

To prove that $y^n = x$ we will show that each of the inequalities $y^n < x$ and $y^n > x$ leads to a contradiction.²⁶

The identity $b^n - a^n = (b-a)(b^{n-1} + b^{n-2}a + \cdots + a^{n-1})$ yields the inequality

$$b^n - a^n < (b-a)nb^{n-1}$$

when $0 < a < b$.



The identity, drawn for $n = 3$: the walk from $a \cdot a \cdot a$ to $b \cdot b \cdot b$ replaces one factor at a time, cutting $[a^3, b^3]$ into three pieces whose widths are the three terms of the identity. Since $a < b$, every piece is shorter than the last one, whence $b^n - a^n < (b-a)nb^{n-1}$. (Drawn for $a = 1$, $b = 1.5$. For $n = 2$ this is the difference of squares; at 3.26 the same identity, with $b = 1$, $a = x$, sums the geometric series.)

Assume $y^n < x$. Choose h so that $0 < h < 1$ and

$$h < \frac{x - y^n}{n(y+1)^{n-1}}.$$

25 Two hypotheses of the LUB property are being discharged, and Rudin spends a sentence on each. This paragraph is nonemptiness; the next is boundedness. Both are required before $\sup E$ may be written down — compare the warning at 1.10.

[26] Note which half of Definition 1.8 each case attacks. Assuming $y^n < x$ will contradict “ y is an upper bound”; assuming $y^n > x$ will contradict “ y is the least upper bound.” Both clauses get used exactly once, which is a reliable sign that E was chosen well.

Put $a = y$, $b = y + h$. Then

$$(y + h)^n - y^n < hn(y + h)^{n-1} < hn(y + 1)^{n-1} < x - y^n.$$

Thus $(y + h)^n < x$ and $y + h \in E$. Since $y + h > y$, this contradicts the fact that y is an upper bound of E .²⁷

Assume $y^n > x$. Put

$$k = \frac{y^n - x}{ny^{n-1}}.$$

Then $0 < k < y$. If $t \geq y - k$, we conclude that

$$y^n - t^n \leq y^n - (y - k)^n < kny^{n-1} = y^n - x.$$

Thus $t^n > x$ and $t \notin E$. It follows that $y - k$ is an upper bound of E . But $y - k < y$, which contradicts the fact that y is the least upper bound of E .

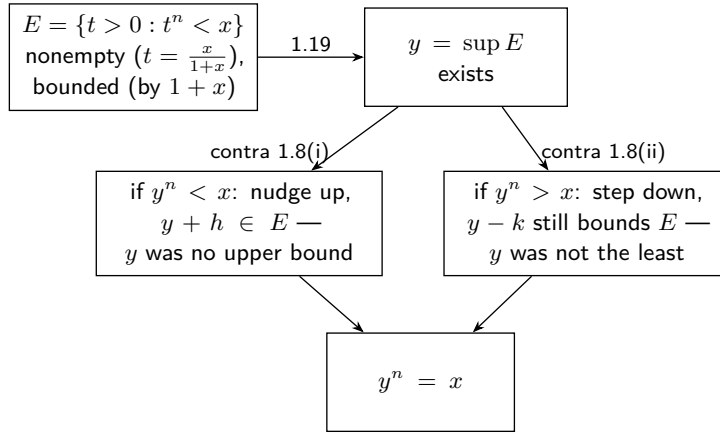
Hence $y^n = x$, and the proof is complete. $\square$

HOW THE PROOF FLOWS

1. **Choose the method.** A number with $y^n = x$ must be *produced*, and the only production tool is the least-upper-bound property — pattern 3 again. So the proof must design a set whose supremum is the root. The natural candidate: everything that undershoots, $E = \{t > 0 : t^n < x\}$.
2. **Buy the supremum.** Both tickets, each bought with one sentence: $x/(1+x)$ lies in E (nonempty), $1+x$ bounds it. Rudin's care here is the warning of 1.10 being heeded, and $y = \sup E$ exists.
3. **Set up the pincer.** By trichotomy (1.5), it suffices to kill $y^n < x$ and $y^n > x$. There is no direct computation of y^n available — y is only known as a supremum — so each case must be turned into a statement *about bounds*, the only language y speaks.
4. **Kill $y^n < x$.** If y undershoots, continuity-style estimates (the binomial identity) produce a slightly larger $y + h$ still in E — contradicting clause (i) of 1.8: y was supposed to be an upper bound. The choice of h is reverse-engineered in the next box.
5. **Kill $y^n > x$.** If y overshoots, the same estimate run downward shows $y - k$ already bounds E — contradicting clause (ii): y was supposed to be the *least* upper bound.
6. **Audit.** Clause (i) died in step 4, clause (ii) in step 5 — each spent exactly once, as in 1.11. When both halves of the supremum's definition are consumed, the set E was the right one; when a proof of this shape never uses one clause, suspect the

²⁷ Do not try to see the choice of h at a glance — nobody does. It was found backwards: write the goal $(y + h)^n < x$, bound the left side by the displayed inequality, and solve for h ; the strange side condition $h < 1$ exists only to get h out of the exponent first. The box after the proof reconstructs the whole derivation, and the k of the next paragraph is found the same way.

set.



The proof as a flow diagram: qualify E , mint $y = \sup E$, then kill both strict inequalities — each contradiction consuming a different clause of Definition 1.8. Trichotomy (1.5) does the rest.

REVERSE-ENGINEERING THE CHOICE OF H

Nobody guesses that h . It is found by writing the goal down and solving backwards, and Rudin has simply deleted the derivation.

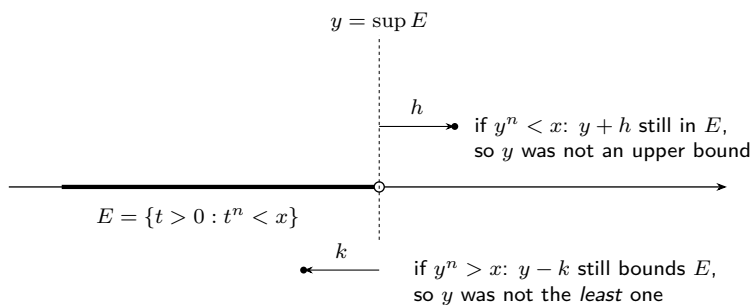
Goal. Find $h > 0$ with $(y + h)^n < x$; equivalently $(y + h)^n - y^n < x - y^n$, whose right-hand side is a fixed positive number because we assumed $y^n < x$.

Bound the left side. We cannot compute $(y + h)^n - y^n$, but the displayed inequality with $a = y$, $b = y + h$ bounds it by $hn(y + h)^{n-1}$. So it suffices to make $hn(y + h)^{n-1} \leq x - y^n$.

Get h out of the exponent. That inequality still has h buried inside $(y + h)^{n-1}$ and cannot be solved as it stands. Kill it by agreeing in advance that $h < 1$; then $(y + h)^{n-1} < (y + 1)^{n-1}$, a constant. This is the *only* reason the condition $0 < h < 1$ appears — it is bookkeeping, not mathematics.

Solve. It now suffices that $hn(y + 1)^{n-1} < x - y^n$, that is, $h < (x - y^n)/(n(y + 1)^{n-1})$, which is Rudin's line. Both conditions on h are upper bounds, so any h below the smaller of the two works, and such an h exists.

The second case is the same argument run downward, and k is found the same way; there the constraint $k < y$ plays the role that $h < 1$ played here.



Both halves of Definition 1.8 get used exactly once — a reliable sign that E was the right set to take a supremum of.

What the proof is really doing

Strip the algebra and the argument reads: $t \mapsto t^n$ is continuous, so if y^n were strictly below x we could nudge y upward and stay below x . The choice of h is a hand-rolled continuity estimate, written out because continuity is not available until 4.4 — and the machinery of Chapter 4 is not available until $\mathbb{R}$ exists, which is what we are in the middle of establishing.

Once Chapter 4 is in place the same theorem falls out of the intermediate value theorem (4.23) in three lines. Rudin proves it here anyway because he needs n th roots long before then, and because it is the first genuine demonstration that the LUB property produces numbers $\mathbb{Q}$ could not.

WHY E IS RESTRICTED TO POSITIVE T

Without positivity the uniqueness claim is simply false: for $n = 2$, $x = 4$, both 2 and -2 are roots. Rudin's "one and only one positive real" is doing real work, and the restriction on E is what licenses the opening line of the proof, $0 < y_1 < y_2 \Rightarrow y_1^n < y_2^n$, which requires both numbers to be positive.

Corollary. *If a and b are positive real numbers and n is a positive integer, then*

$$(ab)^{1/n} = a^{1/n}b^{1/n}.$$

Proof. Put $\alpha = a^{1/n}$, $\beta = b^{1/n}$. Then

$$ab = \alpha^n \beta^n = (\alpha\beta)^n$$

since multiplication is commutative. [Axiom (M2) in Definition 1.12.] The uniqueness assertion of Theorem 1.21 shows therefore that

$$(ab)^{1/n} = \alpha\beta = a^{1/n}b^{1/n}.$$

□

UNIQUENESS DOING THE WORK

Note what carries the argument: not a computation with roots, but the *uniqueness* half of 1.21. We exhibit one positive number, namely $\alpha\beta$, whose n th power is ab ; since 1.21 says there is only one such number, that number must be $(ab)^{1/n}$.

This is the standard way to prove identities about n th roots, and it is worth internalising because the direct approach is hopeless — there is no formula for $a^{1/n}$ to manipulate. Rudin uses the same move at 1.33(c) to get $|zw| = |z||w|$.

1.22 Decimals. We conclude this section by pointing out the relation between real numbers and decimals.

Let $x > 0$ be real. Let n_0 be the largest integer such that $n_0 \leq x$. (Note that the existence of n_0 depends on the archimedean property of $\mathbb{R}$.)²⁸ Having chosen $n_0, n_1, \dots, n_{k-1}$, let n_k denote the largest integer such that

$$n_0 + \frac{n_1}{10} + \dots + \frac{n_k}{10^k} \leq x.$$

Let E be the set of these numbers

$$n_0 + \frac{n_1}{10} + \dots + \frac{n_k}{10^k} \quad (k = 0, 1, 2, \dots). \quad (1.5)$$

Then $x = \sup E$. The decimal expansion of x is

$$n_0.n_1n_2n_3 \dots \quad (1.6)$$

Conversely, for any infinite decimal expansion (1.6) the set E of numbers (1.5) is bounded above and (1.6) is the decimal expansion of $\sup E$.

Since we shall never use decimals, we do not enter into a detailed discussion.²⁹

Why this passage is here at all

It answers the question the Introduction raised and then dropped: the sequence $1, 1.4, 1.41, \dots$ “tends to $\sqrt{2}$ ” — what does it tend to? The answer is now available and is stated in one word: $\sup$. The truncations form a set E , that set is bounded above, and 1.19 hands you its supremum. The number the decimals were approximating is that supremum, and nothing more mysterious.

Note the two directions Rudin states. Every real has a decimal expansion, and every decimal expansion names a real. Together they say the informal picture of $\mathbb{R}$ you arrived with is not wrong — it just could not be a foundation, since E having a supremum is exactly what $\mathbb{Q}$ failed to provide.

28 Rudin flags the dependence himself, which is unusual for him and a signal that it matters. Without 1.20(a) there might be no integer at all below — or above — a given x , and the decimal expansion could not even start.

29 Take that last sentence seriously — it is a warning, not modesty. Decimals are a representation of real numbers, not a definition of them, and reasoning about $\mathbb{R}$ through decimal expansions is a reliable way to go wrong (the two expansions of $1 = 0.999\dots$ are the familiar symptom). Everything in this book is proved from 1.19 and never from digits.

THE EXTENDED REAL NUMBER SYSTEM

A CONVENIENCE, NOT A NUMBER SYSTEM | One definition, no theorems. Rudin is buying a notational convenience: he wants to write

$\sup E = +\infty$ for unbounded E , and $\lim s_n = +\infty$ in Chapter 3, without special-casing every statement. Read it as an agreement about symbols.

The thing to carry away is what he pointedly does *not* do. The extended reals are given an order but not an arithmetic — “does not form a field” is stated flatly — and the conventions in (a)–(c) leave $\infty - \infty$, $0 \cdot \infty$, and ∞/∞ undefined. Those gaps are deliberate and permanent; they are why $\liminf$ and $\limsup$ in 3.16 need care, and why Theorem 3.42 has the hypotheses it does.

1.23 Definition. The *extended real number system* consists of the real field $\mathbb{R}$ and two symbols, $+\infty$ and $-\infty$. We preserve the original order in $\mathbb{R}$ and define

$$-\infty < x < +\infty$$

for every $x \in \mathbb{R}$.

It is then clear that $+\infty$ is an upper bound of every subset of the extended real number system, and that every nonempty subset has a least upper bound.³⁰ If, for example, E is a nonempty subset of real numbers which is not bounded above in $\mathbb{R}$, then $\sup E = +\infty$ in the extended real number system.

Exactly the same remarks apply to lower bounds.

The extended real number system does not form a field, but it is customary to make the following conventions:

(a) If x is real then

$$x + \infty = +\infty, \quad x - \infty = -\infty, \quad \frac{x}{+\infty} = \frac{x}{-\infty} = 0.$$

(b) If $x > 0$ then $x \cdot (+\infty) = +\infty$, $x \cdot (-\infty) = -\infty$.

(c) If $x < 0$ then $x \cdot (+\infty) = -\infty$, $x \cdot (-\infty) = +\infty$.

When it is desired to make the distinction between real numbers on the one hand and the symbols $+\infty$ and $-\infty$ on the other quite explicit, the former are called *finite*.

READ THE CONVENTIONS AS A CLOSED LIST

Every case Rudin lists involves one infinity and one *finite* x . Anything outside that pattern is simply undefined — among them $(+\infty) + (-\infty)$, $0 \cdot (\pm\infty)$, every quotient of two infinities, and every sum or product of two infinite symbols not covered above.

The omissions are deliberate, not oversights, because no assignment preserves the ordinary rules: if $\infty - \infty$ were given a value c , then adding 1 to the first term would force $c = c + 1$. (Division by zero is a separate matter — $x/0$ is undefined in any field, by (M5), and has nothing to do with the extended system.)

The practical consequence: an expression like $\sup E + \sup F$ is not auto-

³⁰ This is the payoff. Of the two hypotheses in Definition 1.10, exactly one becomes vacuous: “bounded above” is now automatic, since $+\infty$ bounds every subset. “Nonempty” is still required, and Rudin keeps it in the sentence alongside. What has been bought, at the cost of field structure, is a sup that exists for every nonempty set whether or not it is bounded.

mathematically meaningful once sup may be infinite, and theorems in Chapter 3 that add limits carry hypotheses excluding these cases. When a later proof says “provided the right-hand side is not of the form $\infty - \infty$,” this is what it is protecting against.

Why not just add ∞ and keep the field axioms?

Because it is impossible, and 1.16(b) shows why in two lines. Suppose $+\infty$ had a multiplicative inverse; convention (a) already forces $x/(+\infty) = 0$, so $1/(+\infty) = 0$, and then $1 = (+\infty) \cdot 0 = 0$ by 1.16(a) — contradicting $1 \neq 0$ in (M4).

So the extended reals are an ordered set with a partial arithmetic, and that is all they can be. Rudin’s phrasing — “it is customary to make the following conventions” — is exact: these are conventions, not theorems, and they are adopted because they make limit statements uniform, not because they are forced.

THE COMPLEX FIELD

EQUIPMENT, WITH ONE EXCEPTION | Nothing in this section is about $\mathbb{R}$, and nothing in it is hard. Rudin is stocking the shelves: Chapters 3, 7, 8 and 11 all work with complex sequences, complex series and complex-valued functions, and he wants $\mathbb{C}$ available with its arithmetic and its absolute value already proved.

The section has a definite shape. Definitions 1.24–1.29 build $\mathbb{C}$ and then recover the familiar notation $a + bi$; 1.30–1.33 introduce the conjugate and the absolute value and establish the triangle inequality; 1.34–1.35 prove the Schwarz inequality. Only two things here have downstream weight: **1.33(e)**, the triangle inequality, which is used on almost every page of Chapters 3–7, and **1.35**, the Schwarz inequality, which is the engine behind the metric on $\mathbb{R}^k$ in the next section and reappears for integrals in Exercise 6.10 and for $\mathcal{L}^2$ functions in 11.35.

Watch one point of method. Rudin defines a complex number as an *ordered pair*, verifies the field axioms by direct computation, and only then — at 1.27 — introduces i . The mysterious square root of -1 is never assumed; it is produced. That ordering is the whole reason the construction is honest.

1.24 Definition. A *complex number* is an ordered pair (a, b) of real numbers. “Ordered” means that (a, b) and (b, a) are regarded as distinct if $a \neq b$.

Let $x = (a, b)$, $y = (c, d)$ be two complex numbers. We write $x = y$ if and only if $a = c$ and $b = d$. (Note that this definition is not entirely superfluous; think of equality of rational numbers, represented

as quotients of integers.) We define

$$\begin{aligned}x + y &= (a + c, b + d) \\ xy &= (ac - bd, ad + bc).\end{aligned}$$

WHERE THE MULTIPLICATION RULE COMES FROM

The addition rule is the obvious one. The multiplication rule is not, and Rudin gives no motivation — but it is forced, not chosen. Suppose you want (a, b) to behave like $a + bi$ with $i^2 = -1$, while keeping the field axioms. Then

$$(a + bi)(c + di) = ac + adi + bci + bd i^2 = (ac - bd) + (ad + bc)i,$$

which is exactly the displayed formula. So the definition is the multiplication $\mathbb{C}$ *must* have if i is to exist at all.

Rudin declines to say this here, because saying it would presuppose i — the very thing he refuses to assume until 1.27. The formula is therefore written down bare, checked to satisfy the axioms in 1.25, and only afterwards revealed to mean what you expected. Read 1.24–1.29 as a single argument with the motivation deliberately withheld until the end.

1.25 Theorem. *These definitions of addition and multiplication turn the set of all complex numbers into a field, with $(0, 0)$ and $(1, 0)$ in the role of 0 and 1.*

Proof. We simply verify the field axioms, as listed in Definition 1.12. (Of course, we use the field structure of $\mathbb{R}$.) Let $x = (a, b)$, $y = (c, d)$, $z = (e, f)$.

(A1) is clear.

(A2) $x + y = (a + c, b + d) = (c + a, d + b) = y + x$.

(A3)

$$\begin{aligned}(x + y) + z &= (a + c, b + d) + (e, f) \\ &= (a + c + e, b + d + f) \\ &= (a, b) + (c + e, d + f) = x + (y + z).\end{aligned}$$

(A4) $x + 0 = (a, b) + (0, 0) = (a, b) = x$.

(A5) Put $-x = (-a, -b)$. Then $x + (-x) = (0, 0) = 0$.

(M1) is clear.

(M2) $xy = (ac - bd, ad + bc) = (ca - db, da + cb) = yx$.

(M3)

$$\begin{aligned}(xy)z &= (ac - bd, ad + bc)(e, f) \\ &= (ace - bde - adf - bcf, acf - bdf + ade + bce) \\ &= (a, b)(ce - df, cf + de) = x(yz).\end{aligned}$$

(M4) $1x = (1, 0)(a, b) = (a, b) = x$.

(M5) If $x \neq 0$ then $(a, b) \neq (0, 0)$, which means that at least one of the real numbers a, b is different from 0. Hence $a^2 + b^2 > 0$, by Proposition 1.18(d), and we can define³¹

$$\frac{1}{x} = \left(\frac{a}{a^2 + b^2}, \frac{-b}{a^2 + b^2} \right).$$

Then

$$x \cdot \frac{1}{x} = (a, b) \left(\frac{a}{a^2 + b^2}, \frac{-b}{a^2 + b^2} \right) = (1, 0) = 1.$$

(D)

$$\begin{aligned} x(y + z) &= (a, b)(c + e, d + f) \\ &= (ac + ae - bd - bf, ad + af + bc + be) \\ &= (ac - bd, ad + bc) + (ae - bf, af + be) \\ &= xy + xz. \end{aligned}$$

□

Nothing here is deep, and that is the point

Every line is arithmetic in $\mathbb{R}$. That is precisely what makes the theorem worth stating: it shows the existence of $\mathbb{C}$ requires no new assumptions whatever — given $\mathbb{R}$, the complex field comes free, as a set of pairs.

Compare the situation for $\mathbb{R}$ itself, where Theorem 1.19 needed a page of Dedekind cuts. The difference is that $\mathbb{C}$ adds no *completeness*; it only adds algebra. That algebra turns out to be enormously powerful — 8.8 will show $\mathbb{C}$ is algebraically closed — but it costs nothing to obtain.

1.26 Theorem. *For any real numbers a and b we have*

$$(a, 0) + (b, 0) = (a + b, 0), \quad (a, 0)(b, 0) = (ab, 0).$$

The proof is trivial.

Theorem 1.26 shows that the complex numbers of the form $(a, 0)$ have the same arithmetic properties as the corresponding real numbers a . We can therefore identify $(a, 0)$ with a . This identification gives us the real field as a subfield of the complex field.³²

The reader may have noticed that we defined the complex numbers without any reference to the mysterious square root of -1 . We now show that the notation (a, b) is equivalent to the more customary $a + bi$.

1.27 Definition. $i = (0, 1)$.

1.28 Theorem. $i^2 = -1$.

Proof. $i^2 = (0, 1)(0, 1) = (-1, 0) = -1$. □

1.29 Theorem. *If a and b are real, then $(a, b) = a + bi$.*

31 Exactly the leverage promised at 1.18(d). Squares are nonnegative and not both a, b vanish, so $a^2 + b^2 > 0$ — and therefore the divisions below are legal. The construction of $\mathbb{C}$ depends on an order property of $\mathbb{R}$, which is worth noticing given that $\mathbb{C}$ itself cannot be ordered (Exercise 1.8).

32 The same manoeuvre as Step 9 of the appendix, where $\mathbb{Q}$ is identified with the rational cuts $\mathbb{Q}^*$. In both cases a copy of the smaller field sits inside the larger one, arithmetic and order preserved, and we agree to stop distinguishing the copy from the original. Rudin explicitly links the two identifications in Step 9; when you read the appendix, come back to this paragraph.

Proof.

$$\begin{aligned} a + bi &= (a, 0) + (b, 0)(0, 1) \\ &= (a, 0) + (0, b) = (a, b). \end{aligned}$$

□

WHAT 1.27–1.29 ACTUALLY ACCOMPLISH

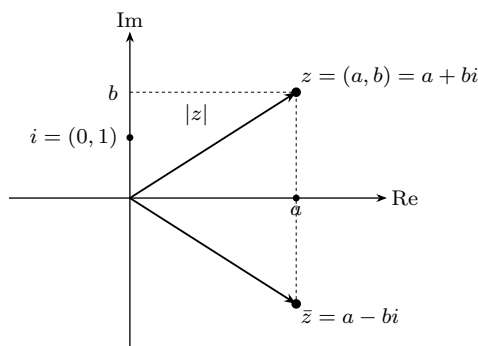
Three lines of computation, and they retire a piece of folklore. The usual story is that i is postulated — an imaginary object adjoined by fiat because we wish $x^2 + 1$ had a root. Rudin's construction assumes no such thing. $\mathbb{C}$ is built from ordered pairs of reals, its arithmetic is defined outright, and then one particular pair, $(0, 1)$, is observed to square to -1 .

So i is not an assumption; it is a *discovery* about a field that already existed. And 1.29 shows the pair notation and the $a + bi$ notation describe the same object, so you may use whichever is convenient — pairs when checking axioms, $a + bi$ when computing.

1.30 Definition. If a, b are real and $z = a + bi$, then the complex number $\bar{z} = a - bi$ is called the *conjugate* of z . The numbers a and b are the *real part* and the *imaginary part* of z , respectively.

We shall occasionally write

$$a = \operatorname{Re}(z), \quad b = \operatorname{Im}(z).$$



Conjugation is reflection in the real axis, and $|z| = \sqrt{a^2 + b^2}$ is distance from the origin. Neither is a new assumption — both are read off Definition 1.24.

1.31 Theorem. If z and w are complex, then

- (a) $\overline{z + w} = \bar{z} + \bar{w}$,
- (b) $\overline{zw} = \bar{z} \cdot \bar{w}$,
- (c) $z + \bar{z} = 2 \operatorname{Re}(z)$, $z - \bar{z} = 2i \operatorname{Im}(z)$,
- (d) $z\bar{z}$ is real and positive (except when $z = 0$).

Proof. (a), (b), and (c) are quite trivial. To prove (d), write $z = a + bi$, and note that $z\bar{z} = a^2 + b^2$. □

Conjugation is a field automorphism

Parts (a) and (b) say that $z \mapsto \bar{z}$ preserves both operations, and since $\bar{\bar{z}} = z$ it is a bijection. In the language of algebra, conjugation is an automorphism of $\mathbb{C}$ fixing $\mathbb{R}$ pointwise — and it is the only nontrivial one.

The practical use is a symmetry principle you will meet repeatedly: any polynomial identity with real coefficients that holds for z also holds for $\bar{z}$. That is why non-real roots of real polynomials come in conjugate pairs, and why 8.8 (the algebraic completeness of $\mathbb{C}$) has the consequences it does for real polynomials.

Part (d) is the one with immediate work to do: it is what makes Definition 1.32 possible, since $z\bar{z} \geq 0$ is exactly the condition needed before taking a square root by 1.21.

1.32 Definition. If z is a complex number, its *absolute value* $|z|$ is the non-negative square root of $z\bar{z}$; that is, $|z| = (z\bar{z})^{1/2}$.

The existence (and uniqueness) of $|z|$ follows from Theorem 1.21 and part (d) of Theorem 1.31.³³

Note that when x is real, then $\bar{x} = x$, hence $|x| = \sqrt{x^2}$. Thus $|x| = x$ if $x \geq 0$, $|x| = -x$ if $x < 0$.

1.33 Theorem. Let z and w be complex numbers. Then

- (a) $|z| > 0$ unless $z = 0$, $|0| = 0$,
- (b) $|\bar{z}| = |z|$,
- (c) $|zw| = |z||w|$,
- (d) $|\operatorname{Re}(z)| \leq |z|$,
- (e) $|z + w| \leq |z| + |w|$.

Proof. (a) and (b) are trivial. Put $z = a + bi$, $w = c + di$, with a, b, c, d real. Then

$$|zw|^2 = (ac - bd)^2 + (ad + bc)^2 = (a^2 + b^2)(c^2 + d^2) = |z|^2|w|^2$$

or $|zw|^2 = (|z||w|)^2$. Now (c) follows from the uniqueness assertion of Theorem 1.21.

To prove (d), note that $a^2 \leq a^2 + b^2$, hence

$$|a| = \sqrt{a^2} \leq \sqrt{a^2 + b^2}.$$

To prove (e), note that $\bar{z}w$ is the conjugate of $z\bar{w}$, so that $z\bar{w} + \bar{z}w = 2\operatorname{Re}(z\bar{w})$. Hence

$$\begin{aligned} |z + w|^2 &= (z + w)(\overline{z + w}) = z\bar{z} + z\bar{w} + \bar{z}w + w\bar{w} \\ &= |z|^2 + 2\operatorname{Re}(z\bar{w}) + |w|^2 \\ &\leq |z|^2 + 2|z\bar{w}| + |w|^2 \\ &= |z|^2 + 2|z||w| + |w|^2 = (|z| + |w|)^2. \end{aligned}$$

Now (e) follows by taking square roots. $\square$

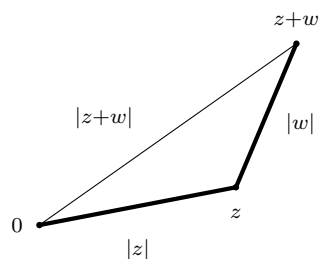
33 Both citations are load-bearing, and note the gap they leave. For $z \neq 0$, 1.31(d) makes $z\bar{z}$ a positive real and 1.21 — which requires a positive input — returns its unique positive square root. For $z = 0$ we have $z\bar{z} = 0$, which 1.21 does not cover; that case is settled separately by the convention $|0| = 0$, recorded in 1.33(a). This is the first place the work of §1.21 gets spent.

THE STANDARD PROOF OF A TRIANGLE INEQUALITY

The shape of this argument recurs throughout analysis, and it is worth extracting because you will need to reproduce it: to bound $|z + w|$, do not work with $|z + w|$ — work with $|z + w|^2$, expand it as $(z + w)(\overline{z + w})$, bound the cross-term, and take square roots at the end.

The cross-term is the only real step, and it goes in two moves: 1.31(c) turns $z\bar{w} + \bar{z}w$ into $2\operatorname{Re}(z\bar{w})$, then (d) bounds Re by modulus and (c) factors it as $|z||w|$.

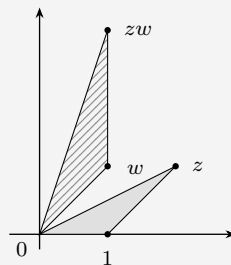
The identical strategy proves 1.37(e) for $\mathbb{R}^k$ in the next section, with the Schwarz inequality playing the part of (d). Recognising the pattern once saves reading the second proof.



Part (e) is about an actual triangle: the straight route from 0 to $z + w$ never beats the detour through z . Equality is the degenerate triangle — z and w pointing the same way, which is exactly $\operatorname{Re}(z\bar{w}) = |z\bar{w}|$, the one inequality in the proof.

Multiplication rotates and scales — (c) drawn instead of computed

Part (c) also has a picture. Multiplying every point of the plane by w rotates it by the angle of w and stretches it by the factor $|w|$: the triangle with vertices $0, w, zw$ is the triangle $0, 1, z$ so rotated and stretched.



The shaded triangle $0, 1, z$ and the hatched triangle $0, w, zw$ are similar, with ratio $|w|$. (Drawn for $z = 2 + i$, $w = 1 + i$; the similarity is general.)

Part (c) is the stretch factor read off the picture: the side $0 \rightarrow zw$ is the side $0 \rightarrow z$ magnified by $|w|$, so $|zw| = |z||w|$. Rudin proves it by the $z\bar{z}$ computation instead, because the rotation–scaling description presupposes angles — sines and cosines — which do not officially exist until Chapter 8. The picture is why the computation was never in doubt; the computation is what makes the picture legal. Taking $w = \bar{z}$ folds the rotation flat onto the real axis: $z\bar{z} = |z|^2$ is

||| this figure with the hatched triangle lying on its back.

1.34 Notation. If $x_1, \dots, x_n$ are complex numbers, we write

$$x_1 + x_2 + \cdots + x_n = \sum_{j=1}^n x_j.$$

We conclude this section with an important inequality, usually known as the *Schwarz inequality*.

1.35 Theorem. If $a_1, \dots, a_n$ and $b_1, \dots, b_n$ are complex numbers, then

$$\left| \sum_{j=1}^n a_j \bar{b}_j \right|^2 \leq \sum_{j=1}^n |a_j|^2 \sum_{j=1}^n |b_j|^2.$$

Proof. Put $A = \sum |a_j|^2$, $B = \sum |b_j|^2$, $C = \sum a_j \bar{b}_j$ (in all sums in this proof, j runs over the values $1, \dots, n$). If $B = 0$, then $b_1 = \cdots = b_n = 0$, and the conclusion is trivial. Assume therefore that $B > 0$. By Theorem 1.31 we have

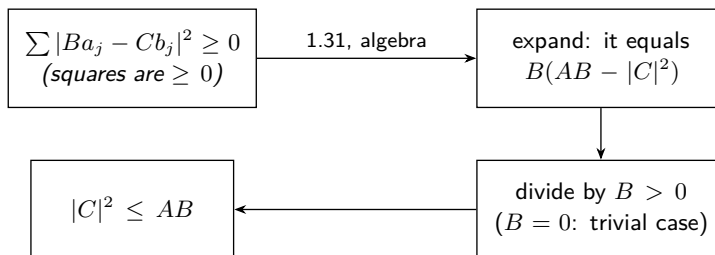
$$\begin{aligned} \sum |Ba_j - Cb_j|^2 &= \sum (Ba_j - Cb_j)(\overline{Ba_j - Cb_j}) \\ &= B^2 \sum |a_j|^2 - B\bar{C} \sum a_j \bar{b}_j \\ &\quad - BC \sum \bar{a}_j b_j + |C|^2 \sum |b_j|^2 \\ &= B^2 A - B|C|^2 \\ &= B(AB - |C|^2). \end{aligned}$$

Since each term in the first sum is nonnegative, we see that

$$B(AB - |C|^2) \geq 0.$$

Since $B > 0$, it follows that $AB - |C|^2 \geq 0$. This is the desired inequality.

□



The proof as a flow diagram: one nonnegative square, expanded and collapsed. All the invention is in choosing which square — the box below recovers that choice.

WHERE $BA_J - CB_J$ COMES FROM

This is the second rabbit of the chapter, after the h of 1.21, and like that one it is reverse-engineered rather than guessed.

The only fact available is that a sum of squared moduli is nonnegative. So the proof must begin with $\sum |\cdot|^2 \geq 0$ for some cleverly chosen expression, and end with the desired inequality. Work backwards: we want $AB - |C|^2 \geq 0$, so we want the sum to evaluate to a positive multiple of $AB - |C|^2$.

Try $\sum |\lambda a_j - \mu b_j|^2$ with constants to be chosen. Expanding gives

$$|\lambda|^2 A - \lambda \bar{\mu} \bar{C} - \bar{\lambda} \mu C + |\mu|^2 B.$$

Choosing $\lambda = B$ (real) and $\mu = C$ makes the middle terms $-B\bar{C}C - BCC\bar{C} = -2B|C|^2$ and the last term $|C|^2 B$, so the whole collapses to $B^2 A - B|C|^2 = B(AB - |C|^2)$. The factor B is then stripped off using $B > 0$.

So the choice is not inspired — it is what you get by writing down the general quadratic and solving for the constants that make it collapse. The hypothesis $B > 0$ and the separate treatment of $B = 0$ exist purely because the division at the last step needs it.

Equality, and what the inequality means

Equality holds precisely when (a_j) and (b_j) are *linearly dependent*. That is Exercise 1.15, and the proof above shows why — with a caveat about how it is usually stated. When $B > 0$, equality forces $\sum |Ba_j - Cb_j|^2 = 0$, and a sum of nonnegative terms vanishes only if every term does, giving $Ba_j = Cb_j$ for all j : the two vectors are proportional. When $B = 0$ every b_j vanishes, both sides of the inequality are 0, and equality holds trivially — but “proportional” now says nothing, since $Ba_j = Cb_j$ reduces to $0 = 0$. Linear dependence is the formulation that covers both cases.

Geometrically, once 1.36 defines the inner product on $\mathbb{R}^k$, this says $|\mathbf{x} \cdot \mathbf{y}| \leq |\mathbf{x}||\mathbf{y}|$, which is the statement that $\cos \theta$ lies between -1 and 1 . Rudin proves the inequality first and lets the geometry follow, rather than the reverse; the inequality is the substance and the picture is the illustration.

Its reach is long. It gives the triangle inequality in $\mathbb{R}^k$ at 1.37(e), hence the metric that all of Chapter 2 is built on; it reappears for integrals in Exercise 6.10 — Hölder’s inequality, of which Rudin there remarks “Theorem 1.35 is a very special case” — and for $\mathcal{L}^2$ functions at 11.35, where it is the same statement about a different inner product.

EUCLIDEAN SPACES

THE LAST PIECE OF EQUIPMENT | Three items, and the section exists for one sentence: Remark 1.38, which observes that $\mathbb{R}^k$ is a metric space. Everything in Chapter 2 — open sets, compactness, connectedness — is developed for metric spaces, and $\mathbb{R}^k$ is the example that motivates all of it. This section supplies the distance function and verifies the three properties Chapter 2 will demand.

Note what is *not* claimed. Rudin gives $\mathbb{R}^k$ a vector space structure and an inner product, but no multiplication of vectors by vectors — $\mathbb{R}^k$ is not a field for $k > 1$. The only exception is $k = 2$, which coincides with $\mathbb{C}$; 1.38 remarks on the coincidence, and it is worth keeping straight that $\mathbb{R}^2$ and $\mathbb{C}$ are the same metric space but only $\mathbb{C}$ has a product.

1.36 Definitions. For each positive integer k , let $\mathbb{R}^k$ be the set of all ordered k -tuples

$$\mathbf{x} = (x_1, x_2, \dots, x_k),$$

where $x_1, \dots, x_k$ are real numbers, called the *coordinates* of $\mathbf{x}$. The elements of $\mathbb{R}^k$ are called *points*, or *vectors*, especially when $k > 1$. We shall denote vectors by boldfaced letters. If $\mathbf{y} = (y_1, \dots, y_k)$ and if α is a real number, put

$$\begin{aligned}\mathbf{x} + \mathbf{y} &= (x_1 + y_1, \dots, x_k + y_k) \\ \alpha \mathbf{x} &= (\alpha x_1, \dots, \alpha x_k)\end{aligned}$$

so that $\mathbf{x} + \mathbf{y} \in \mathbb{R}^k$ and $\alpha \mathbf{x} \in \mathbb{R}^k$. This defines addition of vectors, as well as multiplication of a vector by a real number (a *scalar*). These two operations satisfy the commutative, associative, and distributive laws (the proof is trivial, in view of the analogous laws for the real numbers) and make $\mathbb{R}^k$ into a *vector space over the real field*. The zero element of $\mathbb{R}^k$ (sometimes called the *origin* or the *null vector*) is the point $\mathbf{0}$, all of whose coordinates are 0.

We also define the so-called “inner product” (or scalar product) of $\mathbf{x}$ and $\mathbf{y}$ by

$$\mathbf{x} \cdot \mathbf{y} = \sum_{i=1}^k x_i y_i$$

and the *norm* of $\mathbf{x}$ by

$$|\mathbf{x}| = (\mathbf{x} \cdot \mathbf{x})^{1/2} = \left(\sum_{i=1}^k x_i^2 \right)^{1/2}.$$

The structure now defined (the vector space $\mathbb{R}^k$ with the above inner product and norm) is called *euclidean k -space*.³⁴

1.37 Theorem. Suppose $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{R}^k$, and α is real. Then

34 Three separate structures are being stacked here, and later chapters use them independently: a vector space (addition and scaling, used throughout Chapter 9), an inner product (used for orthogonality and in Chapter 11), and a norm (used to make a metric, which is all Chapter 2 needs). The norm is derived from the inner product; the inner product is not derived from anything.

- (a) $|\mathbf{x}| \geq 0$;
- (b) $|\mathbf{x}| = 0$ if and only if $\mathbf{x} = \mathbf{0}$;
- (c) $|\alpha\mathbf{x}| = |\alpha| |\mathbf{x}|$;
- (d) $|\mathbf{x} \cdot \mathbf{y}| \leq |\mathbf{x}| |\mathbf{y}|$;
- (e) $|\mathbf{x} + \mathbf{y}| \leq |\mathbf{x}| + |\mathbf{y}|$;
- (f) $|\mathbf{x} - \mathbf{z}| \leq |\mathbf{x} - \mathbf{y}| + |\mathbf{y} - \mathbf{z}|$.

Proof. (a), (b), and (c) are obvious, and (d) is an immediate consequence of the Schwarz inequality.³⁵ By (d) we have

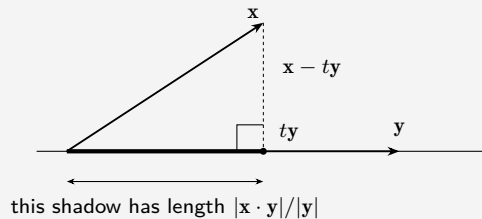
$$\begin{aligned}
 |\mathbf{x} + \mathbf{y}|^2 &= (\mathbf{x} + \mathbf{y}) \cdot (\mathbf{x} + \mathbf{y}) \\
 &= \mathbf{x} \cdot \mathbf{x} + 2\mathbf{x} \cdot \mathbf{y} + \mathbf{y} \cdot \mathbf{y} \\
 &\leq |\mathbf{x}|^2 + 2|\mathbf{x}| |\mathbf{y}| + |\mathbf{y}|^2 \\
 &= (|\mathbf{x}| + |\mathbf{y}|)^2,
 \end{aligned}$$

so that (e) is proved. Finally (f) follows from (e) if we replace $\mathbf{x}$ by $\mathbf{x} - \mathbf{y}$ and $\mathbf{y}$ by $\mathbf{y} - \mathbf{z}$. $\square$

The geometry of 1.37(d) and (e)

Parts (d) and (e) are the two facts of Euclidean geometry that survive into arbitrary dimension, and each has a picture that is also its proof.

(d) Schwarz is a statement about projection. Assume $\mathbf{y} \neq \mathbf{0}$ — otherwise both sides of (d) vanish and there is nothing to prove. Drop a perpendicular from $\mathbf{x}$ onto the line through $\mathbf{y}$. The foot sits at $t\mathbf{y}$ with $t = (\mathbf{x} \cdot \mathbf{y})/|\mathbf{y}|^2$, and the leftover piece $\mathbf{x} - t\mathbf{y}$ is perpendicular to $\mathbf{y}$.



The shadow of $\mathbf{x}$ on $\mathbf{y}$ cannot be longer than $\mathbf{x}$ itself.

Now compute the length of that leftover piece:

$$0 \leq |\mathbf{x} - t\mathbf{y}|^2 = |\mathbf{x}|^2 - \frac{(\mathbf{x} \cdot \mathbf{y})^2}{|\mathbf{y}|^2},$$

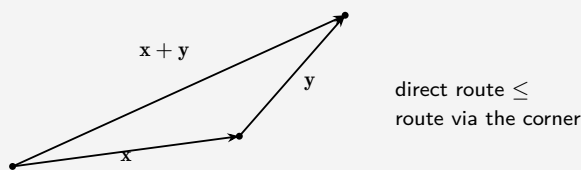
and rearranging gives exactly $(\mathbf{x} \cdot \mathbf{y})^2 \leq |\mathbf{x}|^2 |\mathbf{y}|^2$. That is Schwarz, and no angle was needed — only that a squared length is not negative. Equality holds precisely when the leftover piece vanishes, that is when $\mathbf{x}$ lies on the line through $\mathbf{y}$: the linear dependence of Exercise 1.15.

35 Take 1.35 with all a_j, b_j real; conjugation does nothing, and the statement becomes $(\sum x_i y_i)^2 \leq \sum x_i^2 \sum y_i^2$, which is (d) squared. This is the only place the previous section's main theorem is used, and it is the reason that section had to come first.

This is also Rudin's proof of 1.35 in disguise. His $\sum |Ba_j - Cb_j|^2 \geq 0$ is the same "a squared length is not negative," with $Ba_j - Cb_j$ playing the part of the leftover piece — and his separate treatment of $B = 0$ is exactly the $\mathbf{y} = \mathbf{0}$ case set aside above.

A third way to say it, if you prefer algebra to pictures: the function $q(t) = |\mathbf{x} - t\mathbf{y}|^2$ is a quadratic in t that is never negative, so its discriminant cannot be positive, and $4(\mathbf{x} \cdot \mathbf{y})^2 - 4|\mathbf{y}|^2|\mathbf{x}|^2 \leq 0$. The three proofs are one proof: the projection is where q attains its minimum, and the discriminant condition says that minimum is not below zero.

(e) **The triangle inequality is a triangle.** Lay $\mathbf{y}$ head to tail after $\mathbf{x}$; the direct route to $\mathbf{x} + \mathbf{y}$ is no longer than going via the corner.



Part (f) is the same picture read as three points: going from $\mathbf{x}$ to $\mathbf{z}$ directly beats detouring through $\mathbf{y}$.

But here the picture is not a proof, and it is worth being clear why. For Schwarz the figure could be turned into an argument, because every step was an identity about inner products. Not so here. "The straight path is shortest" is a theorem of Euclidean geometry, and in Rudin's development Euclidean geometry does not yet exist — 1.36 has just *defined* distance by a formula, and 1.37 is where the familiar properties of that formula are established for the first time. Appealing to the picture would assume what is being proved. The figure shows you what 1.37(e) means; the algebra, via Schwarz, is what makes it true.

PART (F) IS THE TRIANGLE INEQUALITY YOU WILL ACTUALLY USE

Parts (e) and (f) say the same thing twice, and it is worth seeing why both are stated. Part (e) is about *vectors*: the length of a sum is at most the sum of the lengths. Part (f) is about *points*: the distance from $\mathbf{x}$ to $\mathbf{z}$ is at most the distance via any detour through $\mathbf{y}$.

The substitution converting one into the other is worth doing once by hand. Setting $\mathbf{u} = \mathbf{x} - \mathbf{y}$ and $\mathbf{v} = \mathbf{y} - \mathbf{z}$, we have $\mathbf{u} + \mathbf{v} = \mathbf{x} - \mathbf{z}$, so (e) reads $|\mathbf{x} - \mathbf{z}| \leq |\mathbf{x} - \mathbf{y}| + |\mathbf{y} - \mathbf{z}|$.

Form (f) is the one that generalises. When Chapter 2 defines a metric space there is no addition available, so vectors cannot be summed and (e) has no meaning — but distances between three

points still make sense, and (f) becomes axiom (c) of Definition 2.15.

[36] This is the entire purpose of the section, stated in one sentence and easy to read past. Compare the three parts he names with Definition 2.15: (a) and (b) give $d(x, y) > 0$ for $x \neq y$ and $d(x, x) = 0$; (f) is the triangle inequality. Symmetry, the remaining axiom, follows from (c) with $\alpha = -1$. Rudin has just verified, in advance, that $\mathbb{R}^k$ is an example of the structure Chapter 2 will study abstractly.

1.38 Remarks. Theorem 1.37(a), (b), and (f) will allow us (see Chap. 2) to regard $\mathbb{R}^k$ as a *metric space*.³⁶

$\mathbb{R}^1$ (the set of all real numbers) is usually called *the line*, or *the real line*. Likewise, $\mathbb{R}^2$ is called *the plane*, or *the complex plane* (compare Definitions 1.24 and 1.36). In these two cases, the norm is just the absolute value of the corresponding real or complex number.

$\mathbb{R}^2$ AND $\mathbb{C}$ ARE THE SAME SPACE BUT NOT THE SAME OBJECT

As metric spaces they are identical: $(a, b) \in \mathbb{R}^2$ and $a + bi \in \mathbb{C}$ have the same norm $\sqrt{a^2 + b^2}$, so every statement about convergence, open sets or compactness transfers word for word.

They differ in algebra. $\mathbb{C}$ has a multiplication making it a field (1.25); $\mathbb{R}^2$ has only the inner product, which produces a *scalar*, not a vector. So zw makes sense for $z, w \in \mathbb{C}$ and $\mathbf{x}\mathbf{y}$ does not for $\mathbf{x}, \mathbf{y} \in \mathbb{R}^2$.

The distinction matters when a theorem is stated for one and you want it for the other. Results resting only on the metric — most of Chapter 2, and the convergence theorems of Chapter 3 — carry across freely. Results using multiplication or division, such as the whole theory of power series in Chapter 8, belong to $\mathbb{C}$ alone.

THE CHAPTER PROPER, IN ONE PARAGRAPH

$\mathbb{Q}$ is unfit for analysis because a bounded set can fail to have a least upper bound (1.1). Name that failure (1.8–1.10), develop just enough algebra to state what $\mathbb{R}$ must be (1.12–1.18), assert a field free of the failure (1.19), and the consequences follow at once: infima exist too (1.11), $\mathbb{N}$ is unbounded and $\mathbb{Q}$ is dense (1.20), and every positive real has every root (1.21). The remaining sections build the objects later chapters will work in — the extended reals for unbounded suprema, $\mathbb{C}$ for Chapters 3 and 8, $\mathbb{R}^k$ for Chapter 2 onward — and prove one inequality of lasting importance, Schwarz’s (1.35). Nothing in the rest of the book uses more about $\mathbb{R}$ than 1.19. When a later proof says “by completeness,” that is what it means.

APPENDIX

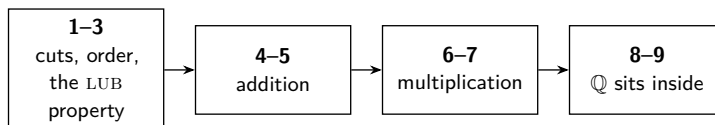
WHAT THE APPENDIX IS DOING, AND WHEN TO READ IT

This is the promissory note from 1.19 being paid. Rudin asserted that an ordered field with the least-upper-bound property exists; here he builds one out of $\mathbb{Q}$ and nothing else.

You may skip it. Nothing in Chapters 2–11 depends on having read it, and Rudin says in the Preface it may be “studied and enjoyed whenever the time seems ripe.” What it buys you is the knowledge that 1.19 is a theorem rather than an article of faith. He is not alone in deferring it: Yu Pin’s Tsinghua notes call real-number theory “very important for the completeness of the knowledge system, but not indispensable for understanding the basic ideas of analysis,” and the USTC lecture notes postpone it to a third volume.

The idea in one sentence. A real number is identified with the set of all rationals below it. That set — a *cut* — is a *subset* of $\mathbb{Q}$, so no new kind of object has to be conjured up; $\sqrt{2}$ becomes $\{p \in \mathbb{Q} : p < 0 \text{ or } p^2 < 2\}$, a perfectly ordinary collection of rationals which exists whether or not $\sqrt{2}$ does.

The nine steps, and which ones matter.



Step 3 is the whole point — it is where the least-upper-bound property appears, and its proof is four lines: the supremum of a set of cuts is their *union*. That is the payoff of representing numbers as sets. Steps 4–7 are bookkeeping, verifying field axioms one at a time, and Rudin openly omits most of Step 6 as too similar to Step 4. Steps 8–9 establish the second sentence of 1.19, that $\mathbb{Q}$ is a subfield.

If you read only two things, read Step 1 (what a cut is, and why property (III) is there) and Step 3.

Theorem 1.19 will be proved in this appendix by constructing $\mathbb{R}$ from $\mathbb{Q}$. We shall divide the construction into several steps.

Step 1 The members of $\mathbb{R}$ will be certain subsets of $\mathbb{Q}$, called *cuts*. A cut is, by definition, any set $\alpha \subseteq \mathbb{Q}$ with the following three properties.

- (I) α is not empty, and $\alpha \neq \mathbb{Q}$.
- (II) If $p \in \alpha$, $q \in \mathbb{Q}$, and $q < p$, then $q \in \alpha$.
- (III) If $p \in \alpha$, then $p < r$ for some $r \in \alpha$.

The letters $p, q, r, \dots$ will always denote rational numbers, and $\alpha, \beta, \gamma, \dots$ will denote cuts.

Note that (III) simply says that α has no largest member; (II) implies two facts which will be used freely:

If $p \in \alpha$ and $q \notin \alpha$, then $p < q$.

If $r \notin \alpha$ and $r < s$, then $s \notin \alpha$.

WHAT THE THREE PROPERTIES ARE FOR

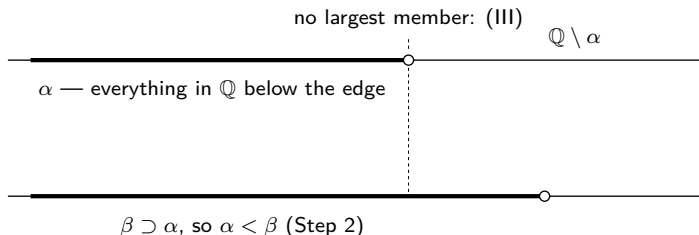
Picture the cut α as “everything strictly below some point on the line.” Each condition rules out a degenerate set that would spoil that picture.

(I) excludes the two extremes. $\emptyset$ and $\mathbb{Q}$ would correspond to $-\infty$ and $+\infty$, which are not real numbers.

(II) says α is downward closed — it is an initial segment of $\mathbb{Q}$, not an arbitrary subset. Without it $\{0, 5\}$ would qualify, and nothing about order would work.

(III) says α has no largest member, which forces the cut to be the set of rationals *strictly* below its edge. This is the delicate one, and its purpose is uniqueness: without it, the point 2 would be represented by two different cuts, $\{p < 2\}$ and $\{p \leq 2\}$, and the map from cuts to numbers would be two-to-one at every rational. Exercise 1.20 asks what breaks if you drop it — the answer is that the additive inverse (A5) fails.

The two consequences of (II) that Rudin records are the workhorses of the whole appendix. Both say the same thing — a cut has no gaps and no stragglers — and he uses them dozens of times without further comment.



A cut is a leftward ray of $\mathbb{Q}$ with no endpoint. Order is containment; one cut is smaller when it is a proper subset.

Step 2 Define “ $\alpha < \beta$ ” to mean: α is a proper subset of β .

Let us check that this meets the requirements of Definition 1.5.

If $\alpha < \beta$ and $\beta < \gamma$ it is clear that $\alpha < \gamma$. (A proper subset of a proper subset is a proper subset.) It is also clear that at most one of the three relations

$$\alpha < \beta, \quad \alpha = \beta, \quad \beta < \alpha$$

can hold for any pair α, β . To show that at least one holds, assume that the first two fail. Then α is not a subset of β . Hence there is a $p \in \alpha$ with $p \notin \beta$. If $q \in \beta$, it follows that $q < p$ (since $p \notin \beta$), hence $q \in \alpha$, by (II). Thus $\beta \subset \alpha$. Since $\beta \neq \alpha$, we conclude $\beta < \alpha$.³⁷

Thus $\mathbb{R}$ is now an ordered set.

Step 3 The ordered set $\mathbb{R}$ has the least upper bound property.

³⁷ Trichotomy is the only part needing work, because containment of arbitrary sets is a partial order — two sets can be incomparable. What rules that out here is (II): downward-closed subsets of a totally ordered set are automatically nested. This is the first place property (II) earns its keep.

To prove this, let A be a nonempty subset of $\mathbb{R}$, and assume that $\beta \in \mathbb{R}$ is an upper bound of A . Define γ to be the union of all $\alpha \in A$. In other words, $p \in \gamma$ if and only if $p \in \alpha$ for some $\alpha \in A$. We shall prove that $\gamma \in \mathbb{R}$ and that $\gamma = \sup A$.

Since A is not empty, there exists an $\alpha_0 \in A$. This α_0 is not empty. Since $\alpha_0 \subset \gamma$, γ is not empty. Next, $\gamma \subset \beta$ (since $\alpha \subset \beta$ for every $\alpha \in A$), and therefore $\gamma \neq \mathbb{Q}$. Thus γ satisfies property (I). To prove (II) and (III), pick $p \in \gamma$. Then $p \in \alpha_1$ for some $\alpha_1 \in A$. If $q < p$, then $q \in \alpha_1$, hence $q \in \gamma$; this proves (II). If $r \in \alpha_1$ is so chosen that $r > p$, we see that $r \in \gamma$ (since $\alpha_1 \subset \gamma$), and therefore γ satisfies (III). Thus $\gamma \in \mathbb{R}$.

It is clear that $\alpha \leq \gamma$ for every $\alpha \in A$.

Suppose $\delta < \gamma$. Then there is an $s \in \gamma$ such that $s \notin \delta$. Since $s \in \gamma$, $s \in \alpha$ for some $\alpha \in A$. Hence $\delta < \alpha$ and δ is not an upper bound of A .

This gives the desired result: $\gamma = \sup A$.

Why both hypotheses of 1.10 appear, and where

Watch where each is consumed. *Nonempty* gives an $\alpha_0 \in A$, which is what makes γ nonempty — half of property (I). *Bounded above* gives a β containing every α , hence $\gamma \subset \beta \neq \mathbb{Q}$ — the other half.

Without either, the union might not be a cut: an empty union is $\emptyset$, and an unbounded union is all of $\mathbb{Q}$. Both are excluded by (I) precisely because they are the cuts that would represent $-\infty$ and $+\infty$. This is the same observation as the warning at 1.10, now visible from the inside: the two hypotheses are not conveniences, they are what keeps the answer inside the system.

The last two paragraphs are the two clauses of Definition 1.8: γ is an upper bound, and nothing smaller is. Note how cheap the second is — if δ is properly contained in the union, it must miss an element that some α contains, so α already exceeds it.

Step 4 If $\alpha \in \mathbb{R}$ and $\beta \in \mathbb{R}$, we define $\alpha + \beta$ to be the set of all sums $r + s$, where $r \in \alpha$ and $s \in \beta$.

We define 0^* to be the set of all negative rational numbers. It is clear that 0^* is a cut. We verify that *the axioms for addition* (see Definition 1.12) *hold in $\mathbb{R}$, with 0^* playing the role of 0.*

(A1) We have to show that $\alpha + \beta$ is a cut. It is clear that $\alpha + \beta$ is a nonempty subset of $\mathbb{Q}$. Take $r' \notin \alpha$, $s' \notin \beta$. Then $r' + s' > r + s$ for all choices of $r \in \alpha$, $s \in \beta$. Thus $r' + s' \notin \alpha + \beta$. It follows that $\alpha + \beta$ has property (I).

Pick $p \in \alpha + \beta$. Then $p = r + s$, with $r \in \alpha$, $s \in \beta$. If $q < p$, then $q - s < r$, so $q - s \in \alpha$, and $q = (q - s) + s \in \alpha + \beta$. Thus (II) holds. Choose $t \in \alpha$ so that $t > r$. Then $p < t + s$ and $t + s \in \alpha + \beta$. Thus (III) holds.

(A2) $\alpha + \beta$ is the set of all $r + s$, with $r \in \alpha$, $s \in \beta$. By the same

definition, $\beta + \alpha$ is the set of all $s + r$. Since $r + s = s + r$ for all $r \in \mathbb{Q}$, $s \in \mathbb{Q}$, we have $\alpha + \beta = \beta + \alpha$.

- (A3) As above, this follows from the associative law in $\mathbb{Q}$.
- (A4) If $r \in \alpha$ and $s \in 0^*$, then $r + s < r$, hence $r + s \in \alpha$. Thus $\alpha + 0^* \subset \alpha$. To obtain the opposite inclusion, pick $p \in \alpha$, and pick $r \in \alpha$, $r > p$. Then $p - r \in 0^*$, and $p = r + (p - r) \in \alpha + 0^*$. Thus $\alpha \subset \alpha + 0^*$. We conclude that $\alpha + 0^* = \alpha$.³⁸
- (A5) Fix $\alpha \in \mathbb{R}$. Let β be the set of all p with the following property:

There exists $r > 0$ such that $-p - r \notin \alpha$.

In other words, some rational number smaller than $-p$ fails to be in α .

We show that $\beta \in \mathbb{R}$ and that $\alpha + \beta = 0^*$.

If $s \notin \alpha$ and $p = -s - 1$, then $-p - 1 \notin \alpha$, hence $p \in \beta$. So β is not empty. If $q \in \alpha$, then $-q \notin \beta$. So $\beta \neq \mathbb{Q}$. Hence β satisfies (I).

Pick $p \in \beta$ and $r > 0$ so that $-p - r \notin \alpha$. If $q < p$, then $-q - r > -p - r$, hence $-q - r \notin \alpha$. Thus $q \in \beta$ and (II) holds. Put $t = p + (r/2)$. Then $t > p$ and $-t - (r/2) = -p - r \notin \alpha$, so that $t \in \beta$. Hence β satisfies (III).

We have proved that $\beta \in \mathbb{R}$.

If $r \in \alpha$ and $s \in \beta$, then $-s \notin \alpha$, hence $r < -s$, $r + s < 0$. Thus $\alpha + \beta \subset 0^*$.

To prove the opposite inclusion, pick $v \in 0^*$, put $w = -v/2$. Then $w > 0$ and there is an integer n such that $nw \in \alpha$ but $(n+1)w \notin \alpha$. (Note that this depends on the fact that $\mathbb{Q}$ has the archimedean property!) Put $p = -(n+2)w$. Then $p \in \beta$, since $-p - w \notin \alpha$, and

$$v = nw + p \in \alpha + \beta.$$

Thus $0^* \subset \alpha + \beta$.

We conclude that $\alpha + \beta = 0^*$. This β will of course be denoted by $-\alpha$.

WHY $-\alpha$ IS DEFINED SO AWKWARDLY

The natural guess is that $-\alpha$ should be $\{-q : q \notin \alpha\}$ — reflect the complement. That is almost right and fails on exactly one point. If α is the cut of a rational r , then $\mathbb{Q} \setminus \alpha$ has a smallest element, namely r itself. Reflecting gives a set with a *largest* element $-r$, violating property (III). So the naive definition does not always produce a cut.

Rudin's “there exists $r > 0$ with $-p - r \notin \alpha$ ” repairs this by de-

38 The reverse inclusion is where property (III) is used, and it is used essentially: we need some $r \in \alpha$ strictly above p so that $p - r$ is negative. If α had a largest member p , no such r would exist and $\alpha + 0^*$ would be missing p .

manding strict room to spare: p qualifies only if something *strictly below* $-p$ already lies outside α . That extra r shaves off the single offending endpoint and nothing else. You can see the repair working in the verification of (III), where $t = p + r/2$ is produced by spending half the slack.

The archimedean appeal at the end is the other subtlety, and Rudin flags it himself with an exclamation mark. To hit an arbitrary $v \in 0^*$ we need integer multiples of w to eventually escape α ; that is exactly 1.20(a), holding in $\mathbb{Q}$. Note the logical order — the archimedean property of $\mathbb{Q}$ is elementary and assumed, and it is being used to build $\mathbb{R}$, not derived from it.

Step 5 Having proved that the addition defined in Step 4 satisfies axioms (A) of Definition 1.12, it follows that Proposition 1.14 is valid in $\mathbb{R}$, and we can prove one of the requirements in Definition 1.17:

If $\alpha, \beta, \gamma \in \mathbb{R}$ and $\beta < \gamma$, then $\alpha + \beta < \alpha + \gamma$.

Indeed, it is obvious from the definition of $+$ in $\mathbb{R}$ that $\alpha + \beta \subset \alpha + \gamma$; if we had $\alpha + \beta = \alpha + \gamma$, the cancellation law (Proposition 1.14) would imply $\beta = \gamma$.³⁹

It also follows that $\alpha > 0^*$ if and only if $-\alpha < 0^*$.

Step 6 Multiplication is a little more bothersome than addition in the present context, since products of negative rationals are positive. For this reason we confine ourselves first to $\mathbb{R}^+$, the set of all $\alpha \in \mathbb{R}$ with $\alpha > 0^*$.

If $\alpha \in \mathbb{R}^+$ and $\beta \in \mathbb{R}^+$, we define $\alpha\beta$ to be the set of all p such that $p \leq rs$ for some choice of $r \in \alpha$, $s \in \beta$, $r > 0$, $s > 0$.⁴⁰

We define 1^* to be the set of all $q < 1$.

Then *the axioms (M) and (D) of Definition 1.12 hold, with $\mathbb{R}^+$ in place of F , and with 1^* in the role of 1 .*

The proofs are so similar to the ones given in detail in Step 4 that we omit them.

Note, in particular, that the second requirement of Definition 1.17 holds: If $\alpha > 0^*$ and $\beta > 0^*$, then $\alpha\beta > 0^*$.

Step 7 We complete the definition of multiplication by setting $\alpha 0^* = 0^* \alpha = 0^*$, and by setting

$$\alpha\beta = \begin{cases} (-\alpha)(-\beta) & \text{if } \alpha < 0^*, \beta < 0^*, \\ -[(-\alpha)\beta] & \text{if } \alpha < 0^*, \beta > 0^*, \\ -[\alpha(-\beta)] & \text{if } \alpha > 0^*, \beta < 0^*. \end{cases}$$

The products on the right were defined in Step 6.

Having proved (in Step 6) that the axioms (M) hold in $\mathbb{R}^+$, it is now perfectly simple to prove them in $\mathbb{R}$, by repeated application of the identity $\gamma = -(-\gamma)$, which is part of Proposition 1.14. (See Step 5.)

[39] Exactly the dividend promised at 1.13(c). Proposition 1.14 was proved once, for an arbitrary set satisfying axioms (A); having checked those axioms in Step 4, Rudin may now invoke cancellation for cuts without reproving anything. The one-time purchase is being collected.

[40] The definition is not “the set of all products rs ” — that would fail property (II), since the products of positive rationals omit everything negative. The clause $p \leq rs$ patches this by throwing in everything below. Contrast Step 4, where the sums $r + s$ already sweep out a downward-closed set on their own.

The proof of the distributive law

$$\alpha(\beta + \gamma) = \alpha\beta + \alpha\gamma$$

breaks into cases. For instance, suppose $\alpha > 0^*$, $\beta < 0^*$, $\beta + \gamma > 0^*$. Then $\gamma = (\beta + \gamma) + (-\beta)$, and (since we already know that the distributive law holds in $\mathbb{R}^+$)

$$\alpha\gamma = \alpha(\beta + \gamma) + \alpha \cdot (-\beta).$$

But $\alpha \cdot (-\beta) = -(\alpha\beta)$. Thus

$$\alpha\beta + \alpha\gamma = \alpha(\beta + \gamma).$$

The other cases are handled in the same way.

We have now completed the proof that $\mathbb{R}$ is an ordered field with the least upper bound property.

Why multiplication needs cases and addition does not

Addition is monotone in each argument on all of $\mathbb{Q}$: if r grows, so does $r + s$. Multiplication is not — multiplying by a negative reverses order. Since a cut is defined by *downward* closure, the union of products $\{rs\}$ points the wrong way as soon as a negative factor appears, and the Step 6 definition simply does not describe a cut.

Rudin's escape is to define multiplication only where it is well behaved, on $\mathbb{R}^+$, and then extend by the sign rules — which is the same device used at 1.16(c),(d) to prove $(-x)(-y) = xy$ from the field axioms. The cases in Step 7 are those rules read as definitions rather than theorems.

This asymmetry is why the appendix is longer than it looks. It is also why the distributive law, which mixes the two operations, must be checked case by case: β and γ can have opposite signs while $\beta + \gamma$ has either.

Step 8 We associate with each $r \in \mathbb{Q}$ the set r^* which consists of all $p \in \mathbb{Q}$ such that $p < r$. It is clear that each r^* is a cut; that is, $r^* \in \mathbb{R}$. These cuts satisfy the following relations:

- (a) $r^* + s^* = (r + s)^*$,
- (b) $r^* s^* = (rs)^*$,
- (c) $r^* < s^*$ if and only if $r < s$.

To prove (a), choose $p \in r^* + s^*$. Then $p = u + v$, where $u < r$, $v < s$. Hence $p < r + s$, which says that $p \in (r + s)^*$.

Conversely, suppose $p \in (r + s)^*$. Then $p < r + s$. Choose t so that $2t = r + s - p$, put

$$r' = r - t, \quad s' = s - t.$$

41 The trick is splitting the shortfall evenly. We know p falls short of $r + s$ by $2t > 0$; docking t from each of r and s keeps both strictly below their targets while making the sum land exactly on p . Halving the slack is the same device as $t = p + r/2$ in Step 4(A5).

Then $r' \in r^*$, $s' \in s^*$, and $p = r' + s'$, so that $p \in r^* + s^*$.⁴¹ This proves (a). The proof of (b) is similar.

If $r < s$ then $r \in s^*$, but $r \notin r^*$; hence $r^* < s^*$.

If $r^* < s^*$, then there is a $p \in s^*$ such that $p \notin r^*$. Hence $r \leq p < s$, so that $r < s$. This proves (c).

Step 9 We saw in Step 8 that the replacement of the rational numbers r by the corresponding “rational cuts” $r^* \in \mathbb{R}$ preserves sums, products, and order. This fact may be expressed by saying that the ordered field $\mathbb{Q}$ is *isomorphic* to the ordered field $\mathbb{Q}^*$ whose elements are the rational cuts. Of course, r^* is by no means the same as r , but the properties we are concerned with (arithmetic and order) are the same in the two fields.

It is this identification of $\mathbb{Q}$ with $\mathbb{Q}^*$ which allows us to regard $\mathbb{Q}$ as a subfield of $\mathbb{R}$.⁴²

The second part of Theorem 1.19 is to be understood in terms of this identification. Note that the same phenomenon occurs when the real numbers are regarded as a subfield of the complex field, and it also occurs at a much more elementary level, when the integers are identified with a certain subset of $\mathbb{Q}$.

It is a fact, which we will not prove here, that any two ordered fields with the least-upper-bound property are isomorphic. The first part of Theorem 1.19 therefore *characterizes* the real field $\mathbb{R}$ completely.⁴³

The books by Landau and Thurston cited in the Bibliography are entirely devoted to number systems. Chapter 1 of Knopp’s book contains a more leisurely description of how $\mathbb{R}$ can be obtained from $\mathbb{Q}$. Another construction, in which each real number is defined to be an equivalence class of Cauchy sequences of rational numbers (see Chap. 3), is carried out in Sec. 5 of the book by Hewitt and Stromberg.

The cuts in $\mathbb{Q}$ which we used here were invented by Dedekind. The construction of $\mathbb{R}$ from $\mathbb{Q}$ by means of Cauchy sequences is due to Cantor. Both Cantor and Dedekind published their constructions in 1872.⁴⁴

WHAT THE APPENDIX ESTABLISHED

$\mathbb{R}$ exists. Concretely, it is the set of downward-closed, unbounded-below, maximum-free subsets of $\mathbb{Q}$, ordered by inclusion, with addition given by elementwise sums and multiplication defined first for positives and then by sign rules. Under those definitions it is an ordered field, its least upper bounds are unions, and it contains an isomorphic copy of $\mathbb{Q}$.

Theorem 1.19 is therefore a theorem. And since complete ordered fields are unique up to isomorphism, this particular construction may now be forgotten — which is precisely what the rest of the book does.

[42] This settles the second sentence of Theorem 1.19, and it is worth being clear about what is and is not claimed. The rational 2 and the cut $2^* = \{p \in \mathbb{Q} : p < 2\}$ are genuinely different objects — one a number, the other an infinite set. What Step 8 establishes is that no arithmetic-or-order question can tell them apart, and we therefore agree to write $\mathbb{Q} \subset \mathbb{R}$. The same courtesy was extended to $\mathbb{R} \subset \mathbb{C}$ after 1.26.

[43] The strongest sentence in the appendix, and it is dropped without proof. It says that Dedekind’s construction was not one choice among many with different outcomes: any route to a complete ordered field — cuts, Cauchy sequences, infinite decimals — lands on the same object up to isomorphism. That is why nothing in the rest of the book ever refers back to cuts. $\mathbb{R}$ is fully determined by its properties, so the properties are all one needs.

[44] Cantor’s route is the one most modern texts take — Tao’s *Analysis I* builds $\mathbb{R}$ as equivalence classes of Cauchy sequences of rationals (his Chapter 5) and explicitly declines Dedekind cuts. It generalises: the same construction completes any metric space (Rudin does exactly this in Exercise 3.24), whereas cuts need an order and so stop at $\mathbb{R}$. Cuts are shorter here, which is why Rudin keeps them. Both books warn off the third route, infinite decimals: it looks familiar but has to legislate $0.999\cdots = 1$, and is the most delicate of the three.

EXERCISES

HOW TO USE THESE

Rudin's exercises are not drill. Several of them — 1.6 and 1.7 in particular — prove results the book will later use, and skipping them leaves a hole. The annotation here gives the *approach*: which theorem to reach for, what shape the argument takes, where the difficulty actually is. It does not give solutions. Read the note only after you are genuinely stuck, and never before attempting the problem.

The ones that matter most. 1.6 and 1.7 construct b^x for real x and its inverse, the logarithm — Rudin defers this to the exercises and then uses exponentials freely from Chapter 3 onward. 1.5 is the inf/sup duality you will use constantly. 1.8 explains why $\mathbb{C}$ is never given an order. 1.20 tests whether you understand why a cut never has a largest member.

The routine ones. 1.1–1.4 and 1.11–1.14 are short consequences of 1.14–1.18 and 1.33; do them quickly. 1.16–1.19 are geometry in $\mathbb{R}^k$ and exercise the Schwarz inequality.

45 Contradiction, and the point is that $\mathbb{Q}$ is closed under the operations. If $r+x$ were rational, then $x = (r+x) - r$ would be a difference of two rationals, hence rational. Same for rx , dividing by r — which is where $r \neq 0$ is needed.

46 The parity argument of 1.1 does not transfer directly, since 12 is not prime. Work with the factor 3 instead of 2: if $m^2 = 12n^2$ then $3 \mid m^2$, hence $3 \mid m$ (this needs 3 prime), and the descent proceeds as before.

47 The hypothesis “nonempty” is the entire exercise, and it is easy to write a proof that never uses it — such a proof is wrong. Pick any $x \in E$; then $\alpha \leq x \leq \beta$. If E were empty, every element would be both a lower and an upper bound and the conclusion would fail.

48 Reflection swaps the two notions: β is a lower bound of A if and only if $-\beta$ is an upper bound of $-A$, and the inequality reverses on negation. Verify both clauses of Definition 1.8 for $-\sup(-A)$. This is a second proof of 1.11 for the special case $S = \mathbb{R}$, and the identity is used without comment throughout Chapters 3 and 6. It is Tao's Exercise 5.5.1.

49 This is a construction, not a computation, and it is the exercise Rudin relies on most. The architecture: (a) shows the rational exponent is well defined — raise both sides to the power nq and use uniqueness in 1.21; (b) is then algebra; (c) is the crucial step, showing the new definition by supremum agrees with the old one on rationals, so that $b^x := \sup B(x)$ extends rather than contradicts; (d) is the payoff. Part (c) is where $b > 1$ matters — it makes $t \mapsto b^t$ increasing, so $B(r)$ has b^r as its largest element. If you get stuck, this construction is carried out in the main text of Tao's Analysis I, §5.6 and §6.7 — what Rudin relegates to an exercise, Tao develops as numbered propositions.

Unless the contrary is explicitly stated, all numbers that are mentioned in these exercises are understood to be real.

1.1. If r is rational ($r \neq 0$) and x is irrational, prove that $r+x$ and rx are irrational.⁴⁵

1.2. Prove that there is no rational number whose square is 12.⁴⁶

1.3. Prove Proposition 1.15.

1.4. Let E be a nonempty subset of an ordered set; suppose α is a lower bound of E and β is an upper bound of E . Prove that $\alpha \leq \beta$.⁴⁷

1.5. Let A be a nonempty set of real numbers which is bounded below. Let $-A$ be the set of all numbers $-x$, where $x \in A$. Prove that⁴⁸

$$\inf A = -\sup(-A).$$

1.6. Fix $b > 1$.⁴⁹

(a) If m, n, p, q are integers, $n > 0$, $q > 0$, and $r = m/n = p/q$, prove that

$$(b^m)^{1/n} = (b^p)^{1/q}.$$

Hence it makes sense to define $b^r = (b^m)^{1/n}$.

- (b) Prove that $b^{r+s} = b^r b^s$ if r and s are rational.
- (c) If x is real, define $B(x)$ to be the set of all numbers b^t , where t is rational and $t \leq x$. Prove that

$$b^r = \sup B(r)$$

when r is rational. Hence it makes sense to define

$$b^x = \sup B(x)$$

for every real x .

- (d) Prove that $b^{x+y} = b^x b^y$ for all real x and y .

1.7. Fix $b > 1$, $y > 0$, and prove that there is a unique real x such that $b^x = y$, by completing the following outline. (This x is called the *logarithm* of y to the base b .)⁵⁰

- (a) For any positive integer n , $b^n - 1 \geq n(b - 1)$.
- (b) Hence $b - 1 \geq n(b^{1/n} - 1)$.
- (c) If $t > 1$ and $n > (b - 1)/(t - 1)$, then $b^{1/n} < t$.
- (d) If w is such that $b^w < y$, then $b^{w+(1/n)} < y$ for sufficiently large n ; to see this, apply part (c) with $t = y \cdot b^{-w}$.
- (e) If $b^w > y$, then $b^{w-(1/n)} > y$ for sufficiently large n .
- (f) Let A be the set of all w such that $b^w < y$, and show that $x = \sup A$ satisfies $b^x = y$.
- (g) Prove that this x is unique.

1.8. Prove that no order can be defined in the complex field that turns it into an ordered field. *Hint:* -1 is a square.⁵¹

1.9. Suppose $z = a + bi$, $w = c + di$. Define $z < w$ if $a < c$, and also if $a = c$ but $b < d$. Prove that this turns the set of complex numbers into an ordered set. (This type of order relation is called a *dictionary order*, or *lexicographic order*, for obvious reasons.) Does this ordered set have the least-upper-bound property?⁵²

1.10. Suppose $z = a + bi$, $w = u + iv$, and

$$a = \left(\frac{|w| + u}{2} \right)^{1/2}, \quad b = \left(\frac{|w| - u}{2} \right)^{1/2}.$$

Prove that $z^2 = w$ if $v \geq 0$ and that $(\bar{z})^2 = w$ if $v \leq 0$. Conclude that every complex number (with one exception!) has two complex square roots.⁵³

50 Compare the whole outline with the proof of 1.21 and the resemblance is exact: form the set of things that undershoot, take its supremum, and rule out both strict inequalities. Parts (a)–(e) are the continuity estimates that Rudin computed by hand there — (d) and (e) are the analogues of choosing h and k . Part (a) is the binomial theorem, or induction. Once you see that 1.21 and 1.7 are the same argument twice, the outline reads itself.

51 Two lines from 1.18(d). That proposition gives $x^2 > 0$ for $x \neq 0$ in any ordered field, so $i^2 = -1 > 0$; but $1 > 0$ also, and then (i) of 1.17 forces $0 = 1 + (-1) > 0$. The point is that no cleverness in choosing the order can help — the obstruction is 1.18(d), which holds for every order whatever.

52 Read alongside 1.8. An order on $\mathbb{C}$ certainly exists; what 1.8 rules out is an order compatible with the arithmetic. Definition 1.6 asks only for trichotomy and transitivity, and lexicographic order supplies both. For the second question, test the set $\{z : \operatorname{Re}(z) < 0\}$ and ask what its least upper bound would have to be.

53 Direct computation: expand $z^2 = a^2 - b^2 + 2abi$ and check the two parts separately. The real part is immediate; for the imaginary part you will need $|w|^2 = u^2 + v^2$ and a square root taken with the right sign, which is exactly where the hypothesis on v enters. Both a and b are well defined because $|w| \geq |u|$ by 1.33(d). The exception is $w = 0$.

54 Take $r = |z|$ and $w = z/|z|$, then handle $z = 0$ separately — and that separate case is the answer to the second question.

- 1.11.** If z is a complex number, prove that there exists an $r \geq 0$ and a complex number w with $|w| = 1$ such that $z = rw$. Are w and r always uniquely determined by z ?⁵⁴

55 Induction on n , with 1.33(e) as the base and the inductive step. Worth doing once carefully: this is the form of the triangle inequality actually used in Chapter 3.

56 The reverse triangle inequality.

Write $x = (x - y) + y$ and apply 1.33(e) to get $|x| - |y| \leq |x - y|$; then swap x and y to get the other direction, and combine. Used constantly later to show that the modulus is continuous.

57 Expand each term as $(1 \pm z)(\overline{1 \pm z})$ using 1.31, and watch the cross terms cancel. The answer is a constant — this is the parallelogram law of Exercise 1.17 in disguise.

- 1.12.** If $z_1, \dots, z_n$ are complex, prove that⁵⁵

$$|z_1 + z_2 + \dots + z_n| \leq |z_1| + |z_2| + \dots + |z_n|.$$

- 1.13.** If x, y are complex, prove that⁵⁶

$$||x| - |y|| \leq |x - y|.$$

- 1.14.** If z is a complex number such that $|z| = 1$, that is, such that $z\bar{z} = 1$, compute⁵⁷

$$|1 + z|^2 + |1 - z|^2.$$

- 1.15.** Under what conditions does equality hold in the Schwarz inequality?

- 1.16.** Suppose $k \geq 3$, $\mathbf{x}, \mathbf{y} \in \mathbb{R}^k$, $|\mathbf{x} - \mathbf{y}| = d > 0$, and $r > 0$. Prove:

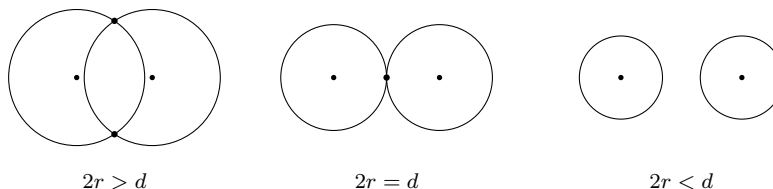
- (a) If $2r > d$, there are infinitely many $\mathbf{z} \in \mathbb{R}^k$ such that

$$|\mathbf{z} - \mathbf{x}| = |\mathbf{z} - \mathbf{y}| = r.$$

- (b) If $2r = d$, there is exactly one such $\mathbf{z}$.

- (c) If $2r < d$, there is no such $\mathbf{z}$.

How must these statements be modified if k is 2 or 1?⁵⁸



The three cases drawn in the plane, where the intersection is two points, one, or none. In $\mathbb{R}^k$ the first case becomes a sphere of dimension $k - 2$ — a circle when $k = 3$, and infinite whenever $k \geq 3$.

- 1.17.** Prove that

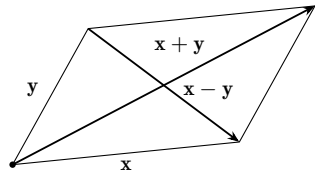
$$|\mathbf{x} + \mathbf{y}|^2 + |\mathbf{x} - \mathbf{y}|^2 = 2|\mathbf{x}|^2 + 2|\mathbf{y}|^2$$

if $\mathbf{x} \in \mathbb{R}^k$ and $\mathbf{y} \in \mathbb{R}^k$. Interpret this geometrically, as a statement about parallelograms.⁵⁹

58 Two spheres of radius r about $\mathbf{x}$ and $\mathbf{y}$: they meet in a sphere of dimension $k - 2$, a point, or nothing.

Algebraically, start from the midpoint $\mathbf{m} = (\mathbf{x} + \mathbf{y})/2$ and look for $\mathbf{z} = \mathbf{m} + \mathbf{v}$ with $\mathbf{v}$ orthogonal to $\mathbf{x} - \mathbf{y}$; Exercise 1.17 gives $|\mathbf{v}|^2 = r^2 - d^2/4$. The condition $k \geq 3$ is what guarantees room for infinitely many such $\mathbf{v}$ — for $k = 2$ there are exactly two, for $k = 1$ none.

59 Expand both sides with the inner product; the cross terms $\pm 2\mathbf{x} \cdot \mathbf{y}$ cancel. Geometrically: the sum of the squares of a parallelogram's diagonals equals the sum of the squares of its four sides. This identity characterises norms that come from an inner product, which is why it fails for the norms of Exercise 1.19 in later chapters.



sum of squares
of the diagonals
= sum of squares
of the four sides

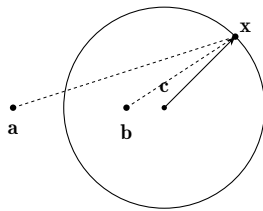
The identity is exactly this, and it fails for norms that do not come from an inner product.

1.18. If $k \geq 2$ and $\mathbf{x} \in \mathbb{R}^k$, prove that there exists $\mathbf{y} \in \mathbb{R}^k$ such that $\mathbf{y} \neq \mathbf{0}$ but $\mathbf{x} \cdot \mathbf{y} = 0$. Is this also true if $k = 1$?⁶⁰

1.19. Suppose $\mathbf{a} \in \mathbb{R}^k$, $\mathbf{b} \in \mathbb{R}^k$. Find $\mathbf{c} \in \mathbb{R}^k$ and $r > 0$ such that

$$|\mathbf{x} - \mathbf{a}| = 2|\mathbf{x} - \mathbf{b}|$$

if and only if $|\mathbf{x} - \mathbf{c}| = r$. (Solution: $3\mathbf{c} = 4\mathbf{b} - \mathbf{a}$, $3r = 2|\mathbf{b} - \mathbf{a}|$.)⁶¹



every $\mathbf{x}$ with
 $|\mathbf{x} - \mathbf{a}| = 2|\mathbf{x} - \mathbf{b}|$
lies on one sphere

The locus of points whose distances to two fixed points keep a constant ratio is a sphere — an Apollonius sphere — not, as one might guess, a plane. It is a plane only when the ratio is 1.

1.20. With reference to the Appendix, suppose that property (III) were omitted from the definition of a cut. Keep the same definitions of order and addition. Show that the resulting ordered set has the least-upper-bound property, that addition satisfies axioms (A1) to (A4) (with a slightly different zero-element!), but that (A5) fails.

On Exercise 1.20

The exercise that explains a definition rather than applying it. Without (III) each rational r has *two* representatives, $\{p < r\}$ and $\{p \leq r\}$; the parenthetical hint is that the identity is now $\{p \leq 0\}$, not $\{p < 0\}$. Inverses are what break. Take $\alpha = \{p < 0\}$, which has no largest member: then $\alpha + \beta$ has no largest member either, whatever β is, so it can never equal the new zero — which has largest member 0. Compare the note at Step 1.

60 Exhibit one: if $\mathbf{x} = (x_1, \dots, x_k)$, try $\mathbf{y} = (-x_2, x_1, 0, \dots, 0)$, and check separately what to do when $x_1 = x_2 = 0$. For $k = 1$ the statement fails in general but not always: if $x \neq 0$ then $xy = 0$ forces $y = 0$ by 1.16(b), so no such y exists — whereas if $x = 0$ then every $y \neq 0$ works. So the honest answer is that it holds precisely when $x = 0$.

61 Square both sides, expand with the inner product, and collect: the $|\mathbf{x}|^2$ terms do not cancel — they combine to $-3|\mathbf{x}|^2$ — so dividing by -3 leaves the equation of a sphere. Completing the square identifies the centre. The locus is an Apollonius sphere; Rudin supplies the answer so you can check your algebra.

Appendix to Chapter 1

HOW TO USE THIS APPENDIX

Nothing in this appendix is Rudin's text. It is imported from the four comparison texts and set apart at every level: the running head names the appendix, section heads carry a reverse-video tag, item numbers are prefixed **S**, and every statement names its source.

Why import anything at all. Chapter 1 is the one chapter where Rudin is deliberately incomplete, and he says so. He assumes $\mathbb{Q}$ outright, exiles the construction of $\mathbb{R}$ to an appendix he tells you to skip, and states his sharpest result — that the complete ordered field is unique — with the words “*which we will not prove here.*” Each omission is defensible; each also leaves a question a reader is entitled to have answered.

What is here.

A. The uniqueness theorem, proved. Rudin's strongest sentence about $\mathbb{R}$ is the one he declines to justify. Proving it is what converts his appendix from “one construction” into “the only possible answer.”

B. How much is the Archimedean property? Rudin gets it free from LUB at 1.20(a), so he never has to show it is not automatic. It is not: there are ordered fields containing infinite elements, and seeing one explains why Yu Pin must assume Archimedes separately.

C. Decimals, and a third route to the supremum. Rudin disposes of decimals in three lines and says he will never use them. ECNU *defines* reals as decimals, which makes $0.999\dots = 1$ a normalisation rule and turns the supremum into something one computes rather than asserts.

D. The toolkit Rudin never states. The real triangle inequality, its reverse form, the band rule for absolute values, and the ε -principle — four facts he uses constantly and states nowhere.

E. Below Rudin's starting line. He assumes $\mathbb{Q}$ in his second sentence. Tao pays that bill in full, and the itemised cost explains what Rudin's appendix is an instance of.

F. $\mathbb{R}$ is not the only completion of $\mathbb{Q}$. “Filling the holes” presumes a way of measuring closeness. Change the absolute value and the holes move.

G. Real exponents, worked. Rudin’s Exercises 1.6–1.7 done properly, including the two steps his hint glosses.

How to read it. Section A belongs with Rudin’s own appendix, and is best read straight after it. Section B can be read as soon as 1.20 is done, C alongside 1.22, D before Chapter 3, and G with Exercises 1.6–1.7. E and F are optional and answer “what did the assumptions cost?” The exercises are the last section.

NOT IN RUDIN

SUPPLEMENT A — THE UNIQUENESS THEOREM, PROVED

WHY THIS SUPPLEMENT BELONGS TO CHAPTER 1

What Rudin does. Theorem 1.19 asserts that an ordered field $\mathbb{R}$ with the least-upper-bound property exists and contains $\mathbb{Q}$ as a subfield; his appendix builds one from Dedekind cuts. At the end of that appendix he writes: “*It is a fact, which we will not prove here, that any two ordered fields with the least-upper-bound property are isomorphic*” — and concludes that 1.19 characterizes $\mathbb{R}$ completely.

The gap. That sentence is the load-bearing one for the entire book, and it arrives unsupported. Everything after Chapter 1 uses only the *properties* of $\mathbb{R}$ and never the cuts; the licence for that is precisely uniqueness. Without it, a sceptic could ask whether the theorems of Chapters 2–11 are theorems about *Dedekind’s* real numbers rather than about the real numbers.

Where the gap bites. Immediately and permanently. It is why Rudin can tell the reader, in his Preface, to skip the appendix; why Tao may build $\mathbb{R}$ from Cauchy sequences and get the same object; why ECNU may define reals as infinite decimals; and why the six equivalent completeness axioms of the Chapter 1 box are equivalent *as descriptions of one thing* rather than as six different subjects.

What this section supplies. The proof. It is not hard — it is the density of $\mathbb{Q}$ used four times — but it is the argument that retires the construction for good.

S1.1 Theorem. *Let F_1 and F_2 be ordered fields with the least-upper-bound property. Then there is exactly one order-preserving field isomor-*

phism $\varphi : F_1 \rightarrow F_2$, and it fixes the rationals.

SOURCE: Nobody's. Rudin states it and declines to prove it; Tao never introduces the notion of an ordered field at all; Yu Pin defers it (p. 35, "as for uniqueness, we will discuss it in passing later") and delivers only the metric version — the completion of a metric space is unique up to isometry (p. 124). The proof below is supplied here.

Proof. Each of F_1, F_2 contains an isomorphic copy of $\mathbb{Q}$, obtained by sending $n \mapsto 1 + \cdots + 1$ and then $m/n \mapsto m \cdot n^{-1}$; the field axioms make this injective and order-preserving, and we identify $\mathbb{Q}$ with its image in each. By 1.20, applied in each field, both are archimedean and $\mathbb{Q}$ is dense in each.

For $x \in F_1$ put

$$Q_x = \{ q \in \mathbb{Q} : q < x \}, \quad \varphi(x) = \sup_{F_2} Q_x.$$

The set Q_x is nonempty and bounded above in F_2 : by the archimedean property there are integers $m < x < n$, so $m \in Q_x$ and n bounds it. Hence the supremum exists in F_2 by hypothesis, and φ is defined.

φ fixes $\mathbb{Q}$ and preserves order. If $r \in \mathbb{Q}$ then $Q_r = \{ q \in \mathbb{Q} : q < r \}$ has supremum r in F_2 , so $\varphi(r) = r$. If $x < y$ in F_1 , density gives rationals with $x < q < r < y$; then every element of Q_x is below q while $r \in Q_y$, so $\varphi(x) \leq q < r \leq \varphi(y)$. In particular φ is injective.

φ is additive. $Q_x + Q_y \subset Q_{x+y}$ gives $\varphi(x) + \varphi(y) \leq \varphi(x+y)$. Conversely, if $q < x+y$ is rational, density supplies rationals $q_1 < x$ and $q_2 < y$ with $q < q_1 + q_2$ — choose q_1 within $\frac{1}{2}(x+y-q)$ of x and q_2 likewise — so $q \leq \varphi(x) + \varphi(y)$. Taking the supremum over such q gives the reverse inequality.

φ is multiplicative. For $x, y > 0$ the same two-sided argument applies to products of positive rationals; the general case follows from additivity, since $\varphi(-x) = -\varphi(x)$.

φ is onto. Given $z \in F_2$, set $x = \sup_{F_1} \{ q \in \mathbb{Q} : q < z \}$, which exists for the same reason as before. Then Q_x and $\{ q \in \mathbb{Q} : q < z \}$ have the same elements, by density in each field, so $\varphi(x) = z$.

Uniqueness of φ . Any order-preserving field isomorphism fixes 1, hence fixes $\mathbb{Q}$; and an order-preserving map fixing a dense set is determined, since $\psi(x)$ must be an upper bound of $\psi(Q_x) = Q_x$ and below every rational above x . $\square$

HOW THE PROOF FLOWS

1. **Choose the method.** An isomorphism must be *built*, and the only structure both fields are known to share is $\mathbb{Q}$ and the supremum operation. So the map must be defined by naming each x with rationals and reassembling it on the other side: pattern 3 of Chapter 1's strategy index, transport to the supremum.
2. **Buy the supremum.** $\varphi(x) = \sup Q_x$ needs Q_x nonempty and bounded — exactly the two tickets 1.10 demands, and exactly what the archimedean property (1.20(a)) supplies. Note this is

the first use of LUB, and it is used to *define* the map, not to prove anything about it.

3. **Get order for free.** Density (1.20(b)) separates any two points by two rationals, and two rationals are enough to separate their images. Injectivity comes with it.
4. **Do the algebra by squeezing.** Additivity and multiplicativity are each proved by two inequalities: one inclusion of sets gives “ $\leq$,” and density gives “ $\geq$.” There is no computation anywhere — only the fact that rationals approximate from below as closely as one likes.
5. **Run the construction backwards for surjectivity.** The same formula with the roles of F_1 and F_2 exchanged produces the preimage. Symmetry of the hypotheses is what makes this free.
6. **Audit.** The least-upper-bound property is spent twice, once in each field, both times to define a map. The archimedean property and density of $\mathbb{Q}$ are spent repeatedly — but both are *consequences* of LUB by 1.20, so the hypothesis list is exactly Rudin’s. Nothing about cuts, decimals or Cauchy sequences appears anywhere, which is the whole point.

What the theorem licenses

Read the theorem as a permission slip. It says the phrase “the real numbers” denotes a single object, so that:

Constructions are disposable. Rudin’s cuts, Tao’s Cauchy sequences, and ECNU’s infinite decimals are three routes to the same destination. Rudin’s advice in the Preface — that beginning with a construction is “pedagogically unsound (though logically correct)” — is only defensible because of this theorem: you may skip the construction precisely because the construction does not matter.

Axiom choices are disposable too. Chapter 1’s box compared four axiomatisations — Rudin’s LUB, Yu Pin’s nested intervals, Tao’s Cauchy completeness, ECNU’s six-cycle. Uniqueness is what makes those four descriptions of one field rather than four fields.

And it is sharp. Drop any hypothesis and the conclusion fails. Without LUB there are many non-isomorphic archimedean ordered fields — $\mathbb{Q}$ itself, and every subfield of $\mathbb{R}$. Without the ordering there are field isomorphisms of $\mathbb{C}$ that are wildly discontinuous. The next section drops the archimedean property, which LUB was silently supplying, and shows what escapes.

SUPPLEMENT B — HOW MUCH IS THE ARCHIMEDEAN PROPERTY?

WHY THIS SUPPLEMENT BELONGS TO CHAPTER 1

What Rudin does. Theorem 1.20(a) derives the archimedean property — for $x > 0$ and any y there is a positive integer n with $nx > y$ — from the least-upper-bound property, in four lines. His margin remark notes that this is a theorem, not an axiom.

The gap. Deriving a property is not the same as showing it has content. Because Rudin obtains Archimedes free of charge, a reader has no way to see that it is a genuine restriction — that ordered fields exist in which some element exceeds every integer, and that such fields therefore cannot have the LUB property.

Where the gap bites. At Chapter 1's own comparison box, where Yu Pin axiomatises $\mathbb{R}$ as field + order + *archimedean* + nested intervals, and the natural question — why does he need the third hypothesis when Rudin does not? — has no answer inside Rudin. It bites again at S2.7 in the Chapter 2 appendix, where recovering sup from Cauchy completeness needs Archimedes as a separate hypothesis, for exactly the reason exhibited here.

What this section supplies. An explicit non-archimedean ordered field, small enough to hold in the hand, together with the proof that no such field can have the least-upper-bound property.

S1.2 Example. (An ordered field with infinite elements.) Let $\mathbb{R}(t)$ be the field of rational functions $p(t)/q(t)$ with real coefficients, and declare $f > 0$ when $f(x) > 0$ for all sufficiently large real x — equivalently, when the leading coefficients of p and q have the same sign. This makes $\mathbb{R}(t)$ an ordered field in which

$$t > n \quad \text{for every positive integer } n.$$

SOURCE: Supplied here: an explicit non-archimedean ordered field is absent from all four comparison texts. Yu Pin's axiom list (field, order, archimedean, nested intervals) is built to expose the possibility — he notes at p. 26 that (F), (O) and (A) alone are satisfied by $\mathbb{Q}$ — but he gives no example in which (A) fails.

CHECKING THE ORDER, AND READING THE ELEMENT T

That the relation is an order needs only that a nonzero rational function has constant sign near $+\infty$ — it has finitely many zeros — and that the sum and product of two eventually-positive functions are eventually positive. Both are immediate, and they are the two clauses of Definition 1.17.

Now look at t . For any integer n the function $t - n$ is eventually positive, so $t > n$: the single element t exceeds every integer. Its

reciprocal $1/t$ is then a positive element smaller than every $1/n$ — a genuine infinitesimal. Neither is a limit of anything; they are simply elements of the field, and the field's arithmetic is entirely ordinary.

The moral is that “ $\mathbb{R}$ has no infinite or infinitesimal elements” is not a consequence of being an ordered field. It has to be imposed, and Rudin imposes it without naming it, by assuming LUB.

S1.3 Theorem. *An ordered field with the least-upper-bound property is archimedean. Consequently $\mathbb{R}(t)$ does not have the least-upper-bound property: the set of positive integers is bounded above in $\mathbb{R}(t)$ but has no supremum.*

Proof. This is Rudin's 1.20(a) restated, and his proof is the argument. Suppose the set $\mathbb{N}$ of positive integers were bounded above in an ordered field with the LUB property, and let $\alpha = \sup \mathbb{N}$. Since $\alpha - 1 < \alpha$, the element $\alpha - 1$ is not an upper bound, so $n > \alpha - 1$ for some $n \in \mathbb{N}$; but then $n + 1 > \alpha$ and $n + 1 \in \mathbb{N}$, contradicting that α is an upper bound.

Hence in such a field $\mathbb{N}$ is unbounded, which is the archimedean property. In $\mathbb{R}(t)$ the element t bounds $\mathbb{N}$ above, so $\mathbb{R}(t)$ cannot have the LUB property. $\square$

Three fields, and what separates them

Line up the examples Chapter 1 now has:

field	arch.	LUB	what it lacks
$\mathbb{Q}$	yes	no	the sup of $\{p : p^2 < 2\}$ (1.1)
$\mathbb{R}(t)$	no	no	any bound on $\mathbb{N}$
$\mathbb{R}$	yes	yes	—

$\mathbb{Q}$ and $\mathbb{R}(t)$ fail to be $\mathbb{R}$ for genuinely different reasons, and the difference is worth holding on to. $\mathbb{Q}$ has the right *scale* — it is archimedean, its integers are unbounded — and is merely full of holes. $\mathbb{R}(t)$ has no holes of $\mathbb{Q}$'s kind at all in the relevant sense; it fails earlier and more crudely, by containing elements no integer can reach.

This is why Yu Pin's axiom list needs four items where Rudin's needs one extra property. Nested intervals alone does not imply Archimedes — one can have a non-archimedean field in which every nested sequence of intervals has a common point — so he must assume it. Rudin's LUB implies both at once, which is economical but conceals that two separate things were bought.

NOT IN RUDIN

SUPPLEMENT C — DECIMALS, AND A THIRD ROUTE TO THE SUPREMUM

WHY THIS SUPPLEMENT BELONGS TO CHAPTER 1

What Rudin does. Theorem 1.22 disposes of decimals in three lines: given $x > 0$, take the largest integer $n_0 \leq x$, then the largest n_1 with $n_0 + n_1/10 \leq x$, and so on; the resulting decimal converges to x . He then writes that “*we shall never use decimals*” and does not mention them again.

The gap. Two, in opposite directions. Forwards: his appendix builds $\mathbb{R}$ from cuts and stops, never reconnecting the constructed field to the notation every reader has used since childhood — so “what *is* $0.999\dots$?” is never answered. Backwards: he presents LUB as an axiom and its construction as a one-off, leaving the impression that a supremum is an object one may assert but not compute.

Where the gap bites. At 1.22 itself, where the reader is told the decimal expansion exists and is unique but not what makes the uniqueness work; at the end of the appendix, where cuts are abandoned without ever being converted into anything familiar; and in every later argument that approximates a real by rationals — 1.20(b), 1.21, 3.20 — each of which rebuilds an approximation scheme that 1.22 could have supplied once.

What this section supplies. ECNU’s development, which runs the other way: reals *are* infinite decimals, the awkward identification $0.999\dots = 1$ is a normalisation rule chosen in advance, and the supremum is produced by an algorithm — one digit at a time — rather than asserted.

S1.4 Definition. (*Reals as decimals.*) A real number is an infinite decimal $a_0.a_1a_2a_3\dots$; it is rational when the expansion terminates or eventually repeats and irrational otherwise. To make the representation unique, every terminating expansion is rewritten with a tail of nines: for $a_n \neq 0$,

$$a_0.a_1\dots a_n = a_0.a_1\dots(a_n - 1)999\dots,$$

and an integer a_0 is written $(a_0 - 1).999\dots$. Order is then lexicographic: $x > y$ if $a_0 > b_0$, or if the digits agree up to place l and $a_{l+1} > b_{l+1}$.

SOURCE: ECNU 定义 1 (bk. pp. 1–2), where this is the definition of $\mathbb{R}$, not a theorem about it.

$0.999\dots = 1$ IS A CONVENTION, NOT A PARADOX

Students meet this identity as a puzzle. In ECNU’s development it is not a puzzle at all: $0.999\dots$ is the *official name* of 1, chosen so that the lexicographic order of Definition S1.4 is a genuine total order.

Why the choice matters: without a convention, 2.001 and $2.000999\dots$ are two names for one number, and lexicographic comparison of names would declare the first larger. Normalising every terminating decimal

to its nine-tail leaves exactly one name per number, and comparison of names then computes comparison of numbers. Note the convention runs *opposite* to intuition — the canonical form is the infinite one.

Rudin sidesteps all of this by refusing to use decimals. His 1.22 asserts existence and uniqueness of the expansion and moves on; the uniqueness there is bought by taking the *largest* n_k at each stage, which produces the terminating form and so is the opposite normalisation. Neither choice is better — but a reader who has seen only one will find the other's $0.999\dots$ baffling.

S1.5 Definition. (*Deficient and excess approximations.*) For $x = a_0.a_1a_2\dots \geq 0$ put

$$x_n = a_0.a_1\dots a_n, \quad \bar{x}_n = x_n + 10^{-n},$$

so that $x_n \leq x < \bar{x}_n$ with $\bar{x}_n - x_n = 10^{-n}$. The x_n increase and the $\bar{x}_n$ decrease.

SOURCE: ECNU 定义 2 (bk. p. 2).

S1.6 Theorem. $x > y$ if and only if $x_n > \bar{y}_n$ for some n . Consequently, if $x < y$ there is a rational r with $x < r < y$.

SOURCE: ECNU 命题 and 例 1 (bk. pp. 2–3).

Proof. If $x_n > \bar{y}_n$ then $x \geq x_n > \bar{y}_n > y$. Conversely, if $x > y$ then $x - y > 0$, and since $\bar{x}_n - x_n = 10^{-n} \rightarrow 0$ the sandwich $x_n \leq x$, $y < \bar{y}_n$ forces $x_n > \bar{y}_n$ once 10^{-n} is smaller than the gap.

For density, apply this with the roles reversed: $x < y$ gives an n with $\bar{x}_n < y_n$, and

$$r = \frac{1}{2}(\bar{x}_n + y_n)$$

is rational — being the average of two terminating decimals — with $x \leq \bar{x}_n < r < y_n \leq y$. $\square$

Two proofs of density, and what each costs

Compare with Rudin's 1.20(b). He proves density from the archimedean property: stretch the interval by an integer n until $n(y - x) > 1$, catch an integer m in the stretched gap, and divide back. The catching step needs that a set of integers bounded below has a least element — well-ordering — which he uses without comment.

The proof above needs none of that. It truncates both numbers at a common depth and averages, and the only fact used is that the truncations sandwich the number within 10^{-n} . It is also *constructive*: it names the rational, whereas Rudin's argument asserts that a suitable m exists.

The trade is that this proof is available only once reals are decimals. Rudin's works in any archimedean ordered field, which is what he needs, since at 1.20 he has no decimals and does not want any.

S1.7 Theorem. (*The supremum, computed.*) Every nonempty set $S \subset \mathbb{R}$ that is bounded above has a least upper bound, and its decimal digits can be generated one at a time.

SOURCE: ECNU 确界原理 定理 1.1 (bk. pp. 5–7), proved from the decimal definition.

Proof. Assume S contains a nonnegative number. Choose a nonnegative integer n such that $x < n + 1$ for every $x \in S$ while some $a_0 \in S$ has $a_0 \geq n$; this is possible because S is bounded above and nonempty.

Divide $[n, n + 1)$ into ten equal parts. Among the ten, there is a largest digit n_1 for which some element of S reaches $n.n_1$, and then

$$x < n.n_1 + 10^{-1} \quad \text{for all } x \in S, \quad a_1 \geq n.n_1 \quad \text{for some } a_1 \in S.$$

Divide $[n.n_1, n.n_1 + 10^{-1})$ into ten parts and repeat. Inductively one obtains digits $n_1, n_2, \dots$ with

$$x < n.n_1 \cdots n_k + 10^{-k} \quad (x \in S), \quad a_k \geq n.n_1 \cdots n_k \quad \text{for some } a_k \in S. \quad (\text{S1.1})$$

Let $\eta = n.n_1 n_2 n_3 \cdots$ be the real number those digits name.

η is an upper bound: if some $x \in S$ had $x > \eta$, then by S1.6 $x_k > \bar{\eta}_k = n.n_1 \cdots n_k + 10^{-k}$ for some k , contradicting the first half of (S1.1).

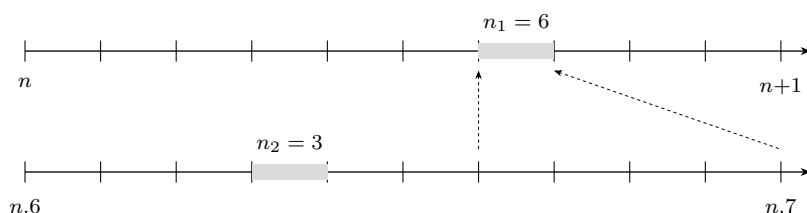
η is least: if $\alpha < \eta$, then by S1.6 there is k with $\eta_k > \bar{\alpha}_k$, and the second half of (S1.1) supplies $a_k \in S$ with $a_k \geq \eta_k > \bar{\alpha}_k \geq \alpha$, so α is not an upper bound. $\square$

HOW THE PROOF FLOWS

1. **Choose the method.** Unlike Rudin, we may not assume the supremum exists — it is the conclusion. So it must be *constructed*, and the only construction material available is decimal digits. The proof is therefore an algorithm: decide the digits one at a time.
2. **Fix the integer part.** Boundedness above gives an integer ceiling; nonemptiness gives a floor. Both hypotheses of 1.10 are spent here, in the first sentence, exactly as in Rudin’s 1.21.
3. **Choose each digit greedily.** At each stage take the largest digit that some element of S still reaches. “Largest that is still attained” is what makes the result least; “still an upper bound” is what makes it an upper bound. The two clauses of (S1.1) are the two halves of Definition 1.8, maintained as loop invariants.
4. **Let the algorithm run forever.** The digits define a real number by S1.4 — and this is the step that would fail over $\mathbb{Q}$, since the digit string need not repeat.
5. **Verify both clauses.** Each is checked by S1.6, which converts a comparison of reals into a comparison of finite truncations.

The invariants were designed so that this conversion closes the argument immediately.

- 6. Audit.** Boundedness and nonemptiness are spent once each, in step 2. The decimal definition of $\mathbb{R}$ is spent once, in step 4. Nothing else enters — in particular no cuts, no Cauchy sequences, and no appeal to LUB, which is what makes this a genuine *third* construction rather than a restatement.



each stage magnifies the chosen tenth and picks the next digit

The supremum computed digit by digit: at each stage the interval is cut into ten, and the algorithm takes the rightmost part that some element of S still reaches. The two conditions (S1.1) — still an upper bound, still attained — are the two clauses of Definition 1.8 carried along as invariants, and the digits they force out spell $\sup S$.

NOT IN RUDIN

SUPPLEMENT D — THE TOOLKIT RUDIN NEVER STATES

WHY THIS SUPPLEMENT BELONGS TO CHAPTER 1

What Rudin does. He proves the triangle inequality twice — at 1.33(e) for complex numbers and at 1.37(e) for $\mathbb{R}^k$ — and defines neighbourhoods once, at 2.18(a), as open balls.

The gap. The real-line versions of both are missing. The real triangle inequality never appears as a standalone statement; the reverse form $||a| - |b|| \leq |a - b|$ appears nowhere at all; the rule that converts $|a| < h$ into a two-sided band is never written down; and the ε -principle — if $a < b + \varepsilon$ for every $\varepsilon > 0$ then $a \leq b$ — is used in almost every proof from Chapter 1 onward without ever being stated.

Where the gap bites. Every ε -argument in the book opens by unpacking an absolute value into a band and closes by letting ε shrink. Rudin performs both moves silently — at 1.20, 1.21, 3.2, 3.3, 4.2 — so a reader learns them by imitation rather than by statement, and the reverse triangle inequality, which is what one needs to bound a difference of norms from below, has to be rediscovered each time it is wanted.

What this section supplies. The four facts, stated once, with

the proof of the triangle inequality that derives it from the band rule rather than by cases.

S1.8 Theorem. (*Absolute value.*) For real a, b and $h > 0$:

- (a) $-|a| \leq a \leq |a|$;
- (b) $|a| < h$ if and only if $-h < a < h$;
- (c) $||a| - |b|| \leq |a \pm b| \leq |a| + |b|$.

SOURCE: ECNU §1.2, six properties of $|\cdot|$ (bk. p. 3).

Proof. (a) and (b) are the definition read in the two cases $a \geq 0$ and $a < 0$.

For (c), add the two instances of (a):

$$-(|a| + |b|) \leq a + b \leq |a| + |b|,$$

and apply (b) in the reverse direction to read that band as $|a + b| \leq |a| + |b|$. Replacing b by $-b$ gives the same bound for $|a - b|$. For the left-hand inequality, write $|a| = |(a - b) + b| \leq |a - b| + |b|$, so $|a| - |b| \leq |a - b|$; exchanging a and b gives $|b| - |a| \leq |a - b|$, and the two together are the claim. $\square$

WHY (B) IS THE LOAD-BEARING ONE

Parts (a) and (c) are the famous ones, but (b) is the one used most often and the one Rudin never states. It is the rule that turns a statement about a *distance* into a statement about *two inequalities*, and back again — and the proof above shows the “back again” direction doing real work: the triangle inequality is proved by producing a band and then reading it as an absolute value, with no case analysis anywhere.

Every ε - δ argument in the book is an application. When Rudin writes “ $|s_n - s| < \varepsilon$ ” and then concludes “ $s - \varepsilon < s_n < s + \varepsilon$,” that is (b), used without comment.

S1.9 Theorem. (*The ε -principle.*) If $a < b + \varepsilon$ for every $\varepsilon > 0$, then $a \leq b$. Equivalently, if $|a - b| < \varepsilon$ for every $\varepsilon > 0$, then $a = b$.

SOURCE: ECNU 例 2 (bk. p. 3) and 习题 1.1 #3.

Proof. Suppose $a > b$. Then $\varepsilon = a - b$ is positive, and the hypothesis applied to this particular ε gives $a < b + (a - b) = a$, which is absurd. $\square$

The proof is three lines and the statement is used everywhere, which is exactly why it deserves a name.⁶²

S1.10 Definition. (*Neighbourhood system.*) For $a \in \mathbb{R}$ and $\delta > 0$ write

$$U(a; \delta) = \{x : |x - a| < \delta\}, \quad U^\circ(a; \delta) = \{x : 0 < |x - a| < \delta\},$$

62 Watch for it in Rudin. The last step of 1.11’s proof, of 1.21, of 3.2(b), and of every uniqueness argument in Chapter 3 is this principle: having shown a quantity is smaller than every positive number, conclude it is zero. Rudin never separates the move from its instances, so it reads as a trick repeated rather than a theorem applied.

and $U_+(a; \delta) = [a, a + \delta)$, $U_-(a; \delta) = (a - \delta, a]$ for the one-sided versions; $U(+\infty) = \{x : x > M\}$, $U(-\infty) = \{x : x < -M\}$ and $U(\infty) = \{x : |x| > M\}$ for M large.

SOURCE: ECNU §2.1 (bk. pp. 4–5).

One notation, and the four definitions it collapses

Rudin introduces neighbourhoods once, at 2.18(a), and only the two-sided ball. Every limit notion he later defines is then written out separately: 4.1 for $\lim_{x \rightarrow p}$, 4.25 for one-sided limits, 4.33 for limits at $\pm\infty$. The three definitions look unrelated on the page.

With the system above they are one sentence. In every case, $\lim f = A$ means

for each $U(A; \varepsilon)$ there is $U^\circ(a; \delta)$ with $f(U^\circ(a; \delta)) \subset U(A; \varepsilon)$,

and the four notions differ only in which neighbourhood of a is used: punctured, right-hand, left-hand, or a neighbourhood of $+\infty$. The Chapter 4 appendix reaches the same unification from the other direction, through Tao’s “limit along a subset” (S4.2); ECNU gets there by naming the neighbourhoods instead of the index sets.

NOT IN RUDIN

SUPPLEMENT E — BELOW RUDIN’S STARTING LINE

WHY THIS SUPPLEMENT BELONGS TO CHAPTER 1

What Rudin does. The second sentence of the book fixes his starting line: *“we shall not, however, enter into any discussion of the axioms that govern the arithmetic of the integers, but assume familiarity with the rational numbers.”* Everything in Chapter 1 is built on $\mathbb{Q}$, taken as given.

The gap. A reader cannot tell whether that assumption is cheap or expensive. Is $\mathbb{Q}$ a few lines of set theory, or a serious construction? And if it is serious, is Rudin’s Dedekind-cut appendix a special trick or an instance of something standard?

Where the gap bites. At the appendix, which looks like a one-off contrivance — why *sets of rationals*, of all things? — and at every use of induction in the book, since Rudin never says what property of $\mathbb{N}$ makes induction legitimate.

What this section supplies. Tao’s construction of $\mathbb{N}$, $\mathbb{Z}$ and $\mathbb{Q}$, priced honestly; and the observation that his three constructions and Rudin’s cuts are the same manoeuvre performed four times.

S1.11 Definition. (*The Peano axioms.*) $\mathbb{N}$ is a set with a distinguished element 0 and a successor operation $n \mapsto n++$ satisfying: 0 is not a

successor; the successor operation is injective; and if $P(0)$ holds and $P(n)$ implies $P(n++)$ for every n , then $P(n)$ holds for all n .

SOURCE: Tao I, Axioms 2.1–2.5 (bk. pp. 16–21).

What Rudin’s one sentence costs

Tao pays the bill in full, and the itemisation is instructive.

Even addition is a theorem first. Before $+$ can be defined by recursion one must prove that recursive definitions are legitimate at all — Tao’s Proposition 2.1.16. Induction is what licenses it. This is the answer to “what makes $\mathbb{N}$ special”: not that its elements are small, but that it supports recursion.

$\mathbb{Z}$ is a quotient. An integer is a formal difference $a - b$ of naturals, with $a - b = c - d$ declared to mean $a + d = c + b$. One then checks that $+$, $\times$ and $<$ are well defined on the classes.

$\mathbb{Q}$ is another quotient. A rational is a formal quotient $a//b$ with $b \neq 0$, and $a//b = c//d$ means $ad = cb$. Same checking again.

So Rudin’s “assume familiarity with the rational numbers” is roughly thirty-five pages, two quotient constructions and a theorem about recursion. His Preface defends skipping it as avoiding what is “pedagogically unsound (though logically correct),” and after S1.1 — uniqueness — that defence is airtight: the construction cannot matter, so it may be skipped.

ONE MANOEUVRE, PERFORMED FOUR TIMES

Look at what $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$ have in common as constructions:

to build	formal symbol	identified when
$\mathbb{Z}$ from $\mathbb{N}$	$a - b$	$a + d = c + b$
$\mathbb{Q}$ from $\mathbb{Z}$	$a//b$	$ad = cb$
$\mathbb{R}$ from $\mathbb{Q}$ (Tao)	$\text{LIM } a_n$	$\lim a_n - b_n = 0$
$\mathbb{R}$ from $\mathbb{Q}$ (Rudin)	a cut α	— (the set <i>is</i> the number)

The first three are identical in shape: introduce a symbol for the operation you wish you could perform, declare when two symbols name the same thing, and prove the arithmetic descends to the classes. Tao says so explicitly — LIM is introduced as “a new formal symbol, similar to $-$ and $//$ ” — and adds that once formal limits are shown to be actual limits, they are never needed again. Rudin’s cuts are the odd one out: a cut is not a symbol standing for something absent but a set that *is* the number. That is why his appendix needs no equivalence classes and why Step 1 spends so long on property (III) — the normalisation that makes the representation unique is built into the definition rather than quotiented out afterwards.

SUPPLEMENT F — $\mathbb{R}$ IS NOT THE ONLY COMPLETION OF $\mathbb{Q}$

WHY THIS SUPPLEMENT BELONGS TO CHAPTER 1

What Rudin does. Section 1.1 exhibits the gap in $\mathbb{Q}$ — the set $\{p : p^2 < 2\}$ with no rational supremum — and 1.19 supplies $\mathbb{R}$ as the field that fills it. The narrative is: $\mathbb{Q}$ has holes, $\mathbb{R}$ is $\mathbb{Q}$ with the holes filled.

The gap. That narrative quietly presumes a notion of “hole,” and a hole is only visible relative to a way of measuring closeness. Rudin uses the ordinary absolute value throughout and never says it is a choice.

Where the gap bites. Nowhere in this book — which is exactly why it is worth one page. A reader who never learns that the choice existed will later meet $\mathbb{Q}_p$ and find it inexplicable, and will also misread the word “complete” in Chapter 3 as a property of a *set* rather than of a *metric*.

What this section supplies. A second absolute value on $\mathbb{Q}$, and the different field it completes to.

S1.12 Definition. (*The p -adic absolute value.*) Fix a prime p . Every nonzero rational x is uniquely $x = p^v m/n$ with m, n coprime to p ; put $v(x) = v$ and

$$|x|_p = p^{-v(x)}, \quad d_p(x, y) = |x - y|_p.$$

Then d_p is a metric on $\mathbb{Q}$, and $(\mathbb{Q}, d_p)$ is not complete; its completion is the field $\mathbb{Q}_p$ of p -adic numbers.

SOURCE: Yu Pin p. 125, as a worked corollary of his completion theorem (Thm 77, pp. 123–124).

What “complete” is a property of

Under d_p a number is *small* when it is divisible by a high power of p . With $p = 5$, the numbers 5, 25, 125, ... march to zero, and $1/5$ is enormous. The metric is genuinely a metric — and it satisfies something stronger than the triangle inequality, $|x+y|_p \leq \max(|x|_p, |y|_p)$, which makes its geometry unrecognisable: every triangle is isosceles, and every point of a ball is its centre.

Complete $(\mathbb{Q}, d_p)$ and you get $\mathbb{Q}_p$, not $\mathbb{R}$. So:

“ $\mathbb{R}$ is $\mathbb{Q}$ with the holes filled” is *incomplete as a description*. The correct statement is that $\mathbb{R}$ is $\mathbb{Q}$ completed *with respect to the ordinary absolute value*. Change the absolute value and the holes move.

This is why Chapter 3 defines completeness for a metric space and not for a set. Rudin’s 3.11 is stated for metric spaces precisely because the property depends on d ; his Exercise 3.24 constructs the completion in general. The Chapter 2 appendix (S2.8) gives

the definition, and Yu Pin’s universal property is what makes “the completion” legitimate language.

There is a theorem lurking here that neither book states: Ostrowski’s, that the ordinary absolute value and the p -adic ones are the *only* absolute values on $\mathbb{Q}$ up to equivalence. So $\mathbb{R}$ and the $\mathbb{Q}_p$ are the complete list of completions — Rudin’s choice was not arbitrary, merely unremarked.

NOT IN RUDIN

SUPPLEMENT G — REAL EXPONENTS, WORKED

WHY THIS SUPPLEMENT BELONGS TO CHAPTER 1

What Rudin does. Theorem 1.21 proves that every positive real has a unique positive n th root. Exercises 1.6 and 1.7 then ask the reader to build b^x for all real x and establish its laws, with a hint.

The gap. The exercises are substantially harder than they look, and two steps in particular are glossed by the hint: that $b^{m/n}$ does not depend on how the fraction m/n is written, and that the limit defining b^x for irrational x exists at all.

Where the gap bites. Every later appearance of x^α , a^x and $\log$ — 3.20’s special limits, 8.6’s exponential — rests on this exercise having been done. A reader who skipped it has a gap under the whole of Chapters 3 and 8.

What this section supplies. Tao’s worked version of Rudin’s exercise, with both glossed steps proved.

S1.13 Theorem. (*Rational exponents.*) For $x > 0$ and n a positive integer put $x^{1/n} = \sup\{y \geq 0 : y^n \leq x\}$. The set is nonempty and bounded above, so the supremum exists; it satisfies $(x^{1/n})^n = x$, and for integers a and positive integers b the definition $x^{a/b} = (x^{1/b})^a$ is independent of the representation of the fraction. With these, $x^{q+r} = x^q x^r$ and $(x^q)^r = x^{qr}$ for all rational q, r .

SOURCE: Tao I, Def. 5.6.4–5.6.7 and Lemmas 5.6.5–5.6.9 (bk. pp. 123–125) — Rudin’s Exercise 1.6 worked out.

Proof. The set $\{y \geq 0 : y^n \leq x\}$ contains 0 and is bounded above by $\max\{1, x\}$, since $y > \max\{1, x\}$ forces $y^n \geq y > x$. So the supremum exists. Because $t \mapsto t^n$ is strictly increasing on $[0, \infty)$, the set is the interval $[0, y]$ where y is the unique positive n th root supplied by 1.21; hence $x^{1/n} = y$ and $(x^{1/n})^n = x$.

For independence, suppose $a/b = c/d$, so $ad = cb$. Both $(x^{1/b})^a$ and $(x^{1/d})^c$ are positive, and raising each to the power bd gives x^{ad} and x^{cb}

respectively — equal numbers. Since 1.21 provides *exactly one* positive bd th root, the two agree.

The laws $x^{q+r} = x^q x^r$ and $(x^q)^r = x^{qr}$ now follow by writing q and r over a common denominator and using the corresponding integer identities together with uniqueness in 1.21. $\square$

Rudin's 1.21 already gives the root; the content here is the *independence*, which needs the uniqueness half of 1.21 to force $(x^{1/bd})^{ad} = (x^{1/b})^a$ whenever $a/b = c/d$.⁶³

S1.14 Theorem. (*Irrational exponents.*) Let $x > 0$ and let α be real. If (q_n) is any sequence of rationals with $q_n \rightarrow \alpha$, then (x^{q_n}) converges, and the limit does not depend on the sequence chosen. One may therefore define $x^\alpha = \lim_n x^{q_n}$.

SOURCE: Tao I, Lemma 6.7.1 (bk. pp. 152–153) — Rudin's Exercise 1.7.

Proof. Take $x \geq 1$ (for $0 < x < 1$ apply the result to $1/x$). If q, r are rationals with $|q|, |r| \leq M$ and $|q - r| \leq 1/K$, then

$$|x^q - x^r| = x^r |x^{q-r} - 1| \leq x^M (x^{1/K} - 1),$$

and $x^{1/K} \rightarrow 1$ as $K \rightarrow \infty$ by 3.20(b). Now let $q_n \rightarrow \alpha$ with each q_n rational. The sequence (q_n) is Cauchy and bounded, so the displayed estimate makes (x^{q_n}) Cauchy, and it converges by completeness of $\mathbb{R}$ (3.11).

If (q'_n) is a second rational sequence tending to α , interleave the two into a single sequence; it also tends to α , so its image converges, and the two original limits — being limits of subsequences of a convergent sequence — coincide. $\square$

THE ESTIMATE THAT MAKES THE DEFINITION LEGAL

Everything rests on one inequality. For rationals q, r with $|q - r| \leq 1/K$ and both at most M in absolute value,

$$|x^q - x^r| \leq x^M (x^{1/K} - 1) \quad (x \geq 1),$$

and $x^{1/K} \rightarrow 1$ as $K \rightarrow \infty$ — which is Rudin's own 3.20(b) in disguise. So (x^{q_n}) is Cauchy whenever (q_n) is, and completeness of $\mathbb{R}$ supplies the limit. Interleaving two sequences with the same limit shows the value is independent of the choice, exactly as in the uniqueness argument of S1.1.

Two things are worth noticing. First, the definition needs completeness of $\mathbb{R}$, so real exponentiation is downstream of 1.19 — it could not have been introduced in $\mathbb{Q}$. Second, this is the same three-step pattern used throughout: *approximate by rationals, show the approximations are Cauchy, pass to the limit*. S1.1 uses it to build an isomorphism, S1.7 to build a supremum, and Chapter 4's S4.17 to extend a uniformly continuous function. Recognising the pattern is worth more than any one instance.

63 Rudin's hint for Exercise 1.6 says to prove $(b^m)^{1/n} = (b^p)^{1/q}$ when $m/n = p/q$ and then define $b^{m/n}$ to be that common value. That single sentence is the independence step, and it is where all the work is — both sides are n th roots, so uniqueness in 1.21 is what identifies them.

APPENDIX EXERCISES

HOW THESE DIFFER FROM RUDIN'S TWENTY

Rudin's own Exercises 1.1–1.20 end the chapter proper. They are almost all verifications inside a structure already given: check a field axiom, compute with cuts, establish an inequality. These fifteen come from the comparison texts and ask two other kinds of question — what the axioms *cost*, and how to actually compute with them.

Group I (1–5): the toolkit. Drills on the four facts of Supplement D, plus the two supremum lemmas every later chapter uses and Rudin never assigns. Exercise 4 is the first place in this book where anyone is asked to *compute* a supremum.

Group II (6–10): what the axioms buy. Where each hypothesis is spent — induction failing on $\mathbb{Z}$, the supremum principle producing Archimedes, and the disconcerting question of whether one can prove any real numbers exist at all.

Group III (11–15): decimals and construction. Exactly which reals have two decimal names, the supremum built by hand, and a game.

Marked ★: harder, or requiring a result from a later chapter.

64 The ε -principle of S1.9 as a drill, and the single most-used move in the book. The proof is one line by contradiction — take $\varepsilon = |a - b|$ — but the habit it builds is what matters: to prove two things equal, show their distance is smaller than every positive number. Rudin uses this at 1.11, 1.21, 3.2(b) and throughout Chapter 3 without ever stating it.

65 Read each expression as a total distance to fixed points on the line. In (a) the sum of distances to 1 and 2 is least — and equal to the gap — exactly when x lies between them. In (b) the middle point is free to be hit exactly, so equality occurs only at $x = 2$: the minimiser of a sum of distances is the median. A geometric habit worth acquiring before Chapter 2 makes distance abstract.

66 Two right triangles sharing the leg a : the hypotenuses cannot differ by more than the other legs do. Algebraically it says $t \mapsto \sqrt{a^2 + t^2}$ never stretches distances — it is what Chapter 4 will call a Lipschitz function with constant 1. Rudin's Chapter 1 exercises contain nothing geometric; this one rewards a picture.

67 Rudin's Chapter 1 never asks anyone to compute a supremum, and computing four of them is the fastest way to internalise that Definition 1.8 has two clauses — an upper bound, and no smaller one. Note (b), where the set is unbounded above and the answer is $+\infty$ only in the extended sense of 1.23; and (c), where the supremum is 1 and is not attained, while every neighbourhood of it is crowded.

S1.1. Let $a, b \in \mathbb{R}$. Prove that if $|a - b| < \varepsilon$ for every $\varepsilon > 0$, then $a = b$.⁶⁴

SOURCE: ECNU 习题 1.1 #3 (bk. p. 4).

S1.2. Prove that for all $x \in \mathbb{R}$: (a) $|x - 1| + |x - 2| \geq 1$; (b) $|x - 1| + |x - 2| + |x - 3| \geq 2$. Determine in each case exactly when equality holds.⁶⁵

SOURCE: ECNU 习题 1.1 #5 (bk. p. 4).

S1.3. Let $a, b, c > 0$. Prove $|\sqrt{a^2 + b^2} - \sqrt{a^2 + c^2}| \leq |b - c|$, and give the inequality a geometric meaning.⁶⁶

SOURCE: ECNU 习题 1.1 #6 (bk. p. 4).

S1.4. For each set, find sup and inf and verify the answer from Definition 1.8: (a) $\{x : x^2 < 2\}$; (b) $\{n! : n \in \mathbb{N}\}$; (c) $\{x \text{ irrational} : 0 < x < 1\}$; (d) $\{1 - 2^{-n} : n \in \mathbb{N}\}$.⁶⁷

SOURCE: ECNU 习题 1.2 #4 (bk. p. 8).

S1.5. Let A, B be nonempty and bounded, and $A + B = \{x + y : x \in$

$A, y \in B\}$. Prove $\sup(A+B) = \sup A + \sup B$ and $\inf(A+B) = \inf A + \inf B$.⁶⁸

SOURCE: ECNU 习题 1.2 #7 (bk. p. 8).

S1.6. Show that the principle of induction does not apply to $\mathbb{Z}$: exhibit a property $P(n)$ of integers such that $P(0)$ holds and $P(n)$ implies $P(n+1)$ for every integer n , yet $P(n)$ fails for some integer.⁶⁹

SOURCE: Tao I, Ex. 4.1.8 (bk. p. 81).

S1.7. Assuming only the field axioms and the order axioms, prove that the least-upper-bound property implies the archimedean property.⁷⁰

SOURCE: Yu Pin, homework A7 (p. 36).

S1.8. ★ Suppose $a > 0$. Can you prove that the open interval $(0, a)$ is nonempty?⁷¹

SOURCE: Yu Pin, 思考题 (p. 22).

S1.9. Prove that the ordered field $\mathbb{R}(t)$ of Supplement B really is ordered — that the sum and product of eventually-positive rational functions are eventually positive, and that trichotomy holds — and then verify directly that the set $\mathbb{N}$ has no least upper bound in it.⁷²

SOURCE: Supplied here; see the provenance note at S1.2.

S1.10. ★ Prove that, given the field and order axioms together with the archimedean property, the least-upper-bound property and the nested-interval property are equivalent. Then explain why the archimedean hypothesis cannot be dropped from this statement.⁷³

SOURCE: Yu Pin, Remark 4 (p. 28) and Thm 5 (pp. 26–27).

S1.11. Prove that a real number has two decimal representations if it is a terminating decimal — that is, if $x = n/10^m$ for integers n, m — and exactly one otherwise. Show also that the only decimal representations of 1 are $1.000\dots$ and $0.999\dots$ ⁷⁴

SOURCE: Tao I, Ex. B.2.2–B.2.3 (bk. p. 338); Yu Pin, homework E1 (p. 506), which states it as a four-way equivalence.

S1.12. Prove the supremum principle by the digit construction of S1.7 for a set of your own choosing — for instance $S = \{x : x^2 < 2\}$ — and carry the algorithm far enough to produce the first four digits of $\sup S$.⁷⁵

SOURCE: ECNU 定理 1.1 (bk. pp. 5–7).

68 The workhorse lemma behind every limit-algebra proof, and it is not in Rudin's exercise set. Both halves are needed: " $\leq$ " because $\sup A + \sup B$ bounds $A + B$, and " $\geq$ " because any smaller bound would fail for some x near $\sup A$ and y near $\sup B$ — which is the second clause of 1.8 doing its work. Compare 3.3, where the same statement for limits gets a whole theorem.

69 Take $P(n)$ to be " $n \geq 0$." One line, and a permanent lesson: induction is not a general logical principle but a specific axiom about $\mathbb{N}$, and what makes it work is that $\mathbb{N}$ has a starting point with nothing before it. Rudin uses induction freely throughout Chapter 1 — at 1.1, at 1.21, in the appendix — and never says what licenses it.

70 This is Rudin's Theorem 1.20(a), but posed inside an axiom system where Archimedes is independent, so that the exercise shows what is being bought rather than merely how. Compare S1.3, and note the shape of the argument: if $\mathbb{N}$ were bounded, $\sup \mathbb{N} - 1$ would fail to be an upper bound, and the integer beating it has a successor beating the supremum.

71 Disconcerting, and meant to be. On the field and order axioms alone the only elements you have names for are 0, 1 and -1 , so producing a real strictly between 0 and a is not the triviality it looks like — you must manufacture new elements, and manufacturing elements in bulk is precisely what the completeness axiom is for. Rudin's 1.19 hands the reader a fully populated $\mathbb{R}$ by fiat, which is why the question cannot even be asked inside his development. Yu Pin's own comment: "you will see the shadow of the Dedekind cut here."

72 The verification behind S1.2 and S1.3. For trichotomy, a nonzero rational function has finitely many zeros, so it has constant sign near $+\infty$. For the second part, show that if f bounded $\mathbb{N}$ above then so would $f - 1$, since $f - 1 > n$ whenever $f > n + 1$ — so no candidate can be least. The example is the one thing Chapter 1 lacks: a field where Archimedes visibly fails.

73 The precise form of Chapter 1's comparison box. One direction is a bisection; the other is the argument of S2.3 in the Chapter 2 appendix. For the last part, look at S1.2: nested intervals alone cannot rule out infinite elements, so it cannot imply Archimedes, whereas LUB implies both at once. This is exactly why Yu Pin lists four axioms where Rudin needs one.

74 The statement Rudin's 1.22 declines to make. He asserts that the expansion exists and is unique, having silently fixed the non-terminating digits by always choosing the largest digit at each stage; ECNU fixes the opposite convention (S1.4). The exercise is to see that the ambiguity is real, is confined to exactly the terminating decimals, and is always a choice between a 0-tail and a 9-tail.

75 The point is to watch Definition 1.8's two clauses act as loop invariants: at each stage one keeps the largest digit that some element still reaches. Doing it once on $\{x : x^2 < 2\}$ produces 1.414 and makes concrete what Rudin's 1.1 says is missing from $\mathbb{Q}$ — the digits are perfectly well defined at every finite stage, and only the infinite string fails to be rational.

76 The same construction used in S2.7 to recover sup from Cauchy completeness, here isolated. Uniqueness of m is where the archimedean property enters — it is what guarantees the set of integers k with k/n an upper bound is nonempty and bounded below. Compare with exercise 12: one construction walks in steps of 10^{-k} , the other in steps of $1/n$.

S1.13. ★ Construct the supremum of a nonempty bounded set by successive rational approximations, *without* invoking the least-upper-bound property: for each n find the unique integer m such that m/n is an upper bound while $(m - 1)/n$ is not, and prove m is unique.⁷⁶

SOURCE: Tao I, Ex. 5.5.2–5.5.3 (bk. p. 121).

S1.14. Prove that $b^{m/n}$ is well defined — that $(b^p)^{1/q} = (b^m)^{1/n}$ whenever $m/n = p/q$ — and deduce the law $b^{r+s} = b^r b^s$ for rational r, s .⁷⁷

SOURCE: Tao I, Lemma 5.6.8 (bk. p. 124); Rudin's Ex. 1.6.

S1.15. ★ (*A game.*) Two players alternately choose nested closed intervals $I_1 \supset I_2 \supset \cdots$, each at most half the length of its predecessor, so that the intersection is a single point x . The first player wins if x is rational, the second if x is irrational. Who has a winning strategy?⁷⁸

SOURCE: Yu Pin, problem G (p. 69), the Banach–Mazur game.

Chapter 2

BASIC TOPOLOGY

WHY THIS CHAPTER EXISTS

Chapter 1 secured a number system with no gaps. That is a statement about $\mathbb{R}$ alone, and almost nothing in analysis is about $\mathbb{R}$ alone — limits, continuity and integration all concern how points sit *near* one another. This chapter builds the language for nearness, and it builds it once, for a general metric space, so that $\mathbb{R}$, $\mathbb{C}$, $\mathbb{R}^k$ and later spaces of functions are all covered by the same theorems.

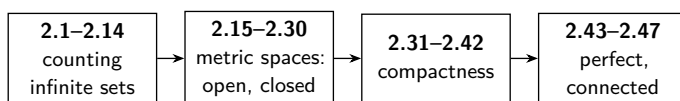
The chapter's centre of gravity is **compactness** (2.31–2.42). It is the property that makes “infinitely many” behave like “finitely many,” and it is what will later force a continuous function on $[a, b]$ to be bounded (4.15), to attain its bounds (4.16), and to be uniformly continuous (4.19). If you take one thing from Chapter 2, take Theorem 2.41.

77 Rudin's Exercise 1.6, and the independence step is the whole of it. Both sides are positive n qth roots of b^{mq} , so the uniqueness half of 1.21 identifies them — one more instance of the pattern flagged at 1.21, where an existence-and- uniqueness theorem is used entirely for its uniqueness. See S1.13.

78 Everything needed lives in Chapter 1: nested intervals, density of $\mathbb{Q}$, and the uncountability that follows from S2.17 in the Chapter 2 appendix. The second player wins, and the reason is the same reason the diagonal argument works — the first player must commit to a countable list of targets while the second retains room to dodge each one in turn. A pleasant way to see that “most” reals are irrational without ever using the word measure.

HOW IT IS ORGANIZED, AND WHY IN THAT ORDER

Five sections, and the first is not topology at all.



The first section is not topology. Countability is set theory, and Rudin puts it here because he needs it in two specific places: to know that $\mathbb{Q}$ is countable (used constantly), and to know that $\mathbb{R}$ is not (2.43, and the Cantor set at 2.44). You could read §§2.15–2.47 first and refer back.

The definitions are chosen for what they enable, not for what they describe. “Open,” “closed,” “limit point” will look like arbitrary vocabulary on first reading. They are not: each is exactly the hypothesis some later theorem needs. Watch where each is used rather than trying to intuit it in advance.

Compactness is defined by open covers (2.32), which is the strangest definition in the book and the one students resist hardest. Rudin's reason appears only at 2.41, where he proves that for subsets of $\mathbb{R}^k$ it coincides with two more familiar

conditions. Until then, treat it as a formal property and watch what it does.

Everything is done for a general metric space where it can be, and for $\mathbb{R}^k$ where it cannot. That division is deliberate — 2.41 and 2.42 are *false* in general metric spaces, and knowing which theorems need $\mathbb{R}^k$ is worth more than knowing the theorems.

FINITE, COUNTABLE, AND UNCOUNTABLE SETS

WHAT THIS SECTION IS FOR | Two facts are needed later and everything here exists to produce them: $\mathbb{Q}$ is countable (Corollary to 2.13) and the set of 0–1 sequences is not (2.14). The first makes $\mathbb{Q}$ small enough to enumerate — spent in Exercises 2.18 and 2.29, in Chapter 3, and wherever a countable dense set is wanted. The second is what makes $\mathbb{R}$ genuinely bigger than $\mathbb{Q}$, and it reappears as 2.43.

The section also fixes notation for functions (2.1–2.2) that the whole book uses. Read 2.2 carefully even if the rest is familiar: Rudin’s f^{-1} is defined for *sets* and does not presuppose that f has an inverse.

We begin this section with a definition of the function concept.

2.1 Definition. Consider two sets A and B , whose elements may be any objects whatsoever, and suppose that with each element x of A there is associated, in some manner, an element of B , which we denote by $f(x)$. Then f is said to be a *function* from A to B (or a *mapping* of A into B). The set A is called the *domain* of f (we also say f is defined on A) and the elements $f(x)$ are called the *values* of f . The set of all values of f is called the *range* of f .⁷⁹

2.2 Definition. Let A and B be two sets and let f be a mapping of A into B . If $E \subset A$, $f(E)$ is defined to be the set of all elements $f(x)$, for $x \in E$. We call $f(E)$ the *image* of E under f . In this notation, $f(A)$ is the range of f . It is clear that $f(A) \subset B$. If $f(A) = B$, we say that f maps A *onto* B . (Note that, according to this usage, *onto* is more specific than *into*.)

If $E \subset B$, $f^{-1}(E)$ denotes the set of all $x \in A$ such that $f(x) \in E$. We call $f^{-1}(E)$ the *inverse image* of E under f .⁸⁰ If $y \in B$, $f^{-1}(y)$ is the set of all $x \in A$ such that $f(x) = y$. If, for each $y \in B$, $f^{-1}(y)$ consists of at most one element of A , then f is said to be a *1-1* (one-to-one) mapping of A into B . This may also be expressed as follows: f is a 1-1 mapping of A into B provided that $f(x_1) \neq f(x_2)$ whenever $x_1 \neq x_2$, $x_1 \in A$, $x_2 \in A$.

79 “Associated, in some manner” is not a definition, and Rudin knows it — he is declining to build set theory, exactly as he declined to build $\mathbb{Z}$ in Chapter 1. One repair is to define a function as a set of ordered pairs in which each x appears exactly once; another, Tao’s (his 3.3.1), keeps functions primitive and defines them by any property $P(x, y)$ passing the “vertical line test,” with the ordered-pair encoding relegated to his Exercise 3.5.10. Nothing in this book depends on the choice.

80 f^{-1} here is an operation on subsets, and it is always defined — for every f , invertible or not. $f^{-1}(E)$ is simply “everything that lands in E ,” possibly empty. Do not read it as an inverse function. The clash of notation is universal and permanent, and the only defence is to check whether the argument is a set or a point.

(The notation $x_1 \neq x_2$ means that x_1 and x_2 are distinct elements; otherwise we write $x_1 = x_2$.)

Why inverse images, and not images, are the useful notion

Nothing yet suggests it, but f^{-1} is the better-behaved of the two operations, and Chapter 4 will turn on the fact. Inverse images respect every set operation:

$$\begin{aligned} f^{-1}(E \cup F) &= f^{-1}(E) \cup f^{-1}(F), \\ f^{-1}(E \cap F) &= f^{-1}(E) \cap f^{-1}(F), \\ f^{-1}(E^c) &= (f^{-1}(E))^c. \end{aligned}$$

Images respect only the first. $f(E \cap F) \subset f(E) \cap f(F)$ can be strict — take f constant and E, F disjoint — and images do not interact with complements at all.

That asymmetry is why Theorem 4.8 characterises continuity by “ f^{-1} of every open set is open” rather than by anything about images — no image-based version could work, since a constant map is continuous yet its images are single points, never open. Even the closure relation survives only one-sidedly for images: Exercise 4.2 proves $f(\overline{E}) \subset \overline{f(E)}$ and asks for an example where the inclusion is proper.

2.3 Definition. If there exists a 1-1 mapping of A onto B , we say that A and B can be put in *1-1 correspondence*, or that A and B have the same *cardinal number*, or, briefly, that A and B are *equivalent*, and we write $A \sim B$. This relation clearly has the following properties:

It is reflexive: $A \sim A$.

It is symmetric: If $A \sim B$, then $B \sim A$.

It is transitive: If $A \sim B$ and $B \sim C$, then $A \sim C$.

Any relation with these three properties is called an *equivalence relation*.

2.4 Definition. For any positive integer n , let $\mathbb{N}_n$ be the set whose elements are the integers $1, 2, \dots, n$; let $\mathbb{N}$ be the set consisting of all positive integers. For any set A , we say:

- (a) A is *finite* if $A \sim \mathbb{N}_n$ for some n (the empty set is also considered to be finite).
- (b) A is *infinite* if A is not finite.
- (c) A is *countable* if $A \sim \mathbb{N}$.
- (d) A is *uncountable* if A is neither finite nor countable.
- (e) A is *at most countable* if A is finite or countable.

Countable sets are sometimes called *enumerable*, or *denumerable*.⁸¹

For two finite sets A and B , we evidently have $A \sim B$ if and only if A and B contain the same number of elements. For infinite sets, however,

81 Rudin’s “countable” means *countably infinite* – it excludes *finite sets*. Many other books use “countable” for what he calls “at most countable,” and then say “countably infinite” for his sense. When you read a theorem elsewhere, check which convention is in force; several of Rudin’s statements become false under the other one.

the idea of “having the same number of elements” becomes quite vague, whereas the notion of 1-1 correspondence retains its clarity.

2.5 Example. Let $\mathbb{Z}$ be the set of all integers. Then $\mathbb{Z}$ is countable. For, consider the following arrangement of the sets $\mathbb{Z}$ and $\mathbb{N}$:

$$\begin{array}{cccccccc} \mathbb{Z}: & 0, & 1, & -1, & 2, & -2, & 3, & -3, & \dots \\ \mathbb{N}: & 1, & 2, & 3, & 4, & 5, & 6, & 7, & \dots \end{array}$$

We can, in this example, even give an explicit formula for a function f from $\mathbb{N}$ to $\mathbb{Z}$ which sets up a 1-1 correspondence:

$$f(n) = \begin{cases} \frac{n}{2} & (n \text{ even}), \\ -\frac{n-1}{2} & (n \text{ odd}). \end{cases}$$

$$\begin{array}{cccccccc} \mathbb{Z} & 0 & 1 & -1 & 2 & -2 & 3 & -3 & \dots \\ \uparrow & \uparrow & \uparrow & \uparrow & \uparrow & \uparrow & \uparrow & \uparrow & \\ \mathbb{N} & 1 & 2 & 3 & 4 & 5 & 6 & 7 & \dots \end{array}$$

Counting the integers by alternating sides of 0. Every integer is reached exactly once, which is all “countable” asks.

2.6 Remark. A finite set cannot be equivalent to one of its proper subsets. That this is, however, possible for infinite sets, is shown by Example 2.5, in which $\mathbb{N}$ is a proper subset of $\mathbb{Z}$.

In fact, we could replace Definition 2.4(b) by the statement: A is infinite if A is equivalent to one of its proper subsets.⁸²

2.7 Definition. By a *sequence*, we mean a function f defined on the set $\mathbb{N}$ of all positive integers. If $f(n) = x_n$, for $n \in \mathbb{N}$, it is customary to denote the sequence f by the symbol $(x_n)_{n=1}^{\infty}$, or more briefly by (x_n) , or sometimes by $x_1, x_2, x_3, \dots$. The values of f , that is, the elements x_n , are called the *terms* of the sequence. If A is a set and if $x_n \in A$ for all $n \in \mathbb{N}$, then (x_n) is said to be a *sequence in A* , or a *sequence of elements of A* .

Note that the terms $x_1, x_2, x_3, \dots$ of a sequence need not be distinct.⁸³

Since every countable set is the range of a 1-1 function defined on $\mathbb{N}$, we may regard every countable set as the range of a sequence of distinct terms. Speaking more loosely, we may say that the elements of any countable set can be “arranged in a sequence.”

Sometimes it is convenient to replace $\mathbb{N}$ in this definition by the set of all nonnegative integers, i.e., to start with 0 rather than 1.

2.8 Theorem. *Every infinite subset of a countable set A is countable.*

Proof. Suppose $E \subset A$, and E is infinite. Arrange the elements x of A in a sequence (x_n) of distinct elements. Construct a sequence (n_k) as follows:

82 This is Dedekind’s definition of infinity, and taking it as primary is attractive because it needs no reference to $\mathbb{N}$ at all. The two definitions agree, but proving they agree requires a form of the axiom of choice — which is why Rudin offers the Dedekind version as an aside rather than adopting it.

83 A sequence is not its range. $(1, 1, 1, \dots)$ is an infinite sequence whose range is the one-element set $\{1\}$. Rudin flags this because the next paragraph slides between the two, and because Chapter 3 will repeatedly distinguish “the sequence converges” from statements about the set of its values.

Let n_1 be the smallest positive integer such that $x_{n_1} \in E$. Having chosen $n_1, \dots, n_{k-1}$ ($k = 2, 3, 4, \dots$), let n_k be the smallest integer greater than n_{k-1} such that $x_{n_k} \in E$.⁸⁴

Putting $f(k) = x_{n_k}$ ($k = 1, 2, 3, \dots$), we obtain a 1-1 correspondence between E and $\mathbb{N}$. $\square$

The theorem shows that, roughly speaking, countable sets represent the “smallest” infinite: No uncountable set can be a subset of a countable set.

WHAT 2.8 RULES OUT

Read as a statement about size, 2.8 says there is nothing strictly between finite and countable. An infinite subset of a countable set cannot be a “smaller infinity” — it is countable, full stop.

That has an immediate consequence used everywhere: to prove some set S is countable it is enough to exhibit a 1-1 map from S into $\mathbb{N}$, or into any countable set, together with the knowledge that S is infinite. You never have to construct a bijection. Rudin uses exactly this shortcut at the end of 2.12, where he obtains $S \sim T$ for some $T \subset \mathbb{N}$ and then invokes 2.8.

2.9 Definition. Let A and Ω be sets, and suppose that with each element α of A there is associated a subset of Ω which we denote by E_α .

The set whose elements are the sets E_α will be denoted by $\{E_\alpha\}$. Instead of speaking of sets of sets, we shall sometimes speak of a *collection* of sets, or a *family* of sets.

The *union* of the sets E_α is defined to be the set S such that $x \in S$ if and only if $x \in E_\alpha$ for at least one $\alpha \in A$. We use the notation

$$S = \bigcup_{\alpha \in A} E_\alpha. \quad (2.1)$$

If A consists of the integers $1, 2, \dots, n$, one usually writes

$$S = \bigcup_{m=1}^n E_m \quad (2.2)$$

or

$$S = E_1 \cup E_2 \cup \dots \cup E_n. \quad (2.3)$$

If A is the set of all positive integers, the usual notation is

$$S = \bigcup_{m=1}^{\infty} E_m. \quad (2.4)$$

The symbol ∞ in (2.4) merely indicates that the union of a countable collection of sets is taken, and should not be confused with the symbols $+\infty$, $-\infty$, introduced in Definition 1.23.⁸⁵

84 “Smallest” is well-ordering again, the principle used silently in 1.1 and in the proof of 1.20(b). Choosing the least admissible index at each stage is what makes the construction canonical — no arbitrary choices are made, so no appeal to the axiom of choice is needed here. Contrast the proof of 2.12.

85 Worth pausing on. The ∞ in $\bigcup_{m=1}^{\infty}$ is not a limit and nothing is approached — it abbreviates “over all $m \in \mathbb{N}$,” and the union is defined outright by the membership condition, not as the limit of finite unions. Rudin flags the clash because Chapter 3 will use the same symbol for genuine limits.

The *intersection* of the sets E_α is defined to be the set P such that $x \in P$ if and only if $x \in E_\alpha$ for every $\alpha \in A$. We use the notation

$$P = \bigcap_{\alpha \in A} E_\alpha, \quad (2.5)$$

or

$$P = \bigcap_{m=1}^n E_m = E_1 \cap E_2 \cap \cdots \cap E_n, \quad (2.6)$$

or

$$P = \bigcap_{m=1}^{\infty} E_m, \quad (2.7)$$

as for unions. If $A \cap B$ is not empty, we say that A and B *intersect*; otherwise they are *disjoint*.

2.10 Examples.

- (a) Suppose E_1 consists of 1, 2, 3 and E_2 consists of 2, 3, 4. Then $E_1 \cup E_2$ consists of 1, 2, 3, 4, whereas $E_1 \cap E_2$ consists of 2, 3.
- (b) Let A be the set of real numbers x such that $0 < x \leq 1$. For every $x \in A$, let E_x be the set of real numbers y such that $0 < y < x$. Then

- (i) $E_x \subset E_z$ if and only if $0 < x \leq z \leq 1$;
- (ii) $\bigcup_{x \in A} E_x = E_1$;
- (iii) $\bigcap_{x \in A} E_x$ is empty.

(i) and (ii) are clear. To prove (iii), we note that for every $y > 0$, $y \notin E_x$ if $x < y$. Hence $y \notin \bigcap_{x \in A} E_x$.⁸⁶

⁸⁶ Example (b)(iii) is the one to remember: an intersection of nonempty nested sets can be empty. Each E_x contains points, and any finitely many of them have a common point, yet nothing survives all of them. Theorem 2.36 will show that compactness is exactly what forbids this, and 2.39 is the version for k -cells.

2.11 Remarks. Many properties of unions and intersections are quite similar to those of sums and products; in fact, the words *sum* and *product* were sometimes used in this connection, and the symbols Σ and Π were written in place of $\bigcup$ and $\bigcap$.

The commutative and associative laws are trivial:

$$A \cup B = B \cup A; \quad A \cap B = B \cap A. \quad (2.8)$$

$$(A \cup B) \cup C = A \cup (B \cup C); \quad (A \cap B) \cap C = A \cap (B \cap C). \quad (2.9)$$

Thus the omission of parentheses in (2.3) and (2.6) is justified.

The distributive law also holds:

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C). \quad (2.10)$$

To prove this, let the left and right members of (2.10) be denoted by E and F , respectively.

Suppose $x \in E$. Then $x \in A$ and $x \in B \cup C$, that is, $x \in B$ or $x \in C$ (possibly both). Hence $x \in A \cap B$ or $x \in A \cap C$, so that $x \in F$. Thus $E \subset F$.

Next, suppose $x \in F$. Then $x \in A \cap B$ or $x \in A \cap C$. That is, $x \in A$, and $x \in B \cup C$. Hence $x \in A \cap (B \cup C)$, so that $F \subset E$.

It follows that $E = F$.⁸⁷

We list a few more relations which are easily verified:

$$A \subset A \cup B, \quad (2.11)$$

$$A \cap B \subset A. \quad (2.12)$$

If $\emptyset$ denotes the empty set, then

$$A \cup \emptyset = A, \quad A \cap \emptyset = \emptyset. \quad (2.13)$$

If $A \subset B$, then

$$A \cup B = B, \quad A \cap B = A. \quad (2.14)$$

2.12 Theorem. Let $\{E_n\}$, $n = 1, 2, 3, \dots$, be a countable collection of countable sets, and put

$$S = \bigcup_{n=1}^{\infty} E_n. \quad (2.15)$$

Then S is countable.

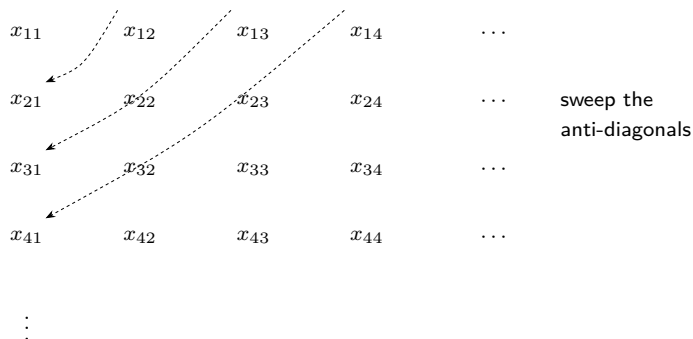
Proof. Let every set E_n be arranged in a sequence (x_{nk}) , $k = 1, 2, 3, \dots$, and consider the infinite array⁸⁸

$$\begin{array}{ccccccc} x_{11} & x_{12} & x_{13} & x_{14} & \cdots & & \\ x_{21} & x_{22} & x_{23} & x_{24} & \cdots & & \\ x_{31} & x_{32} & x_{33} & x_{34} & \cdots & & \\ x_{41} & x_{42} & x_{43} & x_{44} & \cdots & & \\ \dots & \dots & \dots & \dots & \dots & \dots & \end{array} \quad (2.16)$$

in which the elements of E_n form the n th row. The array contains all elements of S . As indicated by the arrows, these elements can be arranged in a sequence

$$x_{11}; x_{21}, x_{12}; x_{31}, x_{22}, x_{13}; x_{41}, x_{32}, x_{23}, x_{14}; \dots \quad (2.17)$$

If any two of the sets E_n have elements in common, these will appear more than once in (2.17). Hence there is a subset T of the set of all positive integers such that $S \sim T$, which shows that S is at most countable (Theorem 2.8). Since $E_1 \subset S$, and E_1 is infinite, S is infinite, and thus countable. $\square$



87 The template for every set identity in the book: prove two inclusions, each by chasing a single element. Rudin writes it out once here and never again — later identities are left as “easily verified.”

88 This opening clause is where the axiom of countable choice enters, and Rudin does not mention it. Each E_n can be arranged in a sequence, but there is no canonical way to do it, and we are choosing one arrangement for each of infinitely many n simultaneously. Without some choice principle the theorem is not provable — it is consistent with ZF that $\mathbb{R}$ is a countable union of countable sets. Compare 2.8, where “smallest index” made every choice canonical and no such principle was needed.

Array (2.16). Each anti-diagonal is finite, so sweeping them in turn reaches every entry after finitely many steps — which is exactly what an enumeration must do.

WHY THE DIAGONAL SWEEP, AND NOT THE OBVIOUS ORDERS

The natural first attempts both fail, and seeing why is the content of the proof.

Row by row never finishes row 1, so x_{21} is never reached. *Column by column* fails the same way. What the anti-diagonals have that rows do not is **finiteness**: the d th anti-diagonal holds exactly d entries, so every entry sits at a finite distance from the start.

That is the whole idea, and it is worth stating in the form used later: a countable union of *finite* sets is at most countable, and the array argument works by exhibiting S as such a union — the anti-diagonals.

Corollary. Suppose A is at most countable, and, for every $\alpha \in A$, B_α is at most countable. Put

$$T = \bigcup_{\alpha \in A} B_\alpha.$$

Then T is at most countable, for T is equivalent to a subset of (2.15).

2.13 Theorem. Let A be a countable set, and let B_n be the set of all n -tuples $(a_1, \dots, a_n)$, where $a_k \in A$ ($k = 1, \dots, n$), and the elements $a_1, \dots, a_n$ need not be distinct. Then B_n is countable.

Proof. That B_1 is countable is evident, since $B_1 = A$. Suppose B_{n-1} is countable ($n = 2, 3, 4, \dots$). The elements of B_n are of the form

$$(b, a) \quad (b \in B_{n-1}, a \in A). \quad (2.18)$$

For every fixed b , the set of pairs (b, a) is equivalent to A , and hence countable. Thus B_n is the union of a countable set of countable sets. By Theorem 2.12, B_n is countable.

The theorem follows by induction. $\square$

Corollary. The set of all rational numbers is countable.

Proof. We apply Theorem 2.13, with $n = 2$, noting that every rational r is of the form b/a , where a and b are integers. The set of pairs (a, b) , and therefore the set of fractions b/a , is countable. $\square$

The corollary is the reason for the section

$\mathbb{Q}$ countable is used more often than any other fact in this chapter. It gives $\mathbb{R}$ a countable dense subset (by 1.20(b)) — which is what Exercise 2.18 uses to build a perfect set avoiding every rational, what the structure theorem for open sets (Exercise 2.29) rests on, and what makes $\mathbb{R}^k$ separable (Exercise 2.22).

Note also the shape of the argument, which recurs: $\mathbb{Q}$ is not obviously countable, but it is the image of a set of *pairs*, and pairs

are handled by 2.13. Whenever an object is specified by finitely many integers, this argument shows there are only countably many of them — which is how Exercise 2.2 gets the algebraic numbers. The strict inequality $|\mathbb{Q}| < |\mathbb{R}|$ that follows from 2.14 is worth pausing on for what it says about 1.20(b): between any two reals there is a rational, yet the rationals are vanishingly rare. Density and abundance are unrelated.

In fact, even the set of all algebraic numbers is countable (see Exercise 2.2). That not all infinite sets are, however, countable, is shown by the next theorem.

2.14 Theorem. *Let A be the set of all sequences whose elements are the digits 0 and 1. This set A is uncountable.*

The elements of A are sequences like $1, 0, 0, 1, 0, 1, 1, 1, \dots$

Proof. Let E be a countable subset of A , and let E consist of the sequences $s_1, s_2, s_3, \dots$. We construct a sequence s as follows. If the n th digit in s_n is 1, we let the n th digit of s be 0, and vice versa. Then the sequence s differs from every member of E in at least one place; hence $s \notin E$. But clearly $s \in A$, so that E is a proper subset of A .

We have shown that every countable subset of A is a proper subset of A . It follows that A is uncountable (for otherwise A would be a proper subset of A , which is absurd).⁸⁹ □

s_1	1	1	0	1	0	...
s_2	0	1	1	0	1	...
s_3	1	0	0	0	1	...
s_4	1	1	1	0	0	...
s_5	0	0	1	1	1	...
.....						
s	0	0	1	1	0	...

flip the boxed
diagonal

89 The last step is where 2.6 is spent. If A were countable then A itself would be one of the countable subsets E , and the argument just showed every such E is a proper subset of A — so $A \subsetneq A$. Rudin's phrasing ("every countable subset is proper") avoids having to fix an enumeration of A in advance and then derive a contradiction, which is the more familiar but slightly clumsier route.

Cantor's diagonal process. The new sequence s disagrees with s_n in place n , so it is not in the list — whatever the list was.

The idea of the above proof was first used by Cantor, and is called *Cantor's diagonal process*; for, if the sequences $s_1, s_2, s_3, \dots$ are placed in an array like (2.16), it is the elements on the diagonal which are involved in the construction of the new sequence.

Readers who are familiar with the binary representation of the real numbers (base 2 instead of 10) will notice that Theorem 2.14 implies that the set of all real numbers is uncountable. We shall give a second proof of this fact in Theorem 2.43.

The map from 0–1 sequences to reals in $[0, 1]$ is *not* a bijection: $0.0111\dots$ and $0.1000\dots$ name the same number, exactly as $0.999\dots = 1$ in base ten. So 2.14 does not immediately transfer.

The repair is cheap. Only the dyadic rationals have two expansions, and there are countably many of them, so the map fails to be injective on a countable set — not enough to make an uncountable set countable. But the gap is real, and it is the same delicacy that made Rudin warn “we shall never use decimals” at 1.22 and made Tao call the decimal construction of $\mathbb{R}$ the most complicated of the three.

Theorem 2.43 avoids the issue entirely, which is presumably why Rudin promises a second proof.

METRIC SPACES

WHY THIS SECTION IS HERE | Chapter 1 ended by noting that $\mathbb{R}^k$ has a distance function satisfying three properties (1.37(a),(b),(f)). This section takes those three properties as the *definition* of a space and develops everything that follows from them alone. The gain is reach: every theorem proved here holds for $\mathbb{R}$, for $\mathbb{C}$, for $\mathbb{R}^k$, for subsets of any of these, and later for spaces of functions (7.14, 11.35).

Definition 2.18 is the hinge of the chapter — ten pieces of vocabulary in one block, arriving with no motivation. Do not try to absorb them all. The two that carry the weight are **limit point** (b) and **open** (f); “closed” is defined from the first, and everything in 2.19–2.30 is bookkeeping about how the two interact. The rest of the list is there so Rudin need not interrupt himself later.

Read 2.29–2.30 with attention even though they look technical. They say that “open” is not a property a set has by itself but one it has *relative to an ambient space*, and that ambiguity causes more confusion in Chapter 4 than any other point in the book.

2.15 Definition. A set X , whose elements we shall call *points*, is said to be a *metric space* if with any two points p and q of X there is associated a real number $d(p, q)$, called the *distance* from p to q , such that

- (a) $d(p, q) > 0$ if $p \neq q$; $d(p, p) = 0$;
- (b) $d(p, q) = d(q, p)$;
- (c) $d(p, q) \leq d(p, r) + d(r, q)$ for any $r \in X$.

Any function with these three properties is called a *distance function*, or a *metric*.⁹⁰

2.16 Examples. The most important examples of metric spaces, from our standpoint, are the euclidean spaces $\mathbb{R}^k$, especially $\mathbb{R}^1$ (the real line)

⁹⁰ Compare 1.37: (a) here is 1.37(a),(b); (b) follows from 1.37(c) with $\alpha = -1$; (c) is 1.37(f). Rudin verified all three for $\mathbb{R}^k$ one chapter ago, which is why 2.16 can assert the example in a single line. Note what is absent: no addition, no scalar multiplication, no inner product. A metric space has distance and nothing else.

and $\mathbb{R}^2$ (the complex plane); the distance in $\mathbb{R}^k$ is defined by

$$d(\mathbf{x}, \mathbf{y}) = |\mathbf{x} - \mathbf{y}| \quad (\mathbf{x}, \mathbf{y} \in \mathbb{R}^k). \quad (2.19)$$

By Theorem 1.37, the conditions of Definition 2.15 are satisfied by (2.19).

It is important to observe that every subset Y of a metric space X is a metric space in its own right, with the same distance function. For it is clear that if conditions (a) to (c) of Definition 2.15 hold for $p, q, r \in X$, they also hold if we restrict p, q, r to lie in Y .⁹¹

Thus, every subset of a euclidean space is a metric space. Other examples are the spaces $\mathcal{C}(K)$ and $\mathcal{L}^2(\mu)$, which are discussed in Chaps. 7 and 11, respectively.

2.17 Definition. By the *segment* (a, b) we mean the set of all real numbers x such that $a < x < b$.

By the *interval* $[a, b]$ we mean the set of all real numbers x such that $a \leq x \leq b$.

Occasionally we shall also encounter “half-open intervals” $[a, b)$ and $(a, b]$; the first consists of all x such that $a \leq x < b$, the second of all x such that $a < x \leq b$.

If $a_i < b_i$ for $i = 1, \dots, k$, the set of all points $\mathbf{x} = (x_1, \dots, x_k)$ in $\mathbb{R}^k$ whose coordinates satisfy the inequalities $a_i \leq x_i \leq b_i$ ($1 \leq i \leq k$) is called a *k-cell*. Thus a 1-cell is an interval, a 2-cell is a rectangle, etc.⁹²

If $\mathbf{x} \in \mathbb{R}^k$ and $r > 0$, the *open* (or *closed*) *ball* B with center at $\mathbf{x}$ and radius r is defined to be the set of all $\mathbf{y} \in \mathbb{R}^k$ such that $|\mathbf{y} - \mathbf{x}| < r$ (or $|\mathbf{y} - \mathbf{x}| \leq r$).

We call a set $E \subset \mathbb{R}^k$ *convex* if

$$\lambda \mathbf{x} + (1 - \lambda) \mathbf{y} \in E$$

whenever $\mathbf{x} \in E$, $\mathbf{y} \in E$, and $0 < \lambda < 1$.

For example, balls are convex. For if $|\mathbf{y} - \mathbf{x}| < r$, $|\mathbf{z} - \mathbf{x}| < r$, and $0 < \lambda < 1$, we have

$$\begin{aligned} |\lambda \mathbf{y} + (1 - \lambda) \mathbf{z} - \mathbf{x}| &= |\lambda(\mathbf{y} - \mathbf{x}) + (1 - \lambda)(\mathbf{z} - \mathbf{x})| \\ &\leq \lambda |\mathbf{y} - \mathbf{x}| + (1 - \lambda) |\mathbf{z} - \mathbf{x}| < \lambda r + (1 - \lambda)r \\ &= r. \end{aligned}$$

The same proof applies to closed balls. It is also easy to see that *k*-cells are convex.

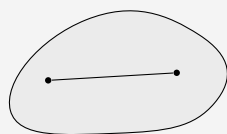
Convexity is not a metric notion

Notice that convexity is defined only for $E \subset \mathbb{R}^k$, never for a general metric space — and it could not be, because $\lambda \mathbf{x} + (1 - \lambda) \mathbf{y}$ requires addition and scalar multiplication, which 2.15 does not provide. Convexity belongs to the vector space structure, not the metric. Rudin introduces it here and then barely uses it in this chapter

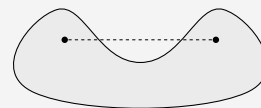
91 This one-sentence observation is what makes 2.29–2.30 necessary. Because every subset is itself a metric space, the words “open” and “closed” become ambiguous the moment more than one ambient space is in play — and from Chapter 4 onward there always is.

92 Rudin’s “segment” means open interval and his “interval” means closed. Almost no other book uses these words this way, and he uses them constantly. A *k*-cell is always closed — the inequalities are $\leq$ — which is why 2.40 can prove they are compact.

— Exercise 2.21(c) below proves convex sets are connected, and the next real use is Theorem 9.19, where the mean value inequality needs the segment joining two points to stay inside the domain. (Convex *functions*, a different notion, appear in Exercise 4.23.) The line “balls are convex” is worth keeping, though: it depends on the triangle inequality alone and is the reason pictures of neighbourhoods may be drawn as round blobs without loss.



convex: every segment stays in



not convex: this one leaves

The definition quantifies over every pair of points and every λ : one escaping chord is enough to fail it.

2.18 Definition. Let X be a metric space. All points and sets mentioned below are understood to be elements and subsets of X .

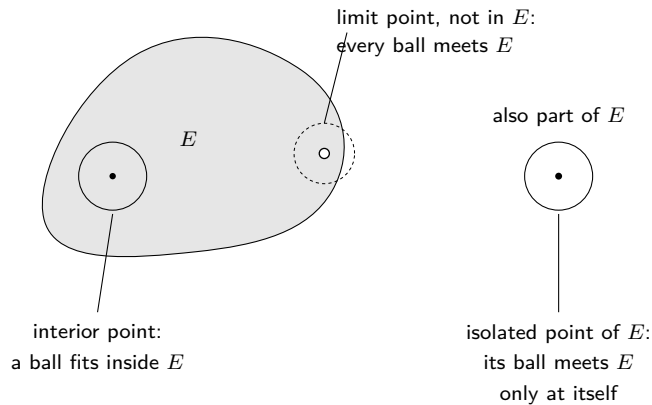
- (a) A *neighborhood* of p is a set $N_r(p)$ consisting of all q such that $d(p, q) < r$ for some $r > 0$. The number r is called the *radius* of $N_r(p)$.
- (b) A point p is a *limit point* of the set E if every neighborhood of p contains a point $q \neq p$ such that $q \in E$.⁹³
- (c) If $p \in E$ and p is not a limit point of E , then p is called an *isolated point* of E .
- (d) E is *closed* if every limit point of E is a point of E .
- (e) A point p is an *interior point* of E if there is a neighborhood N of p such that $N \subset E$.
- (f) E is *open* if every point of E is an interior point of E .
- (g) The *complement* of E (denoted by E^c) is the set of all points $p \in X$ such that $p \notin E$.
- (h) E is *perfect* if E is closed and if every point of E is a limit point of E .
- (i) E is *bounded* if there is a real number M and a point $q \in X$ such that $d(p, q) < M$ for all $p \in E$.
- (j) E is *dense* in X if every point of X is a limit point of E , or a point of E (or both).

Let us note that in $\mathbb{R}^1$ neighborhoods are segments, whereas in $\mathbb{R}^2$ neighborhoods are interiors of circles.

93 The clause $q \neq p$ is the whole definition. Without it every point of E would trivially be a limit point of E , and the notion would be useless. A *limit point* is a point that E crowds from outside itself — and it need not belong to E : see 2.21(e), where 0 is a limit point of $\{1/n\}$ and lies nowhere in the set.

THE TEN DEFINITIONS IN ONE PICTURE

Take E to be a shaded blob in the plane together with one stray point off to the side. Every term in 2.18 can be read off it.



A neighbourhood is a ball. An interior point has one inside E ; a limit point has every one of its balls meeting E elsewhere; an isolated point of E has a ball meeting E only at itself.

Three consequences are worth fixing now, because Rudin uses them without comment.

Open and closed are not opposites. They are defined by unrelated conditions — (f) is about points *of* E , (d) is about limit points, which may lie outside. A set can be both (the whole space X , and $\emptyset$), or neither (the half-open $[0, 1)$ in $\mathbb{R}$). Theorem 2.23 supplies the only link there is: E open $\iff E^c$ closed.

A point of E is either isolated or a limit point of E , by (c), and never both. A perfect set (h) is one with no isolated points at all, closed.

“Dense” means the closure is everything. Once 2.26 defines $\overline{E} = E \cup E'$, clause (j) reads $\overline{E} = X$. Theorem 1.20(b) says exactly that $\mathbb{Q}$ is dense in $\mathbb{R}$.

2.19 Theorem. *Every neighborhood is an open set.*

Proof. Consider a neighborhood $E = N_r(p)$, and let q be any point of E . Then there is a positive real number h such that

$$d(p, q) = r - h.$$

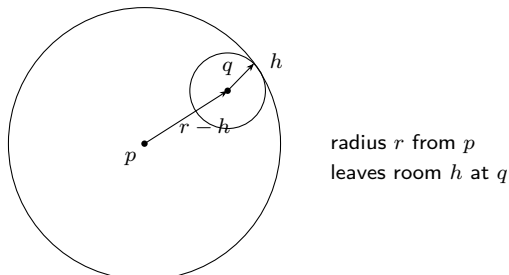
For all points s such that $d(q, s) < h$, we have then

$$d(p, s) \leq d(p, q) + d(q, s) < r - h + h = r,$$

so that $s \in E$. Thus q is an interior point of E . □

A BALL INSIDE A BALL

The statement sounds like a tautology and is not: “neighborhood” is defined by a distance inequality, “open” by the existence of small balls around every point. The theorem says the first implies the second, and the triangle inequality is what does it.



Whatever is left over after travelling $d(p, q)$ is exactly the radius that fits at q .

The number $h = r - d(p, q)$ is positive precisely because q is *inside* the neighbourhood. That is the only thing being used, and it is why closed balls in $\mathbb{R}^k$ are not open: a point on the boundary sphere has $h = 0$ and no room at all.

2.20 Theorem. *If p is a limit point of a set E , then every neighborhood of p contains infinitely many points of E .*

Proof. Suppose there is a neighborhood N of p which contains only a finite number of points of E . Let $q_1, \dots, q_n$ be those points of $N \cap E$ which are distinct from p , and put

$$r = \min_{1 \leq m \leq n} d(p, q_m).$$

[We use this notation to denote the smallest of the numbers $d(p, q_1), \dots, d(p, q_n)$.] The minimum of a finite set of positive numbers is clearly positive, so that $r > 0$.⁹⁴

The neighborhood $N_r(p)$ contains no point q of E such that $q \neq p$, so that p is not a limit point of E . This contradiction establishes the theorem. $\square$

Corollary. *A finite point set has no limit points.*

2.21 Examples. Let us consider the following subsets of $\mathbb{R}^2$:

- (a) The set of all complex z such that $|z| < 1$.
- (b) The set of all complex z such that $|z| \leq 1$.
- (c) A nonempty finite set.
- (d) The set of all integers.
- (e) The set consisting of the numbers $1/n$ ($n = 1, 2, 3, \dots$). Let us note that this set E has a limit point (namely, $z = 0$) but that no

⁹⁴ Here is where finiteness is spent, and it is the entire proof. An infinite set of positive numbers can have infimum 0 — take $\{1/n\}$ — so no such r could be extracted. The same finite-minimum move appears in 2.24(c), and both are the reason those theorems say “finite.”

point of E is a limit point of E ; we wish to stress the difference between having a limit point and containing one.

(f) The set of all complex numbers (that is, $\mathbb{R}^2$).

(g) The segment (a, b) .

Let us note that (d), (e), (g) can be regarded also as subsets of $\mathbb{R}^1$.

Some properties of these sets are tabulated below:

	Closed	Open	Perfect	Bounded
(a)	No	Yes	No	Yes
(b)	Yes	No	Yes	Yes
(c)	Yes	No	No	Yes
(d)	Yes	No	No	No
(e)	No	No	No	Yes
(f)	Yes	Yes	Yes	No
(g)	No		No	Yes

In (g), we left the second entry blank. The reason is that the segment (a, b) is not open if we regard it as a subset of $\mathbb{R}^2$, but it is an open subset of $\mathbb{R}^1$.⁹⁵

READING THE TABLE

Row (f) is worth staring at: $\mathbb{R}^2$ is closed *and* open in itself. So is $\emptyset$. That is not a paradox — it is what the definitions say. In fact connectedness (2.45) amounts to exactly this: a space is connected precisely when $\emptyset$ and the whole space are its only subsets that are both open and closed, as a short argument from Definition 2.45 shows.

Row (e) is the one Rudin flags. $E = \{1/n\}$ has the limit point 0 but does not *contain* it, so E is not closed. No point of E is a limit point of E , so every point of E is isolated. A set can therefore be entirely isolated points and still have a limit point elsewhere.

Rows (b) and (f) are the perfect ones. Perfect means closed with no isolated points, and (c), (d), (e) fail it for having isolated points, while (a) and (g) fail it for not being closed.

95 The blank entry is the most instructive cell in the table. “Open” is not a property of a set; it is a property of a set in an ambient space. As a subset of the line, (a, b) is open; the same points sitting in the plane form a segment with empty interior, since no disc fits inside it. This is what 2.29–2.30 are about, and forgetting it is the standard error in Chapter 4.

2.22 Theorem. Let $\{E_\alpha\}$ be a (finite or infinite) collection of sets E_α . Then

$$\left(\bigcup_{\alpha} E_{\alpha}\right)^c = \bigcap_{\alpha} (E_{\alpha}^c). \quad (2.20)$$

Proof. Let A and B be the left and right members of (2.20). If $x \in A$, then $x \notin \bigcup_{\alpha} E_{\alpha}$, hence $x \notin E_{\alpha}$ for any α , hence $x \in E_{\alpha}^c$ for every α , so that $x \in \bigcap_{\alpha} E_{\alpha}^c$. Thus $A \subset B$.

Conversely, if $x \in B$, then $x \in E_{\alpha}^c$ for every α , hence $x \notin E_{\alpha}$ for any α , hence $x \notin \bigcup_{\alpha} E_{\alpha}$, so that $x \in (\bigcup_{\alpha} E_{\alpha})^c$. Thus $B \subset A$.

It follows that $A = B$. $\square$

2.23 Theorem. *A set E is open if and only if its complement is closed.*

Proof. First, suppose E^c is closed. Choose $x \in E$. Then $x \notin E^c$, and x is not a limit point of E^c . Hence there exists a neighborhood N of x such that $E^c \cap N$ is empty, that is, $N \subset E$. Thus x is an interior point of E , and E is open.

Next, suppose E is open. Let x be a limit point of E^c . Then every neighborhood of x contains a point of E^c , so that x is not an interior point of E . Since E is open, this means that $x \in E^c$. It follows that E^c is closed. $\square$

Corollary. *A set F is closed if and only if its complement is open.*

2.22 and 2.23 are the engine of the section

Every remaining proof about closed sets is obtained by complementing a proof about open sets. 2.22 (De Morgan) turns unions into intersections; 2.23 turns open into closed. Together they let Rudin prove half of 2.24 and get the other half free, which is exactly what he does.

The pattern is worth naming because it recurs throughout the book: prove the statement for open sets, where the definition is local and easy to verify at a point, then complement. Closedness is a statement about limit points, which are harder to handle directly — so one almost never argues with it head on.

2.24 Theorem.

- (a) *For any collection $\{G_\alpha\}$ of open sets, $\bigcup_\alpha G_\alpha$ is open.*
- (b) *For any collection $\{F_\alpha\}$ of closed sets, $\bigcap_\alpha F_\alpha$ is closed.*
- (c) *For any finite collection $G_1, \dots, G_n$ of open sets, $\bigcap_{i=1}^n G_i$ is open.*
- (d) *For any finite collection $F_1, \dots, F_n$ of closed sets, $\bigcup_{i=1}^n F_i$ is closed.*

Proof. Put $G = \bigcup_\alpha G_\alpha$. If $x \in G$ then $x \in G_\alpha$ for some α . Since x is an interior point of G_α , x is also an interior point of G , and G is open. This proves (a).

By Theorem 2.22,

$$\left(\bigcap_\alpha F_\alpha\right)^c = \bigcup_\alpha (F_\alpha^c), \quad (2.21)$$

and F_α^c is open, by Theorem 2.23. Hence (a) implies that (2.21) is open, so that $\bigcap_\alpha F_\alpha$ is closed.

Next, put $H = \bigcap_{i=1}^n G_i$. For any $x \in H$, there exist neighborhoods N_i of x , with radii r_i , such that $N_i \subset G_i$ ($i = 1, \dots, n$). Put

$$r = \min\{r_1, \dots, r_n\}$$

and let N be the neighborhood of x of radius r . Then $N \subset G_i$ for $i = 1, \dots, n$, so that $N \subset H$, and H is open.⁹⁶

⁹⁶ The finite minimum again, exactly as in 2.20. With infinitely many G_i the radii could shrink to zero and leave no room, which is precisely what happens in Example 2.25.

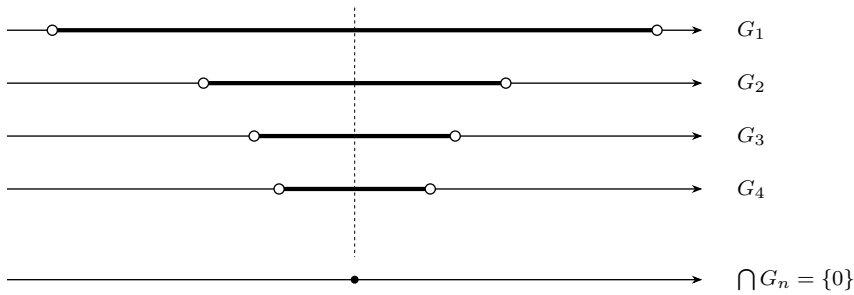
By taking complements, (d) follows from (c):

$$\left(\bigcup_{i=1}^n F_i\right)^c = \bigcap_{i=1}^n (F_i^c).$$

□

2.25 Example. In parts (c) and (d) of the preceding theorem, the finiteness of the collections is essential. For let G_n be the segment $(-1/n, 1/n)$ ($n = 1, 2, 3, \dots$). Then G_n is an open subset of $\mathbb{R}^1$. Put $G = \bigcap_{n=1}^{\infty} G_n$. Then G consists of a single point (namely, $x = 0$) and is therefore not an open subset of $\mathbb{R}^1$.

Thus the intersection of an infinite collection of open sets need not be open. Similarly, the union of an infinite collection of closed sets need not be closed.



Each G_n is open, yet the radii shrink to nothing and the intersection is a single point — not open.

2.26 Definition. If X is a metric space, if $E \subset X$, and if E' denotes the set of all limit points of E in X , then the *closure* of E is the set $\overline{E} = E \cup E'$.

2.27 Theorem. If X is a metric space and $E \subset X$, then

- (a) $\overline{E}$ is closed,
- (b) $E = \overline{E}$ if and only if E is closed,
- (c) $\overline{E} \subset F$ for every closed set $F \subset X$ such that $E \subset F$.

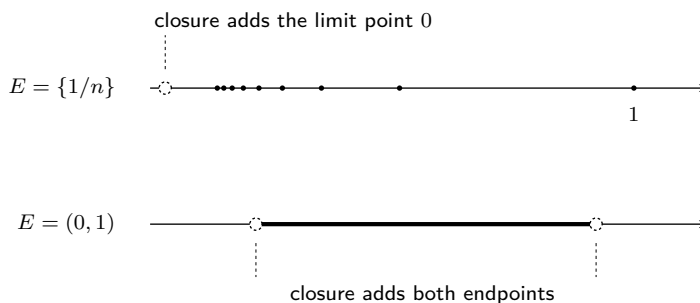
By (a) and (c), $\overline{E}$ is the smallest closed subset of X that contains E .⁹⁷

Proof.

- (a) If $p \in X$ and $p \notin \overline{E}$ then p is neither a point of E nor a limit point of E . Hence p has a neighborhood which does not intersect E . The complement of $\overline{E}$ is therefore open. Hence $\overline{E}$ is closed.
- (b) If $E = \overline{E}$, (a) implies that E is closed. If E is closed, then $E' \subset E$ [by Definitions 2.18(d) and 2.26], hence $E = \overline{E}$.
- (c) If F is closed and $F \supset E$, then $F \supset F'$, hence $F \supset E'$. Thus $F \supset \overline{E}$.

⁹⁷ That last sentence is the content; (a) and (b) are warm-up. “Smallest closed set containing E ” is the characterisation you should carry — it is how closure is defined in most later treatments, with $\overline{E} = E \cup E'$ demoted to a theorem.

□



Closure fills in exactly the limit points (dashed circles); nothing else changes. Compare 2.28: for a bounded set of reals, $\sup E$ is always among the points filled in, if it was missing.

2.28 Theorem. Let E be a nonempty set of real numbers which is bounded above. Let $y = \sup E$. Then $y \in \overline{E}$. Hence $y \in E$ if E is closed.

Compare this with the examples in Sec. 1.9.

Proof. If $y \in E$ then $y \in \overline{E}$. Assume $y \notin E$. For every $h > 0$ there exists then a point $x \in E$ such that $y - h < x < y$, for otherwise $y - h$ would be an upper bound of E . Thus y is a limit point of E . Hence $y \in \overline{E}$. □

CHAPTER 1 MEETS CHAPTER 2

This is the first theorem to connect the order structure of $\mathbb{R}$ with the topology, and the proof is two lines because the work was done in Chapter 1. “For every $h > 0$ there is $x \in E$ with $y - h < x$ ” is precisely clause (ii) of Definition 1.8, in the working form given there — and clause (ii) says exactly that y is crowded from below by E , which is what a limit point is.

So $\sup E$ is either a member of E or a limit point of it, never anything else. The examples at 1.9 now read topologically: $\sup E_1 = 0 \notin E_1$ makes 0 a limit point, while $\sup E_2 = 0 \in E_2$. Theorem 2.28 is what lets Chapter 4 conclude that a continuous function on a compact set attains its supremum (4.16).

2.29 Remark. Suppose $E \subset Y \subset X$, where X is a metric space. To say that E is an open subset of X means that to each point $p \in E$ there is associated a positive number r such that the conditions $d(p, q) < r$, $q \in X$ imply that $q \in E$. But we have already observed (Example 2.16) that Y is also a metric space, so that our definitions may equally well be made within Y . To be quite explicit, let us say that E is *open relative to Y* if to each $p \in E$ there is associated an $r > 0$ such that $q \in E$ whenever $d(p, q) < r$ and $q \in Y$. Example 2.21(g) showed that a set may be open relative to Y without being an open subset of X . However, there is a simple relation between these concepts, which we now state.

2.30 Theorem. Suppose $Y \subset X$. A subset E of Y is open relative to Y if and only if $E = Y \cap G$ for some open subset G of X .

Proof. Suppose E is open relative to Y . To each $p \in E$ there is a positive number r_p such that the conditions $d(p, q) < r_p$, $q \in Y$ imply that $q \in E$. Let V_p be the set of all $q \in X$ such that $d(p, q) < r_p$, and define

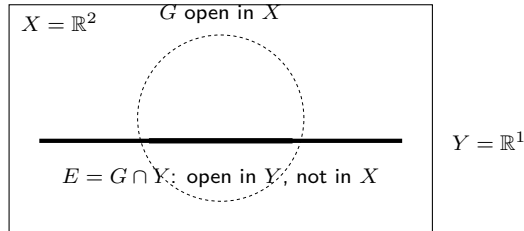
$$G = \bigcup_{p \in E} V_p.$$

Then G is an open subset of X , by Theorems 2.19 and 2.24.

Since $p \in V_p$ for all $p \in E$, it is clear that $E \subset G \cap Y$.

By our choice of V_p , we have $V_p \cap Y \subset E$ for every $p \in E$, so that $G \cap Y \subset E$. Thus $E = G \cap Y$, and one half of the theorem is proved.

Conversely, if G is open in X and $E = G \cap Y$, every $p \in E$ has a neighborhood $V_p \subset G$. Then $V_p \cap Y \subset E$, so that E is open relative to Y . $\square$



The same points form an open set inside the line and a non-open set inside the plane. “Open” always means open in something.

COMPACT SETS

THE CENTRE OF THE CHAPTER | Everything so far has been vocabulary. This section introduces the one property that does real work, and it arrives in a form that looks arbitrary: a set is compact if every open cover has a finite subcover.

Why that definition. Compactness is the hypothesis that converts “for each point there is some local fact” into “there is one fact for the whole set.” Local statements come with a neighbourhood attached; a finite subcover lets you take a minimum over finitely many of them, and a minimum of finitely many positive numbers is positive (the move already used at 2.20 and 2.24(c)). Every application in Chapter 4 — boundedness 4.15, attainment 4.16, uniform continuity 4.19 — is that one manoeuvre.

The plan of the section. 2.33–2.37 develop compactness abstractly, in any metric space. 2.38–2.40 then do the hard work of exhibiting compact sets in $\mathbb{R}^k$, culminating in “every k -cell is compact.” 2.41 collects the payoff: in $\mathbb{R}^k$, compact means closed and bounded. Note the division — 2.33–2.37 are general, 2.38–2.42 are

about $\mathbb{R}^k$ and are *false* in general metric spaces.

2.31 Definition. By an *open cover* of a set E in a metric space X we mean a collection $\{G_\alpha\}$ of open subsets of X such that $E \subset \bigcup_\alpha G_\alpha$.

2.32 Definition. A subset K of a metric space X is said to be *compact* if every open cover of K contains a finite subcover.

More explicitly, this requirement is that if $\{G_\alpha\}$ is an open cover of K , then there are finitely many indices $\alpha_1, \dots, \alpha_n$ such that

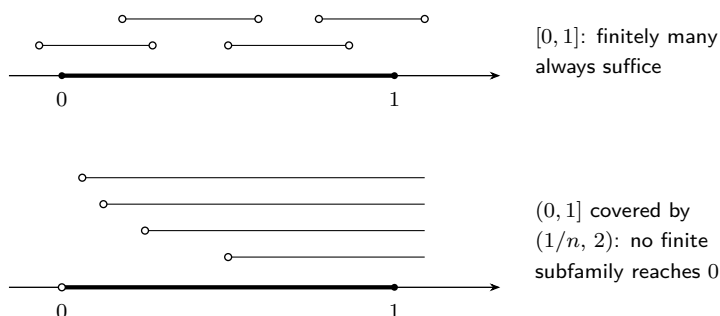
$$K \subset G_{\alpha_1} \cup \dots \cup G_{\alpha_n}.$$

The notion of compactness is of great importance in analysis, especially in connection with continuity (Chap. 4).⁹⁸

It is clear that every finite set is compact. The existence of a large class of infinite compact sets in $\mathbb{R}^k$ will follow from Theorem 2.41.

WHAT A COVER LOOKS LIKE, AND WHAT FAILS

Two examples settle the intuition. Both are covers of a subset of the line by open intervals; one collapses to finitely many, the other cannot.



Compactness fails for $(0, 1]$ because the cover creeps toward a missing endpoint. Any finite subfamily stops short.

The failure is instructive: $(0, 1]$ is bounded but not closed, and the cover exploits exactly the missing point. Theorem 2.41 will say that in $\mathbb{R}^k$ these are the only two ways compactness can fail.

We observed earlier (in Remark 2.29) that if $E \subset Y \subset X$, then E may be open relative to Y without being open relative to X . The property of being open thus depends on the space in which E is embedded. The same is true of the property of being closed.

Compactness, however, behaves better, as we shall now see. To formulate the next theorem, let us say, temporarily, that K is *compact relative to X* if the requirements of Definition 2.32 are met.

2.33 Theorem. Suppose $K \subset Y \subset X$. Then K is compact relative to X if and only if K is compact relative to Y .

98 Read the quantifiers carefully. Every cover must admit a finite subcover — exhibiting one cover that happens to have a finite subcover proves nothing. To show a set is not compact it suffices to produce a single bad cover, which is why non-compactness is usually the easy direction.

By virtue of this theorem we are able, in many situations, to regard compact sets as metric spaces in their own right, without paying any attention to any embedding space. In particular, although it makes little sense to talk of open spaces, or of closed spaces (every metric space X is an open subset of itself, and is a closed subset of itself), it does make sense to talk of *compact metric spaces*.⁹⁹

Proof. Suppose K is compact relative to X , and let $\{V_\alpha\}$ be a collection of sets, open relative to Y , such that $K \subset \bigcup_\alpha V_\alpha$. By Theorem 2.30, there are sets G_α , open relative to X , such that $V_\alpha = Y \cap G_\alpha$, for all α ; and since K is compact relative to X , we have

$$K \subset G_{\alpha_1} \cup \cdots \cup G_{\alpha_n} \quad (2.22)$$

for some choice of finitely many indices $\alpha_1, \dots, \alpha_n$. Since $K \subset Y$, (2.22) implies

$$K \subset V_{\alpha_1} \cup \cdots \cup V_{\alpha_n}. \quad (2.23)$$

This proves that K is compact relative to Y .

Conversely, suppose K is compact relative to Y , let $\{G_\alpha\}$ be a collection of open subsets of X which covers K , and put $V_\alpha = Y \cap G_\alpha$. Then (2.23) will hold for some choice of $\alpha_1, \dots, \alpha_n$; since $V_\alpha \subset G_\alpha$, (2.23) implies (2.22).

This completes the proof. $\square$

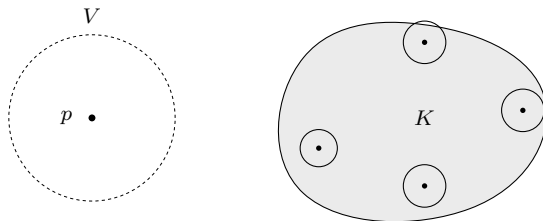
2.34 Theorem. Compact subsets of metric spaces are closed.

Proof. Let K be a compact subset of a metric space X . We shall prove that the complement of K is an open subset of X .

Suppose $p \in X$, $p \notin K$. If $q \in K$, let V_q and W_q be neighborhoods of p and q , respectively, of radius less than $\frac{1}{2}d(p, q)$ [see Definition 2.18(a)]. Since K is compact, there are finitely many points $q_1, \dots, q_n$ in K such that

$$K \subset W_{q_1} \cup \cdots \cup W_{q_n} = W.$$

If $V = V_{q_1} \cap \cdots \cap V_{q_n}$, then V is a neighborhood of p which does not intersect W . Hence $V \subset K^c$, so that p is an interior point of K^c . The theorem follows.¹⁰⁰ $\square$



W : finitely many balls covering K

Each $q \in K$ is separated from p by a pair of balls. Finitely many W_q already cover K , so the matching finitely many V_q may be intersected to give one ball around p missing K .

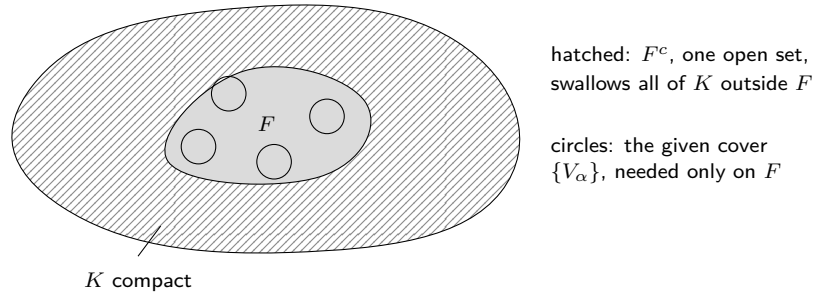
⁹⁹ This is why compactness is the better notion. “Open” and “closed” are relative and must always be qualified; compactness is intrinsic — a property of K with its own metric, independent of where K happens to sit. That is what makes it usable as a hypothesis in Chapter 4, where the ambient space changes from line to line.

¹⁰⁰ Here is the manoeuvre the signpost promised, in its purest form. For each q we get a local separation — two small balls, radius depending on q . Compactness reduces the infinitely many q to finitely many, and only then may we intersect the V_q : an infinite intersection of neighbourhoods need not be a neighbourhood, exactly as 2.25 showed.

2.35 Theorem. *Closed subsets of compact sets are compact.*

Proof. Suppose $F \subset K \subset X$, F is closed (relative to X), and K is compact. Let $\{V_\alpha\}$ be an open cover of F . If F^c is adjoined to $\{V_\alpha\}$, we obtain an open cover Ω of K .¹⁰¹ Since K is compact, there is a finite subcollection Φ of Ω which covers K , and hence F . If F^c is a member of Φ , we may remove it from Φ and still retain an open cover of F . We have thus shown that a finite subcollection of $\{V_\alpha\}$ covers F . $\square$

101 The trick of the proof, and it is worth remembering: to convert a cover of F into a cover of the larger K , throw in the single open set F^c , which mops up everything outside F . It is legitimately open precisely because F is closed — 2.23.



Adjoining the single open set F^c turns a cover of F into a cover of K ; compactness of K then hands back finitely many sets, and discarding F^c leaves a finite cover of F .

Corollary. *If F is closed and K is compact, then $F \cap K$ is compact.*

Proof. Theorems 2.24(b) and 2.34 show that $F \cap K$ is closed; since $F \cap K \subset K$, Theorem 2.35 shows that $F \cap K$ is compact. $\square$

2.36 Theorem. *If $\{K_\alpha\}$ is a collection of compact subsets of a metric space X such that the intersection of every finite subcollection of $\{K_\alpha\}$ is nonempty, then $\bigcap K_\alpha$ is nonempty.*

Proof. Fix a member K_1 of $\{K_\alpha\}$ and put $G_\alpha = K_\alpha^c$. Assume that no point of K_1 belongs to every K_α . Then the sets G_α form an open cover of K_1 ; since K_1 is compact, there are finitely many indices $\alpha_1, \dots, \alpha_n$ such that $K_1 \subset G_{\alpha_1} \cup \dots \cup G_{\alpha_n}$. But this means that

$$K_1 \cap K_{\alpha_1} \cap \dots \cap K_{\alpha_n}$$

is empty, in contradiction to our hypothesis. $\square$

Corollary. *If $\{K_n\}$ is a countable collection of nonempty compact sets such that $K_n \supset K_{n+1}$ ($n = 1, 2, 3, \dots$), then $\bigcap_{n=1}^{\infty} K_n$ is not empty.*

The finite intersection property is compactness turned inside out

2.36 is 2.32 with complements taken, and nothing more. “Every open cover has a finite subcover” says: if no finite subfamily of $\{G_\alpha\}$ covers K , then $\{G_\alpha\}$ does not cover K . Complementing every G turns “covers” into “has empty intersection with,” and the statement becomes 2.36.

The corollary is the form that gets used, and it answers the worry raised at Example 2.10(b)(iii), where nested nonempty sets had empty intersection. There the sets were the segments $(0, x)$ —

bounded but not closed, hence not compact. Compactness is exactly what forbids the nesting from escaping to nothing.

Compare Theorem 2.38, the same statement for closed intervals proved directly from the least-upper-bound property, and Yu Pin's nested interval axiom in Chapter 1. Three routes to one fact, and Rudin uses whichever is cheapest at the point of need.

2.37 Theorem. *If E is an infinite subset of a compact set K , then E has a limit point in K .*

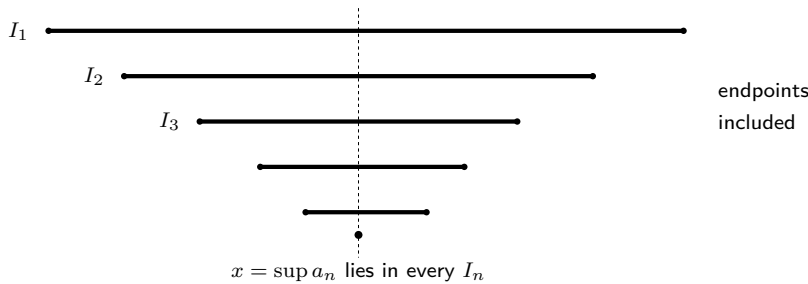
Proof. If no point of K were a limit point of E , then each $q \in K$ would have a neighborhood V_q which contains at most one point of E (namely, q , if $q \in E$). It is clear that no finite subcollection of $\{V_q\}$ can cover E ; and the same is true of K , since $E \subset K$. This contradicts the compactness of K . $\square$

2.38 Theorem. *If $\{I_n\}$ is a sequence of intervals in $\mathbb{R}^1$, such that $I_n \supset I_{n+1}$ ($n = 1, 2, 3, \dots$), then $\bigcap_{n=1}^{\infty} I_n$ is not empty.*

Proof. If $I_n = [a_n, b_n]$, let E be the set of all a_n . Then E is nonempty and bounded above (by b_1). Let x be the sup of E . If m and n are positive integers, then

$$a_n \leq a_{m+n} \leq b_{m+n} \leq b_m,$$

so that $x \leq b_m$ for each m . Since it is obvious that $a_m \leq x$, we see that $x \in I_m$ for $m = 1, 2, 3, \dots$ ¹⁰² $\square$



102 This is Chapter 1 doing the work again — the whole proof is “take a supremum,” which exists by 1.19. Every b_m is an upper bound for the left endpoints, so the least upper bound x sits below all of them and above all the a ’s. Note the closedness of the intervals is essential: for $I_n = (0, 1/n)$ the same construction gives $x = 0$, which lies in no I_n .

Left endpoints increase, right endpoints decrease, and $\sup a_n$ is caught between. Drop the endpoints and the common point can escape.

2.39 Theorem. *Let k be a positive integer. If $\{I_n\}$ is a sequence of k -cells such that $I_n \supset I_{n+1}$ ($n = 1, 2, 3, \dots$), then $\bigcap_{n=1}^{\infty} I_n$ is not empty.*

Proof. Let I_n consist of all points $\mathbf{x} = (x_1, \dots, x_k)$ such that

$$a_{n,j} \leq x_j \leq b_{n,j} \quad (1 \leq j \leq k; n = 1, 2, 3, \dots),$$

and put $I_{n,j} = [a_{n,j}, b_{n,j}]$. For each j , the collection $\{I_{n,j}\}$ satisfies the hypotheses of Theorem 2.38. Hence there are real numbers x_j^* ($1 \leq j \leq k$) such that

$$a_{n,j} \leq x_j^* \leq b_{n,j} \quad (1 \leq j \leq k; n = 1, 2, 3, \dots).$$

103 The k -dimensional statement is k separate one-dimensional statements, because a k -cell is a product of intervals and a point lies in it exactly when each coordinate lies in the corresponding interval. This coordinatewise reduction is used again in 2.40 and throughout Chapter 9.

Setting $\mathbf{x}^* = (x_1^*, \dots, x_k^*)$, we see that $\mathbf{x}^* \in I_n$ for $n = 1, 2, 3, \dots$. The theorem follows.¹⁰³ $\square$

2.40 Theorem. Every k -cell is compact.

Proof. Let I be a k -cell, consisting of all points $\mathbf{x} = (x_1, \dots, x_k)$ such that $a_j \leq x_j \leq b_j$ ($1 \leq j \leq k$). Put

$$\delta = \left\{ \sum_1^k (b_j - a_j)^2 \right\}^{1/2}.$$

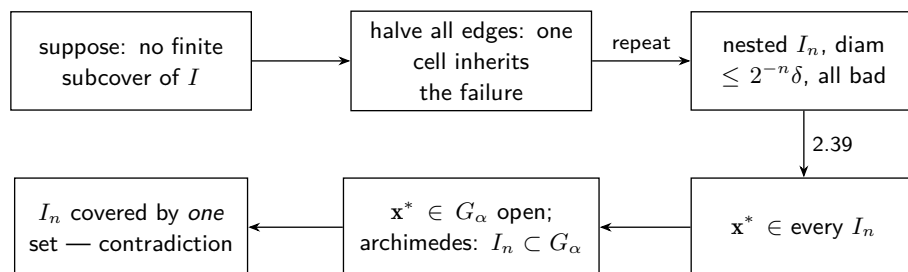
Then $|\mathbf{x} - \mathbf{y}| \leq \delta$, if $\mathbf{x} \in I, \mathbf{y} \in I$.

Suppose, to get a contradiction, that there exists an open cover $\{G_\alpha\}$ of I which contains no finite subcover of I . Put $c_j = (a_j + b_j)/2$. The intervals $[a_j, c_j]$ and $[c_j, b_j]$ then determine 2^k k -cells Q_i whose union is I . At least one of these sets Q_i , call it I_1 , cannot be covered by any finite subcollection of $\{G_\alpha\}$ (otherwise I could be so covered). We next subdivide I_1 and continue the process. We obtain a sequence (I_n) with the following properties:

- (a) $I \supset I_1 \supset I_2 \supset I_3 \supset \dots$;
- (b) I_n is not covered by any finite subcollection of $\{G_\alpha\}$;
- (c) If $\mathbf{x} \in I_n$ and $\mathbf{y} \in I_n$, then $|\mathbf{x} - \mathbf{y}| \leq 2^{-n}\delta$.

By (a) and Theorem 2.39, there is a point $\mathbf{x}^*$ which lies in every I_n . For some α , $\mathbf{x}^* \in G_\alpha$. Since G_α is open, there exists $r > 0$ such that $|\mathbf{y} - \mathbf{x}^*| < r$ implies that $\mathbf{y} \in G_\alpha$. If n is so large that $2^{-n}\delta < r$ (there is such an n , for otherwise $2^n \leq \delta/r$ for all positive integers n , which is absurd since $\mathbb{R}$ is archimedean), then (c) implies that $I_n \subset G_\alpha$, which contradicts (b).

This completes the proof. $\square$

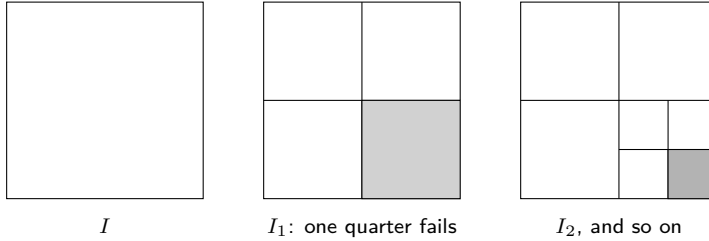


The proof as a flow diagram. Completeness enters once (2.39, via 2.38), the archimedean property once, and the contradiction is that one is finite.

BISECTION: THE SHAPE OF THE ARGUMENT

This is the most important proof in the chapter, and its structure recurs throughout analysis. Assume the conclusion fails, then manufacture a nested sequence of ever-smaller pieces in which it keeps failing, and squeeze them onto a single point where the failure be-

comes absurd.



Halving each edge gives 2^k pieces; at least one inherits the failure. Diameters fall like $2^{-n}\delta$, so the nested pieces close on a point.

Three ingredients, each traceable to Chapter 1.

Something survives the nesting — Theorem 2.39, which rests on 2.38, which rests on the least-upper-bound property. This is where completeness enters, and it is the only place. Over $\mathbb{Q}$ the argument breaks here and k -cells are not compact.

The pieces shrink to nothing — property (c). Diameters are halved each time.

Shrinking beats any fixed radius — the archimedean property, which Rudin flags explicitly in parentheses. Without it $2^{-n}\delta$ might stay above r forever.

Then the contradiction is immediate: I_n fits inside the single set G_α , so one set covers it, contradicting “no finite subcollection covers I_n .” One is finite.

Seen from the comparison at Definition 1.10: this proof *is* the implication “nested intervals $\Rightarrow$ Heine–Borel” from the ECNU text’s cycle of six equivalent completeness statements, specialised to cells. Bisection is the standard bridge between the two.

The equivalence of (a) and (b) in the next theorem is known as the *Heine–Borel theorem*.

2.41 Theorem. *If a set E in $\mathbb{R}^k$ has one of the following three properties, then it has the other two:*

- (a) *E is closed and bounded.*
- (b) *E is compact.*
- (c) *Every infinite subset of E has a limit point in E .*

Proof. If (a) holds, then $E \subset I$ for some k -cell I , and (b) follows from Theorems 2.40 and 2.35. Theorem 2.37 shows that (b) implies (c). It remains to be shown that (c) implies (a).

If E is not bounded, then E contains points $\mathbf{x}_n$ with

$$|\mathbf{x}_n| > n \quad (n = 1, 2, 3, \dots).$$

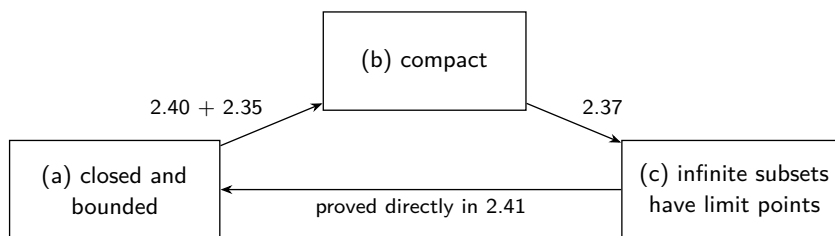
The set S consisting of these points $\mathbf{x}_n$ is infinite and clearly has no limit point in $\mathbb{R}^k$, hence has none in E . Thus (c) implies that E is bounded.

If E is not closed, then there is a point $\mathbf{x}_0 \in \mathbb{R}^k$ which is a limit point of E but not a point of E . For $n = 1, 2, 3, \dots$, there are points $\mathbf{x}_n \in E$ such that $|\mathbf{x}_n - \mathbf{x}_0| < 1/n$. Let S be the set of these points $\mathbf{x}_n$. Then S is infinite (otherwise $|\mathbf{x}_n - \mathbf{x}_0|$ would have a constant positive value, for infinitely many n), S has $\mathbf{x}_0$ as a limit point, and S has no other limit point in $\mathbb{R}^k$. For if $\mathbf{y} \neq \mathbf{x}_0$, then

$$\begin{aligned} |\mathbf{x}_n - \mathbf{y}| &\geq |\mathbf{x}_0 - \mathbf{y}| - |\mathbf{x}_n - \mathbf{x}_0| \\ &\geq |\mathbf{x}_0 - \mathbf{y}| - \frac{1}{n} \geq \frac{1}{2}|\mathbf{x}_0 - \mathbf{y}| \end{aligned}$$

for all but finitely many n ; this shows that $\mathbf{y}$ is not a limit point of S (Theorem 2.20). Thus S has no limit point in E ; hence E must be closed if (c) holds. $\square$

We should remark, at this point, that (b) and (c) are equivalent in any metric space (Exercise 2.26), but that (a) does not, in general, imply (b) and (c). Examples are furnished by Exercise 2.16 and by the space $\mathcal{L}^2$, which is discussed in Chap. 11.



The cycle proved in 2.41. In a general metric space the arrow (a)→(b) is the one that breaks (Exercise 2.16, $\mathcal{L}^2$), while (b)⇔(c) survives everywhere (2.37 and Exercise 2.26).

Where this sits among the completeness theorems

Rudin's remark deserves emphasis: **(a) is the odd one out.** "Closed and bounded" is not equivalent to compactness in a general metric space — it is a fact about $\mathbb{R}^k$ specifically, and the whole apparatus 2.38–2.40 exists to establish it. Exercise 2.16 gives a closed bounded non-compact set inside $\mathbb{Q}$; in $\mathcal{L}^2$ the closed unit ball fails too.

This is also where Chapter 2 rejoins the discussion at Definition 1.10. Of the six equivalent formulations of completeness collected in the ECNU text, three appear in this chapter: **Heine–Borel** is (a)⇔(b) here, **Bolzano–Weierstrass** is 2.42, and the **nested interval** property is 2.38. Rudin proves each separately from the least-upper-bound property rather than deriving them from one another, so their equivalence never becomes visible. It is worth knowing that the four statements — least-upper-bound, nested intervals,

Heine–Borel, Bolzano–Weierstrass — are interderivable, and that Rudin’s ordering is a choice, not a necessity.

Tao (*Analysis II*, 1.5) takes the opposite route: he defines compactness by sequences (every sequence has a convergent subsequence, Rudin’s (c)) and proves the open-cover version afterwards. That ordering is friendlier, and it is why (c) is often the easier property to verify in practice — but it does not generalise past metric spaces, whereas 2.32 does.

2.42 Theorem (Weierstrass). *Every bounded infinite subset of $\mathbb{R}^k$ has a limit point in $\mathbb{R}^k$.*

Proof. Being bounded, the set E in question is a subset of a k -cell $I \subset \mathbb{R}^k$. By Theorem 2.40, I is compact, and so E has a limit point in I , by Theorem 2.37. $\square$

PERFECT SETS

TWO RESULTS, ONE OF THEM A WARNING | Theorem 2.43 is a second, self-contained proof that $\mathbb{R}$ is uncountable, using compactness instead of Cantor’s diagonal — and unlike 2.14 it needs no discussion of binary expansions.

Section 2.44 is the real business. The Cantor set is the standing counterexample of the subject: compact, perfect, uncountable, yet containing no interval and (as Chapter 11 will show) of measure zero. Every intuition of the form “a set that big must contain an interval” dies here. Expect to meet it again in Exercises 2.17–2.18, in Exercise 6.6 (integration against the Cantor set), and in Chapter 11.

2.43 Theorem. *Let P be a nonempty perfect set in $\mathbb{R}^k$. Then P is uncountable.*

Proof. Since P has limit points, P must be infinite. Suppose P is countable, and denote the points of P by $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \dots$. We shall construct a countable collection $\{V_n\}$ of neighborhoods, as follows.¹⁰⁴

Let V_1 be any neighborhood of $\mathbf{x}_1$. If V_1 consists of all $\mathbf{y} \in \mathbb{R}^k$ such that $|\mathbf{y} - \mathbf{x}_1| < r$, then the closure $\overline{V}_1$ of V_1 is the set of all $\mathbf{y} \in \mathbb{R}^k$ such that $|\mathbf{y} - \mathbf{x}_1| \leq r$.

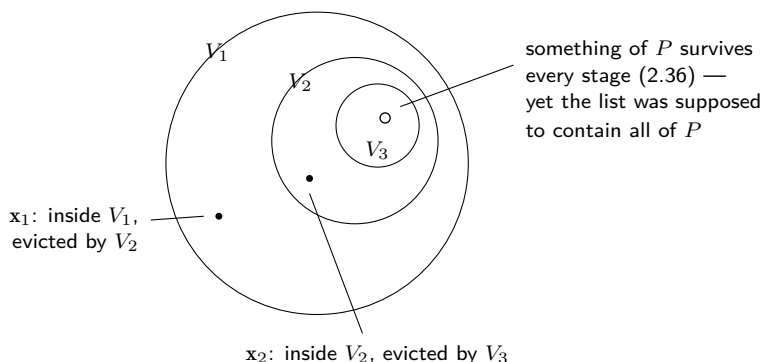
Suppose V_n has been constructed, so that $V_n \cap P$ is not empty. Since every point of P is a limit point of P , there is a neighborhood V_{n+1} such that (i) $\overline{V}_{n+1} \subset V_n$, (ii) $\mathbf{x}_n \notin \overline{V}_{n+1}$, (iii) $V_{n+1} \cap P$ is not empty.¹⁰⁵ By (iii), V_{n+1} satisfies our induction hypothesis, and the construction can proceed.

Put $K_n = \overline{V}_n \cap P$. Since $\overline{V}_n$ is closed and bounded, $\overline{V}_n$ is compact. Since $\mathbf{x}_n \notin K_{n+1}$, no point of P lies in $\bigcap_{n=1}^{\infty} K_n$. Since $K_n \subset P$, this

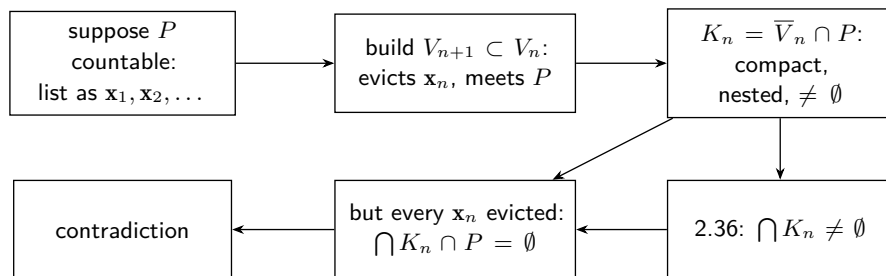
104 The strategy is Cantor’s diagonal in disguise: assume a list of all of P , then trap a point that the list misses. Where 2.14 built the missing object digit by digit, here it is caught by a nested sequence of compact sets — and 2.36 guarantees something survives.

105 Clause (ii) is the diagonal step — it evicts the n th listed point at stage n . Clause (iii) is where perfect is used: because $\mathbf{x}$ is a limit point of P , there are always more points of P nearby to keep the construction alive.

implies that $\bigcap_{n=1}^{\infty} K_n$ is empty. But each K_n is nonempty, by (iii), and $K_n \supset K_{n+1}$, by (i); this contradicts the Corollary to Theorem 2.36. $\square$



Stage $n+1$ shrinks inside stage n and evicts the n th listed point; perfectness keeps each stage nonempty. Compactness forces a survivor, which the list therefore missed — Cantor's diagonal, drawn with balls instead of digits.



The flow: the list drives the construction, perfectness keeps every stage alive, and compactness (2.36) collides with total eviction.

Corollary. Every interval $[a, b]$ ($a < b$) is uncountable. In particular, the set of all real numbers is uncountable.

2.44 The Cantor Set. The set which we are now going to construct shows that there exist perfect sets in $\mathbb{R}^1$ which contain no segment.

Let E_0 be the interval $[0, 1]$. Remove the segment $(\frac{1}{3}, \frac{2}{3})$, and let E_1 be the union of the intervals

$$\left[0, \frac{1}{3}\right], \quad \left[\frac{2}{3}, 1\right].$$

Remove the middle thirds of these intervals, and let E_2 be the union of the intervals

$$\left[0, \frac{1}{9}\right], \quad \left[\frac{2}{9}, \frac{3}{9}\right], \quad \left[\frac{6}{9}, \frac{7}{9}\right], \quad \left[\frac{8}{9}, 1\right].$$

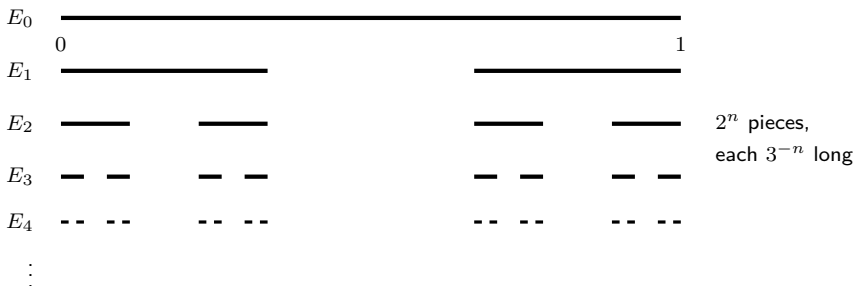
Continuing in this way, we obtain a sequence of compact sets E_n such that

- (a) $E_1 \supset E_2 \supset E_3 \supset \cdots$;
- (b) E_n is the union of 2^n intervals, each of length 3^{-n} .

The set

$$P = \bigcap_{n=1}^{\infty} E_n$$

is called the *Cantor set*. P is clearly compact, and Theorem 2.36 shows that P is not empty.



Removing middle thirds. Total length removed is $\sum 2^{n-1}3^{-n} = 1$, so “nothing” is left — yet what remains is uncountable.

No segment of the form

$$\left(\frac{3k+1}{3^m}, \frac{3k+2}{3^m} \right) \quad (2.24)$$

where k and m are positive integers, has a point in common with P . Since every segment (α, β) contains a segment of the form (2.24), if

$$3^{-m} < \frac{\beta - \alpha}{6},$$

P contains no segment.

To show that P is perfect, it is enough to show that P contains no isolated point. Let $x \in P$, and let S be any segment containing x . Let I_n be that interval of E_n which contains x . Choose n large enough, so that $I_n \subset S$. Let x_n be an endpoint of I_n , such that $x_n \neq x$.

It follows from the construction of P that $x_n \in P$. Hence x is a limit point of P , and P is perfect.¹⁰⁶

One of the most interesting properties of the Cantor set is that it provides us with an example of an uncountable set of measure zero (the concept of measure will be discussed in Chap. 11).

Everything the Cantor set breaks

It is worth listing what P demonstrates, because each item contradicts a plausible belief.

Uncountable but containing no interval. By 2.43 it is uncountable, being perfect and nonempty; yet it contains no segment. So “uncountable” carries no implication of thickness.

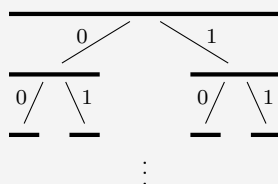
Total length zero. The removed intervals have total length $\frac{1}{3} + \frac{2}{9} + \frac{4}{27} + \cdots = 1$. What remains has measure zero (Chapter 11) — an uncountable set of measure zero.

Every point is a limit of other points, yet no point has

106 The endpoints are the key: they are never removed at any later stage, because removal always takes an open middle third. So every $x \in P$ has other points of P arbitrarily close — the endpoints of the shrinking interval containing it.

neighbours in bulk. P is perfect, so no isolated points; but it is also nowhere dense, so its closure has empty interior. Both at once.

It is not a curiosity. P is homeomorphic to $\{0,1\}^{\mathbb{N}}$, which is why 2.14 and 2.43 measure the same thing — a point of P is exactly a choice of “left third or right third” at every stage, i.e. a 0–1 sequence. That correspondence also makes the uncountability immediate without 2.43.



a point of P = an infinite
path = its address, e.g. 0, 1, 1, 0, ...
— exactly a 0–1 sequence (2.14)

*Left third or right third, forever: the surviving intervals form a binary tree,
and a point of P is an infinite path through it.*

CONNECTED SETS

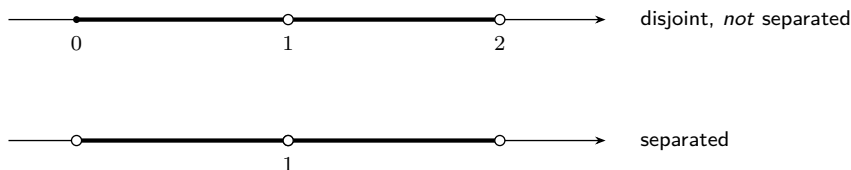
TWO DEFINITIONS AND ONE THEOREM | The shortest section in the chapter, and it exists for the intermediate value theorem (4.23), which will follow in one line from Theorem 4.22: continuous images of connected sets are connected.

The only subtlety is that “connected” is defined via *separated* sets, not disjoint ones. Remark 2.46 is there to stop you conflating the two, and 2.47 shows that on the line the whole notion collapses to something familiar: connected means interval.

2.45 Definition. Two subsets A and B of a metric space X are said to be *separated* if both $A \cap \overline{B}$ and $\overline{A} \cap B$ are empty, i.e., if no point of A lies in the closure of B and no point of B lies in the closure of A .

A set $E \subset X$ is said to be *connected* if E is not a union of two nonempty separated sets.

2.46 Remark. Separated sets are of course disjoint, but disjoint sets need not be separated. For example, the interval $[0, 1]$ and the segment $(1, 2)$ are not separated, since 1 is a limit point of $(1, 2)$. However, the segments $(0, 1)$ and $(1, 2)$ are separated.



*The difference is whether the shared boundary point belongs to one of the sets.
Closures must miss each other, not merely the sets themselves.*

SEPARATED IS STRONGER THAN DISJOINT, AND THE GAP MATTERS

If “connected” were defined by disjointness, *every* set with more than one point would be disconnected — split $[0, 1]$ into $[0, \frac{1}{2}]$ and $(\frac{1}{2}, 1]$ and you have two disjoint nonempty pieces. The definition would be useless.

Requiring the *closures* to miss is what rules this out: $\frac{1}{2}$ lies in the closure of the second piece, so the split is not a separation. In effect, separated sets are ones that do not touch even in the limit.

The connected subsets of the line have a particularly simple structure:

2.47 Theorem. *A subset E of the real line $\mathbb{R}^1$ is connected if and only if it has the following property: If $x \in E$, $y \in E$, and $x < z < y$, then $z \in E$.*

Proof. If there exist $x \in E$, $y \in E$, and some $z \in (x, y)$ such that $z \notin E$, then $E = A_z \cup B_z$ where

$$A_z = E \cap (-\infty, z), \quad B_z = E \cap (z, \infty).$$

Since $x \in A_z$ and $y \in B_z$, A and B are nonempty. Since $A_z \subset (-\infty, z)$ and $B_z \subset (z, \infty)$, they are separated. Hence E is not connected.

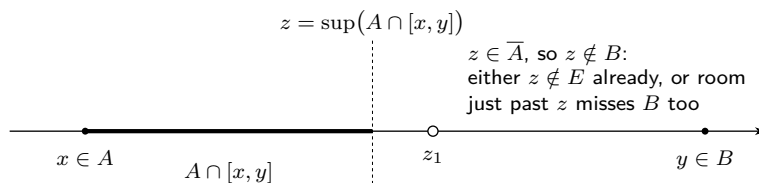
To prove the converse, suppose E is not connected. Then there are nonempty separated sets A and B such that $A \cup B = E$. Pick $x \in A$, $y \in B$, and assume (without loss of generality) that $x < y$. Define

$$z = \sup(A \cap [x, y]).$$

By Theorem 2.28, $z \in \bar{A}$; hence $z \notin B$. In particular, $x \leq z < y$.¹⁰⁷

If $z \notin A$, it follows that $x < z < y$ and $z \notin E$.

If $z \in A$, then $z \notin \bar{B}$, hence there exists z_1 such that $z < z_1 < y$ and $z_1 \notin B$. Then $x < z_1 < y$ and $z_1 \notin E$. $\square$



The same seam picture as Definition 1.8, one chapter later: the supremum lands in $\bar{A}$ and hence off B , and a point missing from E is found at or just past the seam.

CONNECTED ON THE LINE MEANS INTERVAL

The property in 2.47 — if two points are in E , everything between them is too — is exactly the definition of an interval, in the general sense that includes $\emptyset$, single points, half-lines and $\mathbb{R}$ itself. So the theorem reads: *the connected subsets of $\mathbb{R}$ are precisely the intervals*.

That is worth holding because it makes the intermediate value theorem (4.23) transparent. If f is continuous on $[a, b]$ then $f([a, b])$

107 Chapter 1 again. The supremum exists by 1.19, lands in $\bar{A}$ by 2.28, and therefore avoids B because A and B are separated — that is the one place separation is spent. Every step of this converse is an application of something already proved.

is connected (4.22), hence an interval by 2.47, hence contains every value between $f(a)$ and $f(b)$ — the intermediate value theorem. Chapter 1's Theorem 1.21 on n th roots was a hand-rolled special case of exactly this, as the note there observed.

Note also that connectedness, unlike openness, is intrinsic: like compactness it survives change of ambient space, since 2.45 refers only to closures within E 's own metric.

CHAPTER 2 IN ONE PARAGRAPH

Countability sorts infinite sets into two sizes, and $\mathbb{Q}$ and $\mathbb{R}$ land on opposite sides (2.13, 2.43). Metric spaces distill $\mathbb{R}^k$ down to its distance function, and open, closed and limit point are the vocabulary that distance alone supports (2.15–2.30). Compactness — every open cover admits a finite subcover — is the property that turns local facts into global ones, and in $\mathbb{R}^k$ it is exactly closed-and-bounded (2.41), a fact that rests on completeness through the nested-interval theorems (2.38–2.40) and on nothing else. Perfect sets are uncountable (2.43), and the Cantor set shows how thin one can be (2.44). Connected subsets of the line are precisely the intervals (2.47). Chapter 4 will spend all of it: continuity gets defined by inverse images of open sets, compactness becomes boundedness and attainment of extrema, and connectedness becomes the intermediate value theorem.

EXERCISES

HOW TO USE THESE

Chapter 2's exercises are less optional than Chapter 1's. Several supply results the book later assumes, and one chain (2.22–2.26) proves a theorem Rudin states in the text without proof.

The chain that matters. $2.22 \rightarrow 2.23 \rightarrow 2.24 \rightarrow 2.26$ builds up to the claim made in passing after 2.41: that (b) and (c) are equivalent in *any* metric space, not just $\mathbb{R}^k$. Rudin proves (b) $\Rightarrow$ (c) as 2.37 and leaves the converse here. Do them in order; each is a step.

Counterexample exercises. 2.14, 2.15, 2.16 show what fails without compactness, and 2.16 is the one Rudin cites in the text — a closed bounded set that is not compact, living in $\mathbb{Q}$.

Baire. 2.30 is a special case of Baire's theorem, proved by imitating 2.43. It is used throughout functional analysis, and Exercise 3.22 gives the general version.

Routine. 2.1, 2.5–2.9, 2.11 are short checks on the definitions. Do them quickly to make sure the vocabulary of 2.18 has landed.

2.1. Prove that the empty set is a subset of every set.¹⁰⁸

2.2. A complex number z is said to be *algebraic* if there are integers $a_0, \dots, a_n$, not all zero, such that

$$a_0 z^n + a_1 z^{n-1} + \dots + a_{n-1} z + a_n = 0.$$

Prove that the set of all algebraic numbers is countable. *Hint:* For every positive integer N there are only finitely many equations with

$$n + |a_0| + |a_1| + \dots + |a_n| = N.$$

109

2.3. Prove that there exist real numbers which are not algebraic.¹¹⁰

2.4. Is the set of all irrational real numbers countable?¹¹¹

2.5. Construct a bounded set of real numbers with exactly three limit points.¹¹²

2.6. Let E' be the set of all limit points of a set E . Prove that E' is closed. Prove that E and $\overline{E}$ have the same limit points. (Recall that $\overline{E} = E \cup E'$.) Do E and E' always have the same limit points?¹¹³

2.7. Let $A_1, A_2, A_3, \dots$ be subsets of a metric space.

(a) If $B_n = \bigcup_{i=1}^n A_i$, prove that $\overline{B_n} = \bigcup_{i=1}^n \overline{A_i}$ for $n = 1, 2, 3, \dots$

(b) If $B = \bigcup_{i=1}^{\infty} A_i$, prove that $\overline{B} \supset \bigcup_{i=1}^{\infty} \overline{A_i}$.

Show, by an example, that this inclusion can be proper.¹¹⁴

2.8. Is every point of every open set $E \subset \mathbb{R}^2$ a limit point of E ? Answer the same question for closed sets in $\mathbb{R}^2$.¹¹⁵

2.9. Let E° denote the set of all interior points of a set E . [See Definition 2.18(e); E° is called the *interior* of E .]

(a) Prove that E° is always open.

(b) Prove that E is open if and only if $E^\circ = E$.

(c) If $G \subset E$ and G is open, prove that $G \subset E^\circ$.

108 *Vacuous implication:* “every $x \in \emptyset$ satisfies ...” is true because there is no x to check. If you find this uncomfortable, note the alternative would make $\emptyset \subset \emptyset$ false.

109 *The hint is the whole exercise, and it is the anti-diagonal idea of 2.12 again: sort the objects by a size parameter N so that each level is finite, then take a countable union. Each equation has at most n roots, so level N contributes finitely many numbers.*

110 *One line from 2.2 and 2.43: the algebraic numbers are countable, $\mathbb{R}$ is not, so something is left over. Note that this proves transcendentals exist without exhibiting a single one — Cantor’s original observation, and a century younger than the first explicit example.*

111 *No, and the argument is the same as 2.3: if it were, $\mathbb{R} = \mathbb{Q} \cup (\mathbb{R} \setminus \mathbb{Q})$ would be a union of two countable sets, hence countable by 2.12.*

112 *Take three copies of $\{1/n\}$, translated. The point is that limit points can be prescribed at will — there is nothing canonical about them.*

113 *The last question is the interesting one and the answer is no. Take $E = \{1/n\}$: then $E' = \{0\}$, which has no limit points at all, while E has one. Adding limit points can destroy them.*

114 *Finite versus infinite once more — (a) is an equality, (b) only an inclusion. For the proper inclusion enumerate $\mathbb{Q}$ as $A_i = \{q_i\}$: each $\overline{A_i}$ is a point, their union is $\mathbb{Q}$, but $\overline{B} = \mathbb{R}$.*

115 *Yes for open, no for closed. An open set contains a ball about each of its points, and a ball in $\mathbb{R}^2$ has infinitely many points; a closed set may be a single point. So open sets in $\mathbb{R}^k$ ($k \geq 1$) never have isolated points — which is why the perfect sets of 2.44 are interesting: they are closed with no isolated points either, without containing any ball.*

116 Parts (a)–(c) are 2.27 with every inclusion reversed: interior is the largest open set inside, as closure is the smallest closed set outside. Part (d), $(E^\circ)^c = \overline{E^c}$, is the duality that makes the two notions one notion. For (e) and (f) the answer is no both times; $E = \mathbb{Q}$ settles both.

117 The discrete metric, and the standard sanity check on every definition in the chapter. Every subset is open (the ball of radius $\frac{1}{2}$ about p is $\{p\}$), hence every subset is closed, and the compact sets are exactly the finite ones — cover an infinite set by singletons and no finite subfamily suffices. Note the consequence: here every set is closed and bounded, yet almost none is compact. That is Rudin's remark after 2.41, in its cheapest form.

118 Check the three axioms of 2.15 one at a time and the failures are quick: d_1 fails the triangle inequality, d_3 fails $d(p, q) > 0$ for $p \neq q$ (take $x = -y$), d_4 fails symmetry and even $d(p, p) = 0$. The two that work, d_2 and d_5 , are worth remembering — d_5 is the standard way to make any metric bounded without changing which sets are open.

119 The model example, and worth doing by hand. Whatever cover is given, some G_α contains 0; being open it contains all $1/n$ beyond some index, leaving only finitely many points to cover one at a time. Compare 2.32: this is the compactness proof in miniature, and it is why $\{1/n\}$ alone — without 0 — is not compact.

120 Nest the previous example: put a shrunken copy of K near each $1/n$. The limit points are then $\{1/n\} \cup \{0\}$.

121 $\{(1/n, 1)\}$. See the figure at 2.32 — this is exactly the picture drawn there.

122 For “closed” take $F_n = [n, \infty)$; for “bounded” take $(0, 1/n)$. Both are nested with empty intersection, and each fails exactly one half of “closed and bounded.” Compare Example 2.10(b)(iii).

(d) Prove that the complement of E° is the closure of the complement of E .

(e) Do E and $\overline{E}$ always have the same interiors?

(f) Do E and E° always have the same closures?

116

2.10. Let X be an infinite set. For $p \in X$ and $q \in X$, define

$$d(p, q) = \begin{cases} 1 & (\text{if } p \neq q) \\ 0 & (\text{if } p = q). \end{cases}$$

Prove that this is a metric. Which subsets of the resulting metric space are open? Which are closed? Which are compact?¹¹⁷

2.11. For $x \in \mathbb{R}^1$ and $y \in \mathbb{R}^1$, define

$$\begin{aligned} d_1(x, y) &= (x - y)^2, & d_2(x, y) &= \sqrt{|x - y|}, \\ d_3(x, y) &= |x^2 - y^2|, & d_4(x, y) &= |x - 2y|, \\ d_5(x, y) &= \frac{|x - y|}{1 + |x - y|}. \end{aligned}$$

Determine, for each of these, whether it is a metric or not.¹¹⁸

2.12. Let $K \subset \mathbb{R}^1$ consist of 0 and the numbers $1/n$ for $n = 1, 2, 3, \dots$. Prove that K is compact directly from the definition (without using the Heine-Borel theorem).¹¹⁹

2.13. Construct a compact set of real numbers whose limit points form a countable set.¹²⁰

2.14. Give an example of an open cover of the segment $(0, 1)$ which has no finite subcover.¹²¹

2.15. Show that Theorem 2.36 and its Corollary become false (in $\mathbb{R}^1$, for example) if the word “compact” is replaced by “closed” or by “bounded.”¹²²

2.16. Regard $\mathbb{Q}$, the set of all rational numbers, as a metric space, with $d(p, q) = |p - q|$. Let E be the set of all $p \in \mathbb{Q}$ such that $2 < p^2 < 3$. Show that E is closed and bounded in $\mathbb{Q}$, but that E is not compact. Is E open in $\mathbb{Q}$?¹²³

2.17. Let E be the set of all $x \in [0, 1]$ whose decimal expansion contains only the digits 4 and 7. Is E countable? Is E dense in $[0, 1]$? Is E compact? Is E perfect?¹²⁴

2.18. Is there a nonempty perfect set in $\mathbb{R}^1$ which contains no rational number?¹²⁵

2.19. (a) If A and B are disjoint closed sets in some metric space X , prove that they are separated.

(b) Prove the same for disjoint open sets.

(c) Fix $p \in X$, $\delta > 0$, define A to be the set of all $q \in X$ for which $d(p, q) < \delta$, define B similarly, with $>$ in place of $<$. Prove that A and B are separated.

(d) Prove that every connected metric space with at least two points is uncountable. *Hint:* Use (c).

126

2.20. Are closures and interiors of connected sets always connected? (Look at subsets of $\mathbb{R}^2$.)¹²⁷

2.21. Let A and B be separated subsets of some $\mathbb{R}^k$, suppose $\mathbf{a} \in A$, $\mathbf{b} \in B$, and define

$$\mathbf{p}(t) = (1 - t)\mathbf{a} + t\mathbf{b}$$

for $t \in \mathbb{R}^1$. Put $A_0 = \mathbf{p}^{-1}(A)$, $B_0 = \mathbf{p}^{-1}(B)$. [Thus $t \in A_0$ if and only if $\mathbf{p}(t) \in A$.]

(a) Prove that A_0 and B_0 are separated subsets of $\mathbb{R}^1$.

(b) Prove that there exists $t_0 \in (0, 1)$ such that $\mathbf{p}(t_0) \notin A \cup B$.

(c) Prove that every convex subset of $\mathbb{R}^k$ is connected.

128

2.22. A metric space is called *separable* if it contains a countable dense subset. Show that $\mathbb{R}^k$ is separable. *Hint:* Consider the set of points which have only rational coordinates.¹²⁹

2.23. A collection $\{V_\alpha\}$ of open subsets of X is said to be a *base* for X if the following is true: For every $x \in X$ and every open set $G \subset X$ such that $x \in G$, we have $x \in V_\alpha \subset G$ for some α . In other words, every open set in X is the union of a subcollection of $\{V_\alpha\}$.

Prove that every separable metric space has a countable base. *Hint:* Take all neighborhoods with rational radius and center in some countable dense subset of X .¹³⁰

2.24. Let X be a metric space in which every infinite subset has a limit point. Prove that X is separable. *Hint:* Fix $\delta > 0$, and pick $x_1 \in X$. Having chosen $x_1, \dots, x_j \in X$, choose $x_{j+1} \in X$, if

[123] The exercise Rudin cites in the text after 2.41, and the cleanest demonstration that (a) does not imply (b) outside $\mathbb{R}^k$. E is closed in $\mathbb{Q}$ because its would-be boundary points $\pm\sqrt{2}$, $\pm\sqrt{3}$ are not rational — the holes in $\mathbb{Q}$ from Chapter 1 are doing the work. It is also open in $\mathbb{Q}$, so here is a set that is both, in a space that is connected nowhere. Everything that goes wrong traces back to $\mathbb{Q}$ lacking the least-upper-bound property.

124 A Cantor set in disguise: uncountable (diagonalise, 2.14), nowhere dense, compact and perfect. Build it as an intersection of nested closed sets exactly as in 2.44 and every answer follows from that section.

125 Yes. Repeat the construction of 2.44 but at stage n remove an open interval around the n th rational, small enough to leave two closed pieces. Enumerating $\mathbb{Q}$ is possible by the Corollary to 2.13 — this is one of the places countability is actually used.

126 (a) and (b) say that for closed or open sets, disjoint already implies separated — the distinction Remark 2.46 draws only bites for sets that are neither. (d) is the striking one: connectedness forces uncountability. By (c) the space splits at every radius δ unless some point sits at distance exactly δ from p ; so all but countably many δ would disconnect X , and there must be uncountably many distances.

[127] Closures yes, interiors no — and the asymmetry is easy to miss. For the counterexample take two closed discs in the plane touching at a single point: the union is connected, but removing the boundary disconnects the interior into two open discs.

128 The strategy: pull a question about $\mathbb{R}^k$ back to a question about the line, where 2.47 settles everything. Part (b) is 2.47 applied to $A_0 \cup B_0$, which cannot be all of $[0, 1]$ because that would be connected. This is the first appearance of “restrict to a segment,” a device used constantly in Chapter 9.

129 Rational points are countable by 2.13 and dense by 1.20(b) applied coordinatewise. This is the first link of the 2.22–2.26 chain, and it is where Chapter 1’s density theorem is finally spent.

130 A countable base is what lets you replace an arbitrary open cover by a countable one — the crucial reduction in 2.26. Countability of the family comes from 2.13 (pairs: centre and radius).

131 The process must stop because an infinite δ -separated set has no limit point — any two of its points stay δ apart, so no neighbourhood of radius $\delta/2$ can hold two of them. This “greedy separation” construction is worth learning; it appears again in proofs of total boundedness.

132 Direct from 2.32: the balls of radius $1/n$ about every point form an open cover, so finitely many suffice. Taking all n gives a countable family.

133 This completes the claim Rudin makes in the text after 2.41: (b) and (c) are equivalent in any metric space. Only (a) is special to $\mathbb{R}^k$. The exercise is genuinely hard and genuinely important; do 2.22–2.25 first, since each is used. Tao proves the same equivalence as his Theorem 1.5.8, taking the sequential property as the definition and deriving the cover property — the reverse of Rudin’s order.

134 “Limit point” upgraded from “infinitely many nearby” (2.20) to “uncountably many nearby.” The result says an uncountable set is a perfect set plus countable debris — the Cantor–Bendixson theorem, and the reason 2.28 follows.

135 The structure theorem for open subsets of the line, and it has no analogue in $\mathbb{R}^2$. Take the maximal segment inside the set around each point — these are the connected components, by 2.47 — and each contains a rational, so there are at most countably many.

possible, so that $d(x_i, x_{j+1}) \geq \delta$ for $i = 1, \dots, j$. Show that this process must stop after a finite number of steps, and that X can therefore be covered by finitely many neighborhoods of radius δ . Take $\delta = 1/n$ ($n = 1, 2, 3, \dots$), and consider the centers of the corresponding neighborhoods.¹³¹

2.25. Prove that every compact metric space K has a countable base, and that K is therefore separable. *Hint:* For every positive integer n , there are finitely many neighborhoods of radius $1/n$ whose union covers K .¹³²

2.26. Let X be a metric space in which every infinite subset has a limit point. Prove that X is compact. *Hint:* By Exercises 2.23 and 2.24, X has a countable base. It follows that every open cover of X has a countable subcover $\{G_n\}$, $n = 1, 2, 3, \dots$. If no finite subcollection of the $\{G_n\}$ covers X , then the complement F_n of $G_1 \cup \dots \cup G_n$ is nonempty for each n , but $\bigcap_{n=1}^{\infty} F_n$ is empty. If E is a set which contains a point from each F_n , consider a limit point of E , and obtain a contradiction.¹³³

2.27. Define a point p in a metric space X to be a *condensation point* of a set $E \subset X$ if every neighborhood of p contains uncountably many points of E .

Suppose $E \subset \mathbb{R}^k$, E is uncountable, and let P be the set of all condensation points of E . Prove that P is perfect and that at most countably many points of E are not in P . In other words, show that $P^c \cap E$ is at most countable. *Hint:* Let $\{V_n\}$ be a countable base of $\mathbb{R}^k$, let W be the union of those V_n for which $E \cap V_n$ is at most countable, and show that $P = W^c$.¹³⁴

2.28. Prove that every closed set in a separable metric space is the union of a (possibly empty) perfect set and a set which is at most countable. (*Corollary:* Every countable closed set in $\mathbb{R}^k$ has isolated points.) *Hint:* Use Exercise 2.27.

2.29. Prove that every open set in $\mathbb{R}^1$ is the union of an at most countable collection of disjoint segments. *Hint:* Use Exercise 2.22.¹³⁵

2.30. Imitate the proof of Theorem 2.43 to obtain the following result:

If $\mathbb{R}^k = \bigcup_{n=1}^{\infty} F_n$, where each F_n is a closed subset of $\mathbb{R}^k$, then at least one F_n has a nonempty interior.

Equivalent statement: If G_n is a dense open subset of $\mathbb{R}^k$ for $n = 1, 2, 3, \dots$, then $\bigcap_{n=1}^{\infty} G_n$ is not empty (in fact, it is dense in $\mathbb{R}^k$).

(This is a special case of Baire's theorem; see Exercise 3.22 for the general case.)¹³⁶

136 *Baire's theorem, and the most consequential exercise in the chapter. It says $\mathbb{R}^k$ cannot be assembled from countably many "thin" closed pieces — a completeness statement, ultimately resting on 2.36 exactly as 2.43 did. It is the engine behind the uniform boundedness principle and the open mapping theorem in functional analysis, and it gives a one-line proof that $\mathbb{R}$ is uncountable: a countable set is a countable union of points, each closed with empty interior.*

Appendix to Chapter 2

HOW TO USE THIS APPENDIX

Nothing in this appendix is Rudin's text. It is imported from the four comparison texts and set apart at every level: the running head names the appendix, section heads carry a reverse-video tag, item numbers are prefixed **S**, and every statement names its source.

Why import anything at all. Rudin's Chapter 2 is thirty pages of topology with no payoff: every application waits for Chapter 4. He also makes two economies that cost the reader later — he defines *limit point* once and never names the two other formulations he goes on to use, and he proves compactness of a cell by an inlined bisection whose key step is never isolated as a lemma. The Chinese texts organise the same material around *what each theorem buys*, and Tao and Yu Pin carry the metric-space theory past the point where Rudin stops.

What is here.

A. Cluster points. Three inequivalent-looking definitions of “limit point,” proved equivalent, and the collision between the set version and the sequence version that Rudin's notation hides.

B. The completeness theorems as tools. Nested intervals with the corollary Rudin omits; Heine–Borel proved through it instead of by raw bisection; Weierstrass twice; and a construction of the least upper bound *from* Cauchy completeness — the one direction Rudin never runs, since he takes LUB as an axiom.

C. The Heine–Borel Rudin declines. At 2.41 he proves the equivalences *in $\mathbb{R}^k$ only*. The general metric-space theorem exists — compact = complete + totally bounded — and it names exactly which property $\mathbb{R}^k$ was supplying.

D. Two notions of smallness. Cardinality is not the only way a set can be small. Baire's theorem supplies the topological notion, and the two disagree in both directions — with Rudin's own Cantor set as one witness.

E. What survives without a metric. Which of Chapter 2 is really about distance, and which is about open sets alone.

How to read it. Section A is worth reading before Chapter 3, because the confusion it settles is the one that bites at 3.5–3.6.

Section B is best read after 2.41, as a second pass over the same theorems by different routes; C and D after the whole chapter. E is optional and answers the question “why a metric at all?” The exercises are the last section.

NOT IN RUDIN

SUPPLEMENT A — LIMIT POINTS, CLUSTER POINTS, AND A COLLISION

WHY THIS SUPPLEMENT BELONGS TO CHAPTER 2

What Rudin does. Definition 2.18(b) fixes one meaning of *limit point*: every neighbourhood of p meets E in a point other than p . Theorem 2.20 then proves a second — every neighbourhood meets E in *infinitely many* points — and uses it to conclude that a finite set has no limit points.

The gap. There is a third formulation, the one that says p is the limit of a sequence of *distinct* points of E , and Rudin never states it. He nevertheless relies on it constantly from Chapter 3 onward: every argument that begins “choose a sequence in E converging to p ” is using it, and every such argument silently needs the chosen points to be distinct.

Where the gap bites. Twice, and both times painfully. First at 3.5–3.6, where Rudin passes between statements about *sets* (2.42: an infinite subset of a compact set has a limit point) and statements about *sequences* (3.6(b): a bounded sequence in $\mathbb{R}^k$ has a convergent subsequence) as though they were the same theorem. They are not — a sequence has a range, and the range can be finite while the sequence has several subsequential limits. Second at 4.2, where continuity is tested against sequences and one must know which points those sequences may repeat.

What this section supplies. All three formulations, proved equivalent, with the recursion that forces distinctness written out; and the one-line example that separates the set notion from the sequence notion, so that the translation between 2.42 and 3.6(b) can be made carefully instead of tacitly.

S2.1 Theorem. (*Three definitions of a cluster point.*) Let $E \subset \mathbb{R}^1$ and $\xi \in \mathbb{R}^1$. The following are equivalent.

- (a) Every neighbourhood of ξ contains *infinitely many* points of E .
- (b) Every neighbourhood of ξ contains a point of E other than ξ — Rudin’s Definition 2.18(b).

- (c) *There is a sequence (x_n) in E whose terms are **pairwise distinct** with $x_n \rightarrow \xi$.*

SOURCE: ECNU 聚点 Def. 2, 2', 2'' (bk. pp. 151–152). Rudin states (b) and proves (a) as 2.20; (c) he never states.

The one implication with content. That (a) $\Rightarrow$ (b) and (c) $\Rightarrow$ (a) are immediate. For (b) $\Rightarrow$ (c), build the sequence recursively, and *force* distinctness by shrinking the radius past the previous point: having chosen x_{n-1} , apply (b) with

$$\varepsilon_n = \min\left\{\frac{1}{n}, |\xi - x_{n-1}|\right\}$$

to obtain $x_n \in E$ with $0 < |x_n - \xi| < \varepsilon_n$. Then $x_n \rightarrow \xi$, and no point can repeat, since each is strictly nearer ξ than its predecessor. $\square$

WHY THE RECURSION NEEDS THAT PARTICULAR RADIUS

The obvious recursion — take $|x_n - \xi| < 1/n$ — gives convergence but not distinctness: nothing stops the same point being chosen twice. The extra clause $|\xi - x_{n-1}|$ in the minimum makes the distances strictly decreasing, and a strictly decreasing sequence of distances cannot repeat a point.

This is exactly the step a reader of Rudin has to invent unaided. Theorem 2.20 gives him “infinitely many points in every neighbourhood”; every later argument that says “choose a sequence in E converging to ξ ” silently needs those points to be distinct, and that is what the recursion supplies. Rudin uses the fact at 2.20’s corollary (a finite set has no limit points), at 3.6, and throughout Chapter 4.

S2.2 Definition. (*Cluster point of a sequence.*) a is a cluster point of the sequence (x_n) if every neighbourhood of a contains x_n for **infinitely many indices** n — terms counted with repetition. Equivalently, a is the limit of some subsequence.

SOURCE: ECNU §7.2 Def. 1 and the following 注 (bk. p. 156).

THE SET OF TERMS IS NOT THE SEQUENCE

The two notions disagree, and the cleanest witness is one line. Take

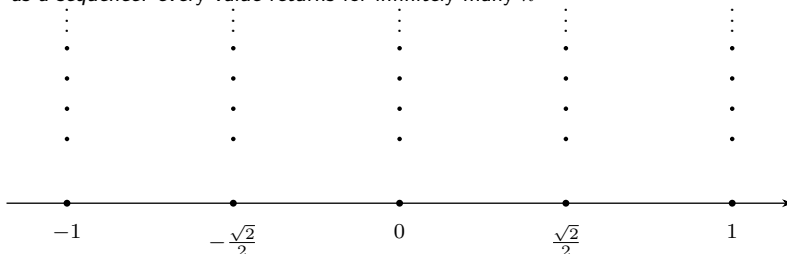
$$x_n = \sin \frac{n\pi}{4}.$$

As a *sequence* it has five cluster points: $-1, -\frac{\sqrt{2}}{2}, 0, \frac{\sqrt{2}}{2}, 1$, each hit infinitely often. As a *set*, its range is the five-element set $\{-1, -\frac{\sqrt{2}}{2}, 0, \frac{\sqrt{2}}{2}, 1\}$, and a finite set has **no** limit points at all (Rudin’s corollary to 2.20).

So “limit point” in Chapter 2 and “subsequential limit” in 3.5–3.6 are different notions that can disagree completely. Rudin uses both words without ever flagging the difference, and 3.6(b) — every bounded sequence in $\mathbb{R}^k$ has a convergent subsequence — is *not* literally 2.42 applied to the range, because the range may be finite. The bridge is:

a subsequential limit of (x_n) is either a limit point of the range, or a value taken infinitely often. Rudin's proof of 3.6(b) quietly splits into exactly those two cases.

as a sequence: every value returns for infinitely many n



as a set: just these five points — no limit point at all

$x_n = \sin(n\pi/4)$ has five cluster points as a sequence and none as a set. Multiplicity is exactly the information the range throws away — which is why Rudin's 2.42 (about sets) does not directly give 3.6(b) (about sequences).

NOT IN RUDIN

SUPPLEMENT B — THE COMPLETENESS THEOREMS AS TOOLS

WHY THIS SUPPLEMENT BELONGS TO CHAPTER 2

What Rudin does. He assumes the least-upper-bound property as an axiom (1.19) and derives everything from it in one direction: 1.20 (Archimedes and density), 2.38 (nested compact sets), 2.40 (k -cells are compact), 2.41 (Heine–Borel in $\mathbb{R}^k$), 3.11 (completeness).

The gap. A one-directional development hides the fact that these statements are *equivalent*. Chapter 1's comparison box lists six formulations of completeness and notes that different books axiomatise different ones; but a list of equivalent statements teaches nothing until you have seen the implications run. Rudin runs them only outward from LUB, so a reader has no way to discover that LUB can be recovered from Cauchy completeness — or what extra hypothesis that recovery needs.

Where the gap bites. At 2.40 the crucial step — that a nested family with vanishing diameters is eventually swallowed by any neighbourhood of its common point — is derived inline, in the middle of a k -dimensional argument, and never named. It is needed again at 2.42, at 3.10, and in every later bisection argument, and each time it is rebuilt from scratch.

What this section supplies. That step, isolated as a lemma with its corollary (S2.3), after which Heine–Borel is three lines

(S2.4) and Weierstrass is four (S2.5); a proof of Bolzano–Weierstrass that uses no bisection at all and secretly computes $\liminf$ (S2.6); and the missing implication, a construction of $\sup$ from Cauchy completeness plus Archimedes (S2.7), which shows precisely why the Archimedean hypothesis cannot be dropped.

S2.3 Theorem. (*Nested intervals, with its corollary.*) Let $[a_{n+1}, b_{n+1}] \subset [a_n, b_n]$ for all n and $b_n - a_n \rightarrow 0$. Then there is exactly one ξ with $\xi \in [a_n, b_n]$ for every n ; and moreover, for every $\varepsilon > 0$ there is an N with

$$[a_n, b_n] \subset (\xi - \varepsilon, \xi + \varepsilon) \quad (n > N).$$

SOURCE: ECNU 区间套定理 7.1 and its 推论 (bk. pp. 150–151).

Proof. The a_n increase and every b_m is an upper bound for them: if $m \leq n$ then $a_n \leq b_n \leq b_m$, and if $m > n$ then $a_n \leq a_m \leq b_m$. So $\{a_n\}$ is nonempty and bounded above, and by 1.19 it has a supremum ξ , with $a_n \leq \xi \leq b_n$ for every n . If ξ' also lay in every $[a_n, b_n]$ then $|\xi - \xi'| \leq b_n - a_n \rightarrow 0$, so $\xi' = \xi$.

For the corollary, given $\varepsilon > 0$ choose N with $b_N - a_N < \varepsilon$. For $n > N$ the interval $[a_n, b_n]$ is contained in $[a_N, b_N]$, has length less than ε , and contains ξ ; hence every one of its points lies within ε of ξ . $\square$

Rudin's 2.38 gives nonempty intersection for nested compact sets and stops there. The corollary — eventually the whole interval, not merely the point, sits inside the neighbourhood — is what does the work in every application below.¹³⁷

CLOSED IS NOT DECORATION

The intervals must be closed. The open intervals $(0, 1/n)$ are nested with lengths tending to 0 and have **empty** intersection. Boundedness matters too: the closed sets $[n, \infty)$ are nested with empty intersection. Rudin's 2.38 requires compactness, which packages both hypotheses at once; seeing the two failures separately is what makes the packaging legible.

S2.4 Theorem. (*Heine–Borel, via nested intervals.*) Every open cover of $[a, b]$ has a finite subcover.

SOURCE: ECNU 有限覆盖定理 7.3 (bk. p. 153).

HOW THIS PROOF DIFFERS FROM RUDIN'S 2.40

1. **Same start.** Suppose no finite subcover exists. Bisect $[a, b]$; at least one half again admits no finite subcover. Iterate, producing nested intervals with lengths $\rightarrow 0$.
2. **The step that is now free.** Let ξ be the common point.

137 The proof of the corollary is one line: $\xi \in [a_n, b_n]$ and $b_n - a_n \rightarrow 0$, so both endpoints are within ε of ξ once $b_n - a_n < \varepsilon$. Trivial to prove, easy to forget you need, and Rudin never names it — so at 2.40 he re-derives it inline.

It lies in some cover element (α, β) . By the *corollary* to S2.3, $[a_n, b_n] \subset (\alpha, \beta)$ for large n — so that interval is covered by a *single* member, contradicting the choice of intervals with no finite subcover.

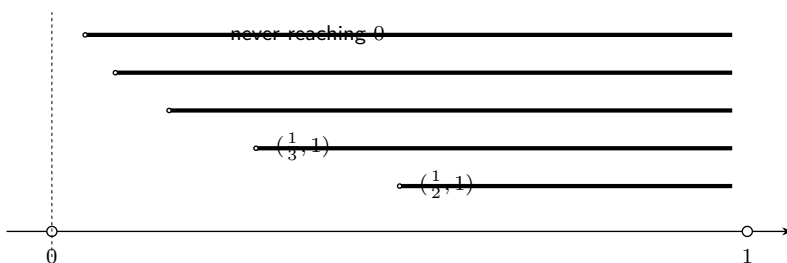
3. **What changed.** Rudin's 2.40 proves the same thing but derives step 2 inline, in the middle of a k -dimensional argument, using $\text{diam } Q_n \rightarrow 0$. Naming the lemma first turns the proof into three lines and makes visible that bisection is used only to *manufacture* a nested family — any other method of manufacturing one would do.
4. **Audit.** Closedness of $[a, b]$ enters once, in S2.3; boundedness enters once, to start the bisection. Both hypotheses are spent exactly where the pitfall above says they must be.

WHERE THE THEOREM FAILS, MADE CONCRETE

On $(0, 1)$ the cover

$$\left\{ \left(\frac{1}{n+1}, 1 \right) : n = 1, 2, 3, \dots \right\}$$

works and no finite subfamily does: the members increase, so a finite subfamily is its largest member, which omits everything below $\frac{1}{N+1}$. Rudin gives the analogous fact at 2.32's examples; the point worth keeping is the mechanism — a totally ordered cover with no largest element can only be finitely reduced if one member already suffices.



The cover of $(0, 1)$ that admits no finite subcover. The members are nested upward, so any finite subfamily equals its largest member and misses a whole interval at the left end. Compactness fails exactly at the point the domain is missing.

S2.5 Theorem. (Weierstrass.) Every bounded infinite subset of $\mathbb{R}^1$ has a cluster point.

SOURCE: ECNU 聚点定理 7.2, with two proofs (bk. pp. 152–153).

Proof. Let E be infinite with $E \subset [-M, M]$. Bisect: at least one of the two halves contains infinitely many points of E , since otherwise E would be finite. Call it $[a_1, b_1]$ and repeat, obtaining nested intervals $[a_n, b_n]$, each containing infinitely many points of E , with $b_n - a_n = 2M/2^n \rightarrow 0$.

By S2.3 there is a point ξ common to all of them. Given $\varepsilon > 0$, the

corollary to S2.3 puts $[a_n, b_n]$ inside $(\xi - \varepsilon, \xi + \varepsilon)$ for all large n ; and $[a_n, b_n]$ contains infinitely many points of E . So every neighbourhood of ξ contains infinitely many points of E , which is S2.1(a). $\square$

Two proofs, landing on two different definitions

First proof (bisection). $E \subset [-M, M]$; bisect, always keeping a half containing infinitely many points of E . By S2.3 and its corollary, every neighbourhood of the common point ξ contains some $[a_n, b_n]$, hence infinitely many points of E . This lands on definition (a) of S2.1.

Second proof (via sequences). Choose pairwise distinct $x_n \in E$; the sequence is bounded, so some subsequence converges, say to x_0 ; then every neighbourhood of x_0 contains infinitely many of the x_n . This lands on definition (c).

The two proofs are the same argument in different costumes, and S2.1 is what lets you move between them. ECNU then states the dictionary Rudin leaves implicit: *the sequence form of Bolzano–Weierstrass is the set form applied to the set of terms, with “infinitely many points” read as “infinitely many indices.”* Rudin’s 2.42 and 3.6(b) sit twenty pages apart with no remark connecting them — and, as the pitfall at S2.2 shows, the translation needs the care that remark would have supplied.

S2.6 Theorem. (*Bolzano–Weierstrass without bisection.*) Every bounded sequence (a_n) of reals has a convergent subsequence.

SOURCE: USTC 定理 8 (bk. p. 12).

The monotone-tail proof. Put $\alpha_n = \inf\{a_n, a_{n+1}, \dots\}$. The sequence (α_n) is increasing and bounded above, hence converges; call the limit α . Now build indices: by the definition of infimum there is n_1 with $\alpha_1 \leq a_{n_1} < \alpha_1 + 1$; having chosen n_k , there is $n_{k+1} > n_k$ with

$$\alpha_{n_k} \leq a_{n_{k+1}} < \alpha_{n_k} + \frac{1}{k+1}.$$

Since $\alpha_{n_k} \rightarrow \alpha$, the squeeze gives $a_{n_k} \rightarrow \alpha$. $\square$

What this proof reveals that Rudin’s route hides

Rudin reaches Bolzano–Weierstrass only through the cell machinery: 2.40 (k -cells are compact) $\rightarrow$ 2.42 (infinite subsets of compact sets have limit points) $\rightarrow$ 3.6(b). That path makes the theorem look as though it *needs* compactness, and in $\mathbb{R}^k$ it is the efficient route. On the line it needs nothing but the monotone convergence theorem. And the auxiliary sequence is not arbitrary: $\alpha_n = \inf_{k \geq n} a_k$ is precisely the quantity whose limit Rudin will call $\liminf a_n$ at 3.16. So this proof silently establishes something stronger than it claims — *the subsequential limit it produces is the lower limit* — which is Rudin’s 3.17(a) arriving a chapter early and for free.

Keep the contrast: bisection halves the *domain*, the monotone-tail argument shaves the *index set*. Bisection generalises to $\mathbb{R}^k$; the tail argument does not, because it needs the order.

S2.7 Theorem. (*The least upper bound, constructed from Cauchy completeness.*) Assume $\mathbb{R}$ is Archimedean and that every Cauchy sequence of reals converges. Then every nonempty set bounded above has a supremum.

SOURCE: ECNU 例 3 (bk. pp. 154–155), the $6 \Rightarrow 1$ step of the completeness cycle.

THE CONSTRUCTION

1. **Manufacture near-misses.** Let $S \neq \emptyset$ be bounded above. For each n , the Archimedean property supplies an integer k_n such that $\lambda_n = k_n/n$ is an upper bound of S while $\lambda_n - 1/n$ is not. (Walk the multiples of $1/n$ upward until they first clear S .)
2. **Show the near-misses are Cauchy.** Since λ_m is an upper bound and $\lambda_n - 1/n$ is not, some $a' \in S$ has $\lambda_m \geq a' > \lambda_n - 1/n$, so $\lambda_n - \lambda_m < 1/n$. By symmetry $|\lambda_m - \lambda_n| < \max\{1/m, 1/n\}$.
3. **Cash in completeness.** So $\lambda_n \rightarrow \lambda$ for some real λ .
4. **Identify the limit.** λ is an upper bound, being a limit of upper bounds. And it is least: given $\delta > 0$ choose n with $1/n < \delta/2$ and $\lambda_n > \lambda - \delta/2$; since $\lambda_n - 1/n$ is not an upper bound, some $a' \in S$ exceeds it, hence exceeds $\lambda - \delta$.
5. **Audit.** The Archimedean property is spent once, in step 1; completeness once, in step 3. Neither is decorative — see the pitfall.

Why Rudin never does this, and why it is worth seeing

Rudin takes the LUB property as an *axiom* (1.19) and derives the Archimedean property from it (1.20a). He therefore never shows the converse direction, and a reader of Rudin alone could reasonably conclude that “complete” has only one meaning.

It has two, and they are not the same. *Cauchy completeness alone does not imply the LUB property*: there are Cauchy-complete ordered fields without it — non-Archimedean ones. That is why step 1 above needs the Archimedean hypothesis, and it is exactly the gap Yu Pin’s axiom list (field + order + Archimedean + nested intervals) is built to expose. Chapter 1’s comparison box lists the equivalences; this is the proof of the one link that Rudin’s development makes structurally invisible.

NOT IN RUDIN

HEINE–BOREL RUDIN DECLINES

WHY THIS SUPPLEMENT BELONGS TO CHAPTER 2

What Rudin does. Theorem 2.41 proves that for a set in $\mathbb{R}^k$, three conditions coincide: compact, closed and bounded, and every infinite subset has a limit point. He states the restriction to $\mathbb{R}^k$ emphatically, and Exercise 2.16 supplies a metric space where closed and bounded fails to give compact.

The gap. The restriction is a warning, not an explanation. A reader finishes 2.41 knowing that “closed and bounded” is the wrong condition in general, but not what the right one is — and therefore not what property $\mathbb{R}^k$ was quietly contributing.

Where the gap bites. Everywhere Rudin later works in a space that is not $\mathbb{R}^k$. At 7.23 (Arzelà–Ascoli) he must characterise the compact subsets of a function space, and the characterisation is exactly the general theorem below — pointwise boundedness and equicontinuity are total boundedness in disguise. Chapter 11’s $\mathcal{L}^2$ is another such space. Without the general statement, each of these looks like a new trick.

What this section supplies. The general theorem — compact = complete + totally bounded — with proof, so that 2.41 decodes: *closed* was supplying completeness and *bounded* was supplying total boundedness, and only the second implication is special to $\mathbb{R}^k$. Also the equivalence of the two compactnesses in a metric space, which Rudin proves only inside $\mathbb{R}^k$.

S2.8 Definition. A metric space X is **complete** if every Cauchy sequence in X converges in X ; it is **totally bounded** if for every $\varepsilon > 0$ finitely many balls of radius ε cover X .

SOURCE: Yu Pin 定义 35 (p.76); Tao II, Ex. 1.5.10 (bk. p.26). Rudin defines completeness at 3.11 and total boundedness nowhere.

S2.9 Theorem. (Heine–Borel for metric spaces.) A metric space is compact if and only if it is complete and totally bounded.

SOURCE: Tao II, Ex. 1.5.10(b),(c).

Proof. Throughout, use S2.10 to read “compact” as “every sequence has a convergent subsequence.”

Compact $\Rightarrow$ totally bounded. Given $\varepsilon > 0$, the balls $B(x, \varepsilon)$, $x \in X$, form an open cover; a finite subcover is a finite family of ε -balls covering X .

Compact $\Rightarrow$ complete. Let (x_n) be Cauchy. Some subsequence converges, say $x_{n_k} \rightarrow x$. Given $\varepsilon > 0$ choose N with $d(x_n, x_m) < \varepsilon/2$ for $n, m \geq N$, and then k with $n_k \geq N$ and $d(x_{n_k}, x) < \varepsilon/2$; for $n \geq N$ the triangle inequality gives $d(x_n, x) < \varepsilon$. So the whole sequence converges.

Complete and totally bounded $\Rightarrow$ compact. Let (x_n) be any sequence. Cover X by finitely many balls of radius 1; one of them contains x_n for infinitely many n , giving a subsequence S_1 lying in a set of diameter at most 2. Cover X by finitely many balls of radius $\frac{1}{2}$; one contains infinitely many terms of S_1 , giving a subsequence $S_2 \subset S_1$ of diameter at most 1. Continuing, we obtain $S_1 \supset S_2 \supset \cdots$, with S_j of diameter at most 2^{2-j} .

Let y_k be the k th term of S_k . For $i, j \geq k$ both y_i and y_j belong to S_k , so $d(y_i, y_j) \leq 2^{2-k}$: the diagonal sequence (y_k) is Cauchy, and by completeness it converges. It is a subsequence of (x_n) , so X is sequentially compact. $\square$

HOW THE PROOF FLOWS

1. **Choose the method.** Two properties must be produced from one, and then one from two. The forward direction is cheap because compactness is a hypothesis about *every* cover and about *every* sequence, so one simply applies it to a well-chosen instance. The converse is where the work is: nothing is compact yet, so a convergent subsequence has to be manufactured by hand.
2. **Spend the cover hypothesis (forward, part one).** Apply compactness to the cover by all ε -balls. There is nothing to choose — the cover is handed to you by the definition of total boundedness itself.
3. **Spend the sequence hypothesis (forward, part two).** A Cauchy sequence needs only *one* convergent subsequence to converge itself; that upgrade is the same three-line estimate as in 3.11 and is the whole content of “compact implies complete.”
4. **Manufacture nested pigeonholes (converse).** Total boundedness gives finitely many balls at each scale, and finitely many boxes holding infinitely many terms means one box holds infinitely many. Halving the radius each time makes the boxes shrink geometrically. This is pattern 3 of Chapter 1’s strategy index — when the hypothesis is finiteness, pigeonhole.
5. **Take the diagonal.** No single S_j converges, so no one of them is the answer; the diagonal belongs to every S_k from the k th place on and is therefore Cauchy. The diagonal is what turns a *nested family* of subsequences into a *single* subsequence.
6. **Cash in completeness.** Cauchy becomes convergent, and only here. Rudin performs the identical manoeuvre at 7.23 to extract a convergent subsequence of functions.
7. **Audit.** Total boundedness is spent once, in step 4, to get the

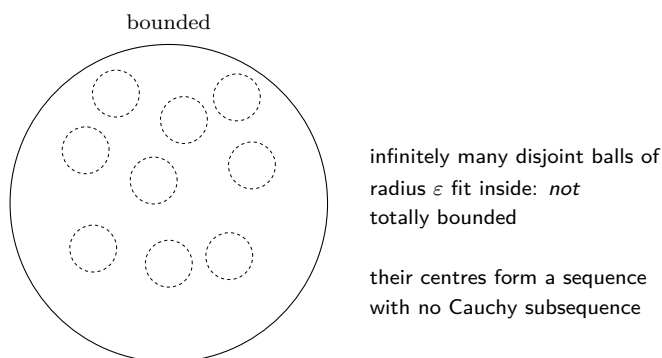
boxes; completeness once, in step 6, to convert the diagonal's Cauchy property into a limit. Neither is decorative: the sequence $(1/n)$ in $(0, 1]$ is totally bounded and not complete, and the unit vectors of the picture below are complete and not totally bounded, and neither space is compact.

Reading Rudin's 2.41 through this

Now Rudin's restriction to $\mathbb{R}^k$ decodes. "Closed and bounded" is doing two separate jobs there:

Closed in $\mathbb{R}^k$ gives **completeness** — a closed subset of a complete space is complete, and $\mathbb{R}^k$ is complete by 3.11. *Bounded* in $\mathbb{R}^k$ gives **total boundedness**, because a bounded set fits in a cell and a cell is chopped into finitely many pieces of any prescribed size.

The second implication is the one that fails elsewhere, and it fails for a reason with a name: in infinite dimensions, bounded does not imply totally bounded. Rudin's Exercise 2.16 — the set $\{p \in \mathbb{Q} : 2 < p^2 < 3\}$, closed and bounded in $\mathbb{Q}$ but not compact — diagnoses the *other* failure, incompleteness. Both diseases exist, and 2.41 is the statement that $\mathbb{R}^k$ has neither.



Bounded but not totally bounded — the failure Rudin's 2.41 rules out by working in $\mathbb{R}^k$. In ℓ^1 the unit vectors $e^{(n)}$ are at mutual distance 2, so the balls of radius 1 about them are disjoint; the set is closed, bounded, and lives in a complete space, yet no subsequence converges. Total boundedness is precisely the hypothesis that forbids this picture.

S2.10 Theorem. *(The two compactnesses agree.) For a metric space X : every open cover of X has a finite subcover if and only if every sequence in X has a convergent subsequence.*

SOURCE: Yu Pin, p. 114 (both directions); Tao II, Thm 1.5.8 and Ex. 1.5.11 — Tao defines compactness sequentially, the mirror image of Rudin.

Proof. *Covers $\Rightarrow$ sequences.* Suppose (x_k) had no convergent subsequence. Then no point x is a subsequential limit, so each x has a radius $r_x > 0$ for which $B(x, r_x)$ contains x_k for only finitely many indices k . These balls cover X ; a finite subcover would then account for only finitely many indices altogether, which is absurd.

Sequences $\Rightarrow$ covers. Let $\{U_\alpha\}$ be an open cover. First, X is totally bounded: if some $\varepsilon > 0$ admitted no finite cover by ε -balls, one could choose x_1 , then x_2 outside $B(x_1, \varepsilon)$, then x_3 outside the first two balls, and so on, producing a sequence with mutual distances at least ε and hence no convergent subsequence.

Second, the cover has a Lebesgue number (S4.19 in the Chapter 4 appendix): if not, then for each n some $B(x_n, 1/n)$ lies in no U_α ; passing to a convergent subsequence $x_{n_k} \rightarrow x$ and choosing β and r with $B(x, r) \subset U_\beta$, we get $B(x_{n_k}, 1/n_k) \subset B(x, r)$ for large k — a contradiction.

Now cover X by finitely many balls of radius δ , a Lebesgue number. Each lies inside a single U_α , and those finitely many U_α cover X . $\square$

Rudin proves one direction inside $\mathbb{R}^k$ (2.41, via 2.37) and never states the general equivalence.¹³⁸

138 *The forward direction is slick: if (x_k) had no convergent subsequence, every tail $\{x_k : k \geq m\}$ would be closed, so the complements $U_m = X \setminus \{x_k : k \geq m\}$ form an open cover with no finite subcover. The reverse direction is the one that needs a uniform scale — the Lebesgue number lemma, stated as S4.19 in the Chapter 4 appendix, where Rudin's 4.19 needs it too.*

NOT IN RUDIN

SUPPLEMENT D — TWO NOTIONS OF SMALLNESS

WHY THIS SUPPLEMENT BELONGS TO CHAPTER 2

What Rudin does. The chapter opens with cardinality (2.1–2.14) and closes with the Cantor set (2.44), an uncountable set that is, in his words, of measure zero. Size is measured by counting.

The gap. Counting is not the only measure of size, and Rudin's own examples show it is not the useful one for topology. $\mathbb{Q}$ is countable — small by counting — yet meets every interval. The Cantor set is uncountable — large by counting — yet contains no interval at all. The notion that separates them is nowhere in Chapter 2, though every ingredient for it is: closure, interior, density.

Where the gap bites. At Exercise 3.22, where Rudin poses Baire's theorem with no indication of what it is for; at 7.18, where the existence of a nowhere-differentiable continuous function looks like a lucky construction rather than the typical case; and in the Chapter 4 appendix, where the question “can a function be continuous exactly on $\mathbb{Q}$?” cannot even be posed without it.

What this section supplies. Cantor's theorem in general form and Schröder–Bernstein, which make cardinality an order relation rather than a collection of tricks; then Baire's theorem with proof, and the table showing that the two notions of smallness disagree in both directions.

S2.11 Theorem. (Cantor.) *For every set X , there is no surjection $X \rightarrow 2^X$; in particular X and its power set never have the same cardinality.*

SOURCE: Tao I, Thm 8.3.1 (bk. p. 195). Rudin's 2.14 is the case $X = \mathbb{N}$.

Proof. Let $f : X \rightarrow 2^X$ be any map and put

$$A = \{x \in X : x \notin f(x)\}.$$

If $A = f(x_0)$ for some x_0 , then $x_0 \in A$ means $x_0 \notin f(x_0) = A$, while $x_0 \notin A$ means $x_0 \in f(x_0) = A$. Both are impossible, so A is not in the range of f . $\square$

The proof is Rudin's 2.14 with the decimal expansions stripped away: given $f : X \rightarrow 2^X$, the set $A = \{x : x \notin f(x)\}$ is not $f(x)$ for any x .¹³⁹

S2.12 Theorem. (Schröder–Bernstein.) *If there are injections $A \rightarrow B$ and $B \rightarrow A$, then there is a bijection $A \rightarrow B$.*

SOURCE: Tao I, Ex. 8.3.3; Yu Pin p. 38 gives a different proof by least fixed point.

Proof. Let $f : A \rightarrow B$ and $g : B \rightarrow A$ be injections. Put

$$C_0 = A \setminus g(B), \quad C_{n+1} = g(f(C_n)), \quad C = \bigcup_{n \geq 0} C_n,$$

and define

$$h(a) = \begin{cases} f(a) & (a \in C), \\ g^{-1}(a) & (a \notin C). \end{cases}$$

The second line makes sense: $a \notin C$ gives $a \notin C_0$, so $a \in g(B)$, and g is injective, so $g^{-1}(a)$ is a single point.

h is injective. Both f and g^{-1} are, so the only risk is $f(a) = g^{-1}(a')$ with $a \in C$, $a' \notin C$. But then $a \in C_n$ for some n and $a' = g(f(a)) \in C_{n+1} \subset C$, a contradiction.

h is onto. Let $b \in B$. If $g(b) \in C$, then $g(b) \in C_{n+1}$ for some n — it cannot lie in C_0 — so $g(b) = g(f(c))$ for some $c \in C_n$, whence $b = f(c) = h(c)$ by injectivity of g . If $g(b) \notin C$, then $h(g(b)) = g^{-1}(g(b)) = b$. $\square$

This is what makes cardinality an order relation rather than a collection of tricks: it is antisymmetry.¹⁴⁰

S2.13 Theorem. (Baire.) *In a complete metric space, the intersection of countably many dense open sets is dense. Equivalently, a countable union of closed sets with empty interior has empty interior.*

SOURCE: Yu Pin 定理 160–161 (pp. 296–297). Rudin gives it as Exercise 3.22.

Proof. Let $U_1, U_2, \dots$ be dense open sets in the complete space X , put $V = \bigcap_n U_n$, and let W be any nonempty open set; we must produce a point of $V \cap W$.

Since U_1 is dense and open, $U_1 \cap W$ is nonempty and open, so it contains a closed ball $\overline{B}(x_1, r_1)$ with $0 < r_1 < 1$. Having chosen $\overline{B}(x_n, r_n)$, the set $U_{n+1} \cap B(x_n, r_n)$ is nonempty and open, so it contains a closed ball $\overline{B}(x_{n+1}, r_{n+1})$ with $0 < r_{n+1} < r_n/2$.

139 Rudin proves $2^{\mathbb{N}}$ is uncountable by a diagonal argument on sequences of 0s and 1s, which is this proof in a costume. Stated for general X it yields an infinite ladder of cardinalities, and with it the question Rudin never raises — whether anything sits strictly between $|\mathbb{N}|$ and $|\mathbb{R}|$. That is the continuum hypothesis, and Gödel and Cohen showed it is undecidable from the usual axioms.

140 Rudin compares cardinalities throughout Chapter 2 — “at most countable,” “ E is equivalent to $\mathbb{N}$ ” — and never needs to state that $|A| \leq |B|$ and $|B| \leq |A|$ force $|A| = |B|$, because his cases are concrete enough to exhibit bijections directly. The general fact is not obvious and the proof is genuinely clever.

The balls are nested, so for $m > n$ we have $x_m \in \overline{B}(x_n, r_n)$ and hence $d(x_m, x_n) \leq r_n < 2^{-n+1}$: the centres form a Cauchy sequence. By completeness $x_n \rightarrow x$. For each n , all x_m with $m \geq n$ lie in the closed set $\overline{B}(x_n, r_n)$, so x does too; therefore $x \in U_n$ for every n , and $x \in \overline{B}(x_1, r_1) \subset W$. $\square$

HOW THE PROOF FLOWS

1. **Choose the method.** “ V is dense” means V meets every nonempty open W , so the goal is to *produce a point* lying in infinitely many sets at once. The only machinery that produces a point from infinitely many conditions is a nested construction followed by completeness — the same shape as S2.3 and 2.38.
2. **Use density to move, openness to get room.** At each stage density guarantees U_{n+1} reaches into the ball we are standing in; openness guarantees the intersection has room for a whole closed ball. Both hypotheses are used at every step, and neither can be dropped: a dense set that is not open, like $\mathbb{Q}$, leaves no room.
3. **Halve the radii.** Forcing $r_{n+1} < r_n/2$ makes the centres Cauchy. Any summable rate would do; halving is the cheapest.
4. **Take *closed* balls.** This is the one design choice with content. The limit x need not lie in any open ball $B(x_n, r_n)$, but it must lie in the closed one, because closed sets contain their limits (2.27). Closing the balls is what makes the intersection catch the limit point.
5. **Cash in completeness.** The centres converge, and the limit lies in every $\overline{B}(x_n, r_n) \subset U_n$, hence in V ; and in $\overline{B}(x_1, r_1) \subset W$.
6. **Audit.** Completeness is spent once, at step 5; density and openness once per stage, at step 2; closedness of the balls once, at step 4. Drop completeness and the theorem is false — in $\mathbb{Q}$, the sets $\mathbb{Q} \setminus \{q_n\}$ are dense and open with empty intersection.

The second notion of smallness, and how it cuts across the first

Call a set *nowhere dense* if its closure has empty interior, and *meagre* if it is a countable union of nowhere dense sets. Baire’s theorem says a complete metric space is not meagre in itself — meagre sets are topologically negligible.

The two notions of smallness are independent, and Chapter 2 already contains both witnesses:

set	cardinality	topologically
$\mathbb{Q} \subset \mathbb{R}$	countable — small	dense, not meagre in itself
the Cantor set (2.44)	uncountable — large	nowhere dense — small

So the Cantor set is large by counting and negligible by topology, while $\mathbb{Q}$ is the reverse. Rudin builds the Cantor set precisely to exhibit an uncountable set of “measure zero” (his phrase at 2.44), which is a third notion again — the one Chapter 11 will develop. Yu Pin’s applications show what the theorem is for: no continuous function on $[a, b]$ can be differentiable with unbounded derivative on every subinterval; the indicator of $\mathbb{Q}$ is not a pointwise limit of continuous functions; and “most” continuous functions are nowhere differentiable, the qualitative form of Rudin’s 7.18. The Chapter 4 appendix uses Baire again, at S4.5, to show no function is continuous exactly on the rationals.

NOT IN RUDIN

SUPPLEMENT E — WHAT SURVIVES WITHOUT A METRIC

WHY THIS SUPPLEMENT BELONGS TO CHAPTER 2

What Rudin does. He defines everything in terms of the distance d : balls, limit points, open sets, compactness, connectedness. Theorem 2.24 then proves that arbitrary unions and finite intersections of open sets are open.

The gap. 2.24 is the moment the metric stops mattering, and Rudin does not say so. From there to the end of the chapter — and through 4.8, 4.14 and 4.22 in the next — the proofs use only the conclusions of 2.24, never the distance that produced them. A reader has no way to tell which results are really about distance and which are about open sets alone.

Where the gap bites. At 2.34 (compact implies closed), which needs a separation hypothesis that $\mathbb{R}^k$ supplies for free; at the equivalence of the two compactnesses (S2.10), which is genuinely metric and fails without it; and at every point where a reader wonders whether uniform continuity could be defined the same way — it cannot, and knowing why saves a false generalisation.

What this section supplies. The definition that takes 2.24 as an axiom, and a table sorting Chapter 2 into what survives and what dies, with the two counterexamples that mark the boundary.

S2.14 Definition. A *topology* on X is a family $\mathcal{F}$ of subsets containing $\emptyset$ and X and closed under arbitrary unions and finite intersections; its members are the open sets. X is **Hausdorff** if distinct points have disjoint neighbourhoods.

SOURCE: Tao II, §2.5 (bk. pp. 39–43); Yu Pin 定义 78 (p. 130).

Which of Rudin's Chapter 2 survives, and which dies

Rudin's 2.24 — arbitrary unions and finite intersections of open sets are open — is not a theorem in this setting but the *definition*. Taking it as the definition, one can ask which of the rest survives.

Survives with no metric

Needs the metric

closure, interior, boundary; the relative topology (2.30); continuity as preimages of open sets (4.8); compactness by finite subcovers; connectedness

Cauchy sequences and completeness; boundedness; uniform continuity; “compact $\Rightarrow$ sequentially compact”

Two of Rudin's results fail outright. *Compact implies closed* (2.34) needs Hausdorff: in the trivial topology $\{\emptyset, X\}$ every subset is compact and only two are closed. And *compact implies sequentially compact* fails even in a Hausdorff space — Tao's Exercise 2.5.8 constructs a compact ordered space containing a sequence with no convergent subsequence, which is exactly why S2.10 above had to say “metric space.”

Uniqueness of limits is the cleanest illustration of what Hausdorff buys: without it a sequence can converge to two different points at once. Rudin gets uniqueness free at 3.2(b) because a metric space is automatically Hausdorff — take balls of radius $d(p, q)/2$.

NOT IN RUDIN

APPENDIX EXERCISES

HOW THESE DIFFER FROM RUDIN'S THIRTY

Rudin's own Exercises 2.1–2.30 end the chapter proper. They drill his vocabulary: closures, condensation points, separability, the distance function. These twenty-one come from the four comparison texts and do something his cannot — they make the machinery *pay*.

Group I (1–10): the completeness theorems, proved with each other. Rudin derives everything from the least-upper-bound axiom in one direction. These run the implications the other way and, in exercise 1, show what fails over $\mathbb{Q}$ — the exercise that turns the equivalence list of Chapter

141 The exercise that gives the equivalence list of Chapter 1 its point. One example does all four: take rationals increasing to $\sqrt{2}$ — say the decimal truncations. The set is bounded above with no rational supremum, the sequence is monotone bounded and does not converge in $\mathbb{Q}$, it is Cauchy without a limit, and its only candidate cluster point is irrational. That a single witness kills all four is the equivalence, seen from below.

142 Three questions probing exactly where finiteness breaks. Note that (c) succeeds while (b) fails, which isolates the mechanism: the trouble is never “the set is infinite” but “the cover degenerates at a point the set does not contain.” Rudin’s cover examples at 2.32 are one-liners with no such sub-questions.

143 The contrapositive direction Rudin never runs. If bounded infinite E had no cluster point, every $x \in [a, b]$ would have a neighbourhood meeting E in finitely many points; these cover $[a, b]$, so finitely many do, making E finite. Compactness converting “locally finite” into “globally finite” is the whole idea, and it recurs at 4.15.

144 A Cauchy sequence is bounded, so it has a cluster point; then show the whole sequence converges to it, using the Cauchy condition to transfer nearness from the subsequence to the tail. Rudin does exactly this at 3.11, without naming the route. Together with exercise 3 this closes the loop of implications.

145 The “creeping along” proof, and a genuinely different mechanism from S2.4. Let E be the set of $x \in [a, b]$ such that $[a, x]$ admits a finite subcover, and put $c = \sup E$. Show $c \in E$ — some cover element contains c , and it also contains points slightly left of c — and then that $c = b$, since otherwise the same element lets you creep past c . Keep the pattern: to prove something holds on all of $[a, b]$, take the supremum of where it holds and show that supremum cannot be interior. It reappears in the proof of the mean value theorem and in Chapter 5’s appendix.

1 into a statement about $\mathbb{R}$ specifically.

Group II (11–16): metric spaces past Rudin’s stopping point. Completion, total boundedness, the contraction mapping principle, and the closed bounded complete set that is still not compact.

Group III (17–20): size. Uncountability without a diagonal argument, Cantor’s theorem, Schröder–Bernstein, and a compact space containing a sequence with no convergent subsequence.

Group IV (21): topology as a tool. Furstenberg’s proof that there are infinitely many primes, using nothing but Rudin’s 2.23.

Marked ★: harder, or requiring a result from a later chapter.

S2.1. Give examples showing that over the rationals, each of the following fails: the least-upper-bound principle; the monotone convergence theorem; the cluster-point (Bolzano–Weierstrass) theorem; and the Cauchy criterion.¹⁴¹

SOURCE: ECNU 习题 7.1 #4 (bk. p. 155).

S2.2. Let $H = \{(\frac{1}{n+2}, \frac{1}{n}) : n = 1, 2, 3, \dots\}$. (a) Does H cover $(0, 1)$? (b) Can finitely many members of H cover $(0, \frac{1}{2})$? (c) Can finitely many cover $(\frac{1}{100}, 1)$?¹⁴²

SOURCE: ECNU 习题 7.1 #5 (bk. p. 155).

S2.3. Prove the cluster-point theorem (S2.5) using the finite covering theorem (S2.4).¹⁴³

SOURCE: ECNU 习题 7.1 #8 (bk. p. 155).

S2.4. Prove the Cauchy criterion using the cluster-point theorem.¹⁴⁴

SOURCE: ECNU 习题 7.1 #9 (bk. p. 155).

S2.5. ★ Prove the finite covering theorem using the least-upper-bound principle, without bisection.¹⁴⁵

SOURCE: ECNU 总练习题 7 #2 (bk. p. 160).

S2.6. Prove that a continuous real function on $[a, b]$ is uniformly continuous, using the finite covering theorem.¹⁴⁶

SOURCE: ECNU 习题 7.1 #11 (bk. p. 155). Companion #10: prove the root-existence theorem by the same method.

S2.7. Let (x_n) be monotone. Prove that if it has a cluster point, that cluster point is unique and equals $\sup_n x_n$ (for increasing (x_n)).¹⁴⁷

SOURCE: ECNU 习题 7.1 #7 (bk. p. 155).

S2.8. Prove that the set of cluster points of the closed interval $[a, b]$ is $[a, b]$ itself.¹⁴⁸

SOURCE: ECNU 习题 7.1 #6 (bk. p. 155).

S2.9. Sharpen the nested interval theorem: if $a_n < a_{n+1} < b_{n+1} < b_n$ strictly for all n and $b_n - a_n \rightarrow 0$, prove the common point ξ satisfies $a_n < \xi < b_n$ strictly for every n .¹⁴⁹

SOURCE: ECNU 习题 7.1 #3 (bk. p. 155).

S2.10. Construct a divergent sequence (a_n) with the property that for every $\varepsilon > 0$ there is an N such that $n > N$ implies $|a_{n+1} - a_n| < \varepsilon$.¹⁵⁰

SOURCE: USTC Ch. 1 suppl. #4 (bk. p. 38).

S2.11. $\star$ (The completion, built by hand.) Let (X, d) be a metric space. Call two Cauchy sequences in X equivalent if $d(x_n, y_n) \rightarrow 0$, write $\bar{X}$ for the set of equivalence classes, and define $\bar{d}([x], [y]) = \lim d(x_n, y_n)$. Prove: (a) the relation is an equivalence relation; (b) the limit defining $\bar{d}$ exists and is independent of representatives, and $\bar{d}$ is a metric; (c) $\bar{X}$ is complete; (d) $x \mapsto [(x, x, x, \dots)]$ embeds X isometrically; (e) the image is dense in $\bar{X}$.¹⁵¹

SOURCE: Tao II, Ex. 1.4.8 (bk. p. 21); Yu Pin 定理 77 (p. 123) adds the universal property.

S2.12. Prove that a metric space is compact if and only if it is complete and totally bounded (S2.9).¹⁵²

SOURCE: Tao II, Ex. 1.5.10 (bk. p. 26).

S2.13. In the space X of absolutely summable sequences with $d(a, b) = \sum_n |a_n - b_n|$, let $e^{(n)}$ be the sequence with 1 in place n and 0 elsewhere. Prove that $\{e^{(n)} : n \in \mathbb{N}\}$ is closed and bounded, that X is complete, and that the set is nevertheless not compact.¹⁵³

SOURCE: Tao II, Ex. 1.5.8 with Ex. 1.1.15 (bk. pp. 26, 10).

S2.14. (Contraction mapping principle.) Let (X, d) be complete and $T : X \rightarrow X$ satisfy $d(Tx, Ty) \leq \gamma d(x, y)$ for some fixed $\gamma \in (0, 1)$. Prove T has exactly one fixed point. Then prove the refinement: if some iterate T^r is a contraction, T still has exactly one fixed point — and check that $x \mapsto e^{-x}$ on $[0, \infty)$ is not a contraction although $f \circ f$ is.¹⁵⁴

SOURCE: Yu Pin 定理 37 (p. 77) and the following exercises.

146 Rudin's 4.19, but assigned here — and the point of assigning it here is that Chapter 2's machinery is already enough. Each x gives δ_x for $\varepsilon/2$; the balls of radius $\delta_x/2$ cover $[a, b]$; take a finite subcover and let δ be the smallest radius. The halving is what makes two nearby points land in a common ball. Compare S4.19 in the Chapter 4 appendix, which names the δ this construction produces.

147 Monotonicity collapses the whole cluster set to a point — the sequence version of the fact that a monotone function has one-sided limits everywhere. Contrast with $\sin(n\pi/4)$ at S2.2, which has five: order is exactly what forbids returning.

148 The concrete instance of “[a, b] is perfect,” which Rudin asserts in passing at 2.44's discussion but never asks anyone to verify. Both inclusions need a word: every point of $[a, b]$ is approached from within, and no point outside is.

149 S2.3 gives only $a_n \leq \xi \leq b_n$, and several later arguments need the strict version — in particular any argument that must place ξ in the interior of every stage. Cheap to prove, easy to assume without noticing.

150 The pseudo-Cauchy counterexample: consecutive terms getting close does not make a sequence Cauchy. The canonical answer is the partial sums of the harmonic series, where $|a_{n+1} - a_n| = 1/(n+1) \rightarrow 0$ while the sums diverge. Rudin states 3.10 with the full $|a_n - a_m|$ and never asks why the weaker condition will not do — but every student proposes it eventually.

151 The construction of $\mathbb{R}$ from $\mathbb{Q}$, done once in generality — Rudin gives it only as Exercise 3.24 and only sketches it. Part (b) is where the work is: you must prove a limit exists before you can use it as a definition, and that is what the Cauchy hypothesis is for. Chapter 1's appendix builds $\mathbb{R}$ by cuts instead; comparing the two is the best available illustration that “the” real numbers are characterised by a property, not a construction.

152 For the hard direction, build nested subsequences: cover by finitely many balls of radius 1 and keep one containing infinitely many terms; refine with radius $1/2$; and so on. The diagonal subsequence is Cauchy, and completeness converts it into a convergent one. The diagonal trick is the same one Rudin uses at 7.23 for Arzelà–Ascoli — worth meeting here first, where it is uncluttered.

153 Sharper than Rudin’s Exercise 2.16, where compactness fails because $\mathbb{Q}$ is incomplete. Here nothing is incomplete: the failure is dimension. Any two of the $e^{(n)}$ are at distance 2, so no subsequence is Cauchy and no finite family of balls of radius 1 covers the set. This is the picture drawn at S2.9.

154 The first theorem in the book that genuinely needs completeness: the iterates form a Cauchy sequence by comparison with a geometric series, and only completeness produces the limit. Rudin proves this at 9.23, five chapters later, for the inverse function theorem; the one-dimensional case is exercise S4.15 in the Chapter 4 appendix.

155 Compactness forcing rigidity. Part (b) is the surprising one — a map that never decreases distances on a compact space cannot increase them either. All three run on the same engine: iterate f , extract a convergent subsequence, and use it to show points that looked separated must be reachable. Nothing in Rudin’s Chapter 2 exercises has this flavour.

156 The universal property of the product, which is what Rudin’s 4.10 (a map into $\mathbb{R}^k$ is continuous iff its components are) is a special case of. The matrix statements are the payoff: openness follows because the determinant is continuous and GL_n is the preimage of $\mathbb{R} \setminus \{0\}$, and continuity of inversion from Cramer’s rule. Chapter 9 will use both without proof.

S2.15. Let (X, d) be a compact metric space and $f : X \rightarrow X$. Prove:
 (a) if $d(f(x), f(y)) = d(x, y)$ for all x, y , then f is a bijection;
 (b) if $d(f(x), f(y)) \geq d(x, y)$ for all x, y , then f is an isometry;
 (c) if $d(f(x), f(y)) \leq d(x, y)$ for all x, y and f is onto, then f is an isometry.¹⁵⁵

SOURCE: Yu Pin, winter problem S (p. 331).

S2.16. On a product $Y \times Z$ of metric spaces put $d((y_1, z_1), (y_2, z_2)) = \sqrt{d_Y(y_1, y_2)^2 + d_Z(z_1, z_2)^2}$. Prove this is a metric, that the projections are continuous, and that $F : X \rightarrow Y \times Z$ is continuous if and only if both components are. Then prove that the invertible $n \times n$ real matrices form an open subset of all matrices and that $A \mapsto A^{-1}$ is continuous on it.¹⁵⁶

SOURCE: Yu Pin, homework 4 A8–A12 (p. 108).

S2.17. Prove that $\mathbb{R}^1$ is uncountable using nested intervals rather than a diagonal argument: (a) given a closed interval J of positive length and any x , find a closed $I \subset J$ of positive length with $x \notin I$; (b) given a sequence (x_n) , build nested closed intervals I_n with $x_n \notin I_n$; (c) conclude.¹⁵⁷

SOURCE: Yu Pin, homework 1 B7 (p. 37).

S2.18. Prove that any family of pairwise disjoint open intervals in $\mathbb{R}^1$ is countable.¹⁵⁸

SOURCE: Yu Pin, homework 1 B11 (p. 37).

S2.19. Prove Cantor’s theorem in general: for any set X there is no surjection $X \rightarrow 2^X$. Then prove the Schröder–Bernstein theorem: if there are injections $A \rightarrow B$ and $B \rightarrow A$, there is a bijection $A \rightarrow B$.¹⁵⁹

SOURCE: Tao I, Thm 8.3.1 and Ex. 8.3.2–8.3.3 (bk. pp. 195, 198).

S2.20. ★ Construct a compact topological space containing a sequence with no convergent subsequence.¹⁶⁰

SOURCE: Tao II, Ex. 2.5.8 with 2.5.5 (bk. pp. 43, 42).

S2.21. ★ (Furstenberg.) On $\mathbb{Z}$ call a set open if it is empty or a union of arithmetic progressions $U_{a,b} = \{ka + b : k \in \mathbb{Z}\}$ with $a \geq 1$. Prove: (a) this is a topology; (b) no nonempty finite set is open, so no set with finite complement is closed; (c) each $U_{a,b}$ is both open and closed; (d) $\mathbb{Z} \setminus \{-1, 1\} = \bigcup_{p \text{ prime}} U_{p,0}$; (e) conclude that there are infinitely many primes.¹⁶¹

SOURCE: Yu Pin, homework 5 娱乐问题 (p. 130).

Chapter 3

NUMERICAL SEQUENCES AND SERIES

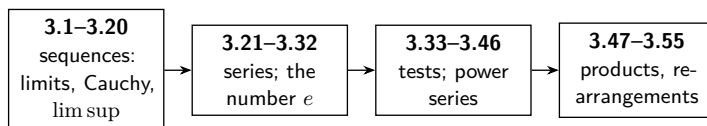
WHY THIS CHAPTER EXISTS

Chapter 1 promised that limits would have somewhere to arrive; Chapter 2 built the spaces they arrive in. This chapter finally defines the arrival — $p_n \rightarrow p$ — and then spends its second half on the single most important special case: series, which are sequences of partial sums.

Two theorems are the chapter's spine. **3.11** says that in $\mathbb{R}^k$ a sequence converges exactly when it is a Cauchy sequence — convergence becomes an *internal* property, checkable without knowing the limit. **3.14** says a monotonic sequence converges exactly when it is bounded. Both are completeness in action, and nearly every convergence test for series (3.24 through 3.44) is one of them wearing different clothes.

HOW IT IS ORGANIZED, AND WHY IN THAT ORDER

Rudin's own account, from the opening paragraph: the chapter “will deal primarily with sequences and series of complex numbers,” but “the basic facts about convergence are just as easily explained in a more general setting” — so the first three sections work in metric spaces, and the theorems proved there are inherited free by $\mathbb{R}$, $\mathbb{C}$ and $\mathbb{R}^k$. The same economy as 1.13(c) and Chapter 2.



Three things to know in advance.

The sequence sections are the load-bearing wall.

Everything about series is a statement about the sequence of partial sums, so 3.21–3.55 consume 3.1–3.20 continuously. In particular $\limsup$ (3.16–3.17), the chapter's hardest definition, exists mainly so that the root test (3.33) and the radius of convergence (3.39) can be stated without exceptions.

Completeness gets its name here. Definition 3.12 finally introduces the word *complete* for “every Cauchy sequence

157 Rudin proves uncountability twice — by diagonalisation at 2.14 and, far more powerfully, as a corollary of “perfect sets are uncountable” at 2.43. This is a third proof, and the cheapest: it uses only S2.3. Note that it is really the same idea as 2.43, with the nested intervals doing what Rudin's shrinking neighbourhoods do.

158 One line if you see it: each interval contains a rational, and disjointness makes the assignment injective. The technique — inject an unknown family into $\mathbb{Q}$ — is the standard way to prove countability, and it is exactly the argument behind Rudin's 4.30 (a monotone function has at most countably many discontinuities) and behind Exercise 2.29.

159 Cantor's proof is Rudin's 2.14 with the decimals removed:

$A = \{x : x \notin f(x)\}$ is missed by every $f(x)$ — and note its kinship with Russell's paradox. Schröder–Bernstein is what makes cardinality an order relation; the slick proof partitions A using $D_0 = B \setminus f(A)$ and $D_{n+1} = f(g(D_n))$.

160 The counterexample proving that S2.10 is a theorem about metric spaces and not about topological ones. Tao's construction well-orders $\mathbb{R}$, takes the union of all countable initial segments, adjoins a maximum, and gives the result its order topology; every point below the maximum has only countably many predecessors, which is what strands the sequence. Rudin never distinguishes the two compactnesses because he never leaves metric spaces.

161 The best advertisement in this chapter for topology as a tool rather than a vocabulary. If there were finitely many primes, the union in (d) would be a finite union of closed sets, hence closed, so $\{-1, 1\}$ would be open — contradicting (b). Every step uses only Rudin's 2.23–2.24, on a space with no metric in sight.

converges,” discharging the promise made in the comparison box at Definition 1.10. Two more of the ECNU text’s six equivalent completeness statements appear as theorems in this chapter: monotone convergence is 3.14, and the Cauchy criterion is 3.11(c).

The last section is a warning. Rearranging a series can change its sum (3.54) unless the convergence is absolute (3.55). The chapter is arranged so that this lands last, as the reason the distinction of 3.45–3.46 was worth drawing at all.

As the title indicates, this chapter will deal primarily with sequences and series of complex numbers. The basic facts about convergence, however, are just as easily explained in a more general setting. The first three sections will therefore be concerned with sequences in euclidean spaces, or even in metric spaces.

CONVERGENT SEQUENCES

3.1 Definition. A sequence (p_n) in a metric space X is said to *converge* if there is a point $p \in X$ with the following property: For every $\varepsilon > 0$ there is an integer N such that $n \geq N$ implies that $d(p_n, p) < \varepsilon$. (Here d denotes the distance in X .)

In this case, we also say that (p_n) *converges to* p , or that p is the *limit* of (p_n) [see Theorem 3.2(b)], and we write $p_n \rightarrow p$, or

$$\lim_{n \rightarrow \infty} p_n = p.$$

If (p_n) does not converge, it is said to *diverge*.¹⁶²

It might be well to point out that our definition of “convergent sequence” depends not only on (p_n) but also on X ; for instance, the sequence $(1/n)$ converges in $\mathbb{R}^1$ (to 0), but fails to converge in the set of all positive real numbers [with $d(x, y) = |x - y|$]. In cases of possible ambiguity, we can be more precise and specify “convergent in X ” rather than “convergent.”¹⁶³

We recall that the set of points p_n ($n = 1, 2, 3, \dots$) is the *range* of (p_n) . The range of a sequence may be a finite set, or it may be infinite. The sequence (p_n) is said to be *bounded* if its range is bounded.

As examples, consider the following sequences of complex numbers (that is, $X = \mathbb{R}^2$):

- (a) If $s_n = 1/n$, then $\lim_{n \rightarrow \infty} s_n = 0$; the range is infinite, and the sequence is bounded.
- (b) If $s_n = n^2$, the sequence (s_n) is unbounded, is divergent, and has infinite range.
- (c) If $s_n = 1 + [(-1)^n/n]$, the sequence (s_n) converges to 1, is bounded, and has infinite range.

162 Read the quantifiers in the stated order: ε first, then N . The challenge comes before the response, and N may depend on ε — almost always does. Every limit argument in this book is a game of respond-to- ε , and the bracketed reference to 3.2(b) is Rudin flagging that “the limit” needs the uniqueness proved there before the phrase is legitimate.

163 The same relativity as “open” in 2.29, and the same cure: name the ambient space. The sequence $(1/n)$ has not changed; what changed is whether the space still contains the point it is crowding toward. Completeness (3.12) is exactly the property of spaces in which this failure mode — converging toward a missing point — cannot occur.

- (d) If $s_n = i^n$, the sequence (s_n) is divergent, is bounded, and has finite range.¹⁶⁴
- (e) If $s_n = 1$ ($n = 1, 2, 3, \dots$), then (s_n) converges to 1, is bounded, and has finite range.

We now summarize some important properties of convergent sequences in metric spaces.

3.2 Theorem. *Let (p_n) be a sequence in a metric space X .*

- (a) *(p_n) converges to $p \in X$ if and only if every neighborhood of p contains p_n for all but finitely many n .*
- (b) *If $p \in X$, $p' \in X$, and if (p_n) converges to p and to p' , then $p' = p$.*
- (c) *If (p_n) converges, then (p_n) is bounded.*
- (d) *If $E \subset X$ and if p is a limit point of E , then there is a sequence (p_n) in E such that $p = \lim_{n \rightarrow \infty} p_n$.*

Proof.

- (a) Suppose $p_n \rightarrow p$ and let V be a neighborhood of p . For some $\varepsilon > 0$, the conditions $d(q, p) < \varepsilon$, $q \in X$ imply $q \in V$. Corresponding to this ε , there exists N such that $n \geq N$ implies $d(p_n, p) < \varepsilon$. Thus $n \geq N$ implies $p_n \in V$.

Conversely, suppose every neighborhood of p contains all but finitely many of the p_n . Fix $\varepsilon > 0$ and let V be the set of all $q \in X$ such that $d(p, q) < \varepsilon$. By assumption, there exists N (corresponding to this V) such that $p_n \in V$ if $n \geq N$. Thus $d(p_n, p) < \varepsilon$ if $n \geq N$; hence $p_n \rightarrow p$.

- (b) Let $\varepsilon > 0$ be given. There exist integers N, N' such that

$$\begin{aligned} n \geq N & \text{ implies } d(p_n, p) < \frac{\varepsilon}{2}, \\ n \geq N' & \text{ implies } d(p_n, p') < \frac{\varepsilon}{2}. \end{aligned}$$

Hence if $n \geq \max\{N, N'\}$, we have

$$d(p, p') \leq d(p, p_n) + d(p_n, p') < \varepsilon.$$

Since ε was arbitrary, we conclude that $d(p, p') = 0$.¹⁶⁵

- (c) Suppose $p_n \rightarrow p$. Then there is an integer N such that $n > N$ implies $d(p_n, p) < 1$. Put

$$r = \max\{1, d(p_1, p), \dots, d(p_N, p)\}.$$

Then $d(p_n, p) \leq r$ for $n = 1, 2, 3, \dots$.¹⁶⁶

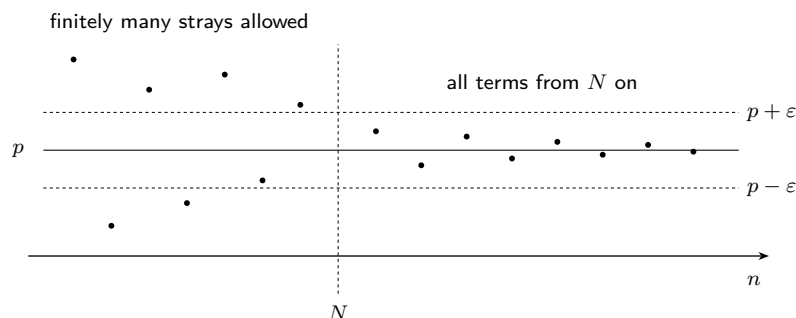
- (d) For each positive integer n , there is a point $p_n \in E$ such that $d(p_n, p) < 1/n$. Given $\varepsilon > 0$, choose N so that $N\varepsilon > 1$. If $n > N$, it follows that $d(p_n, p) < \varepsilon$. Hence $p_n \rightarrow p$.

164 The four powers of i recur forever, so the sequence visits $\{i, -1, -i, 1\}$ in rotation. It diverges even though every point of its range is visited infinitely often — each of the four points is a subsequential limit (3.5), and 3.2(b) says a convergent sequence can have only one.

165 “Since ε was arbitrary” is a standard closing move worth naming: a fixed nonnegative number smaller than every positive ε must be 0 (else take $\varepsilon = d(p, p')/2$). Rudin will use this silently from now on; Exercise 1.5.7 of Chapter 1’s exercises — Tao states it as a lemma — is the same fact.

166 The finite-maximum move of 2.20 and 2.24(c) again, from the other side: the tail is controlled by convergence, and only finitely many early terms remain, so their distances have a maximum. Convergence controls tails; finiteness mops up beginnings.

This completes the proof. $\square$



Convergence, drawn for a real sequence: every ε -tube about p traps the whole tail.

Shrink ε and N moves right; that is the dependence the definition quantifies.

For sequences in $\mathbb{R}^k$ we can study the relation between convergence, on the one hand, and the algebraic operations on the other. We first consider sequences of complex numbers.

3.3 Theorem. Suppose $(s_n), (t_n)$ are complex sequences and

$$\lim_{n \rightarrow \infty} s_n = s, \quad \lim_{n \rightarrow \infty} t_n = t.$$

Then

- (a) $\lim_{n \rightarrow \infty} (s_n + t_n) = s + t$;
- (b) $\lim_{n \rightarrow \infty} cs_n = cs$, $\lim_{n \rightarrow \infty} (c + s_n) = c + s$, for any number c ;
- (c) $\lim_{n \rightarrow \infty} s_n t_n = st$;
- (d) $\lim_{n \rightarrow \infty} (1/s_n) = 1/s$, provided $s_n \neq 0$ ($n = 1, 2, 3, \dots$) and $s \neq 0$.

Proof.

- (a) Given $\varepsilon > 0$, there exist integers N_1, N_2 such that

$$\begin{aligned} n \geq N_1 & \text{ implies } |s_n - s| < \frac{\varepsilon}{2}, \\ n \geq N_2 & \text{ implies } |t_n - t| < \frac{\varepsilon}{2}. \end{aligned}$$

If $N = \max\{N_1, N_2\}$, then $n \geq N$ implies

$$|(s_n + t_n) - (s + t)| \leq |s_n - s| + |t_n - t| < \varepsilon.$$

This proves (a). The proof of (b) is trivial.

- (c) We use the identity

$$s_n t_n - st = (s_n - s)(t_n - t) + s(t_n - t) + t(s_n - s). \quad (3.1)$$

- 167** Given $\varepsilon > 0$, there are integers N_1, N_2 such that

$$\begin{aligned} n \geq N_1 & \text{ implies } |s_n - s| < \sqrt{\varepsilon}, \\ n \geq N_2 & \text{ implies } |t_n - t| < \sqrt{\varepsilon}. \end{aligned}$$

167 The identity is the whole proof, and it is chosen so that every term on the right involves at least one error factor $s_n - s$ or $t_n - t$. Add and subtract st_n (or $s_n t$) and you find it. The same add-and-subtract manufacture drives the proofs of 4.4, 5.3(b) and the product rules everywhere.

If we take $N = \max\{N_1, N_2\}$, $n \geq N$ implies

$$|(s_n - s)(t_n - t)| < \varepsilon,$$

so that

$$\lim_{n \rightarrow \infty} (s_n - s)(t_n - t) = 0.$$

We now apply (a) and (b) to (3.1) and conclude that

$$\lim_{n \rightarrow \infty} (s_n t_n - st) = 0.$$

(d) Choosing m such that $|s_n - s| < |s|/2$ if $n \geq m$, we see that

$$|s_n| > \frac{1}{2}|s| \quad (n \geq m).$$

168 Given $\varepsilon > 0$, there is an integer $N > m$ such that $n \geq N$ implies

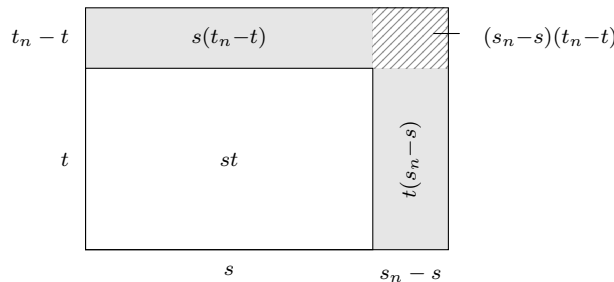
$$|s_n - s| < \frac{1}{2}|s|^2\varepsilon.$$

Hence, for $n \geq N$,

$$\left| \frac{1}{s_n} - \frac{1}{s} \right| = \left| \frac{s_n - s}{s_n s} \right| < \frac{2}{|s|^2} |s_n - s| < \varepsilon.$$

168 The standard “stay away from zero” step: once s_n is within $|s|/2$ of s , it cannot be nearer than $|s|/2$ to 0 (reverse triangle inequality, Exercise 1.13). Without it, $1/s_n$ could blow up even with $s \neq 0$. Every quotient-limit proof in the book opens with this move.

□



Identity (3.1) as areas: the outer rectangle $s_n t_n$ minus the inner st is two thin strips plus one hatched corner — and each of the three pieces carries at least one error factor. The strips are bounded $\times$ small, the corner is small $\times$ small, which is why $\sqrt{\varepsilon}$ suffices for it. (Drawn for real $s_n > s > 0$, $t_n > t > 0$; the identity is pure algebra and holds for complex sequences.)

HOW THESE PROOFS ARE FOUND, NOT JUST CHECKED

All four parts follow one recipe, and it is the recipe for every ε -argument in the book. Write down the quantity that must be small — $|(s_n + t_n) - (s + t)|$, say. Bound it by pieces you control, each going to 0. Then walk the proof backwards: decide what each piece must be smaller than so the total is under ε , and let that dictate the N 's.

The visible fingerprints of working backwards are the odd-looking

thresholds: $\varepsilon/2$ in (a) because two errors will be added; $\sqrt{\varepsilon}$ in (c) because two errors will be *multiplied*; $\frac{1}{2}|s|^2\varepsilon$ in (d) because the error will be divided by $|s_ns| > \frac{1}{2}|s|^2$. None of these is guessed — each is the final inequality, solved for the intermediate one. This is the same reverse-engineering as the choice of h in 1.21.

3.4 Theorem.

(a) Suppose $\mathbf{x}_n \in \mathbb{R}^k$ ($n = 1, 2, 3, \dots$) and

$$\mathbf{x}_n = (\alpha_{1,n}, \dots, \alpha_{k,n}).$$

Then $(\mathbf{x}_n)$ converges to $\mathbf{x} = (\alpha_1, \dots, \alpha_k)$ if and only if

$$\lim_{n \rightarrow \infty} \alpha_{j,n} = \alpha_j \quad (1 \leq j \leq k). \quad (3.2)$$

(b) Suppose $(\mathbf{x}_n), (\mathbf{y}_n)$ are sequences in $\mathbb{R}^k$, (β_n) is a sequence of real numbers, and $\mathbf{x}_n \rightarrow \mathbf{x}$, $\mathbf{y}_n \rightarrow \mathbf{y}$, $\beta_n \rightarrow \beta$. Then

$$\lim_{n \rightarrow \infty} (\mathbf{x}_n + \mathbf{y}_n) = \mathbf{x} + \mathbf{y}, \quad \lim_{n \rightarrow \infty} \mathbf{x}_n \cdot \mathbf{y}_n = \mathbf{x} \cdot \mathbf{y}, \quad \lim_{n \rightarrow \infty} \beta_n \mathbf{x}_n = \beta \mathbf{x}.$$

Proof.

(a) If $\mathbf{x}_n \rightarrow \mathbf{x}$, the inequalities

$$|\alpha_{j,n} - \alpha_j| \leq |\mathbf{x}_n - \mathbf{x}|,$$

which follow immediately from the definition of the norm in $\mathbb{R}^k$, show that (3.2) holds.

Conversely, if (3.2) holds, then to each $\varepsilon > 0$, there corresponds an integer N such that $n \geq N$ implies

$$|\alpha_{j,n} - \alpha_j| < \frac{\varepsilon}{\sqrt{k}} \quad (1 \leq j \leq k).$$

Hence $n \geq N$ implies

$$|\mathbf{x}_n - \mathbf{x}| = \left\{ \sum_{j=1}^k |\alpha_{j,n} - \alpha_j|^2 \right\}^{1/2} < \varepsilon,$$

so that $\mathbf{x}_n \rightarrow \mathbf{x}$. This proves (a).

Part (b) follows from (a) and Theorem 3.3.¹⁶⁹

□

169 Part (a) is the coordinatewise reduction of 2.39 again: convergence in $\mathbb{R}^k$ is k convergences in $\mathbb{R}$, squeezed between the two norm inequalities $|\alpha_j| \leq |\mathbf{x}| \leq \sqrt{k} \max_j |\alpha_j|$. It is why the rest of the chapter can work with complex or real sequences and lose nothing.

SUBSEQUENCES

WHY SUBSEQUENCES, AND WHY NOW | A subsequence discards terms but keeps order, and the notion pays for itself twice. First, it converts Chapter 2's compactness into a statement about sequences: 3.6(a) is Bolzano–Weierstrass in sequence form, the tool that manufactures limits out of mere boundedness. Second, subsequential limits are the raw material for $\limsup$ and $\liminf$ (3.16), which the series tests will need. Theorem 3.7 — the set of subsequential limits is closed — is stocked for 3.17.

3.5 Definition. Given a sequence (p_n) , consider a sequence (n_k) of positive integers such that $n_1 < n_2 < n_3 < \cdots$. Then the sequence (p_{n_i}) is called a *subsequence* of (p_n) . If (p_{n_i}) converges, its limit is called a *subsequential limit* of (p_n) .

It is clear that (p_n) converges to p if and only if every subsequence of (p_n) converges to p . We leave the details of the proof to the reader.¹⁷⁰

3.6 Theorem.

- (a) If (p_n) is a sequence in a compact metric space X , then some subsequence of (p_n) converges to a point of X .
- (b) Every bounded sequence in $\mathbb{R}^k$ contains a convergent subsequence.

Proof.

- (a) Let E be the range of (p_n) . If E is finite then there is a $p \in E$ and a sequence (n_i) with $n_1 < n_2 < n_3 < \cdots$ such that

$$p_{n_1} = p_{n_2} = \cdots = p.$$

The subsequence (p_{n_i}) so obtained converges evidently to p .¹⁷¹

If E is infinite, Theorem 2.37 shows that E has a limit point $p \in X$. Choose n_1 so that $d(p, p_{n_1}) < 1$. Having chosen $n_1, \dots, n_{i-1}$, we see from Theorem 2.20 that there is an integer $n_i > n_{i-1}$ such that $d(p, p_{n_i}) < 1/i$. Then (p_{n_i}) converges to p .¹⁷²

- (b) This follows from (a) since Theorem 2.41 implies that every bounded subset of $\mathbb{R}^k$ lies in a compact subset of $\mathbb{R}^k$.

□

3.6(b) is Bolzano–Weierstrass in sequence form

Compare 2.42: every bounded infinite *set* has a limit point. Part (b) is the same fact retold for sequences, and it is the workhorse version — most of analysis manufactures a convergent subsequence from boundedness and then works with it. The ECNU text lists this (致密性定理, its name for sequential compactness) as one of its six equivalent forms of completeness, and Tao takes exactly this property as his *definition* of compactness (his 1.5.1).

Note what the proof consumed: 2.37, hence compactness of some k -cell, hence — through 2.40 and 2.38 — the least-upper-bound

170 One direction is trivial; the other is Exercise-level but worth doing once: since $n_k \geq k$ (induction on the strictly increasing indices), the tail of a subsequence is trapped by the tail of the sequence. The statement is used constantly in the contrapositive: to show (p_n) diverges, exhibit two subsequences with different limits — as with i^n in 3.1(d).

171 The finite case is not a formality — it is the pigeonhole: infinitely many indices land on finitely many values, so some value is hit infinitely often. Rudin splits by the range, the set-versus-sequence distinction he flagged at 2.7.

172 Theorem 2.20 — every neighbourhood of a limit point contains infinitely many points of E — is exactly what lets the index increase: within distance $1/i$ there are infinitely many terms, so one of them has index beyond n_{i-1} . The $q \neq p$ clause of 2.18(b) has been earning interest since Chapter 2.

property. Over $\mathbb{Q}$ the statement fails: the sequence $1, 1.4, 1.41, \dots$ of Chapter 1 is bounded with no convergent subsequence in $\mathbb{Q}$.

3.7 Theorem. *The subsequential limits of a sequence (p_n) in a metric space X form a closed subset of X .*

Proof. Let E^* be the set of all subsequential limits of (p_n) and let q be a limit point of E^* . We have to show that $q \in E^*$.

Choose n_1 so that $p_{n_1} \neq q$. (If no such n_1 exists, then E^* has only one point, and there is nothing to prove.) Put $\delta = d(q, p_{n_1})$. Suppose $n_1, \dots, n_{i-1}$ are chosen. Since q is a limit point of E^* , there is an $x \in E^*$ with $d(x, q) < 2^{-i}\delta$. Since $x \in E^*$, there is an $n_i > n_{i-1}$ such that $d(x, p_{n_i}) < 2^{-i}\delta$. Thus

$$d(q, p_{n_i}) \leq 2^{1-i}\delta$$

for $i = 1, 2, 3, \dots$. This says that (p_{n_i}) converges to q . Hence $q \in E^*$.¹⁷³

□

173 A two-step chase: q is approximated by subsequential limits x , and each x is approximated by actual terms p_{n_i} — so terms of the sequence approach q through the middlemen, at the price of a doubled error $2 \cdot 2^{-i}\delta$. The geometric budget $2^{-i}\delta$ is chosen so the doubling still goes to zero. This diagonal-through-approximators pattern returns at 7.23.

CAUCHY SEQUENCES

CONVERGENCE WITHOUT THE LIMIT | Definition 3.1 cannot be checked without naming the limit in advance — a severe handicap, since the whole point of many arguments is to *prove* a limit exists. This section removes the handicap. A Cauchy sequence is one whose terms crowd *each other*; Theorem 3.11 says that in $\mathbb{R}^k$ this internal crowding already guarantees a limit. Rudin says exactly this in the note after 3.11, and it is the single most-used fact in the chapter: every series convergence test works by verifying the Cauchy condition, never by producing the sum.

3.8 Definition. A sequence (p_n) in a metric space X is said to be a *Cauchy sequence* if for every $\varepsilon > 0$ there is an integer N such that $d(p_n, p_m) < \varepsilon$ if $n \geq N$ and $m \geq N$.¹⁷⁴

In our discussion of Cauchy sequences, as well as other situations which will arise later, the following geometric concept will be useful.

3.9 Definition. Let E be a nonempty subset of a metric space X and let S be the set of all real numbers of the form $d(p, q)$ with $p \in E$ and $q \in E$. The sup of S is called the *diameter* of E .

If (p_n) is a sequence in X and if E_N consists of the points $p_N, p_{N+1}, p_{N+2}, \dots$, it is clear from the two preceding definitions that (p_n) is a Cauchy sequence if and only if

$$\lim_{N \rightarrow \infty} \text{diam } E_N = 0.$$

3.10 Theorem.

174 Both indices are quantified, independently. The weaker condition $d(p_n, p_{n+1}) \rightarrow 0$ — consecutive terms approaching — does not suffice: $s_n = \sqrt{n}$ has $s_{n+1} - s_n \rightarrow 0$ yet marches to infinity, and the partial sums of the harmonic series (3.28) do the same. Mistaking the consecutive condition for the Cauchy condition is the most common error made with this definition.

(a) If $\overline{E}$ is the closure of a set E in a metric space X , then

$$\text{diam } \overline{E} = \text{diam } E.$$

(b) If K_n is a sequence of compact sets in X such that $K_n \supset K_{n+1}$ ($n = 1, 2, 3, \dots$) and if

$$\lim_{n \rightarrow \infty} \text{diam } K_n = 0,$$

then $\bigcap_{n=1}^{\infty} K_n$ consists of exactly one point.

Proof.

(a) Since $E \subset \overline{E}$, it is clear that

$$\text{diam } E \leq \text{diam } \overline{E}.$$

Fix $\varepsilon > 0$, and choose $p \in \overline{E}$, $q \in \overline{E}$. By the definition of $\overline{E}$, there are points p', q' in E such that $d(p, p') < \varepsilon$, $d(q, q') < \varepsilon$. Hence

$$\begin{aligned} d(p, q) &\leq d(p, p') + d(p', q') + d(q', q) \\ &< 2\varepsilon + d(p', q') \leq 2\varepsilon + \text{diam } E. \end{aligned}$$

It follows that

$$\text{diam } \overline{E} \leq 2\varepsilon + \text{diam } E,$$

and since ε was arbitrary, (a) is proved.

(b) Put $K = \bigcap_{n=1}^{\infty} K_n$. By Theorem 2.36, K is not empty. If K contains more than one point, then $\text{diam } K > 0$. But for each n , $K_n \supset K$ so that $\text{diam } K_n \geq \text{diam } K$. This contradicts the assumption that $\text{diam } K_n \rightarrow 0$.¹⁷⁵

□

175 *Division of labour: 2.36 (compactness) supplies at least one point; shrinking diameters supply at most one. Nested-interval arguments always have these two halves, and it is worth noticing which hypothesis funds which.*

3.11 Theorem.

- (a) In any metric space X , every convergent sequence is a Cauchy sequence.
- (b) If X is a compact metric space and if (p_n) is a Cauchy sequence in X , then (p_n) converges to some point of X .
- (c) In $\mathbb{R}^k$, every Cauchy sequence converges.

Note: The difference between the definition of convergence and the definition of a Cauchy sequence is that the limit is explicitly involved in the former, but not in the latter. Thus Theorem 3.11(b) may enable us to decide whether or not a given sequence converges without knowledge of the limit to which it may converge.

The fact (contained in Theorem 3.11) that a sequence converges in $\mathbb{R}^k$ if and only if it is a Cauchy sequence is usually called the *Cauchy criterion* for convergence.

Proof.

- (a) If $p_n \rightarrow p$ and if $\varepsilon > 0$, there is an integer N such that $d(p, p_n) < \varepsilon$ for all $n \geq N$. Hence

$$d(p_n, p_m) \leq d(p_n, p) + d(p, p_m) < 2\varepsilon$$

as soon as $n \geq N$ and $m \geq N$. Thus (p_n) is a Cauchy sequence.

- (b) Let (p_n) be a Cauchy sequence in the compact space X . For $N = 1, 2, 3, \dots$, let E_N be the set consisting of $p_N, p_{N+1}, p_{N+2}, \dots$. Then

$$\lim_{N \rightarrow \infty} \text{diam } \overline{E}_N = 0, \quad (3.3)$$

by Definition 3.9 and Theorem 3.10(a). Being a closed subset of a compact space X , each $\overline{E}_N$ is compact (Theorem 2.35). Also, $E_N \supset E_{N+1}$, so that $\overline{E}_N \supset \overline{E}_{N+1}$.

Theorem 3.10(b) shows now that there is a unique $p \in X$ which lies in every $\overline{E}_N$.¹⁷⁶

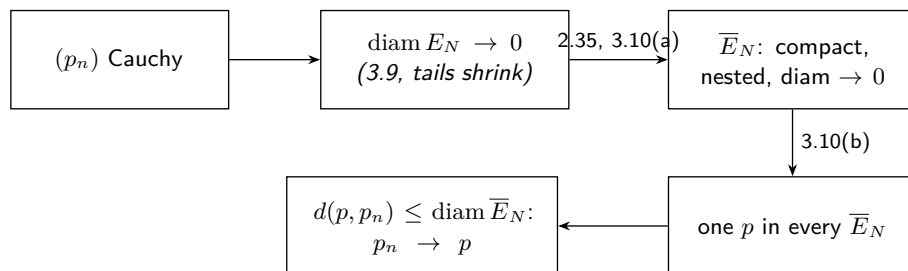
Let $\varepsilon > 0$ be given. By (3.3) there is an integer N_0 such that $\text{diam } \overline{E}_N < \varepsilon$ if $N \geq N_0$. Since $p \in \overline{E}_N$, it follows that $d(p, q) < \varepsilon$ for every $q \in \overline{E}_N$, hence for every $q \in E_N$. In other words, $d(p, p_n) < \varepsilon$ if $n \geq N_0$. This says precisely that $p_n \rightarrow p$.

- (c) Let $(\mathbf{x}_n)$ be a Cauchy sequence in $\mathbb{R}^k$. Define E_N as in (b), with $\mathbf{x}_i$ in place of p_i . For some N , $\text{diam } E_N < 1$. The range of $(\mathbf{x}_n)$ is the union of E_N and the finite set $\{\mathbf{x}_1, \dots, \mathbf{x}_{N-1}\}$. Hence $(\mathbf{x}_n)$ is bounded. Since every bounded subset of $\mathbb{R}^k$ has compact closure in $\mathbb{R}^k$ (Theorem 2.41), (c) follows from (b).¹⁷⁷

□

176 Why the closures? E_N itself need not be compact — it is just a scatter of points — and 3.10(b) needs compact sets. Passing to $\overline{E}_N$ costs nothing precisely because of 3.10(a): closing a set does not increase its diameter. That is the entire reason 3.10(a) was proved.

177 The structure of (c): Cauchy $\Rightarrow$ bounded (cheaply, from one E_N of finite diameter plus finitely many strays — the same split as 3.2(c)), then bounded $\Rightarrow$ inside a compact set (2.41), then apply (b). All the completeness is imported through 2.41.



Theorem 3.11(b) as a flow diagram: the tail-sets carry the whole argument, closures make them compact, and the nested-set theorem hands over the limit. Part (c) bolts “Cauchy $\Rightarrow$ bounded $\Rightarrow$ inside a compact set (2.41)” onto the front.

3.12 Definition. A metric space in which every Cauchy sequence converges is said to be *complete*.

Thus Theorem 3.11 says that all compact metric spaces and all euclidean spaces are complete. Theorem 3.11 implies also that every closed subset E of a complete metric space X is complete. (Every Cauchy sequence in E is a Cauchy sequence in X , hence it converges to some

$p \in X$, and actually $p \in E$ since E is closed.) An example of a metric space which is not complete is the space of all rational numbers, with $d(x, y) = |x - y|$.

The word “complete,” finally — and the full circle

This is the definition promised since Chapter 1. The comparison box at 1.10 said the least-upper-bound property “is what the word *complete* will name in 3.12”; here is the naming, and with it the last of the standard completeness statements enters the book. Of the ECNU text’s six equivalent formulations, five have now appeared: the supremum principle (1.19), nested intervals (2.38), Heine–Borel (2.41), Bolzano–Weierstrass (2.42 and 3.6(b)), and now the Cauchy criterion (3.11(c)); monotone convergence follows as 3.14.

There is also a genuine full circle here. Tao *constructs* $\mathbb{R}$ as the completion of $\mathbb{Q}$ — real numbers are, in his book, equivalence classes of Cauchy sequences of rationals, so “every Cauchy sequence converges” is nearly true by construction. Rudin travelled the other way: $\mathbb{R}$ was characterised by suprema, and Cauchy completeness arrived at the end of a chain through compactness. Exercise 3.24 closes the loop from Rudin’s side, carrying out the Cauchy-sequence completion for an arbitrary metric space.

One caution: completeness and compactness part company here. Compact implies complete (3.11(b)), but $\mathbb{R}^k$ is complete without being compact, and completeness is *not* intrinsic in the sense of 2.33 — $(0, 1)$ and $\mathbb{R}$ are homeomorphic, yet one is complete and the other not. Completeness belongs to the metric, not the topology.

Theorem 3.2(c) and example (d) after Definition 3.1 show that convergent sequences are bounded, but that bounded sequences in $\mathbb{R}^k$ need not converge. However, there is one important case in which convergence is equivalent to boundedness; this happens for monotonic sequences in $\mathbb{R}^1$.

3.13 Definition. A sequence (s_n) of real numbers is said to be

- (a) *monotonically increasing* if $s_n \leq s_{n+1}$ ($n = 1, 2, 3, \dots$);
- (b) *monotonically decreasing* if $s_n \geq s_{n+1}$ ($n = 1, 2, 3, \dots$).

The class of *monotonic* sequences consists of the increasing and the decreasing sequences.

3.14 Theorem. *Suppose (s_n) is monotonic. Then (s_n) converges if and only if it is bounded.*

Proof. Suppose $s_n \leq s_{n+1}$ (the proof is analogous in the other case). Let E be the range of (s_n) . If (s_n) is bounded, let s be the least upper bound of E . Then

$$s_n \leq s \quad (n = 1, 2, 3, \dots).$$

For every $\varepsilon > 0$, there is an integer N such that

$$s - \varepsilon < s_N \leq s,$$

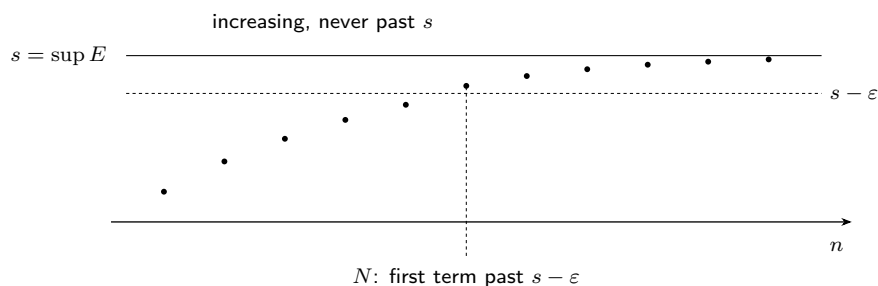
178 Clause (ii) of Definition 1.8 yet again, in its working form — step back from the supremum and some term sticks out past you. Monotonicity then does what no other hypothesis could: one term beyond $s - \varepsilon$ drags the entire tail with it.

for otherwise $s - \varepsilon$ would be an upper bound of E .¹⁷⁸ Since (s_n) increases, $n \geq N$ therefore implies

$$s - \varepsilon < s_n \leq s,$$

which shows that (s_n) converges (to s).

The converse follows from Theorem 3.2(c). □



Monotone convergence: the terms climb, the supremum caps them, and once one term passes $s - \varepsilon$ the rest are trapped in the gap. The limit is the supremum — 3.14 is Definition 1.10 restated for sequences, which is why the ECNU text lists the two as equivalent completeness axioms.

UPPER AND LOWER LIMITS

THE HARDEST DEFINITION IN THE CHAPTER, AND WHAT IT IS FOR | A divergent sequence can still have structure: $(-1)^n$ oscillates between two values rather than wandering. $\limsup$ and $\liminf$ capture the outer envelope of that oscillation — the largest and smallest subsequential limits. Their virtue is *unconditional existence*: every real sequence has both, in the extended reals, with no hypothesis at all. That is why the root test (3.33) and the radius-of-convergence formula (3.39) are stated with $\limsup$ — an ordinary limit might not exist, and the theory would fill up with exceptions. Keep Theorem 3.17 in view as the working characterisation: s^* is the one number that is both *reached* (along some subsequence) and *barely exceeded* (only finitely many terms above any $x > s^*$). The definition via $\sup E$ is for existence; 3.17 is what proofs actually use.

3.15 Definition. Let (s_n) be a sequence of real numbers with the following property: For every real M there is an integer N such that $n \geq N$ implies $s_n \geq M$. We then write

$$s_n \rightarrow +\infty.$$

Similarly, if for every real M there is an integer N such that $n \geq N$ implies $s_n \leq M$, we write

$$s_n \rightarrow -\infty.$$

It should be noted that we now use the symbol $\rightarrow$ (introduced in Definition 3.1) for certain types of divergent sequences, as well as for convergent sequences, but that the definitions of convergence and of limit, given in Definition 3.1, are unchanged.¹⁷⁹

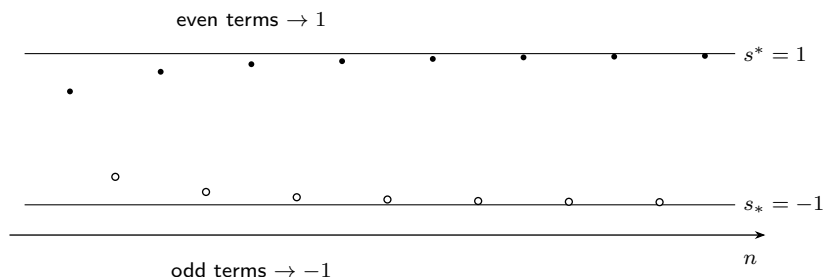
3.16 Definition. Let (s_n) be a sequence of real numbers. Let E be the set of numbers x (in the extended real number system) such that $s_{n_k} \rightarrow x$ for some subsequence (s_{n_k}) . This set E contains all subsequential limits as defined in Definition 3.5, plus possibly the numbers $+\infty$, $-\infty$.

We now recall Definitions 1.8 and 1.23 and put

$$s^* = \sup E, \quad s_* = \inf E.$$

The numbers s^* , s_* are called the *upper* and *lower limits* of (s_n) ; we use the notation

$$\limsup_{n \rightarrow \infty} s_n = s^*, \quad \liminf_{n \rightarrow \infty} s_n = s_*.$$



Example 3.18(b), $s_n = (-1)^n/(1 + 1/n)$: two subsequences, two subsequential limits, $E = \{-1, 1\}$. The upper and lower limits are the envelope the oscillation settles into — not the largest and smallest terms.

3.17 Theorem. Let (s_n) be a sequence of real numbers. Let E and s^* have the same meaning as in Definition 3.16. Then s^* has the following two properties:

- (a) $s^* \in E$.
- (b) If $x > s^*$, there is an integer N such that $n \geq N$ implies $s_n < x$.

Moreover, s^* is the only number with the properties (a) and (b).

Of course, an analogous result is true for s_* .

Proof.

- (a) If $s^* = +\infty$, then E is not bounded above; hence (s_n) is not bounded above, and there is a subsequence (s_{n_k}) such that $s_{n_k} \rightarrow +\infty$.

If s^* is real, then E is bounded above, and at least one subsequential limit exists, so that (a) follows from Theorems 3.7 and 2.28.¹⁸⁰

179 So “ $s_n \rightarrow +\infty$ ” and “ (s_n) diverges” are compatible statements — the arrow has been extended, the word limit has not. This is the extended-real bookkeeping of 1.23 paying its way: $+\infty$ is a destination for notation, not a number a sequence can converge to.

180 Here is where the section’s stockpile is spent, both pieces at once: 3.7 says E is closed, 2.28 says $\sup E$ lies in $\overline{E} = E$. So the supremum of the subsequential limits is itself a subsequential limit — attained, not merely approached. This is the fact that makes $\limsup$ usable.

If $s^* = -\infty$, then E contains only one element, namely $-\infty$, and there is no subsequential limit. Hence, for any real M , $s_n > M$ for at most a finite number of values of n , so that $s_n \rightarrow -\infty$.

This establishes (a) in all cases.

- (b) Suppose there is a number $x > s^*$ such that $s_n \geq x$ for infinitely many values of n . In that case, there is a number $y \in E$ such that $y \geq x > s^*$, contradicting the definition of s^* .¹⁸¹

Thus s^* satisfies (a) and (b). To show the uniqueness, suppose there are two numbers, p and q , which satisfy (a) and (b), and suppose $p < q$. Choose x such that $p < x < q$. Since p satisfies (b), we have $s_n < x$ for $n \geq N$. But then q cannot satisfy (a). $\square$

181 The compressed step: infinitely many terms $\geq x$ form a subsequence in $[x, +\infty]$, and 3.6/3.17(a) applied to it yields a subsequential limit $y \geq x$. “Infinitely many terms above x ” always hands you an element of E at or above x — that conversion, bounded-infinite-to-limit, is Bolzano–Weierstrass 3.6(b) on duty.

THREE WORKING FORMS OF ONE NUMBER

$\limsup$ wears three interchangeable outfits, and fluency means switching freely.

As $\sup E$ (Definition 3.16): best for existence and for symmetry with $\liminf$.

As the two-sided test (Theorem 3.17): s^* is reached along a subsequence, and any $x > s^*$ eventually caps the whole sequence. In ε -form: for every $\varepsilon > 0$, $s_n > s^* + \varepsilon$ only finitely often, while $s_n > s^* - \varepsilon$ infinitely often. This is the form the root test uses.

As a limit of suprema: $s^* = \lim_{N \rightarrow \infty} \sup_{n \geq N} s_n$ — the tail-suprema $\sup_{n \geq N} s_n$ decrease as the tail shrinks, so they converge by 3.14, and their limit is s^* . Rudin leaves this form to Exercise 3.5’s neighbourhood; Tao takes it as the *definition* (his 6.4.6, via “sup of the tail”), and the ECNU text develops it in its §7.2. It explains the name: the *lim* of the *sup*s.

3.18 Examples.

- (a) Let (s_n) be a sequence containing all rationals. Then every real number is a subsequential limit, and

$$\limsup_{n \rightarrow \infty} s_n = +\infty, \quad \liminf_{n \rightarrow \infty} s_n = -\infty.$$

182

- (b) Let $s_n = (-1)^n/[1 + (1/n)]$. Then

$$\limsup_{n \rightarrow \infty} s_n = 1, \quad \liminf_{n \rightarrow \infty} s_n = -1.$$

- (c) For a real-valued sequence (s_n) , $\lim_{n \rightarrow \infty} s_n = s$ if and only if

$$\limsup_{n \rightarrow \infty} s_n = \liminf_{n \rightarrow \infty} s_n = s.$$

183

182 Such a sequence exists because $\mathbb{Q}$ is countable (Corollary to 2.13), and every real is a subsequential limit because $\mathbb{Q}$ is dense (1.20(b)) — the two headline facts about $\mathbb{Q}$, cooperating. Here E is all of the extended line: divergence as total as it gets.

183 The squeeze that makes the pair useful: convergence is exactly “the envelope closes.” Many later proofs show $\limsup \leq c$ and $\liminf \geq c$ separately and conclude $\lim = c$; 3.19 exists to service such arguments.

We close this section with a theorem which is useful and whose proof is quite trivial:

3.19 Theorem. *If $s_n \leq t_n$ for $n \geq N$, where N is fixed, then*

$$\liminf_{n \rightarrow \infty} s_n \leq \liminf_{n \rightarrow \infty} t_n, \quad \limsup_{n \rightarrow \infty} s_n \leq \limsup_{n \rightarrow \infty} t_n.$$

SOME SPECIAL SEQUENCES

A TOOLKIT, ASSEMBLED BY ONE METHOD | Five limits used from here to the end of the book, each proved by the squeeze stated in the opening line: trap x_n between 0 and something already known to vanish. The recurring supplier of the trap is the binomial theorem — expand $(1+x)^n$, keep a single term, and discard the rest. Item (d) is the one to remember: powers beat polynomials, and (e), geometric decay, is its special case $\alpha = 0$. These feed directly into 3.26 and every comparison with a geometric series afterwards.

We shall now compute the limits of some sequences which occur frequently. The proofs will all be based on the following remark: *If $0 \leq x_n \leq s_n$ for $n \geq N$, where N is some fixed number, and if $s_n \rightarrow 0$, then $x_n \rightarrow 0$.*

3.20 Theorem.

- (a) *If $p > 0$, then $\lim_{n \rightarrow \infty} \frac{1}{n^p} = 0$.*
- (b) *If $p > 0$, then $\lim_{n \rightarrow \infty} \sqrt[p]{p} = 1$.*
- (c) $\lim_{n \rightarrow \infty} \sqrt[n]{n} = 1$.
- (d) *If $p > 0$ and α is real, then $\lim_{n \rightarrow \infty} \frac{n^\alpha}{(1+p)^n} = 0$.*
- (e) *If $|x| < 1$, then $\lim_{n \rightarrow \infty} x^n = 0$.*

Proof.

- (a) Take $n > (1/\varepsilon)^{1/p}$. (Note that the archimedean property of the real number system is used here.)¹⁸⁴
- (b) If $p > 1$, put $x_n = \sqrt[p]{p} - 1$. Then $x_n > 0$ and, by the binomial theorem,

$$1 + nx_n \leq (1 + x_n)^n = p,$$

so that

$$0 < x_n \leq \frac{p-1}{n}.$$

Hence $x_n \rightarrow 0$. If $p = 1$, (b) is trivial, and if $0 < p < 1$, the result is obtained by taking reciprocals.

184 And note what else is used: n^p for real p is Exercise 1.6's construction, and the inequality manipulation needs 1.21. The one-line proof stands on the chapter-1 exercises.

(c) Put $x_n = \sqrt[n]{n} - 1$. Then $x_n \geq 0$ and, by the binomial theorem,

$$n = (1 + x_n)^n \geq \frac{n(n-1)}{2} x_n^2.$$

Hence

$$0 \leq x_n \leq \sqrt{\frac{2}{n-1}} \quad (n \geq 2).$$

185 Same trick as (b), one term deeper. Keeping the linear term gives only $x_n \leq (n-1)/n$, which is useless; the quadratic term grows like n^2 and beats the left side. Choosing which binomial term to keep is the entire skill, and (d) makes the choice a third time, with term k .

185

(d) Let k be an integer such that $k > \alpha$, $k > 0$. For $n > 2k$,

$$(1+p)^n > \binom{n}{k} p^k = \frac{n(n-1) \cdots (n-k+1)}{k!} p^k > \frac{n^k p^k}{2^k k!}.$$

Hence

$$0 < \frac{n^\alpha}{(1+p)^n} < \frac{2^k k!}{p^k} n^{\alpha-k} \quad (n > 2k).$$

Since $\alpha - k < 0$, $n^{\alpha-k} \rightarrow 0$, by (a).

(e) Take $\alpha = 0$ in (d). □

SERIES

A SERIES IS A SEQUENCE IN DISGUISE | Nothing new is defined in this section except notation: a series *is* its sequence of partial sums, and Rudin says so twice — in 3.21 (“ s is the limit of a sequence of sums, and is not obtained simply by addition”) and in the remark that every theorem about sequences can be restated for series via $a_n = s_n - s_{n-1}$. The three theorems here are the three big sequence facts passed through that dictionary: 3.22 is the Cauchy criterion 3.11, 3.24 is monotone convergence 3.14, and 3.23 is the trivial-but-vital corollary that terms of a convergent series must die out.

The section ends with the comparison test, 3.25 — the master test. Every test that follows (root, ratio, condensation applications) works by comparing against a series already understood, almost always the geometric series 3.26.

In the remainder of this chapter, all sequences and series under consideration will be complex-valued, unless the contrary is explicitly stated. Extensions of some of the theorems which follow, to series with terms in $\mathbb{R}^k$, are mentioned in Exercise 3.15.

3.21 Definition. Given a sequence (a_n) , we use the notation

$$\sum_{n=p}^q a_n \quad (p \leq q)$$

to denote the sum $a_p + a_{p+1} + \cdots + a_q$. With (a_n) , we associate a sequence (s_n) , where

$$s_n = \sum_{k=1}^n a_k.$$

For (s_n) we also use the symbolic expression

$$a_1 + a_2 + a_3 + \cdots$$

or, more concisely,

$$\sum_{n=1}^{\infty} a_n. \quad (3.4)$$

The symbol in (3.4) we call an *infinite series*, or just a *series*. The numbers s_n are called the *partial sums* of the series. If (s_n) converges to s , we say that the series *converges*, and write

$$\sum_{n=1}^{\infty} a_n = s.$$

The number s is called the *sum* of the series; but it should be clearly understood that s is the limit of a sequence of sums, and is not obtained simply by addition.¹⁸⁶

If (s_n) diverges, the series is said to *diverge*.

Sometimes, for convenience of notation, we shall consider series of the form

$$\sum_{n=0}^{\infty} a_n. \quad (3.5)$$

And frequently, when there is no possible ambiguity, or when the distinction is immaterial, we shall simply write $\sum a_n$ in place of (3.4) or (3.5).

It is clear that every theorem about sequences can be stated in terms of series (putting $a_1 = s_1$ and $a_n = s_n - s_{n-1}$ for $n > 1$), and vice versa. But it is nevertheless useful to consider both concepts.

The Cauchy criterion (Theorem 3.11) can be restated in the following form:

3.22 Theorem. $\sum a_n$ converges if and only if for every $\varepsilon > 0$ there is an integer N such that

$$\left| \sum_{k=n}^m a_k \right| \leq \varepsilon \quad (3.6)$$

if $m \geq n \geq N$.

In particular, by taking $m = n$, (3.6) becomes

$$|a_n| \leq \varepsilon \quad (n \geq N).$$

In other words:

186 This sentence is the guard rail for the whole second half of the chapter. Addition of finitely many numbers is commutative and associative by the field axioms; a “sum” that is really a limit inherits neither property automatically. Theorem 3.54 will show rearrangement genuinely failing, and 3.55 will show what hypothesis buys it back.

3.23 Theorem. *If Σa_n converges, then $\lim_{n \rightarrow \infty} a_n = 0$.*

The condition $a_n \rightarrow 0$ is not, however, sufficient to ensure convergence of Σa_n . For instance, the series

$$\sum_{n=1}^{\infty} \frac{1}{n}$$

187 *Necessary, not sufficient — the single most-cited warning in elementary analysis. The harmonic series is the canonical witness: terms shrink to 0, partial sums climb past every bound. Rudin flags it before anyone can form the converse habit; 3.22 with $m > n$ is what genuine convergence requires, and a block of small terms can still add up.*

188 *Nonnegative terms make the partial sums monotonically increasing, and 3.14 does the rest. This innocent-looking equivalence is the engine of the whole nonnegative theory: comparison (3.25), condensation (3.27), and the root test all reduce convergence to boundedness of partial sums, which is checked by comparing.*

diverges; for the proof, we refer to Theorem 3.28.¹⁸⁷

Theorem 3.14, concerning monotonic sequences, also has an immediate counterpart for series.

3.24 Theorem. *A series of nonnegative¹ terms converges if and only if its partial sums form a bounded sequence.*¹⁸⁸

We now turn to a convergence test of a different nature, the so-called “comparison test.”

3.25 Theorem.

(a) *If $|a_n| \leq c_n$ for $n \geq N_0$, where N_0 is some fixed integer, and if Σc_n converges, then Σa_n converges.*

(b) *If $a_n \geq d_n \geq 0$ for $n \geq N_0$ and if Σd_n diverges, then Σa_n diverges.*

Note that (b) applies only to series of nonnegative terms a_n .

Proof. Given $\varepsilon > 0$, there exists $N \geq N_0$ such that $m \geq n \geq N$ implies

$$\sum_{k=n}^m c_k \leq \varepsilon,$$

by the Cauchy criterion. Hence

$$\left| \sum_{k=n}^m a_k \right| \leq \sum_{k=n}^m |a_k| \leq \sum_{k=n}^m c_k \leq \varepsilon,$$

and (a) follows.

Next, (b) follows from (a), for if Σa_n converges, so must Σd_n [note that (b) also follows from Theorem 3.24]. $\square$

WHAT COMPARISON ACTUALLY PROVES, AND THE ASYMMETRY

Look at the chain of inequalities: it bounds $|\sum a_k|$ through $\sum |a_k|$. So (a) silently proves more than it claims — under its hypotheses $\Sigma |a_n|$ converges too, i.e. the convergence is *absolute* (3.45’s terminology). This is why Rudin will say at 3.46 that comparison, root and ratio “cannot give any information about conditionally convergent series”: all three run through this chain.

The asymmetry between (a) and (b) is not decoration. Domination transfers convergence *downward* and divergence *upward*, never the reverse; and (b) needs $a_n \geq 0$ because a wildly oscillating series can

¹The expression “nonnegative” always refers to real numbers.

dominate a divergent positive one and still converge — cancellations rescue it. Keep the mantra: below a convergent series, convergent; above a divergent nonnegative series, divergent.

The comparison test is a very useful one; to use it efficiently, we have to become familiar with a number of series of nonnegative terms whose convergence or divergence is known.

SERIES OF NONNEGATIVE TERMS

The simplest of all is perhaps the geometric series.

3.26 Theorem. *If $0 \leq x < 1$, then*

$$\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}.$$

If $x \geq 1$, the series diverges.

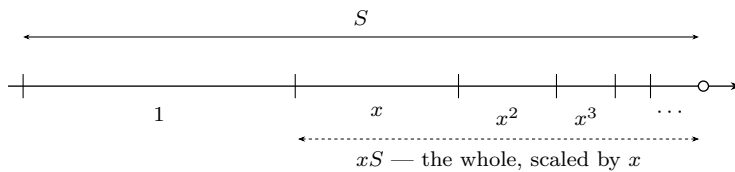
Proof. If $x \neq 1$,

$$s_n = \sum_{k=0}^n x^k = \frac{1 - x^{n+1}}{1 - x}.$$

The result follows if we let $n \rightarrow \infty$. For $x = 1$, we get

$$1 + 1 + 1 + \cdots,$$

which evidently diverges.¹⁸⁹



The segments $1, x, x^2, \dots$ laid end to end (drawn for $x = 0.6$). The tail beyond the first segment is a copy of the entire picture shrunk by the factor x , so $S = 1 + xS$, i.e. $S = 1/(1 - x)$. Rudin's closed form for s_n is this picture stopped after n steps — it is the identity of 1.21, $b^{n+1} - a^{n+1} = (b - a)(b^n + \cdots + a^n)$, with $b = 1$, $a = x$.

In many cases which occur in applications, the terms of the series decrease monotonically. The following theorem of Cauchy is therefore of particular interest. The striking feature of the theorem is that a rather “thin” subsequence of (a_n) determines the convergence or divergence of $\sum a_n$.

3.27 Theorem. *Suppose $a_1 \geq a_2 \geq a_3 \geq \cdots \geq 0$. Then the series $\sum_{n=1}^{\infty} a_n$ converges if and only if the series*

$$\sum_{k=0}^{\infty} 2^k a_{2^k} = a_1 + 2a_2 + 4a_4 + 8a_8 + \cdots \quad (3.7)$$

□ **189** One of only two series in the chapter whose sum is computed rather than merely shown to exist (the other is e). The closed form for s_n — multiply by x , subtract, telescope — is the entire proof, and the limit uses 3.20(e). Every later test converges things by comparing them to this one series; it is the yardstick.

converges.

Proof. By Theorem 3.24, it suffices to consider the boundedness of the partial sums. Let

$$\begin{aligned}s_n &= a_1 + a_2 + \cdots + a_n, \\ t_k &= a_1 + 2a_2 + \cdots + 2^k a_{2^k}.\end{aligned}$$

For $n < 2^k$,

$$\begin{aligned}s_n &\leq a_1 + (a_2 + a_3) + \cdots + (a_{2^k} + \cdots + a_{2^{k+1}-1}) \\ &\leq a_1 + 2a_2 + \cdots + 2^k a_{2^k} \\ &= t_k,\end{aligned}$$

so that

$$s_n \leq t_k. \quad (3.8)$$

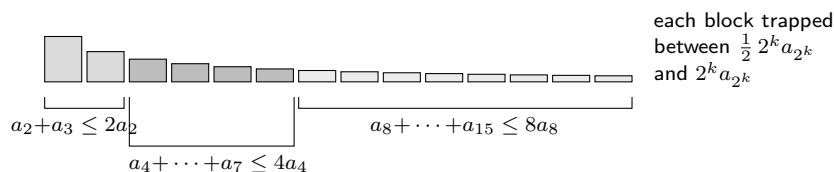
On the other hand, if $n > 2^k$,

$$\begin{aligned}s_n &\geq a_1 + a_2 + (a_3 + a_4) + \cdots + (a_{2^{k-1}+1} + \cdots + a_{2^k}) \\ &\geq \frac{1}{2}a_1 + a_2 + 2a_4 + \cdots + 2^{k-1}a_{2^k} \\ &= \frac{1}{2}t_k,\end{aligned}$$

so that

$$2s_n \geq t_k. \quad (3.9)$$

By (3.8) and (3.9), the sequences (s_n) and (t_k) are either both bounded or both unbounded. This completes the proof. $\square$



Cauchy condensation: chop the series into blocks of doubling length. Monotonicity makes each block at most its first term times its length, and at least its last term times half its length — inequalities (3.8) and (3.9).

Why the sampling is allowed to be so thin

The theorem replaces all of (a_n) by the sample $a_1, a_2, a_4, a_8, \dots$ — vanishingly sparse — and loses nothing. Monotonicity is what pays for it: within a dyadic block the terms cannot vary by more than the endpoints, so the whole block is measured by one sample to within a factor of 2. Without monotonicity the sampled series says nothing (put all the mass off the sample points).

The choice of base 2 is a convenience; any factor works. And the theorem is built for exactly one purpose, visible in 3.28–3.29: it turns $\sum 1/n^p$ into a *geometric* series, and $\sum 1/[n(\log n)^p]$ into $\sum 1/k^p$ —

each application of condensation strips off one logarithm. That is why the borderline series in Rudin's ladder all involve iterated logs.

3.28 Theorem. $\sum \frac{1}{n^p}$ converges if $p > 1$ and diverges if $p \leq 1$.

Proof. If $p \leq 0$, divergence follows from Theorem 3.23. If $p > 0$, Theorem 3.27 is applicable, and we are led to the series

$$\sum_{k=0}^{\infty} 2^k \cdot \frac{1}{2^{kp}} = \sum_{k=0}^{\infty} 2^{(1-p)k}.$$

Now, $2^{1-p} < 1$ if and only if $1 - p < 0$, and the result follows by comparison with the geometric series (take $x = 2^{1-p}$ in Theorem 3.26).¹⁹⁰

□

As a further application of Theorem 3.27, we prove:

3.29 Theorem. If $p > 1$,

$$\sum_{n=2}^{\infty} \frac{1}{n(\log n)^p} \quad (3.10)$$

converges; if $p \leq 1$, the series diverges.

Remark: “ $\log n$ ” denotes the logarithm of n to the base e (compare Exercise 1.7); the number e will be defined in a moment (see Definition 3.30). We let the series start with $n = 2$, since $\log 1 = 0$.

Proof. The monotonicity of the logarithmic function (which will be discussed in more detail in Chap. 8) implies that $(\log n)_{n=1}^{\infty}$ increases. Hence $(1/[n(\log n)^p])_{n=1}^{\infty}$ decreases, and we can apply Theorem 3.27 to (3.10); this leads us to the series

$$\sum_{k=1}^{\infty} 2^k \cdot \frac{1}{2^k (\log 2^k)^p} = \sum_{k=1}^{\infty} \frac{1}{(k \log 2)^p} = \frac{1}{(\log 2)^p} \sum_{k=1}^{\infty} \frac{1}{k^p}, \quad (3.11)$$

and Theorem 3.29 follows from Theorem 3.28. □

This procedure may evidently be continued. For instance,

$$\sum_{n=3}^{\infty} \frac{1}{n \log n \log(\log n)} \quad (3.12)$$

diverges, whereas

$$\sum_{n=3}^{\infty} \frac{1}{n \log n (\log(\log n))^2} \quad (3.13)$$

converges.

We may now observe that the terms of the series (3.12) differ very little from those of (3.13). Still, one diverges, the other converges. If we continue the process which led us from Theorem 3.28 to Theorem 3.29, and then to (3.12) and (3.13), we get pairs of convergent and

¹⁹⁰ With $p = 1$ this settles the harmonic series: condensation turns it into $\sum_k 2^k \cdot 2^{-k} = 1 + 1 + 1 + \cdots$. The classical grouped proof — $\frac{1}{2} + (\frac{1}{3} + \frac{1}{4}) + (\frac{1}{5} + \cdots + \frac{1}{8}) + \cdots \geq \frac{1}{2} + \frac{1}{2} + \cdots$ — is exactly inequality (3.9) written out for this one series.

divergent series whose terms differ even less than those of (3.12) and (3.13). One might thus be led to the conjecture that there is a limiting situation of some sort, a “boundary” with all convergent series on one side, all divergent series on the other side — at least as far as series with monotonic coefficients are concerned. This notion of a “boundary” is of course quite vague. The point we wish to make is this: *No matter how we make this notion precise, the conjecture is false.* Exercises 3.11(b) and 3.12(b) may serve as illustrations.¹⁹¹

We do not wish to go any deeper into this aspect of convergence theory, and refer the reader to Knopp’s “Theory and Application of Infinite Series,” Chap. IX, particularly Sec. 41.

THE NUMBER E

WHY E APPEARS IN A CHAPTER ON SERIES | Partly because it can — the series $\sum 1/n!$ is the book’s first named sum — and partly as a demonstration piece. Theorem 3.31 reconciles the two classical definitions of e using exactly the lim sup/lim inf machinery just built, and Rudin flags it as “a good illustration of operations with limits.” The error estimate (3.16) then delivers the irrationality of e in five lines, the chapter’s most elegant single proof. Yu Pin’s Tsinghua course takes the definition of e (and eventually $e^{i\pi} = -1$) as the organising goal of its entire first semester; Rudin gets the number itself this cheaply because the series technology came first.

3.30 Definition. $e = \sum_{n=0}^{\infty} \frac{1}{n!}$.

Here $n! = 1 \cdot 2 \cdot 3 \cdots n$ if $n \geq 1$, and $0! = 1$. Since

$$\begin{aligned} s_n &= 1 + 1 + \frac{1}{1 \cdot 2} + \frac{1}{1 \cdot 2 \cdot 3} + \cdots + \frac{1}{1 \cdot 2 \cdots n} \\ &< 1 + 1 + \frac{1}{2} + \frac{1}{2^2} + \cdots + \frac{1}{2^{n-1}} < 3, \end{aligned}$$

192 The comparison is with the geometric series, of course: $n! \geq 2^{n-1}$ because each factor past the first is at least 2. Convergence via 3.24 — increasing partial sums, bounded by 3. So $2 < e < 3$ before anything else is known about it.

the series converges, and the definition makes sense.¹⁹² In fact, the series converges very rapidly and allows us to compute e with great accuracy.

It is of interest to note that e can also be defined by means of another limit process; the proof provides a good illustration of operations with limits:

3.31 Theorem. $\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e$.

Proof. Let

$$s_n = \sum_{k=0}^n \frac{1}{k!}, \quad t_n = \left(1 + \frac{1}{n}\right)^n.$$

By the binomial theorem,

$$t_n = 1 + 1 + \frac{1}{2!} \left(1 - \frac{1}{n}\right) + \frac{1}{3!} \left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) + \cdots \\ + \frac{1}{n!} \left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) \cdots \left(1 - \frac{n-1}{n}\right).$$

Hence $t_n \leq s_n$, so that

$$\limsup_{n \rightarrow \infty} t_n \leq e, \quad (3.14)$$

by Theorem 3.19. Next, if $n \geq m$,

$$t_n \geq 1 + 1 + \frac{1}{2!} \left(1 - \frac{1}{n}\right) + \cdots + \frac{1}{m!} \left(1 - \frac{1}{n}\right) \cdots \left(1 - \frac{m-1}{n}\right).$$

Let $n \rightarrow \infty$, keeping m fixed. We get

$$\liminf_{n \rightarrow \infty} t_n \geq 1 + 1 + \frac{1}{2!} + \cdots + \frac{1}{m!},$$

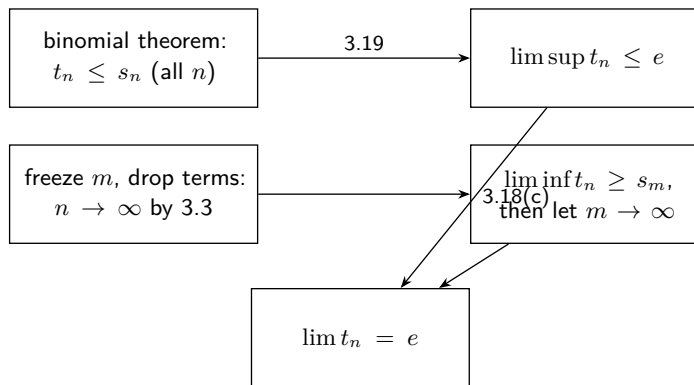
so that

$$s_m \leq \liminf_{n \rightarrow \infty} t_n.$$

Letting $m \rightarrow \infty$, we finally get

$$e \leq \liminf_{n \rightarrow \infty} t_n. \quad (3.15)$$

The theorem follows from (3.14) and (3.15). $\square$



Two independent one-sided bounds, closed by the squeeze of 3.18(c). The upper branch may use all n at once; the lower must freeze m first — the legal order for exchanging two limits.

THE TWO-PARAMETER SQUEEZE, DISSECTED

The proof is a small masterclass in a manoeuvre used constantly from here on. The obstacle: t_n is a sum of n terms, each *changing with n* , so no single limit theorem applies. The solution: cut the dependence in two.

Upper half. Every bracketed factor is ≤ 1 , so term by term $t_n \leq$

s_n ; take $\limsup$ (which exists unconditionally — this is why 3.16 was built) and use 3.19.

Lower half. *Freeze* the number of terms at m , discard the rest — legitimate since all terms are positive — and let only n run. Now there are finitely many factors, each $\rightarrow 1$, so 3.3 applies and gives $\liminf t_n \geq s_m$ for each *fixed* m . Only then release m .

The order of operations is everything: n first with m frozen, m second. Interchanging two limit processes illegally is the classic analysis error, and this proof is the book's first demonstration of how to do it legally — by one-sided bounds and a squeeze. Example 3.18(c) closes the trap: $\limsup = \liminf = e$ forces $\lim = e$.

The rapidity with which the series $\sum 1/n!$ converges can be estimated as follows: If s_n has the same meaning as above, we have

$$\begin{aligned} e - s_n &= \frac{1}{(n+1)!} + \frac{1}{(n+2)!} + \frac{1}{(n+3)!} + \cdots \\ &< \frac{1}{(n+1)!} \left\{ 1 + \frac{1}{n+1} + \frac{1}{(n+1)^2} + \cdots \right\} = \frac{1}{n!n} \end{aligned}$$

so that

$$0 < e - s_n < \frac{1}{n!n}. \quad (3.16)$$

Thus s_{10} , for instance, approximates e with an error less than 10^{-7} . The inequality (3.16) is of theoretical interest as well, since it enables us to prove the irrationality of e very easily.

3.32 Theorem. e is irrational.

Proof. Suppose e is rational. Then $e = p/q$, where p and q are positive integers. By (3.16),

$$0 < q!(e - s_q) < \frac{1}{q}. \quad (3.17)$$

By our assumption, $q!e$ is an integer. Since

$$q!s_q = q! \left(1 + 1 + \frac{1}{2!} + \cdots + \frac{1}{q!} \right)$$

is an integer, we see that $q!(e - s_q)$ is an integer.

Since $q \geq 1$, (3.17) implies the existence of an integer between 0 and 1. We have thus reached a contradiction. $\square$

Actually, e is not even an algebraic number. For a simple proof of this, see page 25 of Niven's book, or page 176 of Herstein's, cited in the Bibliography.

The shape of the irrationality proof

Compare Chapter 1's proof that $\sqrt{2} \notin \mathbb{Q}$: there, rationality forced an infinite descent; here it forces an integer strictly between 0 and 1. Both are impossibility-of-an-integer arguments, and the second

pattern — scale the error by exactly the denominator’s factorial and watch it land in $(0, 1)$ — is the template for most irrationality proofs in analysis.

Everything hinges on *how fast* the series converges: (3.16) says the tail after s_q is smaller than $1/(q!q)$, so multiplying by $q!$ leaves something below $1/q$, still positive. A series converging slower — like $\sum 1/n^2$, whose sum $\pi^2/6$ is also irrational — gives no such one-line proof. Fast convergence is arithmetic leverage: Exercise 2.3 showed transcendental numbers *exist* without naming one, and here analysis pins down an explicit number as irrational.

THE ROOT AND RATIO TESTS

TWO COMPARISONS WITH THE GEOMETRIC SERIES | Neither test is a new idea. Both ask one question — does Σa_n sit below some geometric series $\Sigma \beta^n$ with $\beta < 1$? — and differ only in how they detect it: the root test reads the geometric rate off $\sqrt[n]{|a_n|}$ directly, the ratio test off the quotient of consecutive terms. This is why $\limsup$ was built: the rates need not converge, and 3.17(b) is exactly the tool that converts “ $\limsup < 1$ ” into “eventually below a fixed $\beta < 1$.”

The relationship between the two is settled by 3.37: the root test is strictly stronger, and Example 3.35 exhibits a series the ratio test cannot touch. Use the ratio test for factorials and products, the root test for theory — it is the one the radius of convergence (3.39) is built on.

3.33 Theorem. Given Σa_n , put $\alpha = \limsup_{n \rightarrow \infty} \sqrt[n]{|a_n|}$.

- (a) If $\alpha < 1$, Σa_n converges;
- (b) if $\alpha > 1$, Σa_n diverges;
- (c) if $\alpha = 1$, the test gives no information.

Proof. If $\alpha < 1$, we can choose β so that $\alpha < \beta < 1$, and an integer N such that

$$\sqrt[n]{|a_n|} < \beta$$

for $n \geq N$ [by Theorem 3.17(b)]. That is, $n \geq N$ implies

$$|a_n| < \beta^n.$$

Since $0 < \beta < 1$, $\Sigma \beta^n$ converges. Convergence of Σa_n follows now from the comparison test. ¹⁹³

If $\alpha > 1$, then, again by Theorem 3.17, there is a sequence (n_k) such that

$$\sqrt[n_k]{|a_{n_k}|} \rightarrow \alpha.$$

193 Watch 3.17(b) do precisely its promised job: $\beta > \alpha = s^*$ caps the whole tail. The room between α and 1 is essential — the comparison is with $\Sigma \beta^n$ for a fixed $\beta < 1$, which is why $\alpha = 1$ proves nothing: no room remains.

Hence $|a_n| > 1$ for infinitely many values of n , so that the condition $a_n \rightarrow 0$, necessary for convergence of Σa_n , does not hold (Theorem 3.23).

To prove (c), we consider the series

$$\sum \frac{1}{n}, \quad \sum \frac{1}{n^2}.$$

For each of these series $\alpha = 1$, but the first diverges and the second converges. ¹⁹⁴ □

194 Both α 's are 1 by 3.20(c):

$\sqrt[n]{1/n} = 1/\sqrt[n]{n} \rightarrow 1$, and likewise for $1/n^2$. The p -series live exactly on the geometric tests' blind spot — their terms decay polynomially, slower than every geometric rate, which is why 3.27/3.28 needed a different idea.

3.34 Theorem. The series Σa_n

- (a) converges if $\limsup_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| < 1$,
 (b) diverges if $\left| \frac{a_{n+1}}{a_n} \right| \geq 1$ for all $n \geq n_0$, where n_0 is some fixed integer.

Proof. If condition (a) holds, we can find $\beta < 1$, and an integer N , such that

$$\left| \frac{a_{n+1}}{a_n} \right| < \beta$$

for $n \geq N$. In particular,

$$\begin{aligned} |a_{N+1}| &< \beta |a_N|, \\ |a_{N+2}| &< \beta |a_{N+1}| < \beta^2 |a_N|, \\ &\vdots \\ |a_{N+p}| &< \beta^p |a_N|. \end{aligned}$$

That is,

$$|a_n| < |a_N| \beta^{-N} \cdot \beta^n$$

for $n \geq N$, and (a) follows from the comparison test, since $\Sigma \beta^n$ converges.

If $|a_{n+1}| \geq |a_n|$ for $n \geq n_0$, it is easily seen that the condition $a_n \rightarrow 0$ does not hold, and (b) follows. □

Note: the knowledge that $\lim a_{n+1}/a_n = 1$ implies nothing about the convergence of Σa_n . The series $\sum 1/n$ and $\sum 1/n^2$ demonstrate this.

THE DIVERGENCE HALF OF THE RATIO TEST IS NOT A LIMSUP

Read (b) again: it demands $|a_{n+1}/a_n| \geq 1$ for all $n \geq n_0$ — a pointwise, eventually-everywhere condition. The tempting statement “ $\limsup |a_{n+1}/a_n| > 1$ implies divergence” is **false**, and Example 3.35(b) below is the counterexample: a convergent series with ratio limsup equal to 2. A few large ratios do no harm if they are followed by crashes; only a ratio that *never* again dips below 1 pins the terms away from zero.

Symmetrically, the convergence half really is a limsup, and the liminf-

version “ $\liminf |a_{n+1}/a_n| < 1$ implies convergence” is false too — 3.35(a) has ratio $\liminf 0$ and could easily be built divergent. The test is asymmetric because it detects geometric *domination*, and domination is one-sided.

3.35 Examples.

(a) Consider the series

$$\frac{1}{2} + \frac{1}{3} + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{2^3} + \frac{1}{3^3} + \frac{1}{2^4} + \frac{1}{3^4} + \cdots,$$

for which

$$\begin{aligned}\liminf_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} &= \lim_{n \rightarrow \infty} \left(\frac{2}{3}\right)^n = 0, \\ \liminf_{n \rightarrow \infty} \sqrt[n]{a_n} &= \lim_{n \rightarrow \infty} \sqrt[2n]{\frac{1}{3^n}} = \frac{1}{\sqrt{3}}, \\ \limsup_{n \rightarrow \infty} \sqrt[n]{a_n} &= \lim_{n \rightarrow \infty} \sqrt[2n]{\frac{1}{2^n}} = \frac{1}{\sqrt{2}}, \\ \limsup_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} &= \lim_{n \rightarrow \infty} \frac{1}{2} \left(\frac{3}{2}\right)^n = +\infty.\end{aligned}$$

The root test indicates convergence; the ratio test does not apply.¹⁹⁵

(b) The same is true for the series

$$\frac{1}{2} + 1 + \frac{1}{8} + \frac{1}{4} + \frac{1}{32} + \frac{1}{16} + \frac{1}{128} + \frac{1}{64} + \cdots,$$

where

$$\liminf_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \frac{1}{8}, \quad \limsup_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = 2,$$

but

$$\lim_{n \rightarrow \infty} \sqrt[n]{a_n} = \frac{1}{2}.$$

3.36 Remarks. The ratio test is frequently easier to apply than the root test, since it is usually easier to compute ratios than n th roots. However, the root test has wider scope. More precisely: Whenever the ratio test shows convergence, the root test does too; whenever the root test is inconclusive, the ratio test is too. This is a consequence of Theorem 3.37 and is illustrated by the above examples.

Neither of the two tests is subtle with regard to divergence. Both deduce divergence from the fact that a_n does not tend to zero as $n \rightarrow \infty$.

3.37 Theorem. For any sequence (c_n) of positive numbers,

$$\begin{aligned}\liminf_{n \rightarrow \infty} \frac{c_{n+1}}{c_n} &\leq \liminf_{n \rightarrow \infty} \sqrt[n]{c_n}, \\ \limsup_{n \rightarrow \infty} \sqrt[n]{c_n} &\leq \limsup_{n \rightarrow \infty} \frac{c_{n+1}}{c_n}.\end{aligned}$$

195 The series interleaves two geometric series with different rates. The roots feel only the rates, and both are safely below 1; the ratios feel the jumps between the two strands, which grow without bound. Interleaving is the standard way to break the ratio test.

Proof. We shall prove the second inequality; the proof of the first is quite similar. Put

$$\alpha = \limsup_{n \rightarrow \infty} \frac{c_{n+1}}{c_n}.$$

If $\alpha = +\infty$, there is nothing to prove. If α is finite, choose $\beta > \alpha$. There is an integer N such that

$$\frac{c_{n+1}}{c_n} \leq \beta$$

for $n \geq N$. In particular, for any $p > 0$,

$$c_{N+k+1} \leq \beta c_{N+k} \quad (k = 0, 1, \dots, p-1).$$

Multiplying these inequalities, we obtain

$$c_{N+p} \leq \beta^p c_N,$$

or

$$c_n \leq c_N \beta^{-N} \cdot \beta^n \quad (n \geq N).$$

Hence

$$\sqrt[n]{c_n} \leq \sqrt[n]{c_N \beta^{-N}} \cdot \beta,$$

so that

$$\limsup_{n \rightarrow \infty} \sqrt[n]{c_n} \leq \beta, \quad (3.18)$$

by Theorem 3.20(b). Since (3.18) is true for every $\beta > \alpha$, we have

$$\limsup_{n \rightarrow \infty} \sqrt[n]{c_n} \leq \alpha.$$

196

□

196 The mechanism: a bound on every ratio compounds into a geometric bound on the terms themselves, and taking n th roots flattens the constant $c_N \beta^{-N}$ into 1 by 3.20(b). Roots retain only the exponential rate; ratios carry every local fluctuation. That is the entire content of “root is stronger than ratio,” and it is also why $\lim c_{n+1}/c_n = L$ forces $\lim \sqrt[n]{c_n} = L$ — squeeze both displayed inequalities. Setting $c_n = n!/n^n$ gives the useful byproduct $\sqrt[n]{n!} \sim n/e$.

POWER SERIES

SIX CHAPTERS OF CONSEQUENCES BEGIN HERE | Two items only — a definition and one theorem — but 3.39 is the foundation on which Chapter 8 builds the exponential, the trigonometric functions and eventually $e^{i\pi} = -1$. The theorem says the convergence domain of $\sum c_n z^n$ is always a disc, and computes its radius by the root test. That a single formula answers “for which z ?” is the first sign of how rigid power series are; the second, uniform convergence inside the disc, arrives at 8.1.

3.38 Definition. Given a sequence (c_n) of complex numbers, the series

$$\sum_{n=0}^{\infty} c_n z^n \quad (3.19)$$

is called a *power series*. The numbers c_n are called the *coefficients* of the series; z is a complex number.

In general, the series will converge or diverge, depending on the choice of z . More specifically, with every power series there is associated a circle, the *circle of convergence*, such that (3.19) converges if z is in the interior of the circle and diverges if z is in the exterior of the circle (to cover all cases, we have to consider the plane as the interior of a circle with infinite radius, and a point as a circle with radius zero). The behavior on the circle of convergence is much more varied and cannot be described so simply.

3.39 Theorem. *Given the power series $\sum c_n z^n$, put*

$$\alpha = \limsup_{n \rightarrow \infty} \sqrt[n]{|c_n|}, \quad R = \frac{1}{\alpha}.$$

(If $\alpha = 0$, $R = +\infty$; if $\alpha = +\infty$, $R = 0$.) Then $\sum c_n z^n$ converges if $|z| < R$ and diverges if $|z| > R$.

Proof. Put $a_n = c_n z^n$ and apply the root test:

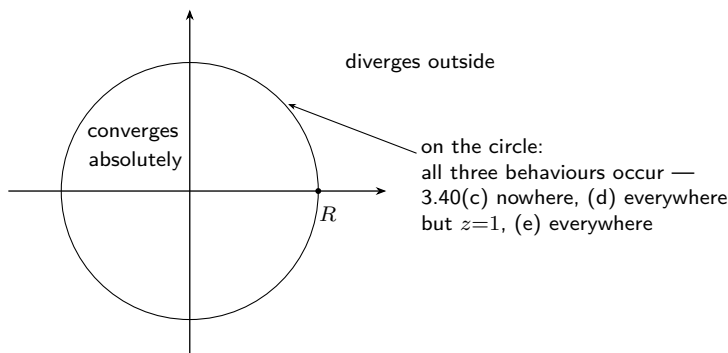
$$\limsup_{n \rightarrow \infty} \sqrt[n]{|a_n|} = |z| \limsup_{n \rightarrow \infty} \sqrt[n]{|c_n|} = \frac{|z|}{R}.$$

□

Note: R is called the *radius of convergence* of $\sum c_n z^n$.

3.40 Examples.

- (a) The series $\sum n^n z^n$ has $R = 0$.
- (b) The series $\sum \frac{z^n}{n!}$ has $R = +\infty$. (In this case the ratio test is easier to apply than the root test.)
- (c) The series $\sum z^n$ has $R = 1$. If $|z| = 1$, the series diverges, since (z^n) does not tend to 0 as $n \rightarrow \infty$.
- (d) The series $\sum \frac{z^n}{n}$ has $R = 1$. It diverges if $z = 1$. It converges for all other z with $|z| = 1$. (The last assertion will be proved in Theorem 3.44.)
- (e) The series $\sum \frac{z^n}{n^2}$ has $R = 1$. It converges for all z with $|z| = 1$, by the comparison test, since $|z^n/n^2| = 1/n^2$.



Theorem 3.39: inside the circle everything converges (absolutely — the proof runs through the root test, hence 3.25), outside everything diverges, and the circle itself carries no general rule.

SUMMATION BY PARTS

THE ONE TOOL FOR CONDITIONAL CONVERGENCE | Rudin will observe at 3.46 that comparison, root and ratio only ever detect *absolute* convergence. This section supplies the single technique in the book that goes further. The identity 3.41 trades $\sum a_n b_n$ for a sum involving the partial sums A_n — profitable exactly when the A_n are bounded though $\sum a_n$ itself diverges, as with $a_n = (-1)^n$ or $a_n = z^n$ on the unit circle. Theorem 3.42 packages the trade; 3.43 (alternating series) and 3.44 (boundary of the disc) are its two famous corollaries.

3.41 Theorem. *Given two sequences (a_n) and (b_n) , put*

$$A_n = \sum_{k=0}^n a_k$$

if $n \geq 0$; put $A_{-1} = 0$. Then, if $0 \leq p \leq q$, we have

$$\sum_{n=p}^q a_n b_n = \sum_{n=p}^{q-1} A_n (b_n - b_{n+1}) + A_q b_q - A_{p-1} b_p. \quad (3.20)$$

Proof.

$$\sum_{n=p}^q a_n b_n = \sum_{n=p}^q (A_n - A_{n-1}) b_n = \sum_{n=p}^q A_n b_n - \sum_{n=p-1}^{q-1} A_n b_{n+1},$$

and the last expression on the right is clearly equal to the right side of (3.20). $\square$

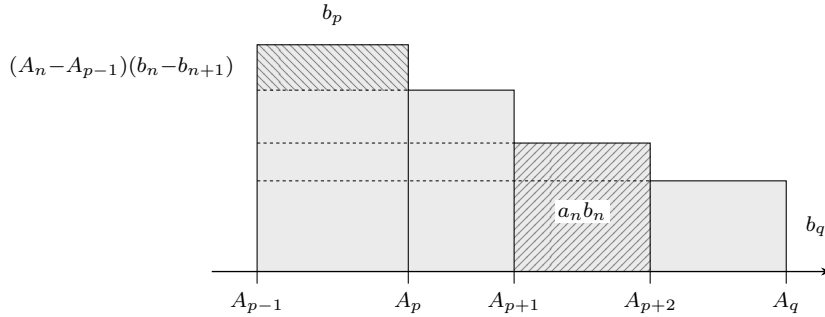
Formula (3.20), the so-called “partial summation formula,” is useful in the investigation of series of the form $\sum a_n b_n$, particularly when (b_n) is monotonic. We shall now give applications.

Integration by parts, one chapter early

Rename the pieces and (3.20) is an old friend: A_n is the “antiderivative” of a_n (partial sums undo differencing, since $a_n = A_n - A_{n-1}$), and $b_n - b_{n+1}$ is the “derivative” of b_n . The formula then reads $\sum a b = - \sum A \Delta b$ + boundary terms, which is $\int u dv = uv - \int v du$ with sums for integrals. Chapter 6 proves the integral version as 6.22, by essentially this computation applied to Riemann–Stieltjes sums.

The strategic content is the same in both settings: move the burden from the factor you cannot control onto the factor that varies slowly.

Here Σa_n may diverge, but only the *bounded* A_n appear on the right, multiplied by the increments of a monotone b_n — and those increments telescope.



Both sides of (3.20) measure this one staircase: columns of height b_n stand over the increments $a_n = A_n - A_{n-1}$. Sliced vertically (hatched sample), the area is $\sum_p^q a_n b_n$. Sliced horizontally, the slab between levels b_{n+1} and b_n has width $A_n - A_{p-1}$, and the base slab, height b_q , runs the full width $A_q - A_{p-1}$; expanding the widths telescopes all the A_{p-1} 's into the single boundary term $-A_{p-1}b_p$ of (3.20). Drawn for $a_n > 0$ and decreasing positive b_n — exactly 3.42's hypotheses; Rudin's two-line proof is the same bookkeeping with signs allowed.

3.42 Theorem. Suppose

- (a) the partial sums A_n of Σa_n form a bounded sequence;
- (b) $b_0 \geq b_1 \geq b_2 \geq \dots$;
- (c) $\lim_{n \rightarrow \infty} b_n = 0$.

Then $\Sigma a_n b_n$ converges.

Proof. Choose M such that $|A_n| \leq M$ for all n . Given $\varepsilon > 0$, there is an integer N such that $b_N \leq \varepsilon/(2M)$. For $N \leq p \leq q$, we have

$$\begin{aligned} \left| \sum_{n=p}^q a_n b_n \right| &= \left| \sum_{n=p}^{q-1} A_n(b_n - b_{n+1}) + A_q b_q - A_{p-1} b_p \right| \\ &\leq M \left[\sum_{n=p}^{q-1} (b_n - b_{n+1}) + b_q + b_p \right] \\ &= 2M b_p \leq 2M b_N \leq \varepsilon. \end{aligned}$$

Convergence now follows from the Cauchy criterion. We note that the first inequality in the above chain depends of course on the fact that $b_n - b_{n+1} \geq 0$.¹⁹⁷ $\square$

3.43 Theorem. Suppose

- (a) $|c_1| \geq |c_2| \geq |c_3| \geq \dots$;
- (b) $c_{2m-1} \geq 0, c_{2m} \leq 0$ ($m = 1, 2, 3, \dots$);
- (c) $\lim_{n \rightarrow \infty} c_n = 0$.

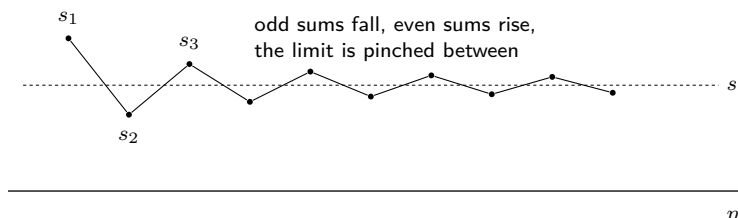
197 The telescope is the payoff: the increments $b_n - b_{n+1}$ are nonnegative, so absolute values cost nothing, and their sum collapses to $b_p - b_q$. All the oscillation of Σa_n has been absorbed into the single constant M ; what remains converges because (b_n) does. Note the shape of the conclusion: convergence via 3.22, no sum ever named.

Then Σc_n converges.

Series for which (b) holds are called “alternating series”; the theorem was known to Leibnitz.

Proof. Apply Theorem 3.42 with $a_n = (-1)^{n+1}$, $b_n = |c_n|$.¹⁹⁸ $\square$

198 The partial sums of $(-1)^{n+1}$ are $1, 0, 1, 0, \dots$ — bounded by $M = 1$ — and the $|c_n|$ decrease to 0. The most famous instance: $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots$ converges — its sum turns out to be $\log 2$, a fact that needs Chapter 8 — while its absolute version, the harmonic series, diverges. The gap between those two facts is the subject of the whole next section.



Alternating series: each swing overshoots the limit and is shorter than the last. The even partial sums increase, the odd ones decrease, and $|s - s_n| \leq |c_{n+1}|$ — the error is at most the first omitted term.

3.44 Theorem. Suppose the radius of convergence of $\Sigma c_n z^n$ is 1, and suppose $c_0 \geq c_1 \geq c_2 \geq \dots$, $\lim_{n \rightarrow \infty} c_n = 0$. Then $\Sigma c_n z^n$ converges at every point on the circle $|z| = 1$, except possibly at $z = 1$.

Proof. Put $a_n = z^n$, $b_n = c_n$. The hypotheses of Theorem 3.42 are then satisfied, since

$$|A_n| = \left| \sum_{m=0}^n z^m \right| = \left| \frac{1 - z^{n+1}}{1 - z} \right| \leq \frac{2}{|1 - z|},$$

if $|z| = 1$ and $z \neq 1$.¹⁹⁹ $\square$

199 The bound degrades as z approaches 1 — $|1 - z|$ in the denominator — which is why $z = 1$ must be excepted; there $A_n = n + 1$ is unbounded and 3.42 has nothing to grip. This settles 3.40(d): $\Sigma z^n/n$ converges everywhere on the unit circle except at $z = 1$, where it is the harmonic series. Alternating series are the case $z = -1$.

ABSOLUTE CONVERGENCE

The series Σa_n is said to *converge absolutely* if the series $\Sigma |a_n|$ converges.

3.45 Theorem. If Σa_n converges absolutely, then Σa_n converges.

Proof. The assertion follows from the inequality

$$\left| \sum_{k=n}^m a_k \right| \leq \sum_{k=n}^m |a_k|,$$

plus the Cauchy criterion. $\square$

3.46 Remarks. For series of positive terms, absolute convergence is the same as convergence.

If Σa_n converges, but $\Sigma |a_n|$ diverges, we say that Σa_n converges *conditionally* [[nonabsolutely]].²⁰⁰ For instance, the series

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$$

converges conditionally (Theorem 3.43).

200 The doubled brackets are the retyped edition's editorial mark, explained in its foreword: Rudin's own word was “nonabsolutely,” silently modernised to “conditionally” throughout, with the original recorded once — here — at the point of first use.

The comparison test, as well as the root and ratio tests, is really a test for absolute convergence, and therefore cannot give any information about conditionally convergent series. Summation by parts can sometimes be used to handle the latter. In particular, power series converge absolutely in the interior of the circle of convergence.

We shall see that we may operate with absolutely convergent series very much as with finite sums. We may multiply them term by term and we may change the order in which the additions are carried out, without affecting the sum of the series. But for conditionally convergent series this is no longer true, and more care has to be taken when dealing with them.²⁰¹

[201] *This paragraph is the thesis statement for the rest of the chapter. “Absolutely convergent” will turn out to mean “behaves like a finite sum” — products work (3.50), rearrangements are harmless (3.55) — while “conditionally convergent” means the sum is an artifact of the order of summation, so much so that rearrangement can produce any sum whatever (3.54). Absolute convergence is not a technicality; it is the boundary between arithmetic and analysis.*

ADDITION AND MULTIPLICATION OF SERIES

WHICH FINITE-SUM RULES SURVIVE THE LIMIT | The chapter’s closing question: which of the ordinary rules of arithmetic survive when “sum” means “limit of partial sums”? Addition survives unconditionally (3.47). Multiplication survives if at least one factor converges absolutely (Mertens, 3.50), can fail for two conditionally convergent factors (3.49), yet can never converge to the *wrong* value (Abel, 3.51). Rearrangement — the commutative law — is the most fragile of all, and the next section is devoted to its spectacular failure.

3.47 Theorem. *If $\Sigma a_n = A$ and $\Sigma b_n = B$, then $\Sigma(a_n + b_n) = A + B$, and $\Sigma ca_n = cA$, for any fixed c .*

Proof. Let

$$A_n = \sum_{k=0}^n a_k, \quad B_n = \sum_{k=0}^n b_k.$$

Then

$$A_n + B_n = \sum_{k=0}^n (a_k + b_k).$$

Since $\lim_{n \rightarrow \infty} A_n = A$ and $\lim_{n \rightarrow \infty} B_n = B$, we see that

$$\lim_{n \rightarrow \infty} (A_n + B_n) = A + B.$$

The proof of the second assertion is even simpler. □

Thus two convergent series may be added term by term, and the resulting series converges to the sum of the two series. The situation becomes more complicated when we consider multiplication of two series. To begin with, we have to define the product. This can be done in several ways; we shall consider the so-called “Cauchy product.”

3.48 Definition. Given Σa_n and Σb_n , we put

$$c_n = \sum_{k=0}^n a_k b_{n-k} \quad (n = 0, 1, 2, \dots)$$

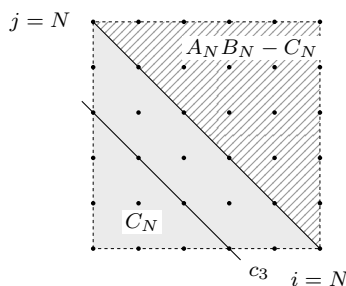
and call Σc_n the *product* of the two given series.

This definition may be motivated as follows. If we take two power series $\Sigma a_n z^n$ and $\Sigma b_n z^n$, multiply them term by term, and collect terms containing the same power of z , we get

$$\begin{aligned} \sum_{n=0}^{\infty} a_n z^n \cdot \sum_{n=0}^{\infty} b_n z^n &= (a_0 + a_1 z + a_2 z^2 + \cdots)(b_0 + b_1 z + b_2 z^2 + \cdots) \\ &= a_0 b_0 + (a_0 b_1 + a_1 b_0)z \\ &\quad + (a_0 b_2 + a_1 b_1 + a_2 b_0)z^2 + \cdots \\ &= c_0 + c_1 z + c_2 z^2 + \cdots \end{aligned}$$

202 The c_n collect the products $a_k b_j$ along the anti-diagonals $k + j = n$ of the multiplication array — the same anti-diagonals as in the countability proof 2.12, and for the same reason: each one is finite. The definition is rigged for power series, and 3.51's proof will return there.

Setting $z = 1$, we arrive at the above definition.²⁰²



The multiplication array $a_i b_j$, one dot per product. Each c_n strings an anti-diagonal (shown: $i + j = 3$), and every anti-diagonal is finite — the definition's whole content. The shaded triangle $i + j \leq N$ sums to C_N ; the dashed square sums to $A_N B_N$. Everything about Σc_n versus AB is the fate of the hatched difference: Mertens (3.50) shows absolute convergence of one factor kills it, while in Example 3.49 it carries mass ≈ 2 forever. These are the anti-diagonals of 2.12's walk, finite for the same reason.

3.49 Example. If

$$A_n = \sum_{k=0}^n a_k, \quad B_n = \sum_{k=0}^n b_k, \quad C_n = \sum_{k=0}^n c_k,$$

and $A_n \rightarrow A$, $B_n \rightarrow B$, then it is not at all clear that (C_n) will converge to AB , since we do not have $C_n = A_n B_n$. The dependence of (C_n) on (A_n) and (B_n) is quite a complicated one (see the proof of Theorem 3.50). We shall now show that the product of two convergent series may actually diverge.

The series

$$\sum_{n=0}^{\infty} \frac{(-1)^n}{\sqrt{n+1}} = 1 - \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} - \frac{1}{\sqrt{4}} + \cdots$$

converges (Theorem 3.43). We form the product of this series with itself and obtain

$$\begin{aligned} \sum_{n=0}^{\infty} c_n &= 1 - \left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} \right) + \left(\frac{1}{\sqrt{3}} + \frac{1}{\sqrt{2}\sqrt{2}} + \frac{1}{\sqrt{3}} \right) \\ &\quad - \left(\frac{1}{\sqrt{4}} + \frac{1}{\sqrt{3}\sqrt{2}} + \frac{1}{\sqrt{2}\sqrt{3}} + \frac{1}{\sqrt{4}} \right) + \cdots, \end{aligned}$$

so that

$$c_n = (-1)^n \sum_{k=0}^n \frac{1}{\sqrt{(n-k+1)(k+1)}}.$$

Since

$$(n-k+1)(k+1) = \left(\frac{n}{2} + 1 \right)^2 - \left(\frac{n}{2} - k \right)^2 \leq \left(\frac{n}{2} + 1 \right)^2,$$

we have

$$|c_n| \geq \sum_{k=0}^n \frac{2}{n+2} = \frac{2(n+1)}{n+2},$$

so that the condition $c_n \rightarrow 0$, which is necessary for the convergence of $\sum c_n$, is not satisfied.²⁰³

In view of the next theorem, due to Mertens, we note that we have here considered the product of two conditionally convergent series.

3.50 Theorem. Suppose

- (a) $\sum_{n=0}^{\infty} a_n$ converges absolutely,
- (b) $\sum_{n=0}^{\infty} a_n = A$,
- (c) $\sum_{n=0}^{\infty} b_n = B$,
- (d) $c_n = \sum_{k=0}^n a_k b_{n-k} \quad (n = 0, 1, 2, \dots)$.

Then

$$\sum_{n=0}^{\infty} c_n = AB.$$

That is, the product of two convergent series converges, and to the right value, if at least one of the two series converges absolutely.

Proof. Put

$$A_n = \sum_{k=0}^n a_k, \quad B_n = \sum_{k=0}^n b_k, \quad C_n = \sum_{k=0}^n c_k, \quad \beta_n = B_n - B.$$

Then

$$\begin{aligned} C_n &= a_0 b_0 + (a_0 b_1 + a_1 b_0) + \cdots + (a_0 b_n + a_1 b_{n-1} + \cdots + a_n b_0) \\ &= a_0 B_n + a_1 B_{n-1} + \cdots + a_n B_0 \\ &= a_0 (B + \beta_n) + a_1 (B + \beta_{n-1}) + \cdots + a_n (B + \beta_0) \\ &= A_n B + a_0 \beta_n + a_1 \beta_{n-1} + \cdots + a_n \beta_0. \end{aligned}$$

203 Why it fails: each c_n averages $n+1$ products of terms of size about $1/\sqrt{n}$ each, so the products are of size about $1/n$ and their sum stays near 2. The factor-series converge only by cancellation, and the Cauchy product lines up terms so that the cancellation is destroyed. Convergence-by-cancellation is not robust under rearrangement-like operations — the theme of the whole final section.

204 The first rewriting is the anti-diagonal picture again: summing $c_0, \dots, c_n$ sweeps the triangle $k+j \leq n$ of the array, which can be sliced as a_k times the partial sum B_{n-k} . Then each B_{n-k} is split into limit plus error, $B + \beta_{n-k}$, isolating the main term $A_n B$. The proof will live or die on the error convolution γ_n .

204 Put

$$\gamma_n = a_0\beta_n + a_1\beta_{n-1} + \cdots + a_n\beta_0.$$

We wish to show that $C_n \rightarrow AB$. Since $A_nB \rightarrow AB$, it suffices to show that

$$\lim_{n \rightarrow \infty} \gamma_n = 0. \quad (3.21)$$

Put

$$\alpha = \sum_{n=0}^{\infty} |a_n|.$$

[It is here that we use (a).] Let $\varepsilon > 0$ be given. By (c), $\beta_n \rightarrow 0$. Hence we can choose N such that $|\beta_n| \leq \varepsilon$ for $n \geq N$, in which case

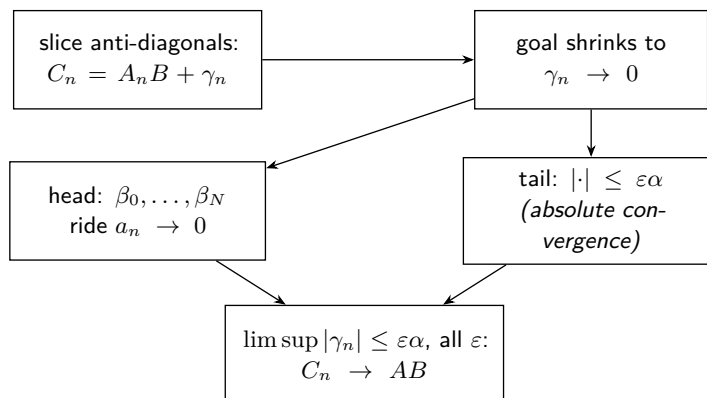
$$\begin{aligned} |\gamma_n| &\leq |\beta_0 a_n + \cdots + \beta_N a_{n-N}| + |\beta_{N+1} a_{n-N-1} + \cdots + \beta_n a_0| \\ &\leq |\beta_0 a_n + \cdots + \beta_N a_{n-N}| + \varepsilon \alpha. \end{aligned}$$

Keeping N fixed and letting $n \rightarrow \infty$, we get

$$\limsup_{n \rightarrow \infty} |\gamma_n| \leq \varepsilon \alpha,$$

since $a_k \rightarrow 0$ as $k \rightarrow \infty$. Since ε is arbitrary, (3.21) follows. ²⁰⁵ □

205 The split-the-sum device, and the same two-parameter discipline as in 3.31: the old errors $\beta_0, \dots, \beta_N$ are finitely many and ride multiples of $a_n \rightarrow 0$; the recent errors are small and ride the absolutely convergent tail, total at most $\varepsilon\alpha$. Freeze N , let n run, then shrink ε . Without (a) the constant α does not exist, and 3.49 shows the theorem genuinely fails.



Mertens as a flow diagram: isolate the error convolution, split it at N , and let each half die for its own reason. Hypothesis (a) funds exactly one box — the tail.

Another question which may be asked is whether the series Σc_n , if convergent, must have the sum AB . Abel showed that the answer is in the affirmative.

3.51 Theorem. If the series Σa_n , Σb_n , Σc_n converge to A , B , C , and $c_n = a_0 b_n + \cdots + a_n b_0$, then $C = AB$.

Here no assumption is made concerning absolute convergence. We shall give a simple proof (which depends on the continuity of power series) after Theorem 8.2. ²⁰⁶

REARRANGEMENTS

206 A promissory note worth remembering when it is paid. The future proof: the three power series $\Sigma a_n x^n$, $\Sigma b_n x^n$, $\Sigma c_n x^n$ converge absolutely for $0 < x < 1$, where Mertens applies and gives $f(x)g(x) = h(x)$; then let $x \rightarrow 1$ and invoke Abel's continuity theorem (8.2) three times. The pattern — prove an identity where convergence is absolute, then reach the boundary by continuity — is a first taste of a standard analytic method.

3.52 Definition. Let (k_n) , $n = 1, 2, 3, \dots$, be a sequence in which every positive integer appears once and only once (that is, (k_n) is a 1-1 function from $\mathbb{N}$ onto $\mathbb{N}$, in the notation of Definition 2.2). Putting

$$a'_n = a_{k_n} \quad (n = 1, 2, 3, \dots),$$

we say that $\Sigma a'_n$ is a *rearrangement* of Σa_n .

If (s_n) , (s'_n) are the sequences of partial sums of Σa_n , $\Sigma a'_n$, it is easily seen that, in general, these two sequences consist of entirely different numbers. We are thus led to the problem of determining under what conditions all rearrangements of a convergent series will converge, and whether the sums are necessarily the same.

3.53 Example. Consider the convergent series

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \dots \quad (3.22)$$

and one of its rearrangements

$$1 + \frac{1}{3} - \frac{1}{2} + \frac{1}{5} + \frac{1}{7} - \frac{1}{4} + \frac{1}{9} + \frac{1}{11} - \frac{1}{6} + \dots \quad (3.23)$$

in which two positive terms are always followed by one negative. If s is the sum of (3.22), then

$$s < 1 - \frac{1}{2} + \frac{1}{3} = \frac{5}{6}.$$

²⁰⁷ Since

$$\frac{1}{4k-3} + \frac{1}{4k-1} - \frac{1}{2k} > 0$$

for $k \geq 1$, we see that $s'_3 < s'_6 < s'_9 < \dots$, where s'_n is the n th partial sum of (3.23).

Hence

$$\limsup_{n \rightarrow \infty} s'_n > s'_3 = \frac{5}{6},$$

so that (3.23) certainly does not converge to s [we leave it to the reader to verify that (3.23) does, however, converge].²⁰⁸

This example illustrates the following theorem, due to Riemann.

3.54 Theorem. Let Σa_n be a series of real numbers which converges, but not absolutely. Suppose

$$-\infty \leq \alpha \leq \beta \leq +\infty.$$

Then there exists a rearrangement $\Sigma a'_n$ with partial sums s'_n such that

$$\liminf_{n \rightarrow \infty} s'_n = \alpha, \quad \limsup_{n \rightarrow \infty} s'_n = \beta. \quad (3.24)$$

Proof. Let

$$p_n = \frac{|a_n| + a_n}{2}, \quad q_n = \frac{|a_n| - a_n}{2} \quad (n = 1, 2, 3, \dots).$$

²⁰⁷ Why this bound: (3.22) is alternating with decreasing terms, so by the zigzag picture at 3.43 the partial sums oscillate around s , and every odd partial sum overestimates it — $s < s_3 = 5/6$.

²⁰⁸ The same numbers, added in a different order, give a different sum: taking positives twice as fast as negatives borrows against the divergent positive half, and each three-term block adds a surplus. This is the concrete disaster; 3.54 shows it is total.

Then $p_n - q_n = a_n$, $p_n + q_n = |a_n|$, $p_n \geq 0$, $q_n \geq 0$. The series Σp_n , Σq_n must both diverge, for if both were convergent, then

$$\Sigma(p_n + q_n) = \Sigma|a_n|$$

would converge, contrary to hypothesis. Since

$$\sum_{n=1}^N a_n = \sum_{n=1}^N (p_n - q_n) = \sum_{n=1}^N p_n - \sum_{n=1}^N q_n,$$

divergence of Σp_n and convergence of Σq_n (or vice versa) implies divergence of Σa_n , again contrary to hypothesis.²⁰⁹

Now let $P_1, P_2, P_3, \dots$ denote the nonnegative terms of Σa_n , in the order in which they occur, and let $Q_1, Q_2, Q_3, \dots$ be the absolute values of the negative terms of Σa_n , also in their original order.

The series ΣP_n , ΣQ_n differ from Σp_n , Σq_n only by zero terms, and are therefore divergent.

We shall construct sequences (m_n) , (k_n) such that the series

$$\begin{aligned} P_1 + \cdots + P_{m_1} - Q_1 - \cdots - Q_{k_1} + P_{m_1+1} + \cdots \\ + P_{m_2} - Q_{k_1+1} - \cdots - Q_{k_2} + \cdots, \end{aligned} \quad (3.25)$$

which clearly is a rearrangement of Σa_n , satisfies (3.24).

Choose real-valued sequences (α_n) , (β_n) such that $\alpha_n \rightarrow \alpha$, $\beta_n \rightarrow \beta$, $\alpha_n < \beta_n$, $\beta_1 > 0$.

Let m_1, k_1 be the smallest positive integers such that

$$\begin{aligned} P_1 + \cdots + P_{m_1} &> \beta_1, \\ P_1 + \cdots + P_{m_1} - Q_1 - \cdots - Q_{k_1} &< \alpha_1; \end{aligned}$$

let m_2, k_2 be the smallest positive integers such that

$$\begin{aligned} P_1 + \cdots + P_{m_1} - Q_1 - \cdots - Q_{k_1} + P_{m_1+1} + \cdots + P_{m_2} &> \beta_2, \\ P_1 + \cdots + P_{m_1} - Q_1 - \cdots - Q_{k_1} + P_{m_1+1} + \cdots + P_{m_2} \\ - Q_{k_1+1} - \cdots - Q_{k_2} &< \alpha_2, \end{aligned}$$

and continue in this way. This is possible since ΣP_n and ΣQ_n diverge.

If x_n, y_n denote the partial sums of (3.25) whose last terms are $P_{m_n}, -Q_{k_n}$, then

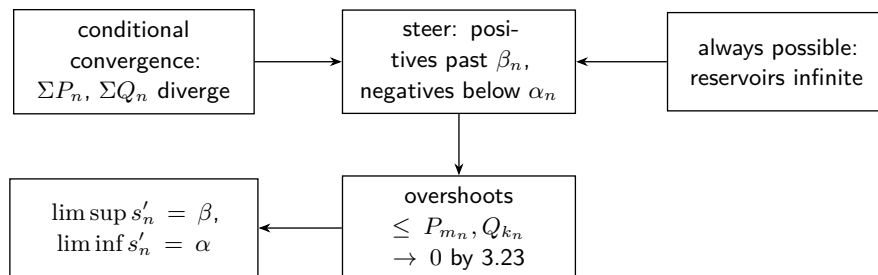
$$|x_n - \beta_n| \leq P_{m_n}, \quad |y_n - \alpha_n| \leq Q_{k_n}.$$

Since $P_n \rightarrow 0$ and $Q_n \rightarrow 0$ as $n \rightarrow \infty$, we see that $x_n \rightarrow \beta$, $y_n \rightarrow \alpha$.²¹⁰

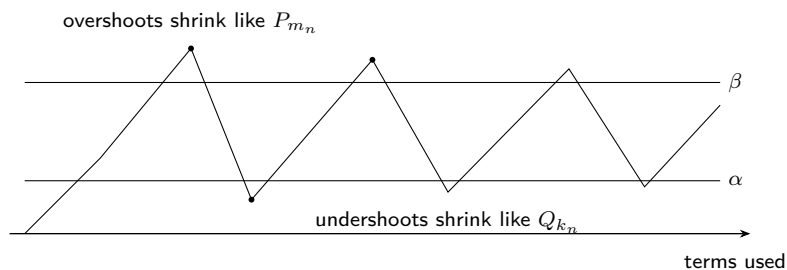
Finally, it is clear that no number less than α or greater than β can be a subsequential limit of the partial sums (3.25). $\square$

209 The anatomy of conditional convergence, and the engine of the whole proof: split the series into its positive part and its negative part, and both halves diverge. The finite sum s is the difference of two infinities kept in balance by the original ordering. That balance is the only thing a rearrangement can destroy — and it can destroy it completely, because each half is an inexhaustible reservoir.

210 The overshoot estimate is where convergence of the original series earns its keep: $a_n \rightarrow 0$ (3.23), so the reservoir terms P_n, Q_n shrink, and each crossing of the target overshoots by at most the last term used. The steering wheel: pay out positives until just past β_n , then negatives until just below α_n — “smallest integer such that” makes each overshoot minimal.



Riemann's theorem as a flow diagram: divergence of both reservoirs makes the steering possible, convergence of the original series ($a_n \rightarrow 0$) makes it accurate. Each hypothesis funds exactly one arrow.



Riemann's steering: positives push the partial sums just past β , negatives pull them just below α , forever. Divergence of both reservoirs keeps the steering possible; $a_n \rightarrow 0$ makes the overshoots die, so the upper and lower limits are exactly β and α . Take $\alpha = \beta$ to hit any prescribed sum.

3.55 Theorem. *If Σa_n is a series of complex numbers which converges absolutely, then every rearrangement of Σa_n converges, and they all converge to the same sum.*

Proof. Let $\Sigma a'_n$ be a rearrangement, with partial sums s'_n . Given $\varepsilon > 0$, there exists an integer N such that $m \geq n \geq N$ implies

$$\sum_{i=n}^m |a_i| \leq \varepsilon. \quad (3.26)$$

Now choose p such that the integers $1, 2, \dots, N$ are all contained in the set $k_1, k_2, \dots, k_p$ (we use the notation of Definition 3.52). Then if $n > p$, the numbers $a_1, \dots, a_N$ will cancel in the difference $s_n - s'_n$, so that $|s_n - s'_n| \leq \varepsilon$, by (3.26). Hence (s'_n) converges to the same sum as (s_n) . ²¹¹ □

CHAPTER 3 IN ONE PARAGRAPH

Convergence of a sequence is defined by the ε - N contract (3.1), and completeness makes it checkable from inside: Cauchy in $\mathbb{R}^k$ means convergent (3.11), monotone-and-bounded means convergent (3.14), and every real sequence has lim sup and lim inf as its extreme subsequential limits (3.17). A series is the sequence of

211 Absolute convergence makes the tail unconditionally negligible: (3.26) bounds any finite batch of far-out terms regardless of order or sign, so once both partial sums have swallowed $a_1, \dots, a_N$, they differ only by scraps from the tail. Compare 3.54, where the tail's two halves were infinite reservoirs; here the tail is just small. This is the precise sense of Rudin's slogan at 3.46 that absolutely convergent series behave like finite sums.

its partial sums (3.21), so those three facts become the Cauchy criterion (3.22), the boundedness criterion for nonnegative series (3.24), and the root test (3.33); every test is ultimately a comparison (3.25) with the geometric series (3.26), except summation by parts (3.41–3.44), which alone reaches conditional convergence. The number e arrives by series, fast enough for irrationality to fall out (3.30–3.32). The chapter’s final lesson is the boundary drawn at 3.46 and dramatised at 3.54–3.55: absolutely convergent series behave like finite sums under products and rearrangements; conditionally convergent series have sums that depend on the order of the terms — so much so that any prescribed value, or none, can be arranged.

EXERCISES

HOW TO USE THESE

Chapter 3’s exercise set contains two of the most important constructions in the book. 3.23–3.24 build the **completion** of an arbitrary metric space out of Cauchy sequences — Tao’s construction of $\mathbb{R}$ itself, carried out here as an exercise, with 3.25 closing the circle by asking what the completion of $\mathbb{Q}$ is. 3.22 is **Baire’s theorem** in its general form, promised at Exercise 2.30.

Technique builders. 3.3 and 3.16–3.18 are the monotone-convergence workhorses: recursive sequences squeezed by 3.14, with 3.16 being Newton’s method for square roots and its blazing quadratic convergence. 3.14 develops Cesàro means — the first taste of summability theory, needed for Fourier series (8.15–8.16) and Tauberian arguments.

Counterexample duty. 3.6–3.8 and 3.11–3.12 sharpen the convergence tests; 3.11(b) and 3.12(b) are the pairs Rudin cited in the text to kill the “boundary between convergence and divergence” conjecture.

Routine. 3.1, 3.2, 3.9, 3.20 are quick checks; do them first.

212 Reverse triangle inequality

(Exercise 1.13): $||s_n| - |s|| \leq |s_n - s|$.

The converse fails at once:

$$s_n = (-1)^n.$$

213 Multiply and divide by the

conjugate $\sqrt{n^2 + n} + n$; the answer is $\frac{1}{2}$. The moral: $\infty - \infty$ expressions are settled by algebra first, limit theorems second — 3.3 applies only after the difference is rewritten as a quotient.

3.1. Prove that convergence of (s_n) implies convergence of $(|s_n|)$. Is the converse true?²¹²

3.2. Calculate $\lim_{n \rightarrow \infty} (\sqrt{n^2 + n} - n)$.²¹³

3.3. If $s_1 = \sqrt{2}$ and

$$s_{n+1} = \sqrt{2 + \sqrt{s_n}} \quad (n = 1, 2, 3, \dots),$$

prove that (s_n) converges, and that $s_n < 2$ for $n = 1, 2, 3, \dots$.²¹⁴

3.4. Find the upper and lower limits of the sequence (s_n) defined by

$$s_1 = 0; \quad s_{2m} = \frac{s_{2m-1}}{2}; \quad s_{2m+1} = \frac{1}{2} + s_{2m}.$$

215

3.5. For any two real sequences (a_n) , (b_n) , prove that

$$\limsup_{n \rightarrow \infty} (a_n + b_n) \leq \limsup_{n \rightarrow \infty} a_n + \limsup_{n \rightarrow \infty} b_n,$$

provided the sum on the right is not of the form $\infty - \infty$.²¹⁶

3.6. Investigate the behavior (convergence or divergence) of Σa_n if

(a) $a_n = \sqrt{n+1} - \sqrt{n}$;

(b) $a_n = \frac{\sqrt{n+1} - \sqrt{n}}{n}$;

(c) $a_n = (\sqrt[n]{n} - 1)^n$;

(d) $a_n = \frac{1}{1+z^n}$, for complex values of z .

217

3.7. Prove that the convergence of Σa_n implies the convergence of

$$\sum_{n=1}^{\infty} \frac{\sqrt{a_n}}{n},$$

if $a_n \geq 0$.²¹⁸

3.8. If Σa_n converges and if (b_n) is monotonic and bounded, prove that $\Sigma a_n b_n$ converges.²¹⁹

3.9. Find the radius of convergence of each of the following power series:

(a) $\sum n^3 z^n$,

(c) $\sum \frac{2^n}{n^2} z^n$,

(b) $\sum \frac{2^n}{n!} z^n$,

(d) $\sum \frac{n^3}{3^n} z^n$.

220

3.10. Suppose that the coefficients of the power series $\Sigma a_n z^n$ are integers, infinitely many of which are distinct from zero. Prove that the radius of convergence is at most 1.²²¹

214 The 3.14 template, used again in 3.16–3.18: prove boundedness ($s_n < 2$) and monotonicity together by induction, then monotone convergence hands you a limit. Only afterwards may you pass to the limit in the recursion to identify it — existence must precede the algebra.

215 Track the two subsequences separately: the even-indexed terms crawl up toward $\frac{1}{2}$, the odd ones toward 1. So $E = \{\frac{1}{2}, 1\}$, $\liminf = \frac{1}{2}$, $\limsup = 1$ — a sequence need not oscillate symmetrically about anything.

216 Inequality, not equality — and strictness is common: $a_n = (-1)^n$, $b_n = (-1)^{n+1}$ gives $0 < 2$. The two sequences need not peak at the same indices. (The excluded case is exactly the arithmetic left undefined at 1.23.) The working proof uses the tail-sup form from the box at 3.17: $\sup_{n \geq N} (a_n + b_n) \leq \sup_{n \geq N} a_n + \sup_{n \geq N} b_n$, then let $N \rightarrow \infty$.

217 (a) telescopes to $\sqrt{N+1} - 1$: diverges. (b) conjugate trick gives $a_n \sim \frac{1}{2}n^{-3/2}$: converges by comparison with 3.28. (c) root test: $\sqrt[n]{a_n} = \sqrt[n]{n} - 1 \rightarrow 0$ by 3.20(c): converges. (d) splits by $|z|$: for $|z| > 1$ comparison with $\Sigma |z|^{-n}$ converges; for $|z| \leq 1$ the terms do not tend to 0 (3.23) — watch $|z| = 1$, where $1 + z^n$ can even vanish.

218 The trick to remember: $\sqrt{a_n}/n \leq \frac{1}{2}(a_n + 1/n^2)$ by the AM–GM inequality $xy \leq \frac{1}{2}(x^2 + y^2)$ — which is Schwarz 1.35 for $n = 1$. Both halves converge, so comparison finishes it.

219 Summation by parts, but not quite 3.42, whose hypothesis $b_n \rightarrow 0$ may fail here. Write $b_n = (b_n - b) + b$ where $b = \lim b_n$ (exists by 3.14): the b -part converges by 3.47, and $(b_n - b)$ decreases (or increases) to 0, so 3.42 handles the rest. Note the partial sums A_n are bounded because Σa_n converges — a weaker demand than 3.42 makes.

220 All four by 3.37's byproduct ($\lim c_{n+1}/c_n$, when it exists, is α): (a) $R = 1$ — polynomial factors never move R , by 3.20(c); (b) $R = \infty$; (c) $R = \frac{1}{2}$; (d) $R = 3$.

221 A nonzero integer has $|a_n| \geq 1$, so $\sqrt[n]{|a_n|} \geq 1$ infinitely often and $\alpha = \limsup \geq 1$. Integrality forces smallness to be impossible — the same arithmetic leverage as in the irrationality proof 3.32.

222 This is the divergent half of Rudin's "no boundary" claim after 3.29: by (b), $\Sigma a_n/s_n$ diverges yet its terms are smaller than a_n by the unbounded factor s_n — every divergent series dominates a slower one. Part (b)'s display is a Cauchy-criterion violation served ready-made: fix N , let k grow, and the blocks refuse to shrink because $s_{N+k} \rightarrow \infty$. Part (c) telescopes. In (d) the first can do either (try $a_n = 1$ vs. a_n supported on sparse blocks); the second always converges, since $a_n/(1 + n^2 a_n) \leq 1/n^2$.

223 The convergent half of "no boundary": by (b), dividing a convergent series by the vanishing factor $\sqrt{r_n}$ leaves it convergent — every convergent series is dominated by a slower one. Together with 3.11(b) this kills the conjecture in both directions. The engine in both parts is the same: r_n telescopes ($a_n = r_n - r_{n+1}$), and (b) is the algebraic identity $\frac{r_n - r_{n+1}}{\sqrt{r_n}} < 2(\sqrt{r_n} - \sqrt{r_{n+1}})$, i.e. $\sqrt{r_n} + \sqrt{r_{n+1}} < 2\sqrt{r_n}$.

224 Compare $\sum_n |c_n|$ against $(\Sigma|a_n|)(\Sigma|b_n|)$: every partial sum of $|c_n|$ is trapped in a triangle of the array of $|a_k b_j|$, which sits inside a full rectangle. With 3.50 this closes the multiplication story: absolute $\times$ absolute is absolute; absolute $\times$ conditional converges to the right value (Mertens); conditional $\times$ conditional can diverge (3.49).

3.11. Suppose $a_n > 0$, $s_n = a_1 + \cdots + a_n$, and Σa_n diverges.

(a) Prove that $\sum \frac{a_n}{1 + a_n}$ diverges.

(b) Prove that

$$\frac{a_{N+1}}{s_{N+1}} + \cdots + \frac{a_{N+k}}{s_{N+k}} \geq 1 - \frac{s_N}{s_{N+k}}$$

and deduce that $\sum \frac{a_n}{s_n}$ diverges.

(c) Prove that

$$\frac{a_n}{s_n^2} \leq \frac{1}{s_{n-1}} - \frac{1}{s_n}$$

and deduce that $\sum \frac{a_n}{s_n^2}$ converges.

(d) What can be said about

$$\sum \frac{a_n}{1 + na_n} \quad \text{and} \quad \sum \frac{a_n}{1 + n^2 a_n} ?$$

222

3.12. Suppose $a_n > 0$ and Σa_n converges. Put

$$r_n = \sum_{m=n}^{\infty} a_m.$$

(a) Prove that

$$\frac{a_m}{r_m} + \cdots + \frac{a_n}{r_n} > 1 - \frac{r_n}{r_m}$$

if $m < n$, and deduce that $\sum \frac{a_n}{r_n}$ diverges.

(b) Prove that

$$\frac{a_n}{\sqrt{r_n}} < 2(\sqrt{r_n} - \sqrt{r_{n+1}})$$

and deduce that $\sum \frac{a_n}{\sqrt{r_n}}$ converges.

223

3.13. Prove that the Cauchy product of two absolutely convergent series converges absolutely.²²⁴

3.14. If (s_n) is a complex sequence, define its arithmetic means σ_n by

$$\sigma_n = \frac{s_0 + s_1 + \cdots + s_n}{n+1} \quad (n = 0, 1, 2, \dots).$$

(a) If $\lim s_n = s$, prove that $\lim \sigma_n = s$.

- (b) Construct a sequence (s_n) which does not converge, although $\lim \sigma_n = 0$.
- (c) Can it happen that $s_n > 0$ for all n and that $\limsup s_n = \infty$, although $\lim \sigma_n = 0$?
- (d) Put $a_n = s_n - s_{n-1}$ for $n \geq 1$. Show that

$$s_n - \sigma_n = \frac{1}{n+1} \sum_{k=1}^n k a_k.$$

Assume that $\lim(na_n) = 0$ and that (σ_n) converges. Prove that (s_n) converges. [This gives a converse of (a), but under the additional assumption that $na_n \rightarrow 0$.]

- (e) Derive the last conclusion from a weaker hypothesis: Assume $M < \infty$, $|na_n| \leq M$ for all n , and $\lim \sigma_n = \sigma$. Prove that $\lim s_n = \sigma$, by completing the following outline:

If $m < n$, then

$$s_n - \sigma_n = \frac{m+1}{n-m}(\sigma_n - \sigma_m) + \frac{1}{n-m} \sum_{i=m+1}^n (s_n - s_i).$$

For these i ,

$$|s_n - s_i| \leq \frac{(n-i)M}{i+1} \leq \frac{(n-m-1)M}{m+2}.$$

Fix $\varepsilon > 0$ and associate with each n the integer m that satisfies

$$m \leq \frac{n-\varepsilon}{1+\varepsilon} < m+1.$$

Then $(m+1)/(n-m) \leq 1/\varepsilon$ and $|s_n - s_i| < M\varepsilon$. Hence

$$\limsup_{n \rightarrow \infty} |s_n - \sigma| \leq M\varepsilon.$$

Since ε was arbitrary, $\lim s_n = \sigma$.

225

- 3.15.** Definition 3.21 can be extended to the case in which the a_n lie in some fixed $\mathbb{R}^k$. Absolute convergence is defined as convergence of $\Sigma |\mathbf{a}_n|$. Show that Theorems 3.22, 3.23, 3.25(a), 3.33, 3.34, 3.42, 3.45, 3.47, and 3.55 are true in this more general setting. (Only slight modifications are required in any of the proofs.)²²⁶

- 3.16.** Fix a positive number α . Choose $x_1 > \sqrt{\alpha}$, and define $x_2, x_3, x_4, \dots$ by the recursion formula

$$x_{n+1} = \frac{1}{2} \left(x_n + \frac{\alpha}{x_n} \right).$$

[225] Cesàro summability, and a first look at an entire subject. Part (a) says averaging never destroys a limit; (b), with $s_n = (-1)^n$, says it can manufacture one — so “Cesàro sum” genuinely extends “sum,” which is how Fourier series will be rescued at 8.15–8.16 (Fejér’s idea). Parts (d) and (e) are Tauberian theorems: conditions under which the extension collapses back to ordinary convergence. Hardy’s condition $|na_n| \leq M$ in (e) is the historical original. The two-parameter bookkeeping in the outline — m tied to n through ε — is 3.31’s freeze-and-release discipline pushed one level up.

226 The checklist of which theory is metric and which is order-theoretic: everything driven by the Cauchy criterion and the triangle inequality survives in $\mathbb{R}^k$ verbatim. What is missing from the list matters too — 3.24, 3.27, and comparison (b) need nonnegativity, and the root/ratio tests survive only because they test the scalar series $\Sigma |\mathbf{a}_n|$.

(a) Prove that (x_n) decreases monotonically and that $\lim x_n = \sqrt{\alpha}$.

(b) Put $\varepsilon_n = x_n - \sqrt{\alpha}$, and show that

$$\varepsilon_{n+1} = \frac{\varepsilon_n^2}{2x_n} < \frac{\varepsilon_n^2}{2\sqrt{\alpha}},$$

so that, setting $\beta = 2\sqrt{\alpha}$,

$$\varepsilon_{n+1} < \beta \left(\frac{\varepsilon_1}{\beta} \right)^{2^n} \quad (n = 1, 2, 3, \dots).$$

(c) This is a good algorithm for computing square roots, since the recursion formula is simple and the convergence is extremely rapid. For example, if $\alpha = 3$ and $x_1 = 2$, show that $\varepsilon_1/\beta < 1/10$ and that therefore

$$\varepsilon_5 < 4 \cdot 10^{-16}, \quad \varepsilon_6 < 4 \cdot 10^{-32}.$$

227 Newton's method on $f(x) = x^2 - \alpha$, and the recursion Chapter 1's note on Example 1.1 promised: it always lands above $\sqrt{\alpha}$ (AM–GM), so the sequence is decreasing and 3.14 applies. Part (b) is quadratic convergence — the error squares each step, doubling the correct digits — against which the linear rate of 1.1's map $q = (2p+2)/(p+2)$, and of 3.17 below, can be measured.

228 The general- α version of Chapter 1's map (there $\alpha = 2$): $q = (2+p)/(1+p)$. Unlike Newton, it alternates sides of $\sqrt{\alpha}$, so the two subsequences must be handled separately by 3.14 and then reunited — the strategy of 3.4. Convergence is linear: the error is multiplied by roughly $(\sqrt{\alpha}-1)/(\sqrt{\alpha}+1)$ each step, one new digit per few iterations, versus Newton's doubling.

229 Newton for $f(x) = x^p - \alpha$: the limit is $\sqrt[p]{\alpha}$, with the same quadratic convergence and the same monotone-decrease-from-above proof. A numerical companion to Theorem 1.21, which proved the p th root exists; this computes it.

230 The address system of Chapter 2's binary-tree figure made exact: digit α_n chooses the left or right third at stage n , and the series is the point the nested thirds close on. Series notation turns the geometric description into a formula — and makes the bijection with $\{0, 2\}^{\mathbb{N}} \sim \{0, 1\}^{\mathbb{N}}$ (hence uncountability, 2.14) immediate.

231 Split $d(p_n, p) \leq d(p_n, p_{n_1}) + d(p_{n_1}, p)$: the first half is small because the sequence is Cauchy, the second because the subsequence converges. Small but used constantly — including in the very next two exercises, and it is how 3.11(b) could have been proved from 3.6(a) directly.

227

3.17. Fix $\alpha > 1$. Take $x_1 > \sqrt{\alpha}$, and define

$$x_{n+1} = \frac{\alpha + x_n}{1 + x_n} = x_n + \frac{\alpha - x_n^2}{1 + x_n}.$$

(a) Prove that $x_1 > x_3 > x_5 > \dots$.

(b) Prove that $x_2 < x_4 < x_6 < \dots$.

(c) Prove that $\lim x_n = \sqrt{\alpha}$.

(d) Compare the rapidity of convergence of this process with the one described in Exercise 3.16.

228

3.18. Replace the recursion formula of Exercise 3.16 by

$$x_{n+1} = \frac{p-1}{p}x_n + \frac{\alpha}{p}x_n^{-p+1},$$

where p is a fixed positive integer, and describe the behavior of the resulting sequences (x_n) .²²⁹

3.19. Associate to each sequence $a = (\alpha_n)$, in which α_n is 0 or 2, the real number

$$x(a) = \sum_{n=1}^{\infty} \frac{\alpha_n}{3^n}.$$

Prove that the set of all $x(a)$ is precisely the Cantor set described in Theorem 2.44.²³⁰

3.20. Suppose (p_n) is a Cauchy sequence in a metric space X , and some subsequence (p_{n_i}) converges to a point $p \in X$. Prove that the full sequence (p_n) converges to p .²³¹

3.21. Prove the following analogue of Theorem 3.10(b): If (E_n) is a sequence of closed, nonempty, and bounded sets in a *complete* metric space X , if $E_n \supset E_{n+1}$, and if

$$\lim_{n \rightarrow \infty} \text{diam } E_n = 0,$$

then $\bigcap_{n=1}^{\infty} E_n$ consists of exactly one point.²³²

3.22. Suppose X is a nonempty complete metric space, and (G_n) is a sequence of dense open subsets of X . Prove Baire's theorem, namely, that $\bigcap_{n=1}^{\infty} G_n$ is not empty. (In fact, it is dense in X .) *Hint:* Find a shrinking sequence of neighborhoods E_n such that $\overline{E_n} \subset G_n$, and apply Exercise 3.21.²³³

3.23. Suppose (p_n) and (q_n) are Cauchy sequences in a metric space X . Show that the sequence $(d(p_n, q_n))$ converges. *Hint:* For any m, n ,

$$d(p_n, q_n) \leq d(p_n, p_m) + d(p_m, q_m) + d(q_m, q_n);$$

it follows that

$$|d(p_n, q_n) - d(p_m, q_m)|$$

is small if m and n are large.²³⁴

3.24. Let X be a metric space.

(a) Call two Cauchy sequences $(p_n), (q_n)$ in X *equivalent* if

$$\lim_{n \rightarrow \infty} d(p_n, q_n) = 0.$$

Prove that this is an equivalence relation.

(b) Let X^* be the set of all equivalence classes so obtained. If $P \in X^*, Q \in X^*, (p_n) \in P, (q_n) \in Q$, define

$$\Delta(P, Q) = \lim_{n \rightarrow \infty} d(p_n, q_n);$$

by Exercise 3.23, this limit exists. Show that the number $\Delta(P, Q)$ is unchanged if (p_n) and (q_n) are replaced by equivalent sequences, and hence that Δ is a distance function in X^* .

(c) Prove that the resulting metric space X^* is complete.

232 *Compactness weakened to closed-and-bounded, compensated by completeness: pick $x_n \in E_n$; shrinking diameters make (x_n) Cauchy, completeness gives the limit, closedness puts it in every E_n . Compare 2.36, where compactness did the work with no diameter hypothesis at all — and note that here, unlike there, “bounded” can even be dropped once diameters shrink.*

[233] *The general Baire theorem, upgrading Exercise 2.30 from $\mathbb{R}^k$ to any complete metric space — with 3.21 replacing compactness as the engine. The construction: density of G_1 gives a ball inside it; density of G_2 gives a smaller ball inside both, radius halved; iterate, and 3.21 yields a point surviving every stage. This theorem underlies uniform boundedness and the open mapping theorem in functional analysis, and Exercise 2.30's consequences (e.g. $\mathbb{R}$ uncountable) now hold in every complete metric space without isolated points.*

234 *Note where the limit lives: the real sequence $(d(p_n, q_n))$ is Cauchy, so it converges by 3.11(c) — completeness of $\mathbb{R}$ — even though (p_n) and (q_n) may converge to nothing in X . That asymmetry is exactly what the next exercise exploits.*

- (d) For each $p \in X$, there is a Cauchy sequence all of whose terms are p ; let P_p be the element of X^* which contains this sequence. Prove that

$$\Delta(P_p, P_q) = d(p, q)$$

for all $p, q \in X$. In other words, the mapping φ defined by $\varphi(p) = P_p$ is an *isometry* (i.e., a distance-preserving mapping) of X into X^* .

- (e) Prove that $\varphi(X)$ is dense in X^* , and that $\varphi(X) = X^*$ if X is complete. By (d), we may identify X and $\varphi(X)$ and thus regard X as embedded in the complete metric space X^* . We call X^* the *completion* of X .

235 *The construction the appendix to Chapter 1 declined: a missing limit is represented by the sequences that reach for it, with two sequences identified when they reach for the same thing. This is Cantor's route to $\mathbb{R}$ and the one Tao's Analysis I takes as its foundation (his Chapter 5: reals are equivalence classes of rational Cauchy sequences). Note what makes it outrun Dedekind cuts, as the appendix's note anticipated: nothing here mentions order — only d — so every metric space acquires a completion, function spaces included. Part (c) has the construction's one subtlety: a Cauchy sequence of equivalence classes is unwound by picking representatives and a diagonal argument, the 3.7-style chase again.*

236 $\mathbb{R}$, of course — and the point of asking is that the completion was built without $\mathbb{R}$: 3.24 used only $\mathbb{Q}$'s own metric. Chapter 1 characterised $\mathbb{R}$ by order (least upper bounds); this exercise constructs it from distance. Same object, two foundations, and the uniqueness-up-to-isomorphism noted in Step 9 of the appendix is what guarantees they agree.

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- 3.25.** Let X be the metric space whose points are the rational numbers, with the metric $d(x, y) = |x - y|$. What is the completion of this space? (Compare Exercise 3.24.)²³⁶

Appendix to Chapter 3

HOW TO USE THIS APPENDIX

Nothing in this appendix is Rudin's text. It is imported from the four comparison texts and set apart at every level: the running head names the appendix, section heads carry a reverse-video tag, item numbers are prefixed **S**, and every statement names its source.

Why import anything at all. Rudin's Chapter 3 is the most complete chapter in the book, and most of what the other texts add is not missing theory but missing *leverage*. He defines $\limsup$ as the supremum of the set of subsequential limits and never gives the formula one actually computes with. He proves the root and ratio tests and then remarks that both fail when the limit is 1 — without saying what to do next. He proves summation by parts but extracts no reusable bound from it. Each of those is one theorem away from being a tool.

What is here.

A. Upper and lower limits, finished. The inf-of-sup formula, the ε -characterisation, and the algebra — subadditivity, products, reciprocals — that Rudin's 3.19 only begins.

B. Past the root and ratio tests. Raabe's test, which compares against the p -series instead of the geometric series; the classical series that forces it; and the reason the hierarchy of tests can never terminate.

C. Summation by parts, quantitatively. Abel's lemma with an explicit constant, Abel's test with bounded rather than null weights, and the Dirichlet-kernel estimate that carries Chapter 3 into Fourier series.

D. Grouping and multiplying. What inserting parentheses is allowed to do, and the form of the product theorem that holds for *every* arrangement of the double array rather than the diagonal one.

E. Summation without an order. Rudin's 3.55 says an absolutely convergent series may be reordered freely. Tao takes that as licence to drop the order entirely and sum over an index set — after which double series and iterated sums are theorems rather than notation.

How to read it. Section A pays for itself immediately — read it with 3.16–3.19. Sections B–D are best read after 3.44 and 3.50 respectively, and E after 3.55. The exercises are the last section.

NOT IN RUDIN

SUPPLEMENT A — UPPER AND LOWER LIMITS, FINISHED

WHY THIS SUPPLEMENT BELONGS TO CHAPTER 3

What Rudin does. Definition 3.16 puts E for the set of subsequential limits (in the extended reals) and defines $\limsup x_n = \sup E$. Theorem 3.17 then proves that $\limsup$ belongs to E and that no larger number does, and 3.19 gives the one comparison $a_n \leq b_n \Rightarrow \limsup a_n \leq \limsup b_n$.

The gap. That definition is superb for proving things and useless for computing. To find $\limsup$ from it you would first have to find every subsequential limit — which is harder than the original problem. The formula everyone actually uses, $\limsup x_n = \lim_n \sup_{k \geq n} x_k$, is never stated. Nor is the algebra: no subadditivity, no product rule, no reciprocal rule.

Where the gap bites. At 3.33 and 3.34, the root and ratio tests, which are stated in terms of $\limsup$ and therefore cannot be applied until one can evaluate a $\limsup$; at 3.39, the radius of convergence, where $1/\limsup \sqrt[n]{|c_n|}$ has to be computed for actual power series; and at 3.37, whose proof compares two upper limits and needs exactly the algebra that 3.19 does not supply.

What this section supplies. The monotone formula, with the proof that it agrees with Rudin's definition; the two-clause ε -characterisation that every later $\limsup$ argument runs on; and the algebra, including the counterexample showing why the inequalities are one-directional.

S3.1 Theorem. (*The formula.*) For any real sequence (x_n) ,

$$\limsup_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} \sup_{k \geq n} x_k, \quad \liminf_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} \inf_{k \geq n} x_k,$$

the two limits existing in the extended reals because the sequences $M_n = \sup_{k \geq n} x_k$ and $m_n = \inf_{k \geq n} x_k$ are monotone.

SOURCE: ECNU 定理 7.9 (bk. p. 158), which notes that many texts take this as the definition; Rudin never states it.

Proof. Since $\{x_k : k \geq n+1\} \subset \{x_k : k \geq n\}$, the suprema satisfy $M_{n+1} \leq M_n$: the sequence (M_n) decreases, so it has a limit M in the extended reals. Write E for Rudin's set of subsequential limits; we show $M = \sup E$.

If $L \in E$, say $x_{n_j} \rightarrow L$, then $x_{n_j} \leq M_n$ for every j with $n_j \geq n$, so $L \leq M_n$ for every n and hence $L \leq M$. Thus $\sup E \leq M$.

Conversely $M \in E$. Choose indices $n_1 < n_2 < \dots$ so that $x_{n_k} > M_{n_k} - 1/k$, which is possible because M_{n_k} is the supremum of the tail beginning at n_k . Since also $x_{n_k} \leq M_{n_k}$ and $M_{n_k} \rightarrow M$, we get $x_{n_k} \rightarrow M$. Hence $M \leq \sup E$. $\square$

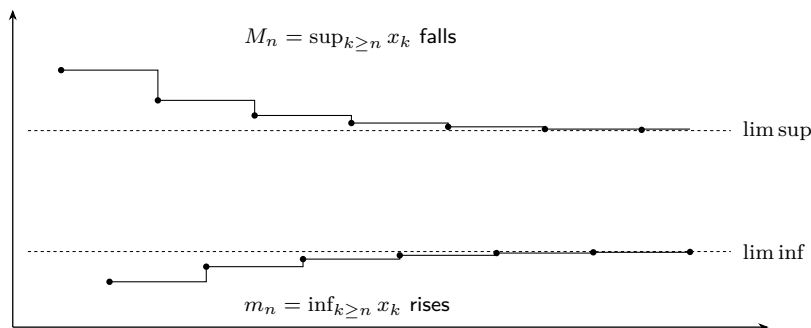
HOW THE PROOF FLOWS

1. **Choose the method.** Two descriptions of one number are given, so the proof is two inequalities. Neither side can be computed, so each must be pinned by an approximation argument.
2. **Get monotonicity free.** Discarding terms can only lower a supremum. That single observation is what makes $\lim M_n$ exist at all — by monotone convergence (3.14), with no boundedness needed once $\pm\infty$ are allowed.
3. **One inequality by domination.** Every subsequential limit is trapped below every tail supremum, so $\sup E \leq M$. Nothing is constructed here.
4. **The other by construction.** To show $M \leq \sup E$ it is not enough that M bounds things — M must *be* a subsequential limit, so a subsequence converging to it has to be built. The recipe is the defining property of a supremum: at stage k pick a term within $1/k$ of M_{n_k} .
5. **Audit.** The least-upper-bound property is spent once, in step 4, to produce each x_{n_k} ; monotone convergence once, in step 2. Note which clause of Definition 1.8 does which job: “upper bound” gives step 3, “least” gives step 4 — the same division of labour as in Rudin’s 1.21.

WHY THIS IS THE VERSION YOU COMPUTE WITH

M_n is a decreasing sequence — taking the supremum over a smaller tail cannot increase it — and it is bounded below whenever (x_n) is. So the limit exists by monotone convergence, with no reference to subsequences at all. To evaluate $\limsup$ you now do two things you know how to do: take a supremum, then take a monotone limit.

Rudin’s E -based definition and this one agree, and the agreement is worth seeing as two inequalities. Every subsequential limit is $\leq M_n$ for all n , hence $\leq \lim M_n$; and $\lim M_n$ is itself a subsequential limit, because for each n one can pick an index $k_n \geq n$ with $x_{k_n} > M_{n_k} - 1/n$. The second half is where the work is, and it is the same construction as ECNU’s proof that the largest cluster point exists.



The two staircases. Discarding a term can only lower a supremum and raise an infimum, so M_n decreases and m_n increases; their limits are the upper and lower limits. The picture also shows why the sequence converges exactly when the staircases meet.

S3.2 Theorem. (ε -characterisation.) $\bar{A} = \limsup x_n$ if and only if for every $\varepsilon > 0$: (i) $x_n < \bar{A} + \varepsilon$ for all sufficiently large n , and (ii) $x_n > \bar{A} - \varepsilon$ for infinitely many n . Equivalently: every $\alpha > \bar{A}$ is exceeded only finitely often, and every $\beta < \bar{A}$ is exceeded infinitely often.

SOURCE: ECNU 定理 7.7 and 7.7' (bk. p. 157). Rudin's 3.17(b) is condition (i) alone.

Proof. Let $\bar{A} = \limsup x_n$ and use S3.1. Given $\varepsilon > 0$, the numbers M_n decrease to $\bar{A}$, so $M_N < \bar{A} + \varepsilon$ for some N , and then $x_n \leq M_N < \bar{A} + \varepsilon$ for all $n \geq N$: that is (i). And $M_n \geq \bar{A}$ for every n , so $\sup_{k \geq n} x_k > \bar{A} - \varepsilon$, which produces some $k \geq n$ with $x_k > \bar{A} - \varepsilon$; as n is arbitrary there are infinitely many such k , which is (ii).

Conversely, suppose A satisfies (i) and (ii). By (i), $M_N \leq A + \varepsilon$ for some N , so $\bar{A} \leq A + \varepsilon$; by (ii), $M_n > A - \varepsilon$ for every n , so $\bar{A} \geq A - \varepsilon$. As ε was arbitrary, $\bar{A} = A$ by the ε -principle (S1.9). $\square$

Rudin's 3.17 says $\limsup$ belongs to E and that nothing above it does. The ε -form is the same content converted into the two statements one actually verifies. ²³⁷

S3.3 Theorem. (Algebra of upper and lower limits.) For bounded real sequences,

- (a) $\liminf x_n = -\limsup(-x_n)$;
- (b) $\limsup(x_n + y_n) \leq \limsup x_n + \limsup y_n$, with equality whenever $\lim x_n$ exists;
- (c) $\liminf x_n + \liminf y_n \leq \liminf(x_n + y_n)$;
- (d) if $x_n, y_n > 0$, then $(\liminf x_n)(\liminf y_n) \leq \liminf(x_n y_n)$;
- (e) if $\liminf x_n > 0$, then $\limsup(1/x_n) = 1/\liminf x_n$.

SOURCE: ECNU 定理 7.8, 例 2, and 习题 7.2 #2 (bk. pp. 158–159).

Proof. All five follow from S3.1 together with elementary facts about suprema over a common tail.

- (a) $\inf_{k \geq n}(-x_k) = -\sup_{k \geq n} x_k$; let $n \rightarrow \infty$.

²³⁷ Read the pair as “eventually below, infinitely often nearly above.” Almost every $\limsup$ argument in later chapters — the root test at 3.33, the radius of convergence at 3.39 — uses exactly these two clauses, one to get an upper bound that eventually holds and one to find infinitely many terms witnessing sharpness.

(b) $\sup_{k \geq n}(x_k + y_k) \leq \sup_{k \geq n} x_k + \sup_{k \geq n} y_k$, since each term on the left is dominated by the right side; let $n \rightarrow \infty$. If $x_n \rightarrow x$ then applying the inequality to $y_n = (x_n + y_n) + (-x_n)$ gives $\limsup y_n \leq \limsup(x_n + y_n) + \limsup(-x_n) = \limsup(x_n + y_n) - x$, which is the reverse inequality.

(c) Apply (b) to $-x_n$ and $-y_n$ and use (a).

(d) For positive terms, $\inf_{k \geq n}(x_k y_k) \geq (\inf_{k \geq n} x_k)(\inf_{k \geq n} y_k)$; let $n \rightarrow \infty$.

(e) If $\liminf x_n > 0$ then $x_k > 0$ for large k and $\sup_{k \geq n}(1/x_k) = 1/\inf_{k \geq n} x_k$; let $n \rightarrow \infty$. $\square$

THE INEQUALITIES ARE ONE-DIRECTIONAL, AND THAT IS THE POINT

It is tempting to expect $\limsup(x_n + y_n) = \limsup x_n + \limsup y_n$. Take

$$x_n = (-1)^n, \quad y_n = (-1)^{n+1}.$$

Then $x_n + y_n = 0$ for every n , so the left side is 0, while the right side is $1 + 1 = 2$. The two sequences achieve their upper limits on *disjoint* sets of indices, and $\limsup$ cannot see that they never peak together. Equality is restored the moment one of the sequences converges, because a convergent sequence peaks everywhere at once. That is why Rudin can get away with 3.19 alone: he never adds two genuinely oscillating sequences. When you do — in the root test applied to a product, say — the inequality is all you have.

NOT IN RUDIN

SUPPLEMENT B — PAST THE ROOT AND RATIO TESTS

WHY THIS SUPPLEMENT BELONGS TO CHAPTER 3

What Rudin does. Theorems 3.33 and 3.34 give the root and ratio tests; the Remark after 3.35 observes that both are silent when the relevant limit equals 1, illustrates with $\Sigma 1/n$ and $\Sigma 1/n^2$, and moves on.

The gap. The remark diagnoses a failure without saying what to do about it, and without explaining *why* the failure is unavoidable. A reader is left with the impression that the borderline is a small blind spot rather than the entrance to an infinite hierarchy.

Where the gap bites. On every series that matters at the boundary. Rudin's own 3.28 — $\Sigma 1/n^p$ — sits exactly there, and he must reach it by a separate device, the condensation test 3.27. The hypergeometric series, the central example of nineteenth-century analysis, is invisible to both of his tests. And

at 3.39 the radius of convergence of a power series is often decided precisely on the circle, where root and ratio give nothing.

What this section supplies. Raabe's test, which replaces the geometric yardstick by the p -series one and so resolves the whole borderline band; the classical series that forces it; and the reason no test can be last — there is no slowest convergent series, so Rudin's remark is a structural fact rather than an omission.

S3.4 Theorem. (*Raabe's test.*) Let $u_n > 0$. If there is $r > 1$ with

$$n\left(1 - \frac{u_{n+1}}{u_n}\right) \geq r \quad (n > N_0),$$

then $\sum u_n$ converges. If $n(1 - u_{n+1}/u_n) \leq 1$ for all $n > N_0$, then $\sum u_n$ diverges. In particular, if the limit $r = \lim n(1 - u_{n+1}/u_n)$ exists, then $r > 1$ gives convergence and $r < 1$ divergence.

SOURCE: ECNU 拉贝判别法 定理 12.10 (下 bk. pp. 14–15).

Proof. *Convergence.* Suppose $n(1 - u_{n+1}/u_n) \geq r > 1$ for $n > N_0$, so that $u_{n+1}/u_n \leq 1 - r/n$. Fix p with $1 < p < r$. Bernoulli's inequality $(1+t)^p \geq 1+pt$, applied with $t = -1/n$, gives

$$\left(\frac{n-1}{n}\right)^p \geq 1 - \frac{p}{n} \geq 1 - \frac{r}{n} \geq \frac{u_{n+1}}{u_n},$$

that is $n^p u_{n+1} \leq (n-1)^p u_n$. So the numbers $(n-1)^p u_n$ decrease for $n > N_0$ and are therefore bounded by some C , giving $u_n \leq C/(n-1)^p$. Since $p > 1$ the series $\sum (n-1)^{-p}$ converges (3.28), and the comparison test 3.25 finishes.

Divergence. If $n(1 - u_{n+1}/u_n) \leq 1$ then $u_{n+1}/u_n \geq 1 - 1/n = (n-1)/n$, so $nu_{n+1} \geq (n-1)u_n$ and the numbers $(n-1)u_n$ increase; hence $(n-1)u_n \geq c > 0$ and $u_n \geq c/(n-1)$. Since $\sum 1/(n-1)$ diverges, so does $\sum u_n$. $\square$

HOW THE PROOF FLOWS

1. **Choose the method.** The ratio test failed because it compares with a geometric series. So compare with a p -series instead — but the hypothesis is about *ratios*, so the comparison must be made between ratios rather than between terms.
2. **Compute the target ratio.** For $v_n = n^{-p}$ the ratio is $v_{n+1}/v_n = ((n)/(n+1))^p$, which Bernoulli bounds below by $1 - p/n$. So “ $u_{n+1}/u_n \leq 1 - p/n$ ” is exactly “the terms decay at least as fast as n^{-p} .”
3. **Buy room with $p < r$.** The hypothesis gives r ; the comparison needs a strict p above 1. Any p between them works, and this is the only place the strictness $r > 1$ is used.

- 4. Convert to a monotone quantity.** The ratio inequality says precisely that $(n-1)^p u_n$ decreases. A decreasing positive sequence is bounded, and boundedness is what the comparison test wants.
- 5. Audit.** Bernoulli is spent once, in step 2; the strict inequality $r > 1$ once, in step 3; the convergence of the p -series once, at the end. Run the same four steps with the inequalities reversed and $p = 1$ and the divergence half falls out — which is why the borderline case $r = 1$ is settled by neither.

WHAT CHANGED: THE YARDSTICK

The root and ratio tests compare Σu_n against a *geometric* series. When $u_{n+1}/u_n \rightarrow 1$ the comparison is vacuous, because the geometric series with ratio 1 is exactly the borderline.

Raabe compares against the p -series instead. Writing $u_{n+1}/u_n \approx 1 - r/n$ and comparing with $(1 - 1/n)^r$ shows that the quantity $n(1 - u_{n+1}/u_n)$ measures which power of n the terms decay like: $r > 1$ means faster than $1/n$, and $r < 1$ slower. So the test is “compare with Σn^{-r} ” with the comparison done through ratios rather than the terms themselves.

That reframing tells you immediately what the next rung must be. Since the p -series is itself borderline at $p = 1$, Raabe fails when $r = 1$ — and the next test compares against $\Sigma 1/(n(\log n)^p)$, which Rudin’s own 3.29 has already introduced.

The series that forces the issue, and why the ladder never ends

The standard example is

$$\sum_{n=1}^{\infty} \left[\frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2 \cdot 4 \cdot 6 \cdots (2n)} \right]^s,$$

whose ratio tends to 1 for *every* s , so ratio and root say nothing whatever. Raabe gives $r = s/2$, hence divergence for $s = 1$ and $s = 2$ and convergence for $s = 3$. The same computation settles the hypergeometric series $F(\alpha, \beta, \gamma, 1)$, which converges precisely when $\gamma > \alpha + \beta$ — historically the problem the whole hierarchy of tests was invented for.

ECNU then states the moral, and it is worth taking seriously: one can always build a finer test, and the process never terminates, because *there is no slowest convergent series*. Given any convergent Σa_n one can manufacture a convergent series whose terms dominate it eventually, so no single comparison series can serve as a universal boundary. Rudin’s 3.35 remark — that when the ratio is 1 “the series may converge or diverge” — is therefore not laziness but a structural fact.

The integral test, and why Rudin cannot state it here

Both Chinese texts prove: if f is nonnegative and decreasing on $[1, \infty)$, then $\Sigma f(n)$ and $\int_1^\infty f(x) dx$ converge together, via the sandwich $\sum_{k=2}^{n+1} f(k) \leq \int_1^{n+1} f \leq \sum_{k=1}^n f(k)$.

It is the fastest route to the p -series and to the iterated-logarithm ladder $\Sigma 1/(n \log n (\log \log n)^p)$. Rudin gets both by *Cauchy condensation* (3.27–3.29) instead, and he has no choice: the integral does not exist until Chapter 6, so a Chapter 3 integral test would be circular in his development. Come back to this box after 6.13, where the comparison becomes available — and notice that condensation and the integral test are the same idea, one summing over dyadic blocks and the other over unit intervals.

NOT IN RUDIN

SUPPLEMENT C — SUMMATION BY PARTS, QUANTITATIVELY

WHY THIS SUPPLEMENT BELONGS TO CHAPTER 3

What Rudin does. Theorem 3.41 is the partial summation formula, and he uses it once, immediately, to prove 3.42 — if the partial sums of Σa_n are bounded and $b_n \downarrow 0$ then $\Sigma a_n b_n$ converges — from which 3.43 (alternating series) and 3.44 follow.

The gap. The formula is consumed on the spot. No *bound* is extracted from it, and no second theorem: Rudin has the Dirichlet test but not the Abel test, which trades the hypotheses the other way round.

Where the gap bites. At 3.44, where $\Sigma z^n/n^\alpha$ on $|z| = 1$ is handled by the Dirichlet route because the Abel route is unavailable; and, more seriously, in Chapter 7, where the uniform versions of these tests are wanted and a uniform conclusion needs an explicit constant, not merely a convergence statement.

What this section supplies. The bound — Abel's lemma, with the constant 3 — from which Rudin's 3.42 and its missing twin both follow in a line; and the Dirichlet kernel estimate, which is the standard bridge from this chapter to Fourier series and which 3.44 stops just short of.

S3.5 Theorem. (*Abel's lemma.*) Let (ε_k) be monotone, let $\sigma_k = v_1 + \cdots + v_k$, and suppose $|\sigma_k| \leq A$ for $k \leq n$. Put $\varepsilon = \max_k |\varepsilon_k|$. Then

$$\left| \sum_{k=1}^n \varepsilon_k v_k \right| \leq 3\varepsilon A.$$

SOURCE: ECNU 阿贝尔引理 (下 bk. pp. 21–22).

Proof. Partial summation (3.41) gives

$$\sum_{k=1}^n \varepsilon_k v_k = \sum_{k=1}^{n-1} (\varepsilon_k - \varepsilon_{k+1}) \sigma_k + \varepsilon_n \sigma_n.$$

Because (ε_k) is monotone, the differences $\varepsilon_k - \varepsilon_{k+1}$ all have one sign, so

$$\sum_{k=1}^{n-1} |\varepsilon_k - \varepsilon_{k+1}| = |\varepsilon_1 - \varepsilon_n| \leq 2\varepsilon.$$

Bounding every $|\sigma_k|$ by A therefore gives $|\sum \varepsilon_k v_k| \leq 2\varepsilon A + \varepsilon A = 3\varepsilon A$.

□

The proof is Rudin's 3.41 followed by one observation: because (ε_k) is monotone, the differences $\varepsilon_k - \varepsilon_{k+1}$ all have the same sign, so $\sum_k |\varepsilon_k - \varepsilon_{k+1}| = |\varepsilon_1 - \varepsilon_n| \leq 2\varepsilon$, and the boundary terms contribute εA more.²³⁸

S3.6 Theorem. (*Abel's test.*) If (a_n) is monotone and **bounded** — not necessarily null — and $\sum b_n$ converges, then $\sum a_n b_n$ converges.

SOURCE: ECNU 定理 12.15 (下 bk. p. 22).

Proof. Being monotone and bounded, (a_n) converges, say to a ; put $b_n = a_n - a$, so (b_n) is monotone with $b_n \rightarrow 0$. Then

$$\sum a_n c_n = a \sum c_n + \sum b_n c_n.$$

The first series converges by hypothesis. For the second, the partial sums of $\sum c_n$ are bounded — a convergent sequence is bounded — and $b_n \downarrow 0$, so Rudin's 3.42 applies. Hence $\sum a_n c_n$ converges. □

Two tests, and which hypothesis each spends

Rudin's 3.42 is the *Dirichlet* test: partial sums of $\sum b_n$ merely bounded, but $a_n \downarrow 0$. Abel's test trades those hypotheses: $\sum b_n$ must actually converge, but (a_n) need only be monotone and bounded. Neither implies the other, and both fall out of Abel's lemma.

	Dirichlet (Rudin 3.42)	Abel (S3.6)
on $\sum b_n$	partial sums bounded	converges
on (a_n)	monotone, $\rightarrow 0$	monotone, bounded
typical use	$\sum \frac{\sin nx}{n}$	multiplying a convergent series by a damping factor

Rudin never states Abel's form, and its absence is felt at 3.44, where he handles $\sum z^n/n^\alpha$ on $|z| = 1$ by the Dirichlet route.

S3.7 Theorem. (*The Dirichlet kernel bound.*) For $x \in (0, 2\pi)$,

$$\frac{1}{2} + \sum_{k=1}^n \cos kx = \frac{\sin((n + \frac{1}{2})x)}{2 \sin(x/2)},$$

238 This is exactly the telescope Rudin performs inside the proof of 3.42, extracted and given a name. Rudin's version is consumed on the spot to prove one convergence theorem; as a standalone inequality it is reusable, and it is what the uniform-convergence versions in Chapter 7 run on.

so the partial sums of $\Sigma \cos kx$ are bounded on any interval $[\delta, 2\pi - \delta]$. Consequently, if $a_1 \geq a_2 \geq \cdots$ with $a_n \rightarrow 0$, then $\Sigma a_n \cos nx$ and $\Sigma a_n \sin nx$ converge for every $x \in (0, 2\pi)$.

SOURCE: ECNU 例 3 (下 bk. pp. 22–23).

Proof. For $x \in (0, 2\pi)$ we have $\sin(x/2) \neq 0$. The product formula $2 \cos kx \sin(x/2) = \sin((k + \frac{1}{2})x) - \sin((k - \frac{1}{2})x)$ makes the sum telescope:

$$2 \sin \frac{x}{2} \left[\frac{1}{2} + \sum_{k=1}^n \cos kx \right] = \sin \frac{x}{2} + \sin((n + \frac{1}{2})x) - \sin \frac{x}{2} = \sin((n + \frac{1}{2})x),$$

which is the identity. Hence $|\sum_{k \leq n} \cos kx| \leq 1/(2|\sin(x/2)|) + \frac{1}{2}$, a bound independent of n and finite for each fixed $x \in (0, 2\pi)$. With $a_n \downarrow 0$, Rudin's 3.42 gives convergence of $\Sigma a_n \cos nx$; the same telescoping with $\sin$ in place of $\cos$ bounds the partial sums of $\Sigma \sin kx$ and settles that series too. $\square$

This is the standard bridge from Chapter 3 to Fourier series: it makes $\Sigma(\sin nx)/n$ and $\Sigma(\cos nx)/n$ convergent everywhere on $(0, 2\pi)$, none of which follows from absolute convergence, since $\Sigma 1/n$ diverges.²³⁹

239 The identity is proved by multiplying the left side by $2 \sin(x/2)$ and telescoping the products $2 \cos kx \sin(x/2) = \sin((k + \frac{1}{2})x) - \sin((k - \frac{1}{2})x)$. Note where the bound degrades: the denominator $2 \sin(x/2)$ tends to 0 as $x \rightarrow 0$, so the partial sums are bounded on each $[\delta, 2\pi - \delta]$ but not uniformly on $(0, 2\pi)$ — which is exactly the phenomenon Chapter 7 will call non-uniform convergence.

NOT IN RUDIN

SUPPLEMENT D — GROUPING AND MULTIPLYING

WHY THIS SUPPLEMENT BELONGS TO CHAPTER 3

What Rudin does. Definition 3.48 defines the product of two series by the anti-diagonals $c_n = \sum_k a_k b_{n-k}$, motivated by power series; 3.49 shows the product of two conditionally convergent series can diverge; 3.50 (Mertens) proves convergence to AB when one factor is absolutely convergent.

The gap. Two operations on series are performed silently throughout the chapter and never licensed. *Grouping* — inserting parentheses — is used inside the condensation proof at 3.27 and in the Remark after 3.44, but never stated as a theorem, so a reader has no way to know it is legal while rearrangement is not. And the choice of the anti-diagonal ordering at 3.48 is presented as forced by power series, leaving open what happens under any other arrangement of the same products.

Where the gap bites. At 3.49 especially, where the divergence of a product looks like a pathology of multiplication rather than the price of insisting on one particular traversal of a doubly-indexed array.

What this section supplies. The grouping theorem with its one-directional caveat, so that the legal manipulation is cleanly

separated from the illegal one; and Cauchy's theorem, that under absolute convergence *every* arrangement of the array converges to AB — which puts 3.50 and 3.49 in their proper places.

S3.8 Theorem. (*Grouping.*) *Inserting parentheses in a convergent series — that is, passing to any subsequence of the partial sums by grouping consecutive terms — changes neither convergence nor the sum.*

SOURCE: ECNU 定理 12.4 (下 bk. pp. 4–5).

Proof. Let s_n be the partial sums of the original series and let the parentheses end at indices $n_1 < n_2 < \dots$. The partial sums of the grouped series are exactly $s_{n_1}, s_{n_2}, \dots$ — a subsequence of (s_n) . A subsequence of a convergent sequence converges to the same limit (3.6(a)). $\square$

GROUPING IS ONE-DIRECTIONAL, UNLIKE REARRANGEMENT

The grouped partial sums form a *subsequence* of (s_n) , and a subsequence of a convergent sequence converges to the same limit — that is the whole proof. The converse fails:

$$(1 - 1) + (1 - 1) + (1 - 1) + \dots = 0$$

converges while $1 - 1 + 1 - 1 + \dots$ diverges. So parentheses may be *inserted* into a convergent series but not *removed* from one.

Keep this distinct from the rearrangement theorem 3.54. Rearranging changes the order of the terms and can change the sum of a conditionally convergent series to anything at all; grouping preserves the order and is always safe in the direction stated. Rudin never states the grouping theorem, although he uses it — in the condensation proof at 3.27 and in the Remark after 3.44.

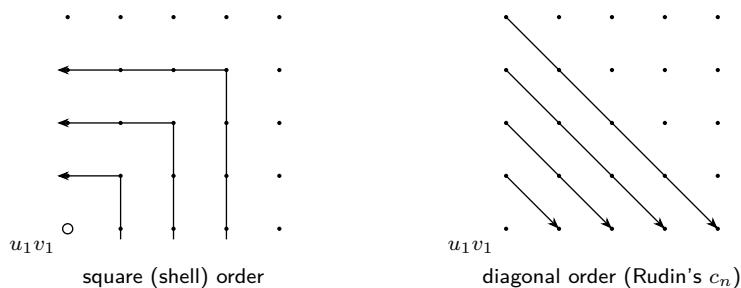
S3.9 Theorem. (*Cauchy's product theorem, any arrangement.*) *If $\sum u_n = A$ and $\sum v_n = B$ both converge absolutely, then the series formed from all products $u_i v_j$, each taken once, **in any order whatever**, converges absolutely to AB .*

SOURCE: ECNU 柯西定理 12.14 (下 bk. pp. 19–21).

Proof. Let (w_k) enumerate the products $u_i v_j$ in any order. Any finite collection of them involves indices $i, j \leq N$ for some N , so

$$\sum_{k \leq K} |w_k| \leq \left(\sum_{i \leq N} |u_i| \right) \left(\sum_{j \leq N} |v_j| \right) \leq \left(\sum_i |u_i| \right) \left(\sum_j |v_j| \right).$$

The partial sums of $\sum |w_k|$ are bounded, so $\sum w_k$ converges absolutely; by 3.55 its sum is the same for every enumeration. Evaluate it along the square ordering, in which the partial sum after the N th shell is exactly $A_N B_N$, and $A_N B_N \rightarrow AB$ by 3.3(c). $\square$



Two ways to enumerate the same array of products $u_i v_j$. Along shells the partial sums are exactly $A_n B_n$, which is why the square order proves the theorem; along anti-diagonals they are Rudin's C_n , which is the order that makes the product a series in the sense of 3.48. Absolute convergence is what makes the two agree.

How this relates to Mertens (3.50)

The two theorems point in opposite directions. Rudin's 3.50 fixes the diagonal ordering — the one that matters for power series — and weakens the hypothesis: only *one* factor need converge absolutely. S3.9 keeps the stronger hypothesis, both absolutely convergent, and buys freedom of arrangement.

The proof of S3.9 is the cleaner of the two, because the square ordering makes the partial sums *exactly* $A_n B_n$ rather than approximately so; all the difficulty in Mertens' theorem comes from the diagonal ordering cutting the array along a line the factors do not respect. Seen this way, Example 3.49 — where the Cauchy product of two conditionally convergent series diverges — is not a pathology but the expected price of insisting on the diagonal.

NOT IN RUDIN

SUPPLEMENT E — SUMMATION WITHOUT AN ORDER

WHY THIS SUPPLEMENT BELONGS TO CHAPTER 3

What Rudin does. Theorem 3.54 shows a conditionally convergent series can be rearranged to any prescribed value, and 3.55 shows an absolutely convergent one cannot: every rearrangement has the same sum.

The gap. 3.55 is stated as a stability property — the sum survives reordering — rather than as what it actually is: a licence to forget the ordering altogether. Rudin never draws that conclusion, so a series remains, for him, a function on $\mathbb{N}$, and anything indexed differently must be converted to one by hand.

Where the gap bites. At 3.48, where the product of two series has to be *defined* by choosing a traversal of a square array, and at 3.49, where the choice turns out to matter. Later, in Chapter 8, power series are multiplied and rearranged repeatedly, and

each manipulation is justified afresh.

What this section supplies. Tao's reading of 3.55: for absolutely summable families the sum is a function of the *index set*, so it may be defined with no order at all. Double series and iterated sums then become theorems — Fubini for series — rather than notational accidents.

S3.10 Definition. (*Sums over arbitrary index sets.*) Let X be a set and $f : X \rightarrow \mathbb{R}$. If X is countable, choose a bijection $g : \mathbb{N} \rightarrow X$ and put $\sum_{x \in X} f(x) = \sum_{n=0}^{\infty} f(g(n))$, provided the latter converges absolutely; the value does not depend on g . If X is uncountable, the same definition applies once one knows that $\{x : f(x) \neq 0\}$ is at most countable.

SOURCE: Tao I, §8.2, Def. 8.2.1–8.2.4 and Lemma 8.2.5 (bk. pp. 187–189).

WHAT RUDIN'S 3.55 BECOMES

Rudin proves that an absolutely convergent series may be rearranged freely (3.55) and that a conditionally convergent one may be rearranged to any sum whatever (3.54). Tao reads the first as a licence: since the value is independent of the enumeration, *the ordering was never part of the data*. A series is then a function on an index set, and $\sum_{x \in X} f(x)$ is well defined with no order in sight. The dividend is that families indexed by things other than $\mathbb{N}$ — pairs, grids, arbitrary countable sets — need no separate theory. Rudin has to manufacture an ordering to define the Cauchy product at 3.48 (his c_n are the anti-diagonals), and Example 3.49's failure is precisely the price of that manufactured ordering.

S3.11 Lemma. If $\sum_{x \in X} f(x)$ is absolutely convergent, then $\{x \in X : f(x) \neq 0\}$ is at most countable — however large X is.

SOURCE: Tao I, Lemma 8.2.5 (bk. p. 188).

So an absolutely summable family over an uncountable index set is really a series in disguise.²⁴⁰

S3.12 Theorem. (*Fubini for series.*) If $f : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{R}$ has $\sum_{(n,m)} f(n,m)$ absolutely convergent, then

$$\sum_{n=0}^{\infty} \left(\sum_{m=0}^{\infty} f(n,m) \right) = \sum_{(n,m) \in \mathbb{N} \times \mathbb{N}} f(n,m) = \sum_{m=0}^{\infty} \left(\sum_{n=0}^{\infty} f(n,m) \right).$$

SOURCE: Tao I, Thm 8.2.2 (bk. p. 188). Rudin states nothing about iterated sums in Chapter 3.

Proof. Write S for the unordered sum, which exists by S3.10. Given $\varepsilon > 0$, absolute convergence supplies a finite set $F \subset \mathbb{N} \times \mathbb{N}$ with $\sum_{(n,m) \notin F} |f(n,m)| < \varepsilon$; choose N with $F \subset [0, N]^2$.

Each inner sum $g(n) = \sum_m f(n,m)$ converges absolutely, and $\sum_{n > N} |g(n)| \leq \sum_{n > N} \sum_m |f(n,m)| < \varepsilon$. Comparing $\sum_{n \leq N} g(n)$ with

240 The proof is the standard countability argument in Rudin's own idiom: for each n the set where $|f| > 1/n$ must be finite, or the sum would exceed any bound; a countable union of finite sets is countable. Compare Exercise S2.18 in the Chapter 2 appendix — disjoint intervals are countable for the same reason.

S , the terms they fail to share all lie outside F , so the two differ by at most ε . Hence $|\sum_{n \leq N} g(n) - S| \leq \varepsilon$ and, adding the tail estimate, $|\sum_n g(n) - S| \leq 2\varepsilon$. As ε was arbitrary, $\sum_n g(n) = S$; the argument with the roles of n and m exchanged gives the other iterated sum. $\square$

Why this is the theorem the Cauchy product wanted

S3.9 said that an absolutely convergent double array may be summed in any arrangement. S3.12 says the two most useful arrangements — by rows and by columns — also agree with each other and with the unordered sum. Together they are the complete story of the array $u_i v_j$, of which Rudin's 3.50 is the single diagonal slice.

Absolute convergence is not decoration here. The standard counterexample sets $f(n, n) = 1$, $f(n, n+1) = -1$, and $f = 0$ elsewhere: summing along rows gives 0 every time, hence 0; summing along columns gives 1 then 0 forever, hence 1. Nothing is infinite or badly behaved — every row and every column is a finite sum — and still the two iterated sums differ, because the family is not absolutely summable. Rudin's Exercise 3.25 asks for this phenomenon in a different guise.

Chapter 8 will need this theorem to multiply and rearrange power series, and Chapter 11 will need its integral analogue.

S3.13 Theorem. (*Means, and the ratio-to-root corollary.*) Let $a_n \rightarrow a$. Then

(a) $\frac{a_1 + \cdots + a_n}{n} \rightarrow a$ (Cesàro);

(b) if moreover $a_n > 0$, then $\sqrt[n]{a_1 a_2 \cdots a_n} \rightarrow a$;

(c) consequently, if $b_n > 0$ and $b_{n+1}/b_n \rightarrow a$, then $\sqrt[n]{b_n} \rightarrow a$.

SOURCE: ECNU 总练习题 2 #3–#4 (bk. p. 39); USTC Ch. 1 suppl. #6–#11 (bk. pp. 38–39).

Proof. (a) Given $\varepsilon > 0$ choose N with $|a_k - a| < \varepsilon/2$ for $k > N$. For $n > N$,

$$\left| \frac{1}{n} \sum_{k \leq n} a_k - a \right| \leq \frac{1}{n} \sum_{k \leq N} |a_k - a| + \frac{\varepsilon}{2},$$

and the first term tends to 0 as $n \rightarrow \infty$ with N fixed, so it is below $\varepsilon/2$ eventually.

(b) If $a > 0$, apply (a) to $\log a_n$ and exponentiate, using continuity of $\exp$. If $a = 0$, given $\varepsilon > 0$ we have $0 < a_k < \varepsilon$ for $k > N$, and the geometric mean is at most $(\prod_{k \leq N} a_k)^{1/n} \varepsilon^{(n-N)/n} \rightarrow \varepsilon$.

(c) Put $a_1 = b_1$ and $a_k = b_k/b_{k-1}$ for $k \geq 2$, so that $a_1 a_2 \cdots a_n = b_n$ and $a_n \rightarrow a$. Now (b) gives $\sqrt[n]{b_n} \rightarrow a$. $\square$

The corollary that explains Rudin's 3.37

Part (c) is the exact statement behind Rudin's Theorem 3.37,

$$\liminf \frac{c_{n+1}}{c_n} \leq \liminf \sqrt[n]{c_n} \leq \limsup \sqrt[n]{c_n} \leq \limsup \frac{c_{n+1}}{c_n},$$

whose corollary he states as “the root test is stronger than the ratio test.” Read through (c): the root test sees a *geometric mean* of the ratios, and geometric means are gentler than the sequence they average — they converge whenever it does, and sometimes when it does not. Rudin's Example 3.35, where the ratios oscillate between 2 and 1/8 while the roots converge, is exactly a case where the mean is tamer than its terms.

Applying (c) to $b_n = n^n/n!$ gives $\lim n/\sqrt[n]{n!} = e$ in one line, and (a) to $\sqrt[n]{n}$ gives $\lim(1 + \sqrt{2} + \sqrt[3]{3} + \cdots + \sqrt[n]{n})/n = 1$. Rudin has the Cesàro mean only as Exercise 3.14(a) and never states the geometric version.

ONE THEOREM YOU WILL LOOK FOR AND NOT FIND

The Stolz–Cesàro theorem — the discrete l'Hospital rule, that $\lim(a_n - a_{n-1})/(b_n - b_{n-1}) = L$ forces $\lim a_n/b_n = L$ for (b_n) strictly increasing and unbounded — appears **in none of the four comparison texts**, and not in Rudin. All four give only the Cesàro special cases of S3.13. The continuous analogue does appear, as Yu Pin's problem A9* (p. 127), where it is attributed to Cauchy rather than to Stolz; it is exercise S4.3 in the Chapter 4 appendix. If you want the discrete theorem you will have to prove it, and S3.13(a) is the case $b_n = n$.

NOT IN RUDIN**APPENDIX EXERCISES****HOW THESE DIFFER FROM RUDIN'S TWENTY-FIVE**

Rudin's own Exercises 3.1–3.25 end the chapter proper. These eighteen come from the four comparison texts.

Group I (1–5): sequences. Coupled recursions — Gauss's arithmetic–geometric mean among them — where Rudin's 3.16–3.17 use a single recursion; plus the means package and bounded variation.

Group II (6–12): series. The decay rate of the terms of a convergent series, Cauchy–Schwarz in ℓ^2 , infinite products, and the positive/negative split that Rudin buries inside the proof of 3.54.

Group III (13–16): upper and lower limits. The algebra of Supplement A, put to work.

241 Gauss's AGM. The engine is the inequality $\sqrt{xy} \leq (x+y)/2$, which gives $b_n < b_{n+1} < a_{n+1} < a_n$: the two sequences are monotone in opposite directions and interleaved, so 3.14 applies to each and the gap $a_{n+1} - b_{n+1} \leq (a_n - b_n)/2$ closes. Rudin's Exercises 3.16–3.17 are single recursions converging to a square root; a coupled recursion is materially different, and the common limit here has no elementary closed form — it is an elliptic integral.

242 The companion to exercise 1, and this time the answer is elementary — because $a_n b_n$ is invariant:

$a_{n+1} b_{n+1} = a_n b_n$ for every n . Spot the invariant and the problem collapses. Compare the strategy index's second pattern in Chapter 1: hunt for the quantity the recursion does not move.

243 Rudin gives only the first part, as Exercise 3.14(a). The deduction is the content of 3.37 — the root test dominates the ratio test — and the last limit is e , obtained by applying (c) to $b_n = n^n/n!$. Show also that the converses all fail: averaging destroys information, so $(-1)^n$ has Cesàro mean 0 without converging.

244 Bounded variation implies convergence — the sequence version of a notion Chapter 6 will develop for functions. The proof is one line once you see it: $a_n = a_1 + \sum_{k \leq n} (a_{k+1} - a_k)$ exhibits (a_n) as the partial sums of an absolutely convergent series. Note the converse fails: a convergent sequence can wander infinitely far, as $a_n = \sum_{k \leq n} (-1)^k/k$ shows.

245 The cleanest packaging of Rudin's 3.6–3.7. Argue by contradiction: if $x_n \not\rightarrow L$, infinitely many terms stay ε away, and that subsequence is bounded, so by 3.6(b) it has a convergent sub-subsequence — whose limit is not L . Note the hypothesis bounded is essential: $x_n = n$ has no convergent subsequence at all, so the condition holds vacuously.

Group IV (17–18): summation over sets. Iterated sums, and the counterexample that shows absolute convergence is not decoration.

Marked ★: harder, or requiring a result from a later chapter.

S3.1. (*Arithmetic–geometric mean.*) Let $a_1 > b_1 > 0$ and put

$$a_{n+1} = \frac{a_n + b_n}{2}, \quad b_{n+1} = \sqrt{a_n b_n}.$$

Prove that $\lim a_n$ and $\lim b_n$ both exist and are equal.²⁴¹

SOURCE: ECNU 习题 2.3 #11 (bk. p. 38).

S3.2. (*Arithmetic–harmonic mean.*) With $a_1, b_1 > 0$ put $a_{n+1} = (a_n + b_n)/2$ and $b_{n+1} = 2a_n b_n / (a_n + b_n)$. Prove both limits exist and equal $\sqrt{a_1 b_1}$.²⁴²

SOURCE: ECNU 总练习题 2 #8 (bk. p. 39).

S3.3. Prove that $a_n \rightarrow a$ implies $(a_1 + \cdots + a_n)/n \rightarrow a$, and that if also $a_n > 0$ then $\sqrt[n]{a_1 \cdots a_n} \rightarrow a$. Deduce that $b_{n+1}/b_n \rightarrow a$ implies $\sqrt[n]{b_n} \rightarrow a$ for $b_n > 0$, and use it to evaluate $\lim n/\sqrt[n]{n!}$.²⁴³

SOURCE: ECNU 总练习题 2 #3–#4 (bk. p. 39); USTC Ch. 1 suppl. #6–#11.

S3.4. Suppose $\sum_{k=1}^n |a_{k+1} - a_k| \leq M$ for every n . Prove that (a_n) converges.²⁴⁴

SOURCE: ECNU 总练习题 2 #6 (bk. p. 39); USTC Ch. 1 suppl. #5.

S3.5. Prove that if every convergent subsequence of a bounded sequence has the same limit L , then the sequence itself converges to L . Deduce that a bounded divergent sequence has two subsequences with different limits.²⁴⁵

SOURCE: USTC §1.2 (bk. p. 15), left there as an exercise.

S3.6. Prove that if $\sum u_n$ is a convergent series of positive terms and (u_n) is monotone, then $\lim n u_n = 0$.²⁴⁶

SOURCE: ECNU 总练习题 12 #1 (下 bk. p. 24).

S3.7. Prove that if $\sum a_n^2$ and $\sum b_n^2$ converge then $\sum a_n b_n$ and $\sum (a_n + b_n)^2$ converge, and

$$\left(\sum a_n b_n\right)^2 \leq \sum a_n^2 \sum b_n^2,$$

$$\left(\sum (a_n + b_n)^2\right)^{1/2} \leq \left(\sum a_n^2\right)^{1/2} + \left(\sum b_n^2\right)^{1/2}.$$

247

SOURCE: ECNU 总练习题 12 #8 (下 bk. p. 24).

S3.8. Let $a_n > 0$. Prove that $\prod_{k=1}^n (1 + a_k)$ converges (to a finite nonzero limit) if and only if $\sum a_n$ converges. Evaluate $\prod_{k=0}^{\infty} (1 + q^{2^k})$ for $|q| < 1$.²⁴⁸

SOURCE: ECNU 习题 12.2 #15 (下 bk. p. 17); USTC 习题 1.1 #8(5).

S3.9. Let $a_n \geq 0$ with (na_n) bounded. Prove $\sum a_n^2$ converges. Prove also that if $\sum u_n$ is a convergent positive series then $\sum \sqrt{u_n u_{n+1}}$ converges.²⁴⁹

SOURCE: ECNU 习题 12.2 #5, #7 (下 bk. p. 16).

S3.10. With $p_n = (|u_n| + u_n)/2$ and $q_n = (|u_n| - u_n)/2$, prove that if $\sum u_n$ converges conditionally then $\sum p_n$ and $\sum q_n$ both diverge.²⁵⁰

SOURCE: ECNU 习题 12.3 #4 (下 bk. p. 23).

S3.11. ★ Prove that $\sum_{n=1}^{\infty} \frac{(-1)^{\lfloor \sqrt{n} \rfloor}}{n}$ converges.²⁵¹

SOURCE: ECNU 习题 12.3 #8 (下 bk. p. 24).

S3.12. Given a convergent series of positive terms $\sum u_n$, must there exist $\varepsilon > 0$ with $u_n/n^{-(1+\varepsilon)} \rightarrow c$ for some $c > 0$?²⁵²

SOURCE: ECNU 总练习题 12 #5(3) (下 bk. p. 24).

S3.13. Prove the parts of S3.3: $\liminf a_n = -\limsup(-a_n)$; $\liminf a_n + \liminf b_n \leq \liminf(a_n + b_n)$; and, for positive sequences with $\liminf a_n > 0$, $\limsup(1/a_n) = 1/\liminf a_n$.²⁵³

SOURCE: ECNU 习题 7.2 #2 (bk. p. 159).

S3.14. Let $a_n > 0$ and suppose $\limsup a_n \cdot \limsup(1/a_n) = 1$. Prove that (a_n) converges.²⁵⁴

SOURCE: ECNU 习题 7.2 #4 (bk. p. 159).

S3.15. ★ Suppose $\liminf x_n = A < B = \limsup x_n$ and $\lim(x_{n+1} - x_n) = 0$. Prove that the set of subsequential limits of (x_n) is exactly $[A, B]$.²⁵⁵

SOURCE: ECNU 总练习题 7 #3 (bk. p. 160), starred there as well.

S3.16. Show by example that $\limsup(a_n b_n) = \limsup a_n \cdot \limsup b_n$ can fail for positive bounded sequences, and prove it holds when one of the sequences converges.²⁵⁶

SOURCE: ECNU 习题 7.2 #2 (bk. p. 159), product part.

246 Sharper than 3.23, which gives only $u_n \rightarrow 0$. The trick is to look at a block: monotonicity makes $u_{n+1} + \cdots + u_{2n} \geq n u_{2n}$, and the left side tends to 0 by the Cauchy criterion. Note what fails without monotonicity — one can have a convergent series with $nu_n \not\rightarrow 0$ by putting spikes at sparse indices. This is the quantitative reason the harmonic series is the boundary.

247 Cauchy–Schwarz and Minkowski in ℓ^2 — Theorem 1.35 and 1.37(e) in infinite dimensions, which Rudin never revisits after Chapter 1. Prove the finite versions and pass to the limit; the only care needed is that the partial sums are increasing and bounded. This is the first appearance of the space ℓ^2 , which Chapter 11 treats seriously.

248 Infinite products reduced entirely to Chapter 3 by taking logarithms: convergence of $\sum \log(1 + a_k)$ is equivalent to convergence of $\sum a_k$, since $\log(1 + x) \sim x$. The evaluation telescopes — multiply by $(1 - q)$ and watch $(1 - q)(1 + q)(1 + q^2)(1 + q^4) \cdots = 1 - q^{2^n}$ — giving $1/(1 - q)$. None of the four texts devotes a section to infinite products; this exercise is the whole theory in miniature.

249 Two decay-rate arguments. The first is comparison with $\sum 1/n^2$; the second is the arithmetic–geometric mean inequality,

$\sqrt{u_n u_{n+1}} \leq (u_n + u_{n+1})/2$. Both illustrate the same habit: when a series resists direct attack, bound its terms by something already known to converge, and prefer the crudest bound that works.

250 The lemma inside Rudin’s proof of 3.54, isolated. If both converged, $\sum |u_n| = \sum(p_n + q_n)$ would converge, contradicting conditionality; if exactly one converged, $\sum u_n = \sum(p_n - q_n)$ would diverge. So both piles are infinite — which is precisely the fuel Riemann’s rearrangement burns. Worth extracting because the statement, not the rearrangement, is what makes 3.54 believable.

251 The signs come in runs of lengths 3, 5, 7, ..., so the alternating series test (3.43) does not apply at all. Group the terms into blocks of constant sign, show the grouped series converges by comparison, and then use the fact that the terms tend to 0 to remove the parentheses — which is legitimate here precisely because the block lengths grow slowly. A genuine stress test of S3.5 and S3.8.

252 No — and the exercise is the concrete form of “there is no slowest convergent series,” the moral of Supplement B. Convergent series can decay arbitrarily slowly, so no single power of n serves as a universal gauge. This is why the Raabe/Gauss hierarchy has no last member, and why Rudin’s remark after 3.35 is a structural fact rather than an omission.

253 All three fall out of the inf-of-sup formula S3.1 plus the elementary facts $\sup(-A) = -\inf A$ and $\inf(A + B) \geq \inf A + \inf B$. Doing them once makes clear why the inequalities point the way they do: taking a supremum over a tail is a sublinear operation, and sublinearity is one-directional.

254 A one-line consequence of S3.3(e) once you see it: the hypothesis says $\limsup a_n = \liminf a_n$, and 3.17 finishes it. The pleasure is that a statement about products of upper limits secretly encodes “the two limits coincide.” Note the hypothesis forces $\liminf a_n > 0$, so the reciprocal rule applies.

255 A beautiful result with no analogue anywhere in Rudin. The hypothesis says the sequence moves in ever-smaller steps while still travelling between A and B infinitely often — so it cannot skip any value in between. Formally: given $c \in (A, B)$ and $\varepsilon > 0$, find a stretch of indices crossing from below c to above c ; because the steps are eventually smaller than ε , some term of that stretch lands within ε of c . Compare S2.10 in the Chapter 2 appendix, where consecutive-terms-close fails to give convergence: here the same hypothesis is turned into positive information.

S3.17. Define $f : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{R}$ by $f(n, n) = 1$, $f(n, n+1) = -1$, and $f = 0$ otherwise. Compute both iterated sums $\sum_n (\sum_m f(n, m))$ and $\sum_m (\sum_n f(n, m))$, and explain why S3.12 does not apply.²⁵⁷

SOURCE: Tao I, §8.2 discussion (bk. p. 188).

S3.18. ★ Let X be any set and $f : X \rightarrow \mathbb{R}$ with $\sum_{x \in X} |f(x)|$ finite. Prove that $\{x : f(x) \neq 0\}$ is at most countable, and that $\sum_{x \in X} f(x)$ is independent of the enumeration used to compute it.²⁵⁸

SOURCE: Tao I, Lemma 8.2.5 and Ex. 8.2.2 (bk. p. 188).

Chapter 4

CONTINUITY

WHY THIS CHAPTER EXISTS

Chapters 2 and 3 built machinery — topology and limits — and this chapter is where it gets spent. Continuity is defined three equivalent ways (ε - δ at 4.5, sequences via 4.2, inverse images at 4.8), and then the two great theorems of Chapter 2 are converted into the two facts every calculus student uses without proof: a continuous function on a compact set is bounded, attains its extrema, and is uniformly continuous (4.14–4.19); a continuous function on an interval takes every intermediate value (4.22–4.23).

The chapter’s centre of gravity is **Theorem 4.8**: continuous means *inverse images of open sets are open*. It converts continuity from a pointwise, quantifier-laden condition into a single structural statement, and it is the form in which every later argument uses the hypothesis.

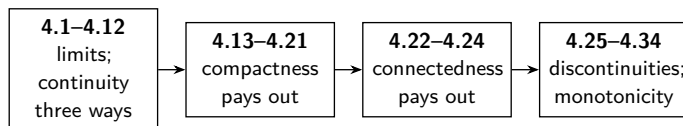
256 *The multiplicative twin of the pitfall at S3.3. Take a_n alternating between 1 and 2 and b_n alternating the other way: each has $\limsup = 2$, but the product is constantly 2. The repair is the same — a convergent factor peaks everywhere at once, so it cannot dodge the other sequence’s peaks.*

257 *The answers are 0 and 1. Every row and every column is a finite sum, so nothing diverges and nothing is subtle — and still the order of summation matters, because $\sum_{(n,m)} |f(n,m)|$ is infinite. The lesson is that absolute convergence is the entire hypothesis of Fubini, not a technical convenience. Rudin’s Exercise 3.25 asks for a relative of this.*

258 *S3.11 and the well-definedness behind S3.10. For the first part, show $\{x : |f(x)| > 1/n\}$ is finite for each n and take the union. For the second, the two enumerations differ by a bijection of $\mathbb{N}$, so this is Rudin’s 3.55 — which is exactly the point: absolute convergence is what makes “the sum over a set” meaningful, and 3.54 says the notion collapses without it.*

HOW IT IS ORGANIZED, AND WHY IN THAT ORDER

Rudin’s own account, from the opening paragraph: the theorems “would not become any easier if we restricted ourselves to real functions,” and it “actually simplifies and clarifies the picture to discard unnecessary hypotheses and to state and prove theorems in an appropriately general context.” So everything is done for metric spaces, exactly as in Chapters 2 and 3, and specialised only when the real line genuinely matters (monotonicity, 4.28–4.30, needs an order; the intermediate value theorem needs intervals).



Three things to know in advance.

The proofs are translations, not inventions. Almost every proof here either translates a sequence theorem through the bridge 4.2 (so Chapter 3 does the work), or pulls a topological property of the target space back through f^{-1} (so Chapter 2 does the work). The strategy index of Chapter 1 gains an eighth

pattern: *transport structure along f or f^{-1} .*

The counterexamples are half the content. 4.20–4.21 show exactly what dies without compactness, and 4.24, 4.27 what the intermediate value property does *not* imply. Rudin places a counterexample after nearly every theorem in this chapter; read them as part of the theorem.

Uniform continuity (4.18–4.19) is the new concept.

Everything else repackages Chapters 2–3; uniformity — one δ serving every point at once — is genuinely new, is the hardest distinction in the chapter, and is the hypothesis integration will need in Chapter 6 (Theorem 6.8).

The function concept and some of the related terminology were introduced in Definitions 2.1 and 2.2. Although we shall (in later chapters) be mainly interested in real and complex functions (i.e., in functions whose values are real or complex numbers) we shall also discuss vector-valued functions (i.e., functions with values in $\mathbb{R}^k$) and functions with values in an arbitrary metric space. The theorems we shall discuss in this general setting would not become any easier if we restricted ourselves to real functions, for instance, and it actually simplifies and clarifies the picture to discard unnecessary hypotheses and to state and prove theorems in an appropriately general context.

The domains of definition of our functions will also be metric spaces, suitably specialized in various instances.

LIMITS OF FUNCTIONS

4.1 Definition. Let X and Y be metric spaces; suppose $E \subset X$, f maps E into Y , and p is a limit point of E . We write $f(x) \rightarrow q$ as $x \rightarrow p$, or

$$\lim_{x \rightarrow p} f(x) = q \quad (4.1)$$

if there is a point $q \in Y$ with the following property: For every $\varepsilon > 0$ there exists a $\delta > 0$ such that

$$d_Y(f(x), q) < \varepsilon \quad (4.2)$$

for all points $x \in E$ for which

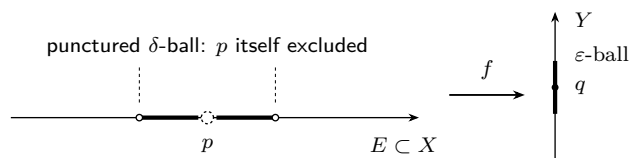
$$0 < d_X(x, p) < \delta. \quad (4.3)$$

The symbols d_X and d_Y refer to the distances in X and Y , respectively. **259**

If X and/or Y are replaced by the real line, the complex plane, or by some euclidean space $\mathbb{R}^k$, the distances d_X , d_Y are of course replaced by absolute values, or by norms of differences (see Example 2.16).

259 Two structural points hide in (4.3). The strict inequality $0 < d_X(x, p)$ deletes the point p itself: the limit ignores what happens at p , and even whether f is defined there. And p must be a limit point of E — otherwise no x satisfies (4.3) for small δ and every q would qualify vacuously. The hypothesis is exactly what makes the limit unique.

It should be noted that $p \in X$, but that p need not be a point of E in the above definition. Moreover, even if $p \in E$, we may very well have $f(p) \neq \lim_{x \rightarrow p} f(x)$.



The contract of 4.1: however small an ε -ball is drawn about q , some punctured δ -ball about p is mapped entirely into it. The value $f(p)$ — if it exists at all — plays no part.

We can recast this definition in terms of limits of sequences:

4.2 Theorem. *Let X, Y, E, f , and p be as in Definition 4.1. Then*

$$\lim_{x \rightarrow p} f(x) = q \quad (4.4)$$

if and only if

$$\lim_{n \rightarrow \infty} f(p_n) = q \quad (4.5)$$

for every sequence (p_n) in E such that

$$p_n \neq p, \quad \lim_{n \rightarrow \infty} p_n = p. \quad (4.6)$$

Proof. Suppose (4.4) holds. Choose (p_n) in E satisfying (4.6). Let $\varepsilon > 0$ be given. Then there exists $\delta > 0$ such that $d_Y(f(x), q) < \varepsilon$ if $x \in E$ and $0 < d_X(x, p) < \delta$. Also, there exists N such that $n > N$ implies $0 < d_X(p_n, p) < \delta$. Thus, for $n > N$, we have $d_Y(f(p_n), q) < \varepsilon$, which shows that (4.5) holds.

Conversely, suppose (4.4) is false. Then there exists some $\varepsilon > 0$ such that for every $\delta > 0$ there exists a point $x \in E$ (depending on δ), for which $d_Y(f(x), q) \geq \varepsilon$ but $0 < d_X(x, p) < \delta$. Taking $\delta_n = 1/n$ ($n = 1, 2, 3, \dots$), we thus find a sequence in E satisfying (4.6) for which (4.5) is false. ²⁶⁰ $\square$

Corollary. *If f has a limit at p , this limit is unique.*

This follows from Theorems 3.2(b) and 4.2.

The bridge, and which way to cross it

Theorem 4.2 is a dictionary: function limits on one side, sequence limits on the other. Its value is strategic, and each direction has a distinct use.

To import theorems, cross left to right. All of Chapter 3's limit arithmetic transfers instantly — that is the entire proof of 4.4, one line long. Anything proved for sequences is proved for function limits, free.

To refute limits, cross right to left. To show $\lim_{x \rightarrow p} f(x)$

260 *The forward direction chains two contracts: ε buys a δ from f , and that δ buys an N from the sequence. The converse is proved contrapositively, and note its shape: negating “for every ε there is δ ” yields one bad ε whose every δ fails — and harvesting one witness x_n for each $\delta_n = 1/n$ assembles the bad sequence. (Silently, a choice is made for each n ; compare the note at 2.12.)*

does not exist, produce *two* sequences approaching p along which f tends to different values — the function-limit analogue of the two-subsequence trick of 3.1(d). Example 4.27(d) will use exactly this on $\sin(1/x)$ -type behaviour.

The quantifier “*every* sequence” is what makes the dictionary faithful, and it is why the refutation direction is cheap while the verification direction through sequences is usually hopeless — one cannot check all sequences one by one. For verifying, ε - δ or 4.8 are the tools.

4.3 Definition. Suppose we have two complex functions, f and g , both defined on E . By $f + g$ we mean the function which assigns to each point x of E the number $f(x) + g(x)$. Similarly we define the difference $f - g$, the product fg , and the quotient f/g of two functions, with the understanding that the quotient is defined only at those points x of E at which $g(x) \neq 0$. If f assigns to each point x of E the same number c , then f is said to be a *constant* function, or simply a *constant*, and we write $f = c$. If f and g are real functions, and if $f(x) \geq g(x)$ for every $x \in E$, we shall sometimes write $f \geq g$, for brevity.

Similarly, if $\mathbf{f}$ and $\mathbf{g}$ map E into $\mathbb{R}^k$, we define $\mathbf{f} + \mathbf{g}$ and $\mathbf{f} \cdot \mathbf{g}$ by

$$(\mathbf{f} + \mathbf{g})(x) = \mathbf{f}(x) + \mathbf{g}(x), \quad (\mathbf{f} \cdot \mathbf{g})(x) = \mathbf{f}(x) \cdot \mathbf{g}(x);$$

and if λ is a real number, $(\lambda \mathbf{f})(x) = \lambda \mathbf{f}(x)$.

4.4 Theorem. Suppose $E \subset X$, a metric space, p is a limit point of E , f and g are complex functions on E , and

$$\lim_{x \rightarrow p} f(x) = A, \quad \lim_{x \rightarrow p} g(x) = B.$$

Then

- (a) $\lim_{x \rightarrow p} (f + g)(x) = A + B$;
- (b) $\lim_{x \rightarrow p} (fg)(x) = AB$;
- (c) $\lim_{x \rightarrow p} \left(\frac{f}{g} \right)(x) = \frac{A}{B}$, if $B \neq 0$.

Proof. In view of Theorem 4.2, these assertions follow immediately from the analogous properties of sequences (Theorem 3.3).²⁶¹ $\square$

Remark: If $\mathbf{f}$ and $\mathbf{g}$ map E into $\mathbb{R}^k$, then (a) remains true, and (b) becomes

$$(b') \lim_{x \rightarrow p} (\mathbf{f} \cdot \mathbf{g})(x) = \mathbf{A} \cdot \mathbf{B}.$$

(Compare Theorem 3.4.)

261 The dictionary in action: not one ε appears, because every ε was already spent in 3.3. This is the chapter's characteristic proof — one sentence of translation — and the same sentence will prove 4.9. When a proof is this short, the work was done elsewhere; the skill is knowing where.

THREE DEFINITIONS CONVERGING | Continuity arrives here in its ε - δ form (4.5), is linked to limits (4.6) and hence to sequences (through 4.2), and is then recast as the inverse-image condition (4.8). All three are the same property; each has its use. ε - δ verifies concrete examples (4.11); sequences refute; 4.8 proves theorems — it is the only form in which the compactness and connectedness arguments of the next two sections can even be stated. The section's quiet workhorses are 4.7 (compositions) and the assembly line of 4.9–4.11, which manufactures the continuity of every rational function once and for all.

4.5 Definition. Suppose X and Y are metric spaces, $E \subset X$, $p \in E$, and f maps E into Y . Then f is said to be *continuous at p* if for every $\varepsilon > 0$ there exists a $\delta > 0$ such that

$$d_Y(f(x), f(p)) < \varepsilon$$

for all points $x \in E$ for which $d_X(x, p) < \delta$.

If f is continuous at every point of E , then f is said to be *continuous on E* .

It should be noted that f has to be defined at the point p in order to be continuous at p . (Compare this with the remark following Definition 4.1.)²⁶²

If p is an isolated point of E , then our definition implies that every function f which has E as its domain of definition is continuous at p . For, no matter which $\varepsilon > 0$ we choose, we can pick $\delta > 0$ so that the only point $x \in E$ for which $d_X(x, p) < \delta$ is $x = p$; then

$$d_Y(f(x), f(p)) = 0 < \varepsilon.$$

263

4.6 Theorem. In the situation given in Definition 4.5, assume also that p is a limit point of E . Then f is continuous at p if and only if $\lim_{x \rightarrow p} f(x) = f(p)$.

Proof. This is clear if we compare Definitions 4.1 and 4.5. □

We now turn to compositions of functions. A brief statement of the following theorem is that a continuous function of a continuous function is continuous.

4.7 Theorem. Suppose X, Y, Z are metric spaces, $E \subset X$, f maps E into Y , g maps the range of f , $f(E)$, into Z , and h is the mapping of E into Z defined by

$$h(x) = g(f(x)) \quad (x \in E).$$

If f is continuous at a point $p \in E$ and if g is continuous at the point $f(p)$, then h is continuous at p .

262 The two definitions differ in exactly two characters: 4.5 requires $p \in E$ where 4.1 required p to be a limit point, and 4.5 drops the $0 <$ from the distance condition — the point p itself is now tested, and the target is $f(p)$, not a free q . Keep the ledger straight: limits ignore p ; continuity is about p .

263 Not a pathology but a consistency check: at an isolated point there is no approach to make, so nothing to fail. It is also why 4.6 needs its extra hypothesis — at an isolated point the limit in 4.6 is undefined (4.1 demands a limit point) while continuity holds vacuously. Every sequence-in- $\mathbb{Z}$, for instance, is a continuous function.

This function h is called the *composition* or the *composite* of f and g . The notation

$$h = g \circ f$$

is frequently used in this context.

Proof. Let $\varepsilon > 0$ be given. Since g is continuous at $f(p)$, there exists $\eta > 0$ such that

$$d_Z(g(y), g(f(p))) < \varepsilon \quad \text{if } d_Y(y, f(p)) < \eta \text{ and } y \in f(E).$$

Since f is continuous at p , there exists $\delta > 0$ such that

$$d_Y(f(x), f(p)) < \eta \quad \text{if } d_X(x, p) < \delta \text{ and } x \in E.$$

It follows that

$$d_Z(h(x), h(p)) = d_Z(g(f(x)), g(f(p))) < \varepsilon$$

if $d_X(x, p) < \delta$ and $x \in E$. Thus h is continuous at p .²⁶⁴ □

4.8 Theorem. A mapping f of a metric space X into a metric space Y is continuous on X if and only if $f^{-1}(V)$ is open in X for every open set V in Y .

(Inverse images are defined in Definition 2.2.) This is a very useful characterization of continuity.

Proof. Suppose f is continuous on X and V is an open set in Y . We have to show that every point of $f^{-1}(V)$ is an interior point of $f^{-1}(V)$. So, suppose $p \in X$ and $f(p) \in V$. Since V is open, there exists $\varepsilon > 0$ such that $y \in V$ if $d_Y(f(p), y) < \varepsilon$; and since f is continuous at p , there exists $\delta > 0$ such that $d_Y(f(x), f(p)) < \varepsilon$ if $d_X(x, p) < \delta$. Thus $x \in f^{-1}(V)$ as soon as $d_X(x, p) < \delta$.

Conversely, suppose $f^{-1}(V)$ is open in X for every open set V in Y . Fix $p \in X$ and $\varepsilon > 0$, let V be the set of all $y \in Y$ such that $d_Y(y, f(p)) < \varepsilon$. Then V is open; hence $f^{-1}(V)$ is open; hence there exists $\delta > 0$ such that $x \in f^{-1}(V)$ as soon as $d_X(p, x) < \delta$. But if $x \in f^{-1}(V)$, then $f(x) \in V$, so that $d_Y(f(x), f(p)) < \varepsilon$.

This completes the proof. □

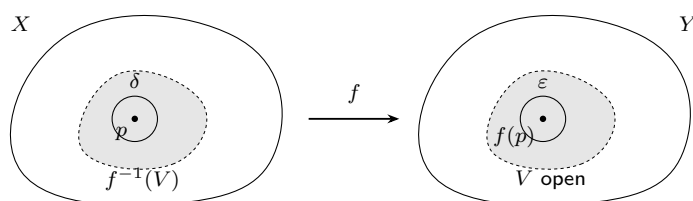
HOW THE PROOF FLOWS

1. **See the two vocabularies.** Continuity speaks in ε - δ ; openness speaks in interior points, i.e. in balls. The theorem claims they say the same thing, so the proof must be a two-way translation, and each direction is found by matching ball to ball.
2. **Forward.** To show $f^{-1}(V)$ open, stand at $p \in f^{-1}(V)$. Openness of V supplies an ε -ball inside V around $f(p)$; continuity at p converts it into a δ -ball around p that f sends into it. That δ -ball certifies p interior. Note the order: V speaks first, conti-

264 A relay race, run backwards from the goal: the final tolerance ε is handed to g , which quotes its price η ; that price becomes the tolerance handed to f , which quotes δ . Chains of quantifiers always compose in this order — last function first — and the middle quantity η is bought from one contract and spent on the other. The same relay reappears whenever composed processes must be estimated (e.g. 7.12).

nuity answers.

- 3. Backward.** To verify continuity at p for a given ε , *manufacture* the open set to pull back: the ε -ball V about $f(p)$ — open by 2.19. The hypothesis opens $f^{-1}(V)$, and any δ -ball inside it around p is exactly the δ that continuity demanded.
- 4. Audit.** Each direction uses its hypothesis once, on the ball handed over by the other side; 2.19 (balls are open) is the only outside fact. The theorem is a change of coordinates, not a computation — and note it is stated for f on all of X , using Remark 4.12's licence to forget ambient sets.



Theorem 4.8: whatever ball fits inside V at $f(p)$, continuity supplies a ball at p mapping into it — so $f^{-1}(V)$ is all interior. The dashed regions correspond under f ; the balls are the currency exchanged.

Corollary. A mapping f of a metric space X into a metric space Y is continuous if and only if $f^{-1}(C)$ is closed in X for every closed set C in Y .

This follows from the theorem, since a set is closed if and only if its complement is open, and since $f^{-1}(E^c) = [f^{-1}(E)]^c$ for every $E \subset Y$.²⁶⁵

We now turn to complex-valued and vector-valued functions, and to functions defined on subsets of $\mathbb{R}^k$.

4.9 Theorem. Let f and g be complex continuous functions on a metric space X . Then $f + g$, fg , and f/g are continuous on X .

In the last case, we must of course assume that $g(x) \neq 0$, for all $x \in X$.

Proof. At isolated points of X there is nothing to prove. At limit points, the statement follows from Theorems 4.4 and 4.6. $\square$

4.10 Theorem.

- (a) Let $f_1, \dots, f_k$ be real functions on a metric space X , and let $\mathbf{f}$ be the mapping of X into $\mathbb{R}^k$ defined by

$$\mathbf{f}(x) = (f_1(x), \dots, f_k(x)) \quad (x \in X); \quad (4.7)$$

then $\mathbf{f}$ is continuous if and only if each of the functions $f_1, \dots, f_k$ is continuous.

- (b) If $\mathbf{f}$ and $\mathbf{g}$ are continuous mappings of X into $\mathbb{R}^k$, then $\mathbf{f} + \mathbf{g}$ and $\mathbf{f} \cdot \mathbf{g}$ are continuous on X .

²⁶⁵ The complement identity is exactly the third displayed law in the box at 2.2 — inverse images respect complements, images do not. Here is the promised payoff: no image-based version of 4.8 exists, and the asymmetry noted in Chapter 2 becomes the reason this whole chapter runs on f^{-1} .

The functions $f_1, \dots, f_k$ are called the *components* of $\mathbf{f}$. Note that $\mathbf{f} + \mathbf{g}$ is a mapping into $\mathbb{R}^k$, whereas $\mathbf{f} \cdot \mathbf{g}$ is a real function on X .

Proof. Part (a) follows from the inequalities

$$|f_j(x) - f_j(y)| \leq |\mathbf{f}(x) - \mathbf{f}(y)| = \left\{ \sum_{i=1}^k |f_i(x) - f_i(y)|^2 \right\}^{1/2},$$

for $j = 1, \dots, k$. Part (b) follows from (a) and Theorem 4.9.²⁶⁶ $\square$

4.11 Examples. If $x_1, \dots, x_k$ are the coordinates of the point $\mathbf{x} \in \mathbb{R}^k$, the functions ϕ_i defined by

$$\phi_i(\mathbf{x}) = x_i \quad (\mathbf{x} \in \mathbb{R}^k) \quad (4.8)$$

are continuous on $\mathbb{R}^k$, since the inequality

$$|\phi_i(\mathbf{x}) - \phi_i(\mathbf{y})| \leq |\mathbf{x} - \mathbf{y}|$$

shows that we may take $\delta = \varepsilon$ in Definition 4.5.

The functions ϕ_i are sometimes called the *coordinate functions*.

Repeated application of Theorem 4.9 then shows that every *monomial*

$$x_1^{n_1} x_2^{n_2} \cdots x_k^{n_k} \quad (4.9)$$

where $n_1, \dots, n_k$ are nonnegative integers, is continuous on $\mathbb{R}^k$. The same is true of constant multiples of (4.9), since constants are evidently continuous. It follows that every *polynomial* P , given by

$$P(\mathbf{x}) = \sum c_{n_1 \dots n_k} x_1^{n_1} \cdots x_k^{n_k} \quad (\mathbf{x} \in \mathbb{R}^k), \quad (4.10)$$

is continuous on $\mathbb{R}^k$. Here the coefficients $c_{n_1 \dots n_k}$ are complex numbers, $n_1, \dots, n_k$ are nonnegative integers, and the sum in (4.10) has finitely many terms.

Furthermore, every *rational function* in $x_1, \dots, x_k$, that is, every quotient of two polynomials of the form (4.10), is continuous on $\mathbb{R}^k$ wherever the denominator is different from zero.²⁶⁷

From the triangle inequality one sees easily that

$$||\mathbf{x}| - |\mathbf{y}|| \leq |\mathbf{x} - \mathbf{y}| \quad (\mathbf{x}, \mathbf{y} \in \mathbb{R}^k). \quad (4.11)$$

Hence the mapping $\mathbf{x} \rightarrow |\mathbf{x}|$ is a continuous real function on $\mathbb{R}^k$.

If now $\mathbf{f}$ is a continuous mapping from a metric space X into $\mathbb{R}^k$, and if ϕ is defined on X by setting $\phi(p) = |\mathbf{f}(p)|$, it follows, by Theorem 4.7, that ϕ is a continuous real function on X .

4.12 Remark. We defined the notion of continuity for functions defined on a subset E of a metric space X . However, the complement of E in X plays no role whatever in this definition (note that the situation was somewhat different for limits of functions). Accordingly, we lose nothing

266 The same two-sided squeeze as 3.4(a): each component error is at most the vector error, and the vector error is at most $\sqrt{k}$ times the worst component. Coordinatewise reduction, third appearance — it will carry Chapter 9's entire differential calculus.

267 An assembly line, and the economical way to prove continuity ever after: two atoms (constants and coordinates, the latter with the cleanest possible modulus, $\delta = \varepsilon$) and three combining rules (sum, product, quotient from 4.9; composition from 4.7). Almost no continuity proof after this page touches ε again — functions are built out of continuous parts. Chapter 8 extends the reach to power series, where the sums become infinite and uniformity must be paid for.

of interest by discarding the complement of the domain of f . This means that we may just as well talk only about continuous mappings of one metric space into another, rather than of mappings of subsets. This simplifies statements and proofs of some theorems. We have already made use of this principle in Theorems 4.8 to 4.10, and will continue to do so in the following section on compactness.²⁶⁸

CONTINUITY AND COMPACTNESS

CHAPTER 2'S INVESTMENT MATURES | Five theorems, one engine. 4.14 shows continuous images of compact sets are compact — a one-paragraph proof, because 4.8 lets the compactness of the domain be spent directly on a pulled-back cover. Everything else in the section is 4.14 plus a Chapter 2 fact: boundedness (4.15, via Heine–Borel), attainment of extrema (4.16, via 2.28), continuity of inverse maps (4.17), and uniform continuity (4.19, where compactness converts infinitely many local δ 's into one global one — the finite-minimum move yet again, and Rudin says so in the proof).

Then the audit: 4.20–4.21 dismantle each theorem when compactness is removed, one counterexample per conclusion. The section is a matched pair — what compactness buys, and the invoice showing nothing was free.

[268] This is 2.16's observation — every subset is a metric space — earning its keep, and it is subtler than it looks. Continuity of $f|_E$ is measured inside E : the function $f(x) = 1$ for rational x , 0 otherwise, is wildly discontinuous on $\mathbb{R}$, yet its restriction to $\mathbb{Q}$ is constant, hence continuous. “Continuous on E ” in Rudin always means continuous as a function on the metric space E — the same relativity as “open in E ” (2.29–2.30), managed the same way: name the space.

4.13 Definition. A mapping $\mathbf{f}$ of a set E into $\mathbb{R}^k$ is said to be *bounded* if there is a real number M such that $|\mathbf{f}(x)| \leq M$ for all $x \in E$.

4.14 Theorem. Suppose f is a continuous mapping of a compact metric space X into a metric space Y . Then $f(X)$ is compact.

Proof. Let $\{V_\alpha\}$ be an open cover of $f(X)$. Since f is continuous, Theorem 4.8 shows that each of the sets $f^{-1}(V_\alpha)$ is open. Since X is compact, there are finitely many indices, say $\alpha_1, \dots, \alpha_n$, such that

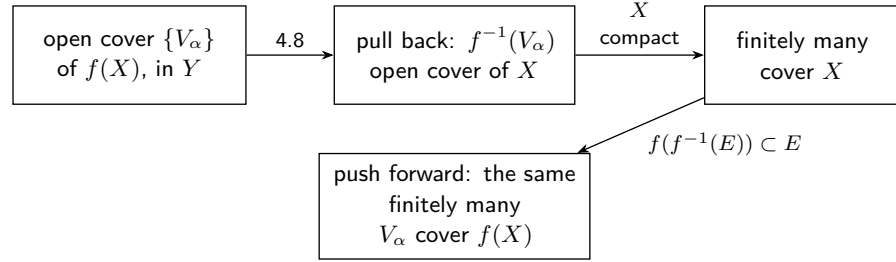
$$X \subset f^{-1}(V_{\alpha_1}) \cup \dots \cup f^{-1}(V_{\alpha_n}). \quad (4.12)$$

Since $f(f^{-1}(E)) \subset E$ for every $E \subset Y$, (4.12) implies that

$$f(X) \subset V_{\alpha_1} \cup \dots \cup V_{\alpha_n}. \quad (4.13)$$

This completes the proof. $\square$

Note: We have used the relation $f(f^{-1}(E)) \subset E$, valid for $E \subset Y$. If $E \subset X$, then $f^{-1}(f(E)) \supset E$; equality need not hold in either case.



Theorem 4.14 as a flow diagram: pull the cover back, spend compactness, push the finite subcover forward. Each arrow is one theorem, and the round trip works because f is defined on all of X — the push-forward inclusion needs nothing else.

We shall now deduce some consequences of Theorem 4.14.

4.15 Theorem. *If f is a continuous mapping of a compact metric space X into $\mathbb{R}^k$, then $f(X)$ is closed and bounded. Thus, f is bounded.*

This follows from Theorem 2.41. The result is particularly important when f is real:

4.16 Theorem. *Suppose f is a continuous real function on a compact metric space X , and*

$$M = \sup_{p \in X} f(p), \quad m = \inf_{p \in X} f(p). \quad (4.14)$$

Then there exist points $p, q \in X$ such that $f(p) = M$ and $f(q) = m$.

The notation in (4.14) means that M is the least upper bound of the set of all numbers $f(p)$, where p ranges over X , and that m is the greatest lower bound of this set of numbers.

The conclusion may also be stated as follows: *There exist points p and q in X such that $f(q) \leq f(x) \leq f(p)$ for all $x \in X$; that is, f attains its maximum (at p) and its minimum (at q).*

Proof. By Theorem 4.15, $f(X)$ is a closed and bounded set of real numbers; hence $f(X)$ contains

$$M = \sup f(X) \quad \text{and} \quad m = \inf f(X),$$

by Theorem 2.28. ²⁶⁹ □

4.17 Theorem. *Suppose f is a continuous 1-1 mapping of a compact metric space X onto a metric space Y . Then the inverse mapping f^{-1} defined on Y by*

$$f^{-1}(f(x)) = x \quad (x \in X)$$

is a continuous mapping of Y onto X .

Proof. Applying Theorem 4.8 to f^{-1} in place of f , we see that it suffices to prove that $f(V)$ is an open set in Y for every open set V in X . Fix such a set V .

The complement V^c of V is closed in X , hence compact (Theorem 2.35); hence $f(V^c)$ is a compact subset of Y (Theorem 4.14) and so is

269 The extreme value theorem of calculus, in three citations: 4.14 makes the image compact, 2.41 reads that as closed and bounded, 2.28 puts the supremum of a closed bounded set inside it. Every hard step was prepaid in Chapter 2 — exactly as the note at 2.28 promised. Notice the theorem gives no way to find p and q ; pure existence, from compactness alone.

270 A ricochet proof: to open $f(V)$, close its complement. The chain closed $\rightarrow$ compact $\rightarrow$ compact image $\rightarrow$ closed converts domain-compactness into the openness 4.8 demands of f^{-1} — and each link is a Chapter 2 theorem (2.35, 4.14, 2.34). Bijectivity enters exactly once, to say image-of-complement is complement-of-image. Example 4.21 shows the whole chain snaps without compactness.

closed in Y (Theorem 2.34). Since f is one-to-one and onto, $f(V)$ is the complement of $f(V^c)$. Hence $f(V)$ is open.²⁷⁰ $\square$

4.18 Definition. Let f be a mapping of a metric space X into a metric space Y . We say that f is *uniformly continuous* on X if for every $\varepsilon > 0$ there exists $\delta > 0$ such that

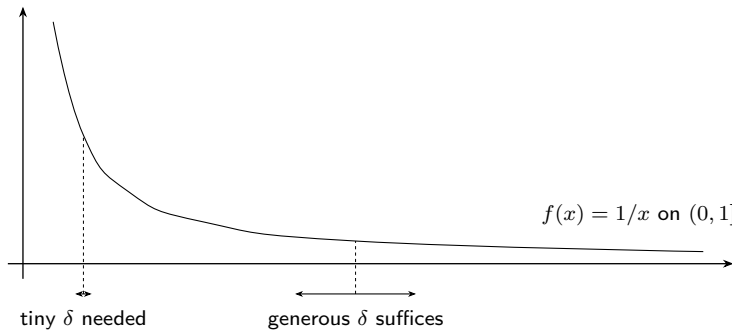
$$d_Y(f(p), f(q)) < \varepsilon \quad (4.15)$$

for all p and q in X for which $d_X(p, q) < \delta$.

Let us consider the differences between the concepts of continuity and of uniform continuity. First, uniform continuity is a property of a function on a set, whereas continuity can be defined at a single point. To ask whether a given function is uniformly continuous at a certain point is meaningless. Second, if f is continuous on X , then it is possible to find, for each $\varepsilon > 0$ and *for each point p of X* , a number $\delta > 0$ having the property specified in Definition 4.5. This δ depends on ε *and on p* . If f is, however, uniformly continuous on X , then it is possible, for each $\varepsilon > 0$, to find *one* number $\delta > 0$ which will do for *all* points p of X .²⁷¹

Evidently, every uniformly continuous function is continuous. That the two concepts are equivalent on compact sets follows from the next theorem.

271 *Quantifier order is the entire difference: continuity is $\forall p \forall \varepsilon \exists \delta$, uniform continuity is $\forall \varepsilon \exists \delta \forall p$. Moving $\forall p$ past $\exists \delta$ strips δ of its dependence on p . This is the book's sharpest lesson in reading quantifiers, and the figure below is the picture to attach to it.*



Continuity that is not uniform: the same ε of vertical tolerance costs ever-smaller δ as the curve steepens toward the missing endpoint. No single δ serves every point — Theorem 4.20(c) makes this the general phenomenon on bounded non-compact sets.

4.19 Theorem. Let f be a continuous mapping of a compact metric space X into a metric space Y . Then f is uniformly continuous on X .

Proof. Let $\varepsilon > 0$ be given. Since f is continuous, we can associate to each point $p \in X$ a positive number $\phi(p)$ such that

$$q \in X, d_X(p, q) < \phi(p) \quad \text{implies} \quad d_Y(f(p), f(q)) < \frac{\varepsilon}{2}. \quad (4.16)$$

Let $J(p)$ be the set of all $q \in X$ for which

$$d_X(p, q) < \frac{1}{2}\phi(p). \quad (4.17)$$

Since $p \in J(p)$, the collection of all sets $J(p)$ is an open cover of X ; since X is compact, there is a finite set of points $p_1, \dots, p_n$ in X , such that

$$X \subset J(p_1) \cup \dots \cup J(p_n). \quad (4.18)$$

We put

$$\delta = \frac{1}{2} \min\{\phi(p_1), \dots, \phi(p_n)\}. \quad (4.19)$$

Then $\delta > 0$. (This is one point where the finiteness of the covering, inherent in the definition of compactness, is essential. The minimum of a finite set of positive numbers is positive, whereas the inf of an infinite set of positive numbers may very well be 0.)

Now let q and p be points of X such that $d_X(p, q) < \delta$. By (4.18), there is an integer m , $1 \leq m \leq n$, such that $p \in J(p_m)$; hence

$$d_X(p, p_m) < \frac{1}{2}\phi(p_m), \quad (4.20)$$

and we also have

$$d_X(q, p_m) \leq d_X(p, q) + d_X(p, p_m) < \delta + \frac{1}{2}\phi(p_m) \leq \phi(p_m).$$

Finally, (4.16) shows that therefore

$$d_Y(f(p), f(q)) \leq d_Y(f(p), f(p_m)) + d_Y(f(q), f(p_m)) < \varepsilon.$$

This completes the proof. $\square$

An alternative proof is sketched in Exercise 4.10.

HOW THE PROOF FLOWS

- 1. Name the enemy.** Continuity hands out a private δ -radius $\phi(p)$ at each point; uniformity demands one public δ . The natural attempt — take $\inf_p \phi(p)$ — fails because an infimum of infinitely many positive numbers can be 0 (Rudin says exactly this in the parenthesis at (4.19)). The proof exists to make that infimum finite.
- 2. Turn the private radii into a cover.** Each point owns the ball $J(p)$ of *half* its radius; together they cover X . Compactness reduces to finitely many centers $p_1, \dots, p_n$ — and now the minimum in (4.19) is over a finite set, hence positive. This is the local-to-global conversion promised in Chapter 2's signpost, performed literally.
- 3. Why half?** The safety margin is what makes the public δ work for pairs. A point p lands in some $J(p_m)$, within $\frac{1}{2}\phi(p_m)$ of the center; its partner q is within $\delta \leq \frac{1}{2}\phi(p_m)$ more, so *both* lie inside the full radius $\phi(p_m)$. Without the halving, q could fall just outside.
- 4. Close through the center.** Neither p nor q is compared with

the other directly — both are compared with p_m , each within $\varepsilon/2$, and the triangle inequality adds the halves. That is why (4.16) was set up with $\varepsilon/2$: working backwards, pattern 4.

- 5. Audit.** Compactness spent once (at the finite subcover), continuity spent once per point (to define ϕ), triangle inequality twice. The halving trick — shrink each local radius so that overlapping neighbours still see each other — recurs in Chapter 7 and in every partition-of-unity argument (Chapter 10).

We now proceed to show that compactness is essential in the hypotheses of Theorems 4.14, 4.15, 4.16, and 4.19.

4.20 Theorem. *Let E be a noncompact set in $\mathbb{R}^1$. Then*

- (a) *there exists a continuous function on E which is not bounded;*
- (b) *there exists a continuous and bounded function on E which has no maximum.*

If, in addition, E is bounded, then

- (c) *there exists a continuous function on E which is not uniformly continuous.*

Proof. Suppose first that E is bounded, so that there exists a limit point x_0 of E which is not a point of E .²⁷² Consider

$$f(x) = \frac{1}{x - x_0} \quad (x \in E). \quad (4.21)$$

This is continuous on E (Theorem 4.9), but evidently unbounded. To see that (4.21) is not uniformly continuous, let $\varepsilon > 0$ and $\delta > 0$ be arbitrary, and choose a point $x \in E$ such that $|x - x_0| < \delta$. Taking t close enough to x_0 , we can then make the difference $|f(t) - f(x)|$ greater than ε , although $|t - x| < \delta$. Since this is true for every $\delta > 0$, f is not uniformly continuous on E .

The function g given by

$$g(x) = \frac{1}{1 + (x - x_0)^2} \quad (x \in E) \quad (4.22)$$

is continuous on E and is bounded, since $0 < g(x) < 1$. It is clear that

$$\sup_{x \in E} g(x) = 1,$$

whereas $g(x) < 1$ for all $x \in E$. Thus g has no maximum on E .

Having proved the theorem for bounded sets E , let us now suppose that E is unbounded. Then $f(x) = x$ establishes (a), whereas

$$h(x) = \frac{x^2}{1 + x^2} \quad (x \in E) \quad (4.23)$$

272 Here 2.41 is being run in reverse: noncompact-but-bounded forces not-closed, which hands the proof its weapon — a missing limit point x_0 . Every counterexample in the bounded case is built by blowing up at x_0 ; the theorem is really about how continuity on E cannot see the hole just outside E .

establishes (b), since

$$\sup_{x \in E} h(x) = 1$$

and $h(x) < 1$ for all $x \in E$.

Assertion (c) would be false if boundedness were omitted from the hypotheses. For, let E be the set of all integers. Then every function defined on E is uniformly continuous on E . To see this, we need merely take $\delta < 1$ in Definition 4.18. ²⁷³ $\square$

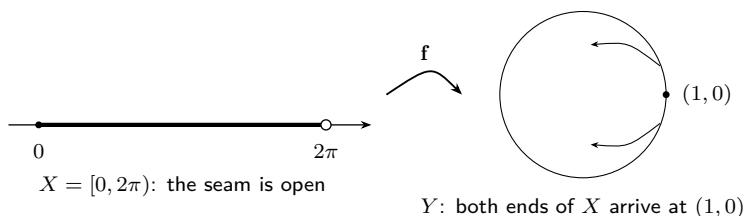
We conclude this section by showing that compactness is also essential in Theorem 4.17.

4.21 Example. Let X be the half-open interval $[0, 2\pi)$ on the real line, and let $\mathbf{f}$ be the mapping of X onto the circle Y consisting of all points whose distance from the origin is 1, given by

$$\mathbf{f}(t) = (\cos t, \sin t) \quad (0 \leq t < 2\pi). \quad (4.24)$$

The continuity of the trigonometric functions cosine and sine, as well as their periodicity properties, will be established in Chap. 8. These results show that $\mathbf{f}$ is a continuous 1-1 mapping of X onto Y .

However, the inverse mapping (which exists, since $\mathbf{f}$ is one-to-one and onto) fails to be continuous at the point $(1, 0) = \mathbf{f}(0)$. Of course, X is not compact in this example. (It may be of interest to observe that $\mathbf{f}^{-1}$ fails to be continuous in spite of the fact that Y is compact!)



Example 4.21: wrapping the half-open interval onto the circle glues the missing endpoint to 0. Points just below 2π land next to $(1, 0)$, but $\mathbf{f}^{-1}$ throws them back near 2π — a jump of almost 2π across the seam. Continuity of the inverse is a property the domain's compactness must fund; the target's cannot.

CONTINUITY AND CONNECTEDNESS

ONE THEOREM, ONE COROLLARY, ONE WARNING | The connectedness investment of 2.45–2.47 pays out in a single exchange: continuous images of connected sets are connected (4.22), and since the connected subsets of $\mathbb{R}$ are exactly the intervals (2.47), the intermediate value theorem falls out in three citations (4.23). The warning is 4.24: the intermediate value *property* does not characterise continuity — $\sin(1/x)$ has it while being badly discontinuous. The converse a beginner expects is false.

273 So the converse pattern is not clean: noncompactness kills boundedness and attainment outright, but uniform continuity survives on $\mathbb{Z}$ — an unbounded, closed, noncompact set where every function is uniformly continuous because points cannot get close. The moral: (a) and (b) fail through escape (to a missing point or to infinity), (c) fails only through crowding against a missing point. Different failure modes, different hypotheses. Exercise 4.8 probes the same boundary.

4.22 Theorem. *If f is a continuous mapping of a metric space X into a metric space Y , and if E is a connected subset of X , then $f(E)$ is connected.*

Proof. Assume, on the contrary, that $f(E) = A \cup B$, where A and B are nonempty separated subsets of Y . Put $G = E \cap f^{-1}(A)$, $H = E \cap f^{-1}(B)$.

Then $E = G \cup H$, and neither G nor H is empty.

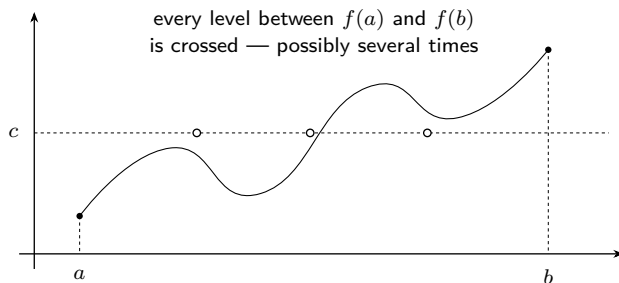
Since $A \subset \bar{A}$ (the closure of A), we have $G \subset f^{-1}(\bar{A})$; the latter set is closed, since f is continuous; hence $\bar{G} \subset f^{-1}(\bar{A})$. It follows that $f(\bar{G}) \subset \bar{A}$. Since $f(H) = B$ and $\bar{A} \cap B$ is empty, we conclude that $\bar{G} \cap H$ is empty.

The same argument shows that $G \cap \bar{H}$ is empty. Thus G and H are separated. This is impossible if E is connected.²⁷⁴ $\square$

4.23 Theorem. *Let f be a continuous real function on the interval $[a, b]$. If $f(a) < f(b)$ and if c is a number such that $f(a) < c < f(b)$, then there exists a point $x \in (a, b)$ such that $f(x) = c$.*

A similar result holds, of course, if $f(a) > f(b)$. Roughly speaking, the theorem says that a continuous real function assumes all intermediate values on an interval.

Proof. By Theorem 2.47, $[a, b]$ is connected; hence Theorem 4.22 shows that $f([a, b])$ is a connected subset of $\mathbb{R}^1$, and the assertion follows if we appeal once more to Theorem 2.47.²⁷⁵ $\square$



The intermediate value theorem: a connected image cannot skip the level c . The theorem asserts at least one crossing and says nothing about how many, nor where.

4.24 Remark. At first glance, it might seem that Theorem 4.23 has a converse. That is, one might think that if for any two points $x_1 < x_2$ and for any number c between $f(x_1)$ and $f(x_2)$ there is a point x in (x_1, x_2) such that $f(x) = c$, then f must be continuous.

That this is not so may be concluded from Example 4.27(d).

274 The same transport strategy as 4.14, run contrapositively: a separation of the image pulls back to a separation of the domain. The only technical point is that “separated” involves closures, and this is where the corollary to 4.8 earns its place — $f^{-1}(\bar{A})$ is closed, so it swallows $\bar{G}$, keeping the pulled-back pieces apart. Images transport connectedness and compactness; inverse images transport openness and closedness. Keep the direction ledger straight.

275 Three citations and no computation: 2.47 converts “interval” to “connected,” 4.22 pushes connectedness through f , and 2.47 converts back to “interval” — which by definition contains everything between $f(a)$ and $f(b)$. Chapter 1’s Theorem 1.21 was this argument done by hand, before any of the three tools existed; compare the box there. Exercise 4.14 (fixed points) is the standard application.

DISCONTINUITIES

A TAXONOMY, THEN A ZOO | Failure of continuity comes in two grades. If both one-sided limits exist, the failure is a bookkeeping

error — a jump, or a wrong value — and is called *simple* (first kind). If a one-sided limit fails to exist, the function oscillates or scatters, and the discontinuity is of the second kind. Example 4.27 exhibits all the species; (a) and (b) are the standard pathologies built from $\mathbb{Q}$, and (d) is the oscillator that killed the converse of the intermediate value theorem in 4.24. The taxonomy exists for the next section: monotonic functions will be shown to admit only the first kind, and only countably many.

If x is a point in the domain of definition of the function f at which f is not continuous, we say that f is *discontinuous at x* , or that f has a *discontinuity at x* . If f is defined on an interval or on a segment, it is customary to divide discontinuities into two types. Before giving this classification, we have to define the *right-hand* and the *left-hand limits* of f at x , which we denote by $f(x+)$ and $f(x-)$, respectively.

4.25 Definition. Let f be defined on (a, b) . Consider any point x such that $a \leq x < b$. We write

$$f(x+) = q$$

if $f(t_n) \rightarrow q$ as $n \rightarrow \infty$, for all sequences (t_n) in (x, b) such that $t_n \rightarrow x$. To obtain the definition of $f(x-)$, for $a < x \leq b$, we restrict ourselves to sequences (t_n) in (a, x) .

It is clear that for any point x of (a, b) , $\lim_{t \rightarrow x} f(t)$ exists if and only if

$$f(x+) = f(x-) = \lim_{t \rightarrow x} f(t).$$

4.26 Definition. Let f be defined on (a, b) . If f is discontinuous at a point x , and if $f(x+)$ and $f(x-)$ exist, then f is said to have a *discontinuity of the first kind*, or a *simple discontinuity*, at x . Otherwise the discontinuity is said to be of the *second kind*.

There are two ways in which a function can have a simple discontinuity: either $f(x+) \neq f(x-)$ [in which case the value $f(x)$ is immaterial], or $f(x+) = f(x-) \neq f(x)$.

4.27 Examples.

(a) Define

$$f(x) = \begin{cases} 1 & (x \text{ rational}), \\ 0 & (x \text{ irrational}). \end{cases}$$

Then f has a discontinuity of the second kind at every point x , since neither $f(x+)$ nor $f(x-)$ exists.²⁷⁶

(b) Define

$$f(x) = \begin{cases} x & (x \text{ rational}), \\ 0 & (x \text{ irrational}). \end{cases}$$

²⁷⁶ The Dirichlet function, and the standard witness that “discontinuous everywhere” is possible. Both $\mathbb{Q}$ and its complement are dense (1.20(b) and Exercise 1.1), so every one-sided approach can be made along either set, producing both 0 and 1 as subsequential limits — the refutation direction of the bridge 4.2. It returns in Chapter 6 as the function that is not Riemann-integrable (Exercise 6.4).

Then f is continuous at $x = 0$ and has a discontinuity of the second kind at every other point.²⁷⁷

(c) Define

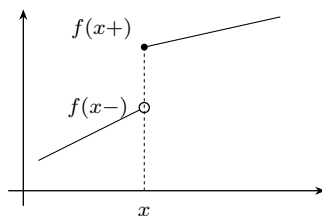
$$f(x) = \begin{cases} x + 2 & (-3 < x < -2), \\ -x - 2 & (-2 \leq x < 0), \\ x + 2 & (0 \leq x < 1). \end{cases}$$

Then f has a simple discontinuity at $x = 0$ and is continuous at every other point of $(-3, 1)$.

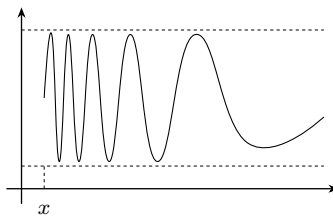
(d) Define

$$f(x) = \begin{cases} \sin \frac{1}{x} & (x \neq 0), \\ 0 & (x = 0). \end{cases}$$

Since neither $f(0+)$ nor $f(0-)$ exists, f has a discontinuity of the second kind at $x = 0$. We have not yet shown that $\sin x$ is a continuous function. If we assume this result for the moment, Theorem 4.7 implies that f is continuous at every point $x \neq 0$.²⁷⁸



first kind: both one-sided limits exist, disagree



second kind: oscillation denser and denser, no limit

The two kinds. Left, a jump (4.27(c)): the function arrives at different platforms from the two sides. Right, $\sin(1/x)$ near 0 (4.27(d)): the one-sided limits do not exist at all, because every neighbourhood of x sweeps the whole band.

277 Continuity at exactly one point — worth a pause, since beginners often believe continuity somewhere forces decency nearby. The squeeze $|f(x)| \leq |x|$ gives continuity at 0 no matter how wild the values elsewhere.

[278] The oscillator that closes 4.24: on any interval $(0, \delta)$ it sweeps all of $[-1, 1]$, so it has the full intermediate value property, yet it is discontinuous at 0. The intermediate value property constrains what a function skips, not how it settles — and settling is what continuity is. Exercise 4.14's fixed-point result survives for such functions; their integrability will be delicate in Chapter 6.

MONOTONIC FUNCTIONS

We shall now study those functions which never decrease (or never increase) on a given segment.

4.28 Definition. Let f be real on (a, b) . Then f is said to be *monotonically increasing* on (a, b) if $a < x < y < b$ implies $f(x) \leq f(y)$. If the last inequality is reversed, we obtain the definition of a *monotonically decreasing* function. The class of *monotonic* functions consists of both the increasing and the decreasing functions.

4.29 Theorem. Let f be monotonically increasing on (a, b) . Then $f(x+)$ and $f(x-)$ exist at every point x of (a, b) . More precisely,

$$\sup_{a < t < x} f(t) = f(x-) \leq f(x) \leq f(x+) = \inf_{x < t < b} f(t). \quad (4.25)$$

Furthermore, if $a < x < y < b$, then

$$f(x+) \leq f(y-). \quad (4.26)$$

Analogous results evidently hold for monotonically decreasing functions.

Proof. By hypothesis, the set of numbers $f(t)$, where $a < t < x$, is bounded above by the number $f(x)$, and therefore has a least upper bound which we shall denote by A . Evidently $A \leq f(x)$. We have to show that $A = f(x-)$.

Let $\varepsilon > 0$ be given. It follows from the definition of A as a least upper bound that there exists $\delta > 0$ such that $a < x - \delta < x$ and

$$A - \varepsilon < f(x - \delta) \leq A. \quad (4.27)$$

Since f is monotonic, we have

$$f(x - \delta) \leq f(t) \leq A \quad (x - \delta < t < x). \quad (4.28)$$

Combining (4.27) and (4.28), we see that

$$|f(t) - A| < \varepsilon \quad (x - \delta < t < x).$$

Hence $f(x-) = A$.²⁷⁹

The second half of (4.25) is proved in precisely the same way.

Next, if $a < x < y < b$, we see from (4.25) that

$$f(x+) = \inf_{x < t < b} f(t) = \inf_{x < t < y} f(t). \quad (4.29)$$

The last equality is obtained by applying (4.25) to (a, y) in place of (a, b) . Similarly,

$$f(y-) = \sup_{a < t < y} f(t) = \sup_{x < t < y} f(t). \quad (4.30)$$

Comparison of (4.29) and (4.30) gives (4.26). $\square$

Corollary. *Monotonic functions have no discontinuities of the second kind.*

This corollary implies that every monotonic function is discontinuous at a countable set of points at most. Instead of appealing to the general theorem whose proof is sketched in Exercise 4.17, we give here a simple proof which is applicable to monotonic functions.

4.30 Theorem. *Let f be monotonic on (a, b) . Then the set of points of (a, b) at which f is discontinuous is at most countable.*

Proof. Suppose, for the sake of definiteness, that f is increasing, and let E be the set of points at which f is discontinuous.

With every point x of E we associate a rational number $r(x)$ such that

$$f(x-) < r(x) < f(x+).$$

279 The bounded-implies-sup pattern of the strategy index (pattern 3), in its function-limit costume: monotonicity makes the left values bounded, the supremum A exists by 1.19, and clause (ii) of 1.8 plants a term within ε ; monotonicity then drags the whole approach into the gap — exactly the mechanism of 3.14, with $t \uparrow x$ in place of $n \rightarrow \infty$.

Since $x_1 < x_2$ implies $f(x_1+) \leq f(x_2-)$, we see that $r(x_1) \neq r(x_2)$ if $x_1 \neq x_2$.

We have thus established a 1-1 correspondence between the set E and a subset of the set of rational numbers. The latter, as we know, is countable.²⁸⁰ $\square$

4.31 Remark. It should be noted that the discontinuities of a monotonic function need not be isolated. In fact, given any countable subset E of (a, b) , which may even be dense, we can construct a function f , monotonic on (a, b) , discontinuous at every point of E , and at no other point of (a, b) .

To show this, let the points of E be arranged in a sequence (x_n) , $n = 1, 2, 3, \dots$. Let (c_n) be a sequence of positive numbers such that $\sum c_n$ converges. Define

$$f(x) = \sum_{x_n < x} c_n \quad (a < x < b). \quad (4.31)$$

The summation is to be understood as follows: Sum over those indices n for which $x_n < x$. If there are no points x_n to the left of x , the sum is empty; following the usual convention, we define it to be zero. Since (4.31) converges absolutely, the order in which the terms are arranged is immaterial.²⁸¹

We leave the verification of the following properties of f to the reader:

- (a) f is monotonically increasing on (a, b) ;
- (b) f is discontinuous at every point of E ; in fact,

$$f(x_n+) - f(x_n-) = c_n.$$

- (c) f is continuous at every other point of (a, b) .

Moreover, it is not hard to see that $f(x-) = f(x)$ at all points of (a, b) . If a function satisfies this condition, we say that f is *continuous from the left*. If the summation in (4.31) were taken over all indices n for which $x_n \leq x$, we would have $f(x+) = f(x)$ at every point of (a, b) ; that is, f would be *continuous from the right*.

Functions of this sort can also be defined by another method; for an example we refer to Theorem 6.16.

280 A counting proof with real geometry in it: by 4.29 every discontinuity of an increasing f is a jump, the jump intervals $(f(x-), f(x+))$ are pairwise disjoint by (4.26), and disjoint intervals can each be tagged with their own rational — density of $\mathbb{Q}$ (1.20(b)) supplies the tag, countability of $\mathbb{Q}$ (2.13) does the counting. The same disjoint-intervals-are-countable argument proved Exercise 2.29. (Choosing $r(x)$ for each x is another silent appeal to choice; here it can be avoided — take the rational with least index in a fixed enumeration.)

[281] Chapter 3's rearrangement theorem (3.55) quietly holding up the definition: the indices with $x_n < x$ arrive in no particular order, so the sum is well defined only because the convergence is absolute. A conditionally convergent (c_n) would make (4.31) meaningless. This jump function is the prototype for the integrators of Chapter 6 (Rudin points to 6.16), where its jumps become point masses.

INFINITE LIMITS AND LIMITS AT INFINITY

THE EXTENDED REALS, CASHED IN | Two definitions and a theorem, closing the chapter by making $\lim_{x \rightarrow \infty}$ and “ $f(t) \rightarrow +\infty$ ” legitimate. The device is the one hinted at in 1.23: give $\pm\infty$ *neighbourhoods*, and Definition 4.1 restates itself with neighbourhoods in place of distances — no new theory needed, and 4.34 lists which limit laws survive, with precisely the exceptions catalogued as un-

defined at 1.23.

To enable us to operate in the extended real number system, we shall now enlarge the scope of Definition 4.1 by reformulating it in terms of neighborhoods.

For any real number x , we have already defined a neighborhood of x to be any segment $(x - \delta, x + \delta)$.

4.32 Definition. For any real c , the set of real numbers x such that $x > c$ is called a *neighborhood of $+\infty$* and is written $(c, +\infty)$. Similarly, the set $(-\infty, c)$ is a neighborhood of $-\infty$.

4.33 Definition. Let f be a real function defined on $E \subset \mathbb{R}$. We say that

$$f(t) \rightarrow A \text{ as } t \rightarrow x,$$

where A and x are in the extended real number system, if for every neighborhood U of A there is a neighborhood V of x such that $V \cap E$ is not empty, and such that $f(t) \in U$ for all $t \in V \cap E$, $t \neq x$.²⁸²

A moment's consideration will show that this coincides with Definition 4.1 when A and x are real.

The analogue of Theorem 4.4 is still true, and the proof offers nothing new. We state it, for the sake of completeness.

4.34 Theorem. Let f and g be defined on $E \subset \mathbb{R}$. Suppose

$$f(t) \rightarrow A, \quad g(t) \rightarrow B \quad \text{as } t \rightarrow x.$$

Then

- (a) $f(t) \rightarrow A'$ implies $A' = A$,
- (b) $(f + g)(t) \rightarrow A + B$,
- (c) $(fg)(t) \rightarrow AB$,
- (d) $(f/g)(t) \rightarrow A/B$,

provided the right members of (b), (c), and (d) are defined.

Note that $\infty - \infty$, $0 \cdot \infty$, ∞ / ∞ , and $A/0$ are not defined (see Definition 1.23).

282 One definition now covers nine cases (x and A each finite, $+\infty$, or $-\infty$), because “neighbourhood” absorbs the difference: an ε -ball around a real number, a ray beyond c for $\pm\infty$. This is the topologist’s trick — rephrase metrically-stated definitions in terms of neighbourhoods and they generalise by themselves. Theorem 3.15’s sequence version was the same move.

CHAPTER 4 IN ONE PARAGRAPH

Continuity is one property in three costumes: ε - δ at a point (4.5), preservation of sequential limits (4.2 with 4.6), and open inverse images (4.8) — the last being the form that proves theorems. What continuity transports is the content of the chapter: compactness pushes forward (4.14), and with it boundedness, attained extrema, continuous inverses, and uniform continuity (4.15–4.19); connectedness pushes forward (4.22), and with it the intermediate

value theorem (4.23). The counterexamples 4.20, 4.21, 4.24 and 4.27 mark the exact boundaries: nothing survives the loss of compactness except by accident, and the intermediate value property is weaker than continuity. Monotonic functions are the tame class — one-sided limits always exist (4.29), so only jumps occur, and only countably many (4.30), though possibly on a dense set (4.31). Chapter 6 will integrate against exactly such jump functions, and Chapter 5's derivative inherits every tool built here.

EXERCISES

HOW TO USE THESE

Chapter 4's exercises divide into a uniform-continuity core, a counterexample gallery, and a toolbox of functions used throughout the rest of the book.

The chain that matters. $4.9 \rightarrow 4.11 \rightarrow 4.13$ develops what uniform continuity is really for: it preserves Cauchy sequences, and therefore lets a function be extended from a dense subset to the whole space — the mechanism by which e^x , defined on $\mathbb{Q}$ in Exercise 1.6, becomes a function on $\mathbb{R}$ in Chapter 8.

The toolbox. 4.20 – 4.22 build the distance function ρ_E and use it to separate closed sets (normality) — machinery reused in Chapters 7 and 10. 4.23 – 4.24 are convex functions, promised in the box at 2.17.

The gallery. 4.1 , 4.7 , 4.16 , 4.18 , $4.21(b)$, $4.25(b)$ are counterexamples to remember; 4.18 (Thomae's function) and 4.7 (continuity along every line without continuity) are famous.

Alternative proofs. 4.10 reproves uniform continuity on compact sets by sequences — compare it with the cover proof of 4.19 to see the two faces of compactness side by side.

4.1. Suppose f is a real function defined on $\mathbb{R}^1$ which satisfies

$$\lim_{h \rightarrow 0} [f(x+h) - f(x-h)] = 0$$

for every $x \in \mathbb{R}^1$. Does this imply that f is continuous?²⁸³

4.2. If f is a continuous mapping of a metric space X into a metric space Y , prove that

$$f(\overline{E}) \subset \overline{f(E)}$$

for every set $E \subset X$. ($\overline{E}$ denotes the closure of E .) Show, by an example, that $f(\overline{E})$ can be a proper subset of $\overline{f(E)}$.²⁸⁴

283 No. The condition compares the two sides with each other, never with $f(x)$ — so any function with matching one-sided limits but the wrong value passes: $f = 0$ except $f(0) = 1$. A symmetric condition cannot see an asymmetric failure. First lesson of the set: check what a limit condition actually quantifies over.

284 Pull back instead of pushing forward: $\overline{f(E)}$ is closed, so $f^{-1}(\overline{f(E)})$ is closed (corollary to 4.8) and contains E , hence contains $\overline{E}$. For proper inclusion, let the closure point escape to infinity: $f(x) = 1/(1+x^2)$ on $E = \mathbb{N}$ gives $f(\overline{E}) = f(E)$ but $0 \in \overline{f(E)}$. This is the exercise promised in Chapter 2's box on why f^{-1} outranks f .

285 One line with the right tool:

$Z(f) = f^{-1}(\{0\})$, and $\{0\}$ is closed.

The 4.8 corollary is doing what it was built for — and 4.22 below will prove the converse, that every closed set is some function's zero set.

286 The rigidity principle. For the second part, take $p \in X$, approach it from E (density plus 3.2(d)), and let continuity carry the agreement to the limit. It is why knowing a continuous function on $\mathbb{Q}$ determines it on $\mathbb{R}$ — the fact 4.13 will exploit for extensions, and Chapter 8 for e^x .

287 Exercise 2.29 is the structural input: the open complement of E is a countable union of disjoint segments, and bridging each with a chord leaves continuity intact. Without closedness the failure is immediate — $1/x$ on $(0, 1)$ has nowhere to send 0. The vector-valued case is componentwise (4.10).

288 A characterisation worth carrying: for compact domains, continuity of the function equals compactness of the graph. The forward direction is 4.14 applied to $x \mapsto (x, f(x))$ (continuous by 4.10). The converse is where the work is — from a sequence $x_n \rightarrow x$, compactness of the graph forces a convergent subsequence of $(x_n, f(x_n))$ whose limit must be $(x, f(x))$, and 3.20-style bookkeeping upgrades subsequences to the sequence. Graphs recur at 4.7's pathologies and in Chapter 9's implicit function theorem.

289 The classic warning for Chapter 9: continuity along every line through a point does not give continuity at the point. Approach f along the parabola $x = y^2$ and it sits at $\frac{1}{2}$; along lines it dies to 0. Directions are not enough in dimension two — curves can sneak in. Remember this when partial derivatives appear (9.20): their existence is a line-by-line statement, and this is why it proves so little.

4.3. Let f be a continuous real function on a metric space X . Let $Z(f)$ (the zero set of f) be the set of all $p \in X$ at which $f(p) = 0$. Prove that $Z(f)$ is closed.²⁸⁵

4.4. Let f and g be continuous mappings of a metric space X into a metric space Y , and let E be a dense subset of X . Prove that $f(E)$ is dense in $f(X)$. If $g(p) = f(p)$ for all $p \in E$, prove that $g(p) = f(p)$ for all $p \in X$. (In other words, a continuous function is determined by its values on a dense subset of its domain.)²⁸⁶

4.5. If f is a real continuous function defined on a closed set $E \subset \mathbb{R}^1$, prove that there exist continuous real functions g on $\mathbb{R}^1$ such that $g(x) = f(x)$ for all $x \in E$. (Such functions g are called *continuous extensions* of f from E to $\mathbb{R}^1$.) Show that the result becomes false if the word “closed” is omitted. Extend the result to vector-valued functions. *Hint:* Let the graph of g be a straight line on each of the segments which constitute the complement of E (compare Exercise 2.29). The result remains true if $\mathbb{R}^1$ is replaced by any metric space, but the proof is not so simple.²⁸⁷

4.6. If f is defined on E , the *graph* of f is the set of points $(x, f(x))$, for $x \in E$. In particular, if E is a set of real numbers and f is real-valued, the graph of f is a subset of the plane.

Suppose E is compact, and prove that f is continuous on E if and only if its graph is compact.²⁸⁸

4.7. If $E \subset X$ and if f is a function defined on X , the *restriction* of f to E is the function g whose domain of definition is E , such that $g(p) = f(p)$ for $p \in E$. Define f and g on $\mathbb{R}^2$ by: $f(0, 0) = g(0, 0) = 0$, $f(x, y) = xy^2/(x^2 + y^4)$, $g(x, y) = xy^2/(x^2 + y^6)$ if $(x, y) \neq (0, 0)$. Prove that f is bounded on $\mathbb{R}^2$, that g is unbounded in every neighborhood of $(0, 0)$, and that f is not continuous at $(0, 0)$; nevertheless, the restrictions of both f and g to every straight line in $\mathbb{R}^2$ are continuous!²⁸⁹

4.8. Let f be a real uniformly continuous function on the bounded set E in $\mathbb{R}^1$. Prove that f is bounded on E . Show that the conclusion is false if the boundedness of E is omitted from the hypothesis.²⁹⁰

4.9. Show that the requirement in the definition of uniform continuity can be rephrased as follows, in terms of diameters of sets: To every $\varepsilon > 0$ there exists a $\delta > 0$ such that $\text{diam } f(E) < \varepsilon$ for all $E \subset X$ with $\text{diam } E < \delta$.²⁹¹

4.10. Complete the details of the following alternative proof of Theo-

rem 4.19: If f is not uniformly continuous, then for some $\varepsilon > 0$ there are sequences $(p_n), (q_n)$ in X such that $d_X(p_n, q_n) \rightarrow 0$ but $d_Y(f(p_n), f(q_n)) > \varepsilon$. Use Theorem 2.37 to obtain a contradiction.²⁹²

4.11. Suppose f is a uniformly continuous mapping of a metric space X into a metric space Y , and prove that $(f(x_n))$ is a Cauchy sequence in Y for every Cauchy sequence (x_n) in X . Use this result to give an alternative proof of the theorem stated in Exercise 4.13.²⁹³

4.12. A uniformly continuous function of a uniformly continuous function is uniformly continuous. State this more precisely and prove it.²⁹⁴

4.13. Let E be a dense subset of a metric space X , and let f be a uniformly continuous real function defined on E . Prove that f has a continuous extension from E to X (see Exercise 4.5 for terminology). (Uniqueness follows from Exercise 4.4.) *Hint:* For each $p \in X$ and each positive integer n , let $V_n(p)$ be the set of all $q \in E$ with $d(p, q) < 1/n$. Use Exercise 4.9 to show that the intersection of the closures of the sets $f(V_1(p)), f(V_2(p)), \dots$ consists of a single point, say $g(p)$, of $\mathbb{R}^1$. Prove that the function g so defined on X is the desired extension of f .

Could the range space $\mathbb{R}^1$ be replaced by $\mathbb{R}^k$? By any compact metric space? By any complete metric space? By any metric space?²⁹⁵

4.14. Let $I = [0, 1]$ be the closed unit interval. Suppose f is a continuous mapping of I into I . Prove that $f(x) = x$ for at least one $x \in I$.²⁹⁶

4.15. Call a mapping of X into Y *open* if $f(V)$ is an open set in Y whenever V is an open set in X . Prove that every continuous open mapping of $\mathbb{R}^1$ into $\mathbb{R}^1$ is monotonic.²⁹⁷

4.16. Let $[x]$ denote the largest integer contained in x , that is, $[x]$ is the integer such that $x - 1 < [x] \leq x$, and let $\{x\} = x - [x]$ denote the fractional part of x . What discontinuities do the functions $[x]$ and $\{x\}$ have?²⁹⁸

4.17. Let f be a real function defined on (a, b) . Prove that the set of points at which f has a simple discontinuity is at most countable. *Hint:* Let E be the set on which $f(x-) < f(x+)$. With each point x of E , associate a triple (p, q, r) of rational numbers such that

290 Uniform continuity buys finitely many δ -steps across a bounded set: cover $\bar{E}$ (compact!) by finitely many δ -balls, or argue directly — each step changes f by at most ε , and boundedness of E caps the number of steps. Compare 4.20(a): plain continuity could not do this. For the failure, $f(x) = x$ on $\mathbb{R}$.

291 Uniform continuity as a statement about sets, with the points quantified away — 3.9's diameter doing for functions what it did for Cauchy sequences. This is the form the extension theorem 4.13 runs on: small sets have small images, uniformly.

292 Worth writing out in full, as the mirror image of 4.19's cover proof. Negate the uniform statement (one bad ε , witnesses for every $\delta_n = 1/n$ — the same contrapositive harvest as 4.2), compactify: some subsequence $p_{n_k} \rightarrow p$, drag q_{n_k} along since the distance dies, and continuity at the single point p contradicts the standing gap ε . Covers versus sequences is the same fork as 2.32 versus 2.41(c) — Rudin chose covers for the text, Tao's compactness definition would make this the primary proof.

293 The heart of the uniform/pointwise distinction: a Cauchy sequence's terms crowd each other at unpredictable locations, so only a location-independent δ can convert crowding in X to crowding in Y . Plain continuity fails the task: $f(x) = 1/x$ on $(0, 1)$ maps the Cauchy sequence $(1/n)$ to the divergent (n) .

294 The relay of 4.7 again, with both contracts now point-free: ε buys η from g uniformly, η buys δ from f uniformly. Nothing in the composition argument ever used the base point — which is the observation this exercise wants made.

295 The most consequential exercise of the chapter. The hint is 3.10(b) recycled: shrinking sets with shrinking diameters (by 4.9!) whose closures nest — their intersection is the value the extension must take. The closing questions locate the true hypothesis: everything works in any complete range space (the nested-closed-sets argument is Exercise 3.21), and completeness is exactly what is needed — the limit must have somewhere to live. This machine builds e^x from its rational values (Chapter 8) and, in functional analysis, extends densely-defined operators.

296 The one-dimensional fixed point theorem, and the standard first application of 4.23: $g(x) = f(x) - x$ has $g(0) \geq 0$, $g(1) \leq 0$, so the intermediate value theorem plants a zero. One line — but only because both endpoints are trapped in I . Brouwer's theorem is this statement in $\mathbb{R}^k$, and needs far heavier tools.

297 Contrapositive: a continuous non-monotonic function has an interior extremum on some $[a, b]$ (attained by 4.16), and then $f((a, b))$ contains its own maximum — a closed endpoint — so is not open. Openness of images turns out to be a strong, rigidity-inducing property; compare the box at 2.2 on why it is not the definition of continuity.

298 Simple jumps at every integer, of size 1, both functions continuous from the right — the floor function is 4.31's jump construction in its simplest form, with $E = \mathbb{Z}$ and all $c_n = 1$. It returns as the integrator of choice in Chapter 6 (6.16), where its jumps turn integrals into sums.

299 4.30's tagging trick without monotonicity: one rational is no longer enough (jumps can overlap), so each jump is tagged with a triple — its level plus a rational buffer on each side — and 2.13 counts triples. The general principle: to prove a set countable, inject it into $\mathbb{Q}^n$; the injectivity is where the analysis lives.

- (a) $f(x-) < p < f(x+)$,
- (b) $a < q < t < x$ implies $f(t) < p$,
- (c) $x < t < r < b$ implies $f(t) > p$.

The set of all such triples is countable. Show that each triple is associated with at most one point of E . Deal similarly with the other possible types of simple discontinuities.²⁹⁹

- 4.18.** Every rational x can be written in the form $x = m/n$, where $n > 0$, and m and n are integers without any common divisors. When $x = 0$, we take $n = 1$. Consider the function f defined on $\mathbb{R}^1$ by

$$f(x) = \begin{cases} 0 & (x \text{ irrational}), \\ \frac{1}{n} & \left(x = \frac{m}{n}\right). \end{cases}$$

Prove that f is continuous at every irrational point, and that f has a simple discontinuity at every rational point.³⁰⁰

- 4.19.** Suppose f is a real function with domain $\mathbb{R}^1$ which has the intermediate value property: If $f(a) < c < f(b)$, then $f(x) = c$ for some x between a and b . Suppose also, for every rational r , that the set of all x with $f(x) = r$ is closed. Prove that f is continuous.

Hint: If $x_n \rightarrow x_0$ but $f(x_n) > r > f(x_0)$ for some r and all n , then $f(t_n) = r$ for some t_n between x_0 and x_n ; thus $t_n \rightarrow x_0$. Find a contradiction. (N. J. Fine, *Amer. Math. Monthly*, vol. 73, 1966, p. 782.)³⁰¹

- 4.20.** If E is a nonempty subset of a metric space X , define the distance from $x \in X$ to E by

$$\rho_E(x) = \inf_{z \in E} d(x, z).$$

- (a) Prove that $\rho_E(x) = 0$ if and only if $x \in \overline{E}$.
- (b) Prove that ρ_E is a uniformly continuous function on X , by showing that

$$|\rho_E(x) - \rho_E(y)| \leq d(x, y)$$

for all $x \in X$, $y \in X$.

Hint: $\rho_E(x) \leq d(x, z) \leq d(x, y) + d(y, z)$, so that

$$\rho_E(x) \leq d(x, y) + \rho_E(y).$$

- 4.21.** Suppose K and F are disjoint sets in a metric space X , K is compact, F is closed. Prove that there exists $\delta > 0$ such that $d(p, q) > \delta$ if $p \in K$, $q \in F$. *Hint:* ρ_F is a continuous positive function on K .

Show that the conclusion may fail for two disjoint closed sets if neither is compact.³⁰³

- 4.22.** Let A and B be disjoint nonempty closed sets in a metric space X , and define

$$f(p) = \frac{\rho_A(p)}{\rho_A(p) + \rho_B(p)} \quad (p \in X).$$

Show that f is a continuous function on X whose range lies in $[0, 1]$, that $f(p) = 0$ precisely on A and $f(p) = 1$ precisely on B . This establishes a converse of Exercise 4.3: *Every closed set $A \subset X$ is $Z(f)$ for some continuous real f on X .* Setting

$$V = f^{-1}\left(\left[0, \frac{1}{2}\right)\right), \quad W = f^{-1}\left(\left(\frac{1}{2}, 1\right]\right),$$

show that V and W are open and disjoint, and that $A \subset V$, $B \subset W$. (Thus pairs of disjoint closed sets in metric spaces can be covered by pairs of disjoint open sets. This property of metric spaces is called *normality*.)³⁰⁴

- 4.23.** A real-valued function f defined in (a, b) is said to be *convex* if

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y)$$

whenever $a < x < b$, $a < y < b$, $0 < \lambda < 1$. Prove that every convex function is continuous. Prove that every increasing convex function of a convex function is convex. (For example, if f is convex, so is e^f .)

If f is convex in (a, b) and if $a < s < t < u < b$, show that

$$\frac{f(t) - f(s)}{t - s} \leq \frac{f(u) - f(s)}{u - s} \leq \frac{f(u) - f(t)}{u - t}.$$

305

- 4.24.** Assume that f is a continuous real function defined in (a, b) such that

$$f\left(\frac{x + y}{2}\right) \leq \frac{f(x) + f(y)}{2}$$

for all $x, y \in (a, b)$. Prove that f is convex.³⁰⁶

- 4.25.** If $A \subset \mathbb{R}^k$ and $B \subset \mathbb{R}^k$, define $A + B$ to be the set of all sums $\mathbf{x} + \mathbf{y}$ with $\mathbf{x} \in A$, $\mathbf{y} \in B$.

[300] *Thomae's function — continuous exactly on the irrationals. The key: near any point, rationals with small denominators are scarce, so $f(t) < \varepsilon$ once t avoids the finitely many fractions with $n \leq 1/\varepsilon$ in a bounded window. The asymmetry with 4.27(b) is instructive, and one can ask for the reverse: a function continuous exactly on $\mathbb{Q}$. None exists — the continuity set of any function is a countable intersection of open sets, and $\mathbb{Q}$ is not one, by Baire (Exercise 3.22). That is a satisfying loop: Baire's theorem answering a question about a single function.*

301 *The repaired converse of 4.23: the intermediate value property alone fails (4.24, 4.27(d)), but adding closed rational level sets restores continuity. In the hint, the level t_n 's accumulate at x_0 , so closedness forces $f(x_0) = r$ — contradiction. Note how the hypothesis is used only through sequences: the corollary to 4.8 in its sequential clothing.*

302 *The workhorse function of metric topology. Part (a) says ρ_E sees exactly the closure; part (b) says it is 1-Lipschitz — the strongest, cleanest form of uniform continuity, with $\delta = \varepsilon$. The hint's move (bound by an over-estimate, then take the inf) is how one always handles infima of distances.*

303 *4.16 and 4.20 cooperating: ρ_F is positive on K (by 4.20(a), since F is closed and disjoint), and a positive continuous function on a compact set attains a positive minimum. For the failure take $F = \mathbb{Z}$ and $K = \{n + 1/n\}$: disjoint, closed, distance 0. Two closed sets can approach each other at infinity; a compact set has no infinity to escape to.*

[304] *Urysohn's lemma for metric spaces, built from Exercise 4.20's bricks in one display. The denominator never vanishes — ρ_A and ρ_B cannot both be zero, by 4.20(a) and disjointness — and the two preimages under f separate A from B openly, with 4.8 certifying openness. In general topology normality is hard-won; metrics give it for free, which is one reason this book never needs spaces beyond metric ones.*

305 The slope inequality at the end is the real content — chords steepen from left to right — and continuity falls out of it: near any interior point the difference quotients are wedged between two fixed chords' slopes, so f is locally Lipschitz. Note “interior” is essential; on $[a, b]$ a convex function may jump at the endpoints. These inequalities return for the differentiable case in Chapter 5's exercises.

306 Midpoint convexity plus continuity gives full convexity: iterate midpoints to reach all dyadic $\lambda = k/2^n$, then let continuity fill in the rest — density doing its usual work (Exercise 4.4's principle applied to λ). Without continuity the statement fails, but the counterexamples need a Hamel basis, far beyond this book.

307 Part (b) is the famous fact that $\{m + n\alpha\}$ is dense for irrational α — Weyl's equidistribution in embryo. The proof is pigeonhole: the fractional parts $(n\alpha)$ are infinitely many points in $[0, 1)$, two must fall within ε , and their difference generates an ε -mesh. Against (a), it shows compactness of one summand is not a removable convenience: closed + closed can be dense-and-not-closed.

308 A cancellation theorem: composing with an injective continuous g can be undone — because 4.17 makes g^{-1} continuous, indeed uniformly so by 4.19, and 4.12 composes the uniformities. Every hypothesis of this chapter's compactness section is consumed in one exercise; the counterexample without compact Y is 4.21's wrap-around map playing the role of g .

- (a) If K is compact and C is closed in $\mathbb{R}^k$, prove that $K + C$ is closed.

Hint: Take $\mathbf{z} \notin K + C$, put $F = \mathbf{z} - C$, the set of all $\mathbf{z} - \mathbf{y}$ with $\mathbf{y} \in C$. Then K and F are disjoint. Choose δ as in Exercise 4.21. Show that the open ball with center $\mathbf{z}$ and radius δ does not intersect $K + C$.

- (b) Let α be an irrational real number. Let C_1 be the set of all integers, let C_2 be the set of all $n\alpha$ with $n \in C_1$. Show that C_1 and C_2 are closed subsets of $\mathbb{R}^1$ whose sum $C_1 + C_2$ is not closed, by showing that $C_1 + C_2$ is a countable dense subset of $\mathbb{R}^1$.³⁰⁷

- 4.26.** Suppose X, Y, Z are metric spaces, and Y is compact. Let f map X into Y , let g be a continuous one-to-one mapping of Y into Z , and put $h(x) = g(f(x))$ for $x \in X$. Prove that f is uniformly continuous if h is uniformly continuous. *Hint:* g^{-1} has compact domain $g(Y)$, and $f(x) = g^{-1}(h(x))$.

Prove also that f is continuous if h is continuous. Show (by modifying Example 4.21, or by finding a different example) that the compactness of Y cannot be omitted from the hypotheses, even when X and Z are compact.³⁰⁸

Appendix to Chapter 4

HOW TO USE THIS APPENDIX

Nothing in this appendix is Rudin's text. It is imported from the four comparison texts and set apart at every level: the running head names the appendix, section heads carry a reverse-video tag, item numbers are prefixed **S**, and every statement names its source.

Why import anything at all. Rudin's Chapter 4 is a chapter about *structure*: what continuity does to compactness and connectedness. He buys that generality by postponing every concrete function — $\sin$, $\exp$, $\log$, x^α do not exist until Chapter 8 — and by leaving the working criteria for limits and uniform continuity to the exercises. The Chinese and Tao texts make the opposite trade. Reading them alongside supplies five things Rudin's chapter genuinely lacks:

1. Criteria (A). A Cauchy criterion for function limits; one definition of limit that specialises to all of Rudin's; a squeeze law; and oscillation, which measures *how* discontinuous a function is at a point.

2. Asymptotics (B). The o/O calculus and equivalent infinitesimals — the machinery every later computation runs on, and which Rudin never introduces at all.

3. The postponed limits (C). $\lim_{x \rightarrow 0} (\sin x)/x = 1$ and $\lim_{x \rightarrow \infty} (1 + 1/x)^x = e$, with the proofs the other books give.

4. Tools left in Rudin's exercises (D, E). Inverse functions without compactness; the sequential criterion for uniform continuity; gluing; extension to the closure. Rudin asks for several of these as Exercises 4.11–4.13; the other texts prove them in the running text and then use them.

5. Finer structure (F). Rudin's two kinds of discontinuity refine to four; connectedness gains a path-connected cousin and a decomposition into components.

How to read it. None of this is needed to follow Rudin, which is why it sits in an appendix rather than in his chapter. Sections C and D in particular are *logically* out of order in his development — they use functions he has not built yet — and each says so where it matters. Read the appendix as a second

pass over Chapter 4, or as the answer to “how would I actually compute with this?” The exercises begin on the last section.

NOT IN RUDIN

SUPPLEMENT A — CRITERIA FOR LIMITS THAT RUDIN OMITS

WHY THIS SUPPLEMENT BELONGS TO CHAPTER 4

What Rudin does. Definition 4.1 defines $\lim_{x \rightarrow p} f(x) = q$ by ε - δ , and 4.2 characterises it by sequences: $f(p_n) \rightarrow q$ for every sequence $p_n \rightarrow p$ in E with $p_n \neq p$.

The gap. Both forms presuppose the candidate q . Neither can be used to decide whether a limit *exists* when no value suggests itself, and Rudin supplies no criterion that does — no Cauchy condition, no squeeze, and no way of measuring how badly a limit fails to exist.

Where the gap bites. At Exercise 4.13, where a uniformly continuous function must be extended to the closure and the limit at each new point has to be produced from nothing; Rudin’s hint amounts to proving a Cauchy criterion by hand in that special case. It bites again at 4.26–4.28, where discontinuities are classified but never quantified, so “how discontinuous” has no answer.

What this section supplies. The Cauchy criterion for function limits; one definition of limit that specialises to all four of Rudin’s; the squeeze law he uses but never states; and oscillation, which turns “continuous at p ” into the numerical statement $\omega_f(p) = 0$ and makes the set of continuity points a computable object.

S4.1 Theorem. (*Cauchy criterion for function limits.*) Let X be a metric space, $E \subset X$, p a limit point of E , and $f : E \rightarrow Y$ with Y complete (for instance $Y = \mathbb{R}^k$). Then $\lim_{x \rightarrow p} f(x)$ exists if and only if: for every $\varepsilon > 0$ there is a $\delta > 0$ such that

$$d_Y(f(x'), f(x'')) < \varepsilon$$

for all $x', x'' \in E$ with $0 < d_X(x', p) < \delta$ and $0 < d_X(x'', p) < \delta$.

SOURCE: ECNU 定理 3.11 (bk. p. 52); USTC 定理 9 (bk. pp. 29–30). Rudin has no counterpart in Chapter 4.

Proof sketch. *Necessity* is the triangle inequality: if $f(x) \rightarrow q$, take δ for $\varepsilon/2$ and add.

Sufficiency routes through sequences, exactly as 4.2 does. Let (p_n)

be any sequence in $E \setminus \{p\}$ with $p_n \rightarrow p$. The condition makes $(f(p_n))$ a Cauchy sequence, hence convergent by completeness (3.11). If (p'_n) is a second such sequence with a different limit, interleave the two into one sequence; its image is Cauchy, so all its subsequential limits agree, and the two limits coincide. Now 4.2 applies. $\square$

Why Rudin can do without it, and you cannot

Rudin never needs S4.1 because he almost never has to *produce* a function limit: his theorems take a convergent hypothesis and transport it. The one place he does need it, he writes it out by hand — Exercise 4.13, extending a uniformly continuous function to the closure, is precisely “Cauchy criterion + completeness” done in a special case. Supplement E below (S4.15) states the general fact the exercise is a shadow of.

Note what the criterion costs: completeness of the *target*. Over $\mathbb{Q}$ it is false, and the failure is the same one that motivates all of Chapter 1.

S4.2 Definition. (*Limits along a subset.*) Let $E \subset X$, let p be adherent to E — that is, $p \in \overline{E}$ — and let f be defined on E . Write

$$\lim_{x \rightarrow p, x \in E} f(x) = q$$

if for every $\varepsilon > 0$ there is $\delta > 0$ with $d(f(x), q) < \varepsilon$ for all $x \in E$ satisfying $d(x, p) < \delta$.

SOURCE: Tao I, Def. 9.3.6 (bk. p. 221), with adherent points Def. 9.1.8.

ONE DEFINITION IN PLACE OF RUDIN'S FOUR

Tao's single notion specialises to everything Rudin defines separately. Fix a function f on a set D and a point p :

Take $E =$	and you get
$D \setminus \{p\}$	Rudin's $\lim_{x \rightarrow p} f(x)$ (4.1)
$D \cap (p, \infty)$	the right-hand limit $f(p+)$ (4.25)
$D \cap (-\infty, p)$	the left-hand limit $f(p-)$
D itself	continuity at p , when the value is $f(p)$
D , with $p = +\infty$	$\lim_{x \rightarrow +\infty} f(x)$ (4.33)
$D \cap A$	the limit of the restriction $f _A$ (4.12)

The fourth row is the interesting one. Rudin's limit at p is *deleted*: p is thrown out of the domain, which is why 4.1 requires p to be a limit point and why $\lim_{x \rightarrow p} f(x)$ can differ from $f(p)$ — the whole content of Definition 4.5 and of the discontinuity classification. Tao's undeleted version with $E = D$ makes “continuous at p ” literally the statement “the limit along the domain is $f(p)$.” The two conventions disagree, and each is chosen to make its own theorems cleanest; knowing which one is in force is a standing source of confusion between textbooks.

S4.3 Theorem. (Squeeze.) If $f \leq g \leq h$ on E and $\lim_{x \rightarrow p, x \in E} f = \lim_{x \rightarrow p, x \in E} h = L$, then $\lim_{x \rightarrow p, x \in E} g = L$.

SOURCE: Tao I, Ex. 9.3.5; ECNU 定理 3.2 (bk. p. 46).

Proof. Let (p_n) be any sequence in E with $p_n \rightarrow p$. Then $f(p_n) \rightarrow L$ and $h(p_n) \rightarrow L$, and $f(p_n) \leq g(p_n) \leq h(p_n)$, so $g(p_n) \rightarrow L$ by the sequence squeeze (3.19 applied to $g(p_n) - f(p_n) \leq h(p_n) - f(p_n)$). Since the sequence was arbitrary, 4.2 gives the conclusion. $\square$

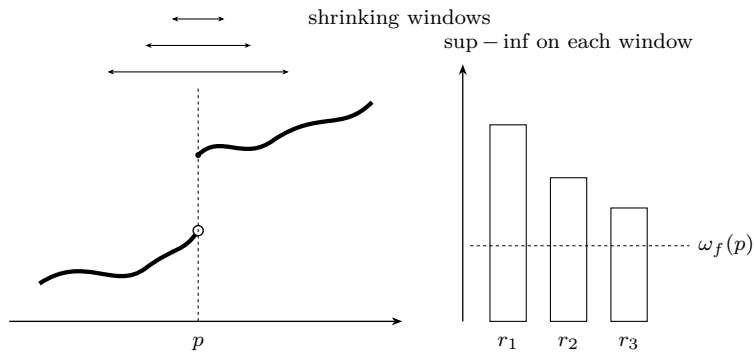
Rudin proves the sequence version (3.19 is its cousin) and never states the function version, though he uses the reasoning freely.³⁰⁹

S4.4 Definition. (Oscillation.) For f real and bounded near p , put

$$\omega_f(p) = \lim_{r \rightarrow 0^+} \left[\sup_{d(x,p) < r} f(x) - \inf_{d(x,p) < r} f(x) \right].$$

The bracket decreases as r decreases, so the limit exists.

SOURCE: Tao I, Ex. 9.3.4 and Tao II, Ex. 2.2.9 (as $\limsup - \liminf$); Yu Pin, Lecture 21 Lemma 131 (p. 224), who defines it for maps of metric spaces as $\omega(f, p) = \inf_{U \ni p} \text{diam } f(U)$ over open U . USTC and ECNU introduce 振幅 only later, for integrability.



Oscillation: measure $\sup - \inf$ over a window about p and let the window close. The bars fall but stall at the size of the jump. The function is continuous at p exactly when they fall to 0.

S4.5 Theorem. f is continuous at p if and only if $\omega_f(p) = 0$. Moreover $\Omega_\varepsilon = \{x : \omega_f(x) \geq \varepsilon\}$ is closed for each $\varepsilon > 0$; consequently the set of points at which f is discontinuous is $\bigcup_n \Omega_{1/n}$, a countable union of closed sets, and the set where f is continuous is a countable intersection of open sets.

SOURCE: Yu Pin, Lemmas 131–132 and p. 227, for arbitrary maps of metric spaces; Tao I, Ex. 9.3.4.

Proof. If $\omega_f(p) = 0$ then given $\varepsilon > 0$ some window has $\sup - \inf < \varepsilon$, so $|f(x) - f(p)| < \varepsilon$ throughout it: f is continuous at p . Conversely continuity makes both $\sup$ and $\inf$ over a small enough window lie within ε of $f(p)$, so their difference is at most 2ε .

For the second claim, let $\omega_f(x_0) < \varepsilon$. Then some ball $B(x_0, r)$ has $\sup - \inf < \varepsilon$ on it, and every $x \in B(x_0, r)$ has a smaller ball inside

³⁰⁹ Cheap to prove and constantly needed — Supplement C below cannot get off the ground without it. The mechanism is 4.2: pull back to sequences and use the sequence squeeze.

$B(x_0, r)$, so $\omega_f(x) < \varepsilon$ too. Hence $\{\omega_f < \varepsilon\}$ is open and its complement Ω_ε is closed. Finally f is discontinuous at x exactly when $\omega_f(x) > 0$, that is when $x \in \Omega_{1/n}$ for some n . $\square$

The payoff: why Thomae's function has no mirror image

Rudin's Exercise 4.18 produces a function continuous at every irrational and discontinuous at every rational. The obvious next question — can a function be continuous at every *rational* and discontinuous at every irrational? — Rudin never asks, and with his Chapter 4 tools he cannot answer it. S4.5 answers it: no.

The continuity set is always $\bigcap_n \{x : \omega_f(x) < 1/n\}$, an intersection of countably many open sets. If some f were continuous exactly on $\mathbb{Q}$, then $\mathbb{Q}$ would be such an intersection, say $\mathbb{Q} = \bigcap_n U_n$ with each U_n open and dense (each contains $\mathbb{Q}$). Enumerate $\mathbb{Q} = \{q_1, q_2, \dots\}$; then

$$\emptyset = \left(\bigcap_n U_n \right) \cap \left(\bigcap_k (\mathbb{R} \setminus \{q_k\}) \right)$$

exhibits $\emptyset$ as a countable intersection of dense open sets, which Baire's theorem forbids. So Thomae's function has no mirror image, and the asymmetry between $\mathbb{Q}$ and its complement — the theme running from 2.44 to Chapter 3's rearrangements — shows up once more.

Rudin proves Baire's theorem in Exercise 3.22; this is one of its first real uses.

NOT IN RUDIN

SUPPLEMENT B — ORDERS OF MAGNITUDE: THE CALCULUS OF LIMITS

WHY THIS SUPPLEMENT BELONGS TO CHAPTER 4

What Rudin does. Nothing — and that is the point. There is no o , no O , no notion of one quantity being negligible beside another anywhere in Chapters 1–4. Limits are proved to exist and are then named.

The gap. Analysis outside this book is conducted almost entirely in asymptotic notation, and the notation is not decoration: it is how one keeps track of which error terms matter. Rudin's abstract line never needs it because he never computes a limit that is not already a theorem.

Where the gap bites. At 5.15, where Taylor's theorem is really a statement that a remainder is $o((x-a)^n)$ and has to be phrased instead with an explicit intermediate point; at 3.20, where the special limits are exactly statements about relative

rates of growth; and in every computation a reader will meet outside these covers.

What this section supplies. The definitions, the substitution theorem that makes them usable, and — most importantly — the pitfall that substitution is legal in factors and illegal in summands, which is the error the notation invites.

S4.6 Definition. Let f, g be defined near p (with $g \neq 0$ there). As $x \rightarrow p$:

- (a) $f = o(g)$ means $f/g \rightarrow 0$;
- (b) $f = O(g)$ means $|f| \leq C|g|$ near p , for some constant C ;
- (c) $f \sim g$ (equivalent) means $f/g \rightarrow 1$.

SOURCE: ECNU 无穷小量的阶, bk. pp. 57–59; USTC 等价无穷小替换, bk. pp. 32–35.

The equality sign in $f = o(g)$ is an abuse: $o(g)$ names a *class* of functions and the statement is membership.³¹⁰

S4.7 Theorem. (Substitution of equivalents.) Suppose $\alpha \sim \beta$ as $x \rightarrow p$. If $\lim \alpha u$ exists, then $\lim \beta u$ exists and they are equal; if $\lim v/\alpha$ exists, then $\lim v/\beta$ exists and they are equal.

SOURCE: ECNU 定理 3.12; USTC 定理 12.

Proof. Write $\beta = \alpha \cdot (\beta/\alpha)$ where $\beta/\alpha \rightarrow 1$. Then $\beta u = (\alpha u)(\beta/\alpha)$, and the limit of a product is the product of the limits (4.4), so $\lim \beta u = (\lim \alpha u) \cdot 1$. The quotient case is identical with $v/\beta = (v/\alpha)(\alpha/\beta)$. $\square$

SUBSTITUTION IS LEGAL IN PRODUCTS AND QUOTIENTS — NOWHERE ELSE

Since $\tan x \sim x$ and $\sin x \sim x$ as $x \rightarrow 0$, one is tempted to replace both in a *difference*:

$$\tan x - \sin x \overset{?}{\sim} x - x = 0,$$

which is not merely imprecise but wrong. The truth is

$$\tan x - \sin x = \sin x \frac{1 - \cos x}{\cos x} \sim x \cdot \frac{x^2}{2} = \frac{x^3}{2},$$

so $(\tan x - \sin x)/\sin^3 x \rightarrow \frac{1}{2}$, not 0. The two equivalences are each accurate to first order, and the difference cancels precisely the order they are accurate to; what survives is third order, about which they say nothing.

The rule to carry: equivalents may be substituted into a *factor* of a product or quotient, never into a summand. When a difference is what you want, expand to enough orders — which is what Taylor’s theorem (5.15) will industrialise.

S4.8 Definition. f is an infinitely large quantity as $x \rightarrow p$ if $|f(x)| \rightarrow$

310 Read every “=” involving o or O left-to-right and never reverse it. From $f = o(x)$ and $h = o(x)$ it does not follow that $f = h$; both are merely in the same class. ECNU makes this explicit by writing $o(g) = \{f : f/g \rightarrow 0\}$.

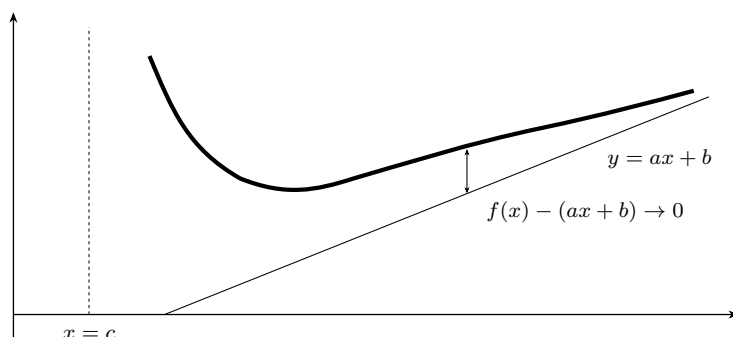
$+\infty$.³¹¹

S4.9 Definition. (*Asymptotes.*) A line L is an asymptote of the curve $y = f(x)$ if the distance from $(x, f(x))$ to L tends to 0 as the point recedes to infinity. The line $y = ax + b$ is an asymptote as $x \rightarrow +\infty$ if and only if

$$a = \lim_{x \rightarrow +\infty} \frac{f(x)}{x}, \quad b = \lim_{x \rightarrow +\infty} [f(x) - ax],$$

both limits existing finitely; and $x = c$ is a vertical asymptote if some one-sided limit of f at c is infinite.

SOURCE: ECNU 曲线的渐近线, bk. pp. 61–62; USTC Ex. 1.2.14. Absent from Rudin entirely.



An oblique asymptote. Because the distance from $(x, f(x))$ to the line is $|f(x) - ax - b|/\sqrt{1 + a^2}$, the geometric condition “distance $\rightarrow 0$ ” is the algebraic condition $f(x) - ax - b \rightarrow 0$ — which is why dividing by x recovers a first, and subtracting ax then recovers b . A vertical asymptote is the separate case where f itself blows up.

WORKED EXAMPLE

For $f(x) = \frac{x^3}{x^2 + 2x - 3}$ division gives

$$f(x) = x - 2 + \frac{7x - 6}{x^2 + 2x - 3},$$

and the remainder tends to 0, so $y = x - 2$ is an asymptote in both directions. Since $x^2 + 2x - 3 = (x + 3)(x - 1)$, the lines $x = -3$ and $x = 1$ are vertical asymptotes. Note that the recipe found a and b without any division: $f(x)/x \rightarrow 1$ and $f(x) - x \rightarrow -2$.

NOT IN RUDIN

SUPPLEMENT C — THE TWO LIMITS RUDIN POSTPONES

READ THIS SECTION AS A LOAN AGAINST CHAPTER 8

Rudin has no $\sin$, $\cos$, $\exp$ or $\log$ at this point: they are *defined* in 8.7 by power series, and every property of them is derived there. So the two theorems below are not available in his development, and the

311 Not the same as unbounded, and the difference is the point of the definition. On $(0, \infty)$ the function $x \sin x$ is unbounded, yet it does not tend to ∞ : at $x = n\pi$ it is 0. ECNU isolates this as the standard beginner’s error — Rudin’s 4.33 defines the notion correctly but never contrasts it with unboundedness.

geometric proof of the first uses the area of a circular sector — that is, integration (Chapter 6) and arc length (8.7 again).

They are included because every other text proves them here, because every computation in a first course needs them, and because the arguments are worth seeing. But the logical order is Rudin's, and Yu Pin — who also defines $\sin$ by its series — shows what that costs: he proves the same limit in one line from the estimate

$$\left| \frac{\sin x}{x} - 1 \right| \leq |x|^2 e^{|x|},$$

read straight off the series with no geometry at all. The picture below is then an illustration of a fact already proved, not the proof.

S4.10 Theorem. $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1.$

SOURCE: ECNU 定理 3.14 (bk. p. 53); USTC 定理 10 (bk. p. 30). Rudin obtains it from 8.7.

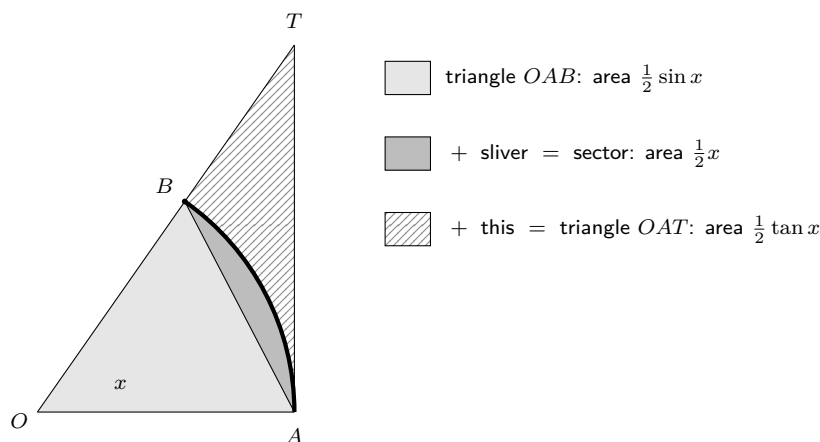
The geometric proof. Let $0 < x < \pi/2$ and work in the unit circle. Comparing the areas of the triangle OAB , the sector OAB , and the triangle OAT gives

$$\frac{1}{2} \sin x < \frac{1}{2} x < \frac{1}{2} \tan x.$$

Dividing by $\frac{1}{2} \sin x > 0$ yields $1 < x/\sin x < 1/\cos x$, that is

$$\cos x < \frac{\sin x}{x} < 1.$$

Since $\cos x \rightarrow 1$ as $x \rightarrow 0^+$, the squeeze S4.3 gives the limit from the right; the quotient is even, so the two-sided limit follows. $\square$



Three nested regions in the unit circle. Containment of the regions is the chain $\sin x < x < \tan x$ — the only inequality the proof needs. The sector's area being $\frac{1}{2}x$ is what makes radian measure the right measure, and is also what this argument borrows from Chapter 6.

S4.11 Theorem. $\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x = e$, in both directions; equivalently $\lim_{t \rightarrow 0} (1+t)^{1/t} = e$.

SOURCE: ECNU bk. p. 54; USTC 定理 11. Rudin proves only the sequence case, 3.31.

Proof. Let $x > 0$ and $n = \lfloor x \rfloor$, so $n \leq x < n+1$. Increasing the base increases the power, and increasing the exponent does too, so

$$\left(1 + \frac{1}{n+1}\right)^n < \left(1 + \frac{1}{x}\right)^x < \left(1 + \frac{1}{n}\right)^{n+1}.$$

The left side is $\left(1 + \frac{1}{n+1}\right)^{n+1}$ divided by $\left(1 + \frac{1}{n+1}\right)$ and the right side is $\left(1 + \frac{1}{n}\right)^n$ times $\left(1 + \frac{1}{n}\right)$; by 3.31 both tend to e . Since $n \rightarrow \infty$ as $x \rightarrow +\infty$, the squeeze S4.3 gives the limit. For $x \rightarrow -\infty$ put $x = -(y+1)$ and compute $\left(1 + \frac{1}{x}\right)^x = \left(1 + \frac{1}{y}\right)^{y+1} \rightarrow e$. $\square$

FROM RUDIN'S SEQUENCE TO A CONTINUOUS VARIABLE

Rudin's 3.31 establishes $\left(1 + \frac{1}{n}\right)^n \rightarrow e$ for integers n . To pass to a real variable, trap x between consecutive integers: with $n = \lfloor x \rfloor$ and $x > 0$,

$$\left(1 + \frac{1}{n+1}\right)^n < \left(1 + \frac{1}{x}\right)^x < \left(1 + \frac{1}{n}\right)^{n+1},$$

because increasing the base increases the power and so does increasing the exponent. Both outer terms are the sequence limit multiplied by a factor tending to 1 — the left is $\left(1 + \frac{1}{n+1}\right)^{n+1}$ divided by $\left(1 + \frac{1}{n+1}\right)$, the right is $\left(1 + \frac{1}{n}\right)^n$ times $\left(1 + \frac{1}{n}\right)$ — so both tend to e , and S4.3 closes the gap. The case $x \rightarrow -\infty$ follows from the substitution $x = -(y+1)$.

This bracketing is the standard way to lift *any* sequence limit to a continuous one when the function is monotone between integers, and it is worth keeping for that reason alone.

NOT IN RUDIN

SUPPLEMENT D — MONOTONICITY, INVERSES, AND THE ELEMENTARY FUNCTIONS

WHY THIS SUPPLEMENT BELONGS TO CHAPTER 4

What Rudin does. Theorem 4.17 proves that a continuous bijection of a *compact* metric space has a continuous inverse, and Example 4.21 shows compactness cannot simply be dropped. Monotone functions appear separately at 4.28–4.30, and the elementary functions do not appear at all — exp, log, sin are built in Chapter 8 from power series.

The gap. On the line there is a second route to the same conclusion that uses no compactness whatever, only order — and

it reaches domains 4.17 cannot, such as open and unbounded intervals. Rudin's two topics, inverse functions and monotone functions, are in fact one topic, and he never connects them.

Where the gap bites. At 4.21 itself, where the circle-wrap counterexample is offered as evidence that compactness is *needed*, when what it really shows is that the domain was not ordered; and at Exercise 4.14, which is about monotone open maps and sits next to a theorem that would explain it.

What this section supplies. The equivalence of injectivity and strict monotonicity for continuous functions on an interval; the inverse function theorem that follows, with no compactness; and the catalogue of elementary functions that this route makes available four chapters early.

S4.12 Theorem. *If f is continuous and one-to-one on an interval $I \subset \mathbb{R}$, then f is strictly monotone on I .*

SOURCE: Tao I, Ex. 9.8.3; USTC 定理 3 states the equivalence (continuous f on $[a, b]$ is invertible iff strictly monotone).

Proof. Suppose f is not strictly monotone. Then there are $u < v < w$ in I with $f(v)$ not strictly between $f(u)$ and $f(w)$ — say $f(v) \geq f(u)$ and $f(v) \geq f(w)$, the other case being symmetric. Pick c with $\max\{f(u), f(w)\} < c < f(v)$, possible unless $f(v)$ equals one of them, which injectivity forbids. The intermediate value theorem 4.23 applied on $[u, v]$ and on $[v, w]$ produces $s \in (u, v)$ and $t \in (v, w)$ with $f(s) = f(t) = c$, contradicting injectivity. $\square$

The proof is the intermediate value theorem used three times: if f were neither increasing nor decreasing, some triple $u < v < w$ would have $f(v)$ outside the interval with endpoints $f(u)$, $f(w)$, and 4.23 would then produce a repeated value on one side or the other.³¹²

S4.13 Theorem. *(Inverse functions without compactness.) Let f be continuous and strictly increasing on an interval I . Then $J = f(I)$ is an interval, $f : I \rightarrow J$ is a bijection, and $f^{-1} : J \rightarrow I$ is continuous and strictly increasing.*

SOURCE: ECNU 定理 4.8 (bk. pp. 73–74); USTC 定理 3; Tao I, Prop. 9.8.3.

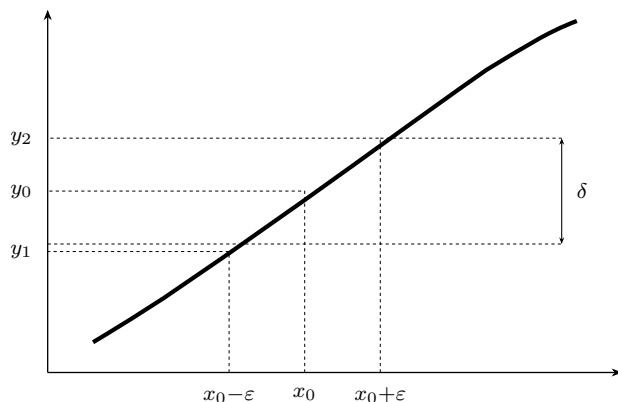
The δ that monotonicity hands you. Let $y_0 = f(x_0)$ with x_0 interior to I , and let $\varepsilon > 0$ be small enough that $[x_0 - \varepsilon, x_0 + \varepsilon] \subset I$. Put $y_1 = f(x_0 - \varepsilon)$ and $y_2 = f(x_0 + \varepsilon)$; strict increase gives $y_1 < y_0 < y_2$. Take

$$\delta = \min\{y_0 - y_1, y_2 - y_0\} > 0.$$

If $|y - y_0| < \delta$ then $y_1 < y < y_2$, and applying the increasing map f^{-1} gives $x_0 - \varepsilon < f^{-1}(y) < x_0 + \varepsilon$. Endpoints of I are handled one-sidedly.

$\square$

³¹² The converse is trivial, so this says: for continuous functions on an interval, injectivity and strict monotonicity are the same hypothesis. Rudin's Exercise 4.14 (open maps are monotone) is a relative of this; the equivalence itself he never states.



Theorem S4.13: the ϵ -interval about x_0 is carried by f onto an interval about y_0 , and δ is simply the smaller of its two halves. Monotonicity is doing all the work — it is what makes the image of an interval an interval, and what lets f^{-1} preserve the trapping. No compactness enters, so I may be open or unbounded.

Two theorems about inverses, neither containing the other

Rudin 4.17 and S4.13 are genuinely different results:

	Rudin 4.17		S4.13
domain	compact	metric	interval in $\mathbb{R}$
hypothesis	continuous bijection		continuous, strictly monotone
mechanism	compactness	(4.14, 2.35)	order + IVT
reaches	$\mathbb{R}^k$, function spaces		$(0, \infty)$, $\mathbb{R}$, half-open
misses	$(0, 1) \rightarrow (0, 1)$		anything not ordered

Example 4.21 — the circle-wrap $t \mapsto (\cos t, \sin t)$ on $[0, 2\pi)$, whose inverse is discontinuous — shows 4.17 fails without compactness; but $f(x) = x/(1-x^2)$ on $(-1, 1)$ is a continuous increasing bijection onto $\mathbb{R}$ with continuous inverse, on a domain that is not compact. Order supplies what compactness supplied.

S4.14 Theorem. (*Continuity of the elementary functions.*) For $a > 0$ the function a^x , defined for irrational x as $\sup\{a^r : r \in \mathbb{Q}, r \leq x\}$, is continuous on $\mathbb{R}$ and satisfies $a^\alpha a^\beta = a^{\alpha+\beta}$ and $(a^\alpha)^\beta = a^{\alpha\beta}$. Its inverse $\log_a$ is continuous on $(0, \infty)$ by S4.13, the power $x^\alpha = e^{\alpha \ln x}$ is continuous on $(0, \infty)$, and $\sin, \cos$ are continuous because $|\sin x - \sin y| \leq |x - y|$. Consequently every function built from these by finitely many algebraic operations, compositions, and inversions is continuous on its domain of definition.

SOURCE: ECNU 定理 4.10–4.13 (bk. pp. 78–80); USTC 定理 4 (bk. pp. 45–46).

Proof sketch. Continuity of a^x at 0 follows from $a^{1/n} \rightarrow 1$ (Rudin 3.20(b)) and monotonicity; the identity $a^{x+h} = a^x a^h$ then transports

continuity from 0 to every point, since $|a^{x+h} - a^x| = a^x|a^h - 1|$. The laws $a^\alpha a^\beta = a^{\alpha+\beta}$ and $(a^\alpha)^\beta = a^{\alpha\beta}$ hold for rational exponents by S1.13 and extend to real exponents by density and continuity (Rudin's Exercise 4.4). Continuity of $\log_a$ is S4.13, that of $x^\alpha = e^{\alpha \ln x}$ is composition, and $|\sin x - \sin y| \leq |x - y|$ gives the trigonometric cases. Closure under sums, products, quotients and composition is 4.9 and 4.7. $\square$

The same destination by two routes

Rudin builds these functions in Chapter 8 from power series, and their continuity is then a corollary of the theorem that a power series is continuous inside its disc of convergence (8.1). The Chinese texts build them here, from the supremum definition of a^x that Rudin sets as Exercises 1.6–1.7, and get continuity by hand.

The trade is visible: Rudin's route is shorter and yields differentiability and the addition formulas at the same stroke, but it defers every concrete function by four chapters; the elementary route has functions available immediately but must prove each property separately. Neither is more rigorous. The Chinese texts' order is why their students compute limits in week three and Rudin's readers do not.

NOT IN RUDIN

SUPPLEMENT E — UNIFORM CONTINUITY: THE WORKING TOOLKIT

WHY THIS SUPPLEMENT BELONGS TO CHAPTER 4

What Rudin does. Definition 4.18 gives uniform continuity, 4.19 proves that continuity on a compact set implies it, and 4.20 exhibits the standard failures. Everything else about the notion is set as Exercises 4.8–4.13.

The gap. Those exercises contain the working theory — uniform continuity preserves Cauchy sequences, preserves boundedness, permits extension to the closure — and as exercises they are unavailable to a reader who has not done them, and unquotable to one who has. There is also no *criterion*: to show a function is not uniformly continuous, Rudin negates the definition by hand each time.

Where the gap bites. In Chapter 8, where b^x is defined on $\mathbb{Q}$ and must be extended to $\mathbb{R}$ — an appeal to Exercise 4.13; and at 7.16 and beyond, where uniform continuity is used as a known tool.

What this section supplies. The sequential criterion, which turns every non-example into a two-line computation; the gluing

lemma; the extension theorem in its general form; the algebra of what preserves uniform continuity; and the Lebesgue number lemma, which is what Rudin's own proof of 4.19 manufactures without naming.

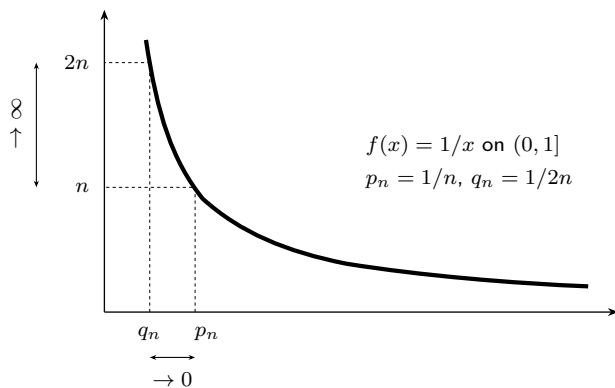
S4.15 Theorem. (Sequential criterion.) $f : E \rightarrow Y$ is uniformly continuous on E if and only if for every pair of sequences $(p_n), (q_n)$ in E with $d(p_n, q_n) \rightarrow 0$ one has $d(f(p_n), f(q_n)) \rightarrow 0$.

SOURCE: Tao I, Prop. 9.9.8 (via “equivalent sequences”); ECNU 例 10 (bk. p. 75).

Proof. If f is uniformly continuous and $d(p_n, q_n) \rightarrow 0$, then given $\varepsilon > 0$ take δ from the definition; eventually $d(p_n, q_n) < \delta$, so eventually $d(f(p_n), f(q_n)) < \varepsilon$.

Conversely, suppose f is not uniformly continuous. Then some $\varepsilon > 0$ resists every δ ; taking $\delta = 1/n$ produces points p_n, q_n with $d(p_n, q_n) < 1/n$ but $d(f(p_n), f(q_n)) \geq \varepsilon$. This pair of sequences violates the stated condition. $\square$

The value is in the negation: to prove a function *not* uniformly continuous, exhibit one pair of sequences that close up while their images do not.³¹³



313 Compare Rudin's own method at 4.20: he argues directly from the ε - δ negation each time. The criterion packages that argument once. Note the parallel it draws: continuity preserves convergent sequences (4.2), uniform continuity preserves pairs that close up.

The criterion applied to $1/x$: the two arguments close up ($p_n - q_n \rightarrow 0$) while their values pull apart (n versus $2n$). One pair of sequences settles it — no ε - δ negation needed. For x^2 on $\mathbb{R}$ take $p_n = n, q_n = n + 1/n$: the gap tends to 0 and the images differ by $2 + 1/n^2$.

S4.16 Theorem. (Gluing.) If $a < c < b$ and f is uniformly continuous on $[a, c]$ and on $[c, b]$, then f is uniformly continuous on $[a, b]$. The same holds for two intervals sharing one endpoint, bounded or not.

SOURCE: ECNU 例 12 (bk. p. 76).

Proof. Given $\varepsilon > 0$, take δ_1 and δ_2 for $\varepsilon/2$ on $[a, c]$ and on $[c, b]$, and put $\delta = \min\{\delta_1, \delta_2\}$. Let $|x - y| < \delta$. If x, y lie in the same piece there is nothing to prove. Otherwise $x \leq c \leq y$, so $|x - c| < \delta$ and $|c - y| < \delta$, and

$$|f(x) - f(y)| \leq |f(x) - f(c)| + |f(c) - f(y)| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

□

WHY GLUING NEEDS AN ARGUMENT AT ALL, AND WHAT IT BUYS

Uniform continuity is not a local property, so the union of two sets on which f is uniformly continuous is not automatically another. The proof works because the two pieces meet: given ε , take δ_1, δ_2 for $\varepsilon/2$ on the two pieces and set $\delta = \min\{\delta_1, \delta_2\}$. If $|x - y| < \delta$ with $x \leq c \leq y$, then $|x - c| < \delta$ and $|c - y| < \delta$, so

$$|f(x) - f(y)| \leq |f(x) - f(c)| + |f(c) - f(y)| < \varepsilon.$$

The shared point c is the whole mechanism, and it is why two intervals with a *gap* between them cannot be glued.

The payoff is a standard trick: to prove f uniformly continuous on $[0, \infty)$, prove it on the compact $[0, 1]$ by 4.19 and on $[1, \infty)$ by a direct estimate. For $f(x) = \sqrt{x}$ the second is $|\sqrt{x} - \sqrt{y}| = |x - y|/(\sqrt{x} + \sqrt{y}) \leq \frac{1}{2}|x - y|$. Rudin's Exercise 4.8 asks for the boundedness half of this story and never supplies the gluing tool.

S4.17 Theorem. (*Extension to the closure.*) Let $f : E \rightarrow Y$ be uniformly continuous with Y complete. Then $\lim_{x \rightarrow p} f(x)$ exists at every $p \in \overline{E}$, and f extends uniquely to a uniformly continuous function on $\overline{E}$.

SOURCE: Tao I, Cor. 9.9.14 and Prop. 9.9.12. Rudin poses the special case as Exercise 4.13.

Proof. Let $p \in \overline{E}$ and let (p_n) be a sequence in E with $p_n \rightarrow p$. It is Cauchy, and a uniformly continuous map sends Cauchy sequences to Cauchy sequences: given ε , take δ from uniform continuity and then N with $d(p_n, p_m) < \delta$ for $n, m \geq N$. Completeness of Y makes $(f(p_n))$ converge. Interleaving two such sequences shows the limit does not depend on the choice, so $\lim_{x \rightarrow p} f(x)$ exists by 4.2.

Defining $\tilde{f}(p)$ to be that limit extends f to $\overline{E}$; the extension is uniformly continuous because the defining inequality passes to limits, and it is unique because two continuous maps agreeing on the dense set E agree on $\overline{E}$ (Rudin's Exercise 4.4). □

HOW THE PROOF FLOWS

1. **Choose the method.** A *value* has to be produced at a point where f is not defined, so something must manufacture it. The only manufacturing device available is completeness of the target, and the only way to reach it is through Cauchy sequences.
2. **Convert the hypothesis into a Cauchy statement.** Uniform continuity says one δ works everywhere, which is exactly what lets a Cauchy sequence in the domain be pushed to a Cauchy sequence in the target. Ordinary continuity cannot do this — the δ would move with the point.

3. **Spend completeness.** Cauchy becomes convergent. This is the only step that uses anything about Y .
4. **Remove the dependence on the sequence.** Interleaving two approximating sequences is the standard device; it converts “every sequence gives a limit” into “the limit exists,” which is 4.2.
5. **Audit.** Uniform continuity is spent once, in step 2; completeness once, in step 3; density of E in $\overline{E}$ once, to know the approximating sequences exist at all. Drop uniformity and the theorem fails — $\sin(1/x)$ on $(0, 1)$ is continuous, sends a Cauchy sequence to a divergent one, and extends to no value at 0.

This is the theorem Rudin’s Exercise 4.13 is a special case of, and the reason uniform continuity matters at all in Chapter 8: it is what lets a function defined on $\mathbb{Q}$ — like b^x for rational x in Exercise 1.6 — become a function on $\mathbb{R}$.³¹⁴

S4.18 Theorem. (*What preserves uniform continuity.*) If f, g are uniformly continuous on E then so are $f+g$, $f-g$, cf , $\max(f, g)$, $\min(f, g)$, and $g \circ f$ where defined. The product fg need not be, but is if f and g are both bounded.

SOURCE: Tao I, Ex. 9.9.6; Tao II, Ex. 2.3.4–2.3.6.

Proof. For $f+g$ take δ to be the smaller of the two δ ’s for $\varepsilon/2$; for cf use $\varepsilon/|c|$; for $g \circ f$ feed the δ of g into f as its ε . The identity $\max(f, g) = \frac{1}{2}(f+g+|f-g|)$ and $||u|-|v|| \leq |u-v|$ handle max and min.

For products with both factors bounded by M ,

$$\begin{aligned} |f(p)g(p) - f(q)g(q)| &\leq |f(p)| |g(p) - g(q)| + |g(q)| |f(p) - f(q)| \\ &\leq M(|g(p) - g(q)| + |f(p) - f(q)|), \end{aligned}$$

so δ for $\varepsilon/2M$ in each factor works. Without boundedness the conclusion fails: $f(x) = g(x) = x$ are uniformly continuous on $\mathbb{R}^1$ and $fg = x^2$ is not. $\square$

The counterexample is the smallest one available: $f(x) = g(x) = x$ is uniformly continuous on $\mathbb{R}$, and $fg = x^2$ is not — the same function Rudin uses at 4.20 for a different purpose.³¹⁵

S4.19 Theorem. (*Lebesgue number.*) Let K be a compact metric space and $\{U_\alpha\}$ an open cover of K . Then there is a $\delta > 0$ — a Lebesgue number of the cover — such that every ball $B(x, \delta)$ with $x \in K$ lies inside a single U_α .

SOURCE: Yu Pin, Prop. 66 (p. 112) and the metric version p. 114.

Proof. Suppose no such δ exists. Then for each n there is $x_n \in K$ with $B(x_n, 1/n)$ contained in no single U_α . Compactness gives a subsequence

314 Chain of reasoning: uniformly continuous maps send Cauchy sequences to Cauchy sequences (Rudin’s Exercise 4.11), the target is complete, so the images converge; S4.1 then upgrades “every sequence works” to “the limit exists.” Plain continuity is not enough: $1/x$ on $(0, 1)$ is continuous, sends the Cauchy sequence $(1/n)$ to an unbounded one, and does not extend to 0.

315 The boundedness repair is the usual splitting: $|fg(p) - fg(q)| \leq |f(p)| |g(p) - g(q)| + |g(q)| |f(p) - f(q)|$, and if both factors are bounded by M each term is small. Compare 3.3(c), where exactly this decomposition proves the product rule for limits — the strips-and-corner picture there is the same algebra.

$x_{n_k} \rightarrow x$, and x lies in some U_β , which contains a ball $B(x, r)$. For k large enough that $d(x_{n_k}, x) < r/2$ and $1/n_k < r/2$, the triangle inequality gives $B(x_{n_k}, 1/n_k) \subset B(x, r) \subset U_\beta$ — contradicting the choice of x_{n_k} . $\square$

HOW THE PROOF FLOWS

1. **Choose the method.** The claim is that a *single* scale works everywhere, and no candidate scale is in sight. So assume none exists and let the failure build a sequence — the standard move when a uniform quantity is wanted and only pointwise data is given.
2. **Let the counterexamples index themselves.** Failure at scale $1/n$ produces a point x_n . The sequence is free: it is the negation of the conclusion, written out.
3. **Spend compactness.** A convergent subsequence appears, and its limit lies in some cover element — which finally supplies a ball of a definite radius r , the first fixed scale in the argument.
4. **Collide the two scales.** For large k the tiny ball around x_{n_k} sits inside $B(x, r)$, hence inside one U_β — contradicting how x_{n_k} was chosen. Note both smallnesses are needed: $d(x_{n_k}, x)$ small and $1/n_k$ small.
5. **Audit.** Compactness is spent once, in step 3; openness of the cover once, also in step 3, to get r . The lemma is what Rudin's own 4.19 constructs by hand — his halving of $\phi(p)$ and taking the minimum over a finite subcover produces exactly this δ .

THE LEMMA HIDING INSIDE RUDIN'S PROOF OF 4.19

Rudin's uniform continuity proof does not name this, but it manufactures it: he takes the balls $J(p)$ of radius $\frac{1}{2}\phi(p)$, extracts a finite subcover, and sets $\delta = \frac{1}{2} \min \phi(p_i)$. That δ is a Lebesgue number of the cover $\{J(p)\}$, and the halving trick is exactly the device that produces one.

Isolate the lemma and 4.19 becomes three lines. Given $\varepsilon > 0$, cover K by the sets $U_x = f^{-1}(B(f(x), \varepsilon/2))$, open by 4.8. Let δ be a Lebesgue number. If $d(p, q) < \delta$ then both lie in one U_x , so $f(p)$ and $f(q)$ are both within $\varepsilon/2$ of $f(x)$, hence within ε of each other. The gain is not brevity but reuse: the same lemma proves that compact and sequentially compact agree, and it is the standard tool whenever a *single* scale must be extracted from a cover. Yu Pin runs Heine–Borel itself off it. Keep it as the answer to “what does compactness actually hand me?” — a uniform scale.

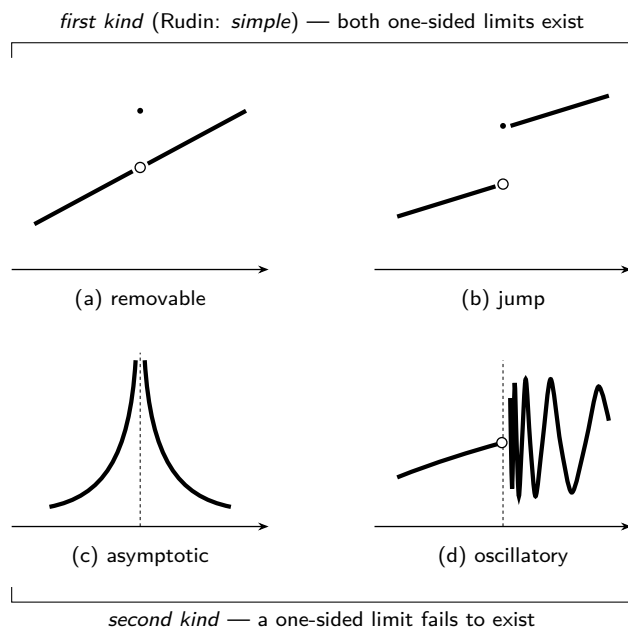
SUPPLEMENT F — FINER STRUCTURE: DISCONTINUITIES AND COMPONENTS

S4.20 Definition. (Four kinds of discontinuity.) Let f be discontinuous at x . The discontinuity is

- (a) **removable** if $f(x+)$ and $f(x-)$ exist and are equal (necessarily $\neq f(x)$);
- (b) **a jump** if both exist and differ;
- (c) **asymptotic** if a one-sided limit fails because $|f|$ is unbounded near x on that side;
- (d) **oscillatory** if a one-sided limit fails while f stays bounded there.

SOURCE: Tao I, Remark 9.5.4; USTC 可去间断点/跳跃点, bk. pp. 42–43.

Rudin's Definition 4.26 stops at two boxes: (a) and (b) together are his *simple* discontinuities, and (c) and (d) together are his *second kind*.³¹⁶



316 The refinement earns its keep twice. Removable discontinuities are exactly the ones a change of a single value repairs, so they are not really properties of the function but of its bookkeeping. And the asymptotic/oscillatory split is the difference between $1/x$ and $\sin(1/x)$ at 0 — two failures that behave nothing alike, which Rudin's 4.27(d) illustrates without naming.

Rudin's two boxes opened into four. His 4.27(c) is case (b) and his 4.27(d) is case (d); cases (a) and (c) he never names. The classification is per side — a function may be removable from the left and asymptotic from the right — so these are the four pure types, not a partition of all discontinuities.

S4.21 Definition. $E \subset X$ is path-connected if for all $p, q \in E$ there is a continuous $\gamma : [0, 1] \rightarrow E$ with $\gamma(0) = p$, $\gamma(1) = q$. Every path-connected set is connected; the converse is false.

SOURCE: Tao II, Ex. 2.4.7 and §2.4.

The counterexample is Rudin's own $\sin(1/x)$

That path-connected implies connected is Rudin 4.22 applied to γ : if $E = A \cup B$ were a separation, the sets $\gamma^{-1}(A)$ and $\gamma^{-1}(B)$ would separate $[0, 1]$, which 2.47 forbids.

For the converse, take the closure of

$$S = \{(x, \sin(1/x)) : 0 < x \leq 1\},$$

which adds the segment $\{0\} \times [-1, 1]$. It is connected — S is connected as a continuous image of $(0, 1]$, and adding limit points of a connected set preserves connectedness — but no path joins a point of the segment to a point of S : a path would have to traverse infinitely much oscillation in finite time, and one shows its first coordinate cannot become positive. So Rudin's favourite pathological function, introduced at 4.27(d) purely as a discontinuity, turns out to also separate two notions of connectedness he never distinguishes.

S4.22 Theorem. (*Components.*) Define $p \sim q$ if some connected subset of X contains both. This is an equivalence relation; its classes, the connected components of X , are connected and closed, and they partition X .

SOURCE: Tao II, Ex. 2.4.6, 2.4.8, 2.4.9.

Proof. Reflexivity holds because $\{p\}$ is connected, and symmetry is trivial. For transitivity, if A is connected containing p, q and B is connected containing q, r , then $A \cup B$ is connected — a separation of $A \cup B$ would separate one of A, B , since q lies in both — and it contains p and r .

The class of p is the union of all connected sets containing p ; that union is connected by the same argument, so each class is connected and is maximal among connected sets meeting it. Finally the closure of a connected set is connected, so a class equals its own closure and is therefore closed. $\square$

Transitivity holds because a union of connected sets with a point in common is connected, and closedness because the closure of a connected set is connected. ³¹⁷

S4.23 Theorem. (*Convexity, five ways.*) For f on an interval I the following are equivalent: (i) $f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y)$ for $\lambda \in (0, 1)$; (ii) for $x_1 < x_2 < x_3$ in I ,

$$\frac{f(x_2) - f(x_1)}{x_2 - x_1} \leq \frac{f(x_3) - f(x_1)}{x_3 - x_1} \leq \frac{f(x_3) - f(x_2)}{x_3 - x_2};$$

(iii) for each fixed a , the slope $x \mapsto (f(x) - f(a))/(x - a)$ is increasing; (iv) the 3×3 determinant with rows $(1, 1, 1)$, (x_1, x_2, x_3) , $(f(x_1), f(x_2), f(x_3))$ is ≥ 0 ; (v) the epigraph $\{(x, y) : y \geq f(x)\}$ is a convex subset of $\mathbb{R}^2$.

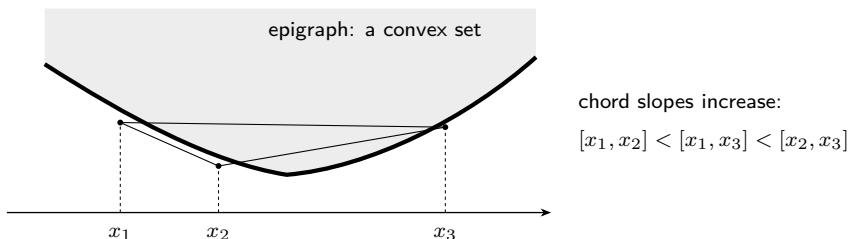
SOURCE: Yu Pin, Thm 106 (pp. 175–177); ECNU 凸函数 §6.5 gives (i)–(iii).

Proof of (i) $\Leftrightarrow$ (ii). Given $x_1 < x_2 < x_3$, write $x_2 = \lambda x_1 + (1 - \lambda)x_3$ with $\lambda = (x_3 - x_2)/(x_3 - x_1) \in (0, 1)$. Then (i) says $f(x_2) \leq \lambda f(x_1) + (1 - \lambda)f(x_3)$, and subtracting $f(x_1)$ from both sides and dividing by $x_2 - x_1 = (1 - \lambda)(x_3 - x_1) > 0$ turns this into the first inequality of (ii);

³¹⁷ Rudin never decomposes a space this way, though 2.47 is the statement that the components of a subset of $\mathbb{R}$ are intervals, and Exercise 2.29 — every open set in $\mathbb{R}$ is a countable union of disjoint segments — is the component decomposition of an open set. Naming the notion makes both results one result.

subtracting instead from $f(x_3)$ and dividing by $x_3 - x_2$ gives the second. Each step is reversible, so (ii) implies (i).

Statement (iii) is (ii) with x_1 held fixed, (iv) is (ii) cleared of denominators, and (v) says that the chord joining two points of the graph lies in the region above the graph — which is (i) verbatim. $\square$



The chord slopes over $[x_1, x_2]$, $[x_1, x_3]$, $[x_2, x_3]$ increase from left to right — statement (ii), which is (i) rewritten with $\lambda = (x_3 - x_2)/(x_3 - x_1)$. Statement (v) says the same thing without picking any points at all: everything on or above the graph is a convex set, which is what “the chord lies above the graph” means.

Why five formulations instead of one

Rudin’s Exercises 4.23–4.24 give the definition and ask for continuity on the open interval plus the midpoint characterisation. The equivalences above are what make those exercises routine, and each is the right tool for a different job. Form (ii) is the one that proves continuity: the slopes from a fixed interior point are bounded on both sides, so f is Lipschitz on any compact subinterval of I° — which incidentally shows why Rudin’s exercise is stated on the *open* interval, since an endpoint can carry a jump upward and stay convex.

Form (v) is the conceptual one: it converts a statement about a function into a statement about a *set*, so that convexity of functions and convexity of sets stop being two ideas. Form (iv) is the computational one, and form (iii) is what one differentiates in Chapter 5 to get “ f' increasing” (Theorem 5.11’s setting).

CONTINUITY ALONG LINES, SEPARATELY AND JOINTLY

Three conditions on $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ at a point, each strictly weaker than the next:

- (a) continuous in each variable separately (two lines);
- (b) continuous along every straight line through the point;
- (c) jointly continuous.

The function $xy/(x^2 + y^2)$, with value 0 at the origin, separates (a) from (b): it vanishes on both axes but equals $m/(1 + m^2)$ along $y = mx$, so it is separately continuous and not even constant along lines. Rudin’s Exercise 4.7 does the harder separation of (b) from (c): $xy^2/(x^2 + y^4)$ is continuous along every line yet approaches $\frac{1}{2}$ along the parabola $x = y^2$.

Keeping the hierarchy straight matters in Chapter 9, where partial derivatives are a statement of type (a) — which is why 9.21 needs continuity of the partials, not merely their existence.

WHAT THE SUPPLEMENTS CHANGED

Take away four things.

Criteria produce limits; Rudin's theorems transport them. S4.1 and S4.17 are the existence tools, and both need completeness of the target — the same property that made Chapter 1 necessary.

o , O , $\sim$ are the working language of limits, and they are legal in factors, never in summands (S4.7). Rudin does without them because he never computes; you will not have that luxury.

Compactness is not the only source of good behaviour. Order does some of the same work: monotone functions have continuous inverses on any interval (S4.13), where Rudin's 4.17 needs compactness. Neither theorem contains the other.

Rudin's classifications are coarse on purpose. Two kinds of discontinuity become four (S4.20), connectedness sprouts a path-connected refinement (S4.21), and continuity at a point becomes a *number*, $\omega_f(p)$, which turns “where is f continuous?” into a question with a theorem for an answer (S4.5).

NOT IN RUDIN

APPENDIX EXERCISES

HOW THESE DIFFER FROM RUDIN'S TWENTY-SIX

Rudin's own twenty-six exercises, which build *his* machinery — the uniform-continuity chain, the ρ_E toolbox, the counterexample gallery — end the chapter proper. These twenty-five come from the four comparison texts and fill the other half of the subject.

Group I (1–6): limits and asymptotics. Computational problems in the $o/\sim$ language of Supplement B, plus three rigidity problems — a periodic function that dies at infinity is dead everywhere, and its relatives. Nothing in Rudin's set resembles these.

Group II (7–11): monotonicity, inverses, functional equations. Where order replaces compactness. Exercise 9 (Cauchy's functional equation) and 10 are the standard route from “continuous at one point” to “continuous, and linear.”

Group III (12–16): what the intermediate value theorem really gives. The discrete mean value property, the universal chord theorem, and a contraction mapping five chapters before Rudin needs one (9.23).

Group IV (17–21): uniform continuity as a working tool. Prove the criteria of Supplement E, then use them. Compare with Rudin's 4.8–4.13, which prove the same facts one at a time by hand.

Group V (22–25): structure. Oscillation, prescribed discontinuity sets, path-connectedness, and one spectacular payoff of “continuous plus dense determines everything” — a proof of the Cayley–Hamilton theorem that uses no algebra beyond the diagonalisable case.

Marked ★: harder, or requiring a result from a later chapter.

S4.1. Let f be periodic on $\mathbb{R}^1$ with $\lim_{x \rightarrow +\infty} f(x) = 0$. Prove that f is identically zero. No continuity is assumed.³¹⁸

SOURCE: ECNU 习题 3.3 #7; USTC Ch. 1 suppl. #13.

S4.2. Suppose f is defined on $(0, +\infty)$, satisfies $f(x^2) = f(x)$ for all x , and $\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow +\infty} f(x) = f(1)$. Prove that f is constant.³¹⁹

SOURCE: ECNU 总练习题 3 #13.

S4.3. ★ Let f be defined on $(a, +\infty)$ and bounded on every finite interval (a, b) , and suppose $\lim_{x \rightarrow +\infty} [f(x+1) - f(x)] = A$. Prove that $\lim_{x \rightarrow +\infty} f(x)/x = A$.³²⁰

SOURCE: ECNU 总练习题 3 #14; Yu Pin, HW5 A9* (p. 127), where it is attributed to Cauchy rather than to Stolz.

S4.4. Show that $f(x) = x \sin x$ is unbounded on $(0, +\infty)$ but is *not* an infinitely large quantity as $x \rightarrow +\infty$; that is, $\lim_{x \rightarrow +\infty} f(x) \neq \infty$.³²¹

SOURCE: ECNU bk. p. 60; USTC Ex. 1.2.12.

S4.5. Using equivalent infinitesimals, evaluate $\lim_{x \rightarrow 0} \frac{\tan x - \sin x}{\sin^3 x}$ and $\lim_{x \rightarrow 0} \frac{\sqrt{1+x^2} - 1}{1 - \cos x}$. Then explain precisely why substituting $\tan x \sim x$ and $\sin x \sim x$ into the numerator of the first gives the wrong answer 0.³²²

SOURCE: ECNU 例 2 and note, bk. p. 59.

318 Fix x and push it to infinity by its own period: $f(x) = f(x + nT)$ for every n , and the right side tends to 0. The whole content is that periodicity lets one evaluate a function arbitrarily far out without moving it. Compare Rudin's Exercise 4.4: there density does the transporting, here periodicity does.

319 Iterate the hypothesis backwards: $f(x) = f(x^{1/2}) = f(x^{1/4}) = \dots = f(x^{1/2^n})$, and $x^{1/2^n} \rightarrow 1$ for every $x > 0$. The two limit hypotheses are there only to handle x where the iteration escapes. This is the shape of every functional-equation rigidity argument: find the orbit, find its limit point, evaluate there.

320 The continuous Stolz–Cesàro theorem. Write $f(x) = f(x_0) + \sum_{k < n} [f(x_0 + k + 1) - f(x_0 + k)]$ where $x = x_0 + n$ with x_0 in a fixed unit window, then average: the increments are eventually near A , and the finitely many early ones are diluted by $1/x$. It is 3.20's averaging idea run on a continuous variable.

321 Two sequences settle it: $x_n = 2n\pi$ gives $f = 0$, while $x_n = 2n\pi + \pi/2$ gives $f \rightarrow +\infty$. The point of Supplement B's S4.8 — “unbounded” is a statement about the range, “tends to infinity” is a statement about the tail, and only the second forbids returning.

322 Answers $\frac{1}{2}$ and 1. The explanation is the content of the pitfall at S4.7: each equivalence is accurate to first order, the subtraction annihilates exactly the first order, and what survives lives at third order where neither equivalence says anything. Factor first —
 $\tan x - \sin x = \sin x(1 - \cos x)/\cos x$ —
 and every factor may then be replaced.

323 Long division is the whole computation; the definition check is the observation that the distance to $y = ax + b$ is $|f(x) - ax - b|/\sqrt{1 + a^2}$, a fixed multiple of the vertical gap. Worth doing once to see that “vertical gap” and “perpendicular distance” differ only by that constant — which is why the limits defining a and b may be taken vertically.

324 Suppose not: then some $u < v < w$ has $f(v)$ outside the interval with endpoints $f(u)$ and $f(w)$. Apply the intermediate value theorem on $[u, v]$ and on $[v, w]$ to produce two points with the same value. The deduction is S4.13. Contrast Rudin 4.17, which gets the same conclusion from compactness and works in any metric space — neither theorem contains the other.

325 Modify Dirichlet: $f = 1$ on the rationals, -1 elsewhere. Then $|f| \equiv 1$. The lesson is that $|\cdot|$ destroys information — the implication “ f continuous $\Rightarrow |f|$ continuous” has no converse, and every proof that passes to absolute values gives something away.

326 Cauchy’s functional equation. Continuity everywhere follows because $f(x + h) - f(x) = f(h)$ — the equation makes the increment independent of position, so continuity at one point is continuity everywhere. Then the equation forces $f(q) = f(1)q$ for rational q by induction, and Rudin’s Exercise 4.4 (agreement on a dense set) finishes it. Without some regularity the conclusion is false, but the counterexamples need a Hamel basis and hence the axiom of choice.

327 Sharper than S4.2: continuity at exactly two points suffices. Note f is even, so it is enough to work on $x \geq 0$; for $x > 0$ the iteration $x^{1/2^n} \rightarrow 1$ uses continuity at 1, and $f(0) = \lim_{x \rightarrow 0} f(x)$ uses continuity at 0. Observe which points the iteration funnels into — those are exactly the points where continuity is required.

S4.6. Find all asymptotes of the curve $y = x^3/(x^2 + 2x - 3)$, and prove that your oblique asymptote satisfies the definition (the distance from $(x, f(x))$ to the line tends to 0).³²³

SOURCE: ECNU 例 6, bk. pp. 61–62.

S4.7. Let f be continuous and one-to-one on an interval I . Prove that f is strictly monotone on I . Deduce that a continuous injection of $[a, b]$ into $\mathbb{R}^1$ has a continuous inverse, without invoking compactness.³²⁴

SOURCE: Tao I, Ex. 9.8.3; USTC 定理 3; Yu Pin, Thm 54.

S4.8. Prove that there exists a function $f : \mathbb{R}^1 \rightarrow \mathbb{R}^1$ that is discontinuous at every point while $|f|$ is continuous at every point.³²⁵

SOURCE: USTC Ex. 2.1.5.

S4.9. Suppose $f : \mathbb{R}^1 \rightarrow \mathbb{R}^1$ satisfies $f(x + y) = f(x) + f(y)$ for all x, y , and is continuous at the single point 0. Prove that f is continuous on all of $\mathbb{R}^1$ and that $f(x) = f(1)x$.³²⁶

SOURCE: ECNU 总练习题 4 #9.

S4.10. Let f be defined on $\mathbb{R}^1$, continuous at the two points 0 and 1 only (as far as is assumed), and satisfy $f(x^2) = f(x)$ for all x . Prove that f is constant.³²⁷

SOURCE: ECNU 总练习题 4 #10.

S4.11. Determine, with proof, which of the following are possible for a continuous real function: domain $[0, 1]$ with range $(0, \infty)$; domain $[0, 1]$ with range $(0, 1)$; domain $[0, 1]$ with range $[0, 1] \cup [2, 4]$; domain $(0, 1)$ with range $[2, \infty)$.³²⁸

SOURCE: USTC Ex. 2.2.10.

S4.12. Let f be continuous on $[a, b]$ and let $x_1, \dots, x_n \in [a, b]$. Prove that there is a $\xi \in [a, b]$ with

$$f(\xi) = \frac{f(x_1) + \dots + f(x_n)}{n}.$$

More generally, prove the same for any weighted average $\sum q_i f(x_i)$ with $q_i > 0$ and $\sum q_i = 1$.³²⁹

SOURCE: ECNU 习题 4.2 #19; USTC Ex. 2.2.7.

S4.13. (Universal chord theorem.) Let f be continuous on $[0, 1]$ with $f(0) = f(1)$. Prove that for every positive integer n there is a $\xi \in [0, 1 - 1/n]$ with

$$f\left(\xi + \frac{1}{n}\right) = f(\xi).$$

Show by example that the conclusion can fail for a chord length that is not of the form $1/n$.³³⁰

SOURCE: ECNU 总练习题 4 #8; USTC Ch. 2 suppl. #5.

S4.14. Prove that for every positive integer n the equation

$$x^n + x^{n-1} + \cdots + x = 1$$

has exactly one positive root x_n ; prove that (x_n) converges, and find its limit.³³¹

SOURCE: USTC Ch. 2 suppl. #9.

S4.15. (*Contraction mapping on an interval.*) Let f map $[a, b]$ into $[a, b]$ and suppose there is a constant k with $0 < k < 1$ such that

$$|f(x) - f(y)| \leq k|x - y| \quad (x, y \in [a, b]).$$

Prove that f has a unique fixed point x_0 , and that for every starting value $x_1 \in [a, b]$ the sequence $x_{n+1} = f(x_n)$ converges to x_0 .³³²

SOURCE: USTC Ch. 2 suppl. #8.

S4.16. $\star$ (*Croft.*) Let f be continuous on $\mathbb{R}^1$ and suppose that for every $\delta > 0$,

$$\lim_{n \rightarrow \infty} f(n\delta) = 0.$$

Prove that $\lim_{x \rightarrow +\infty} f(x) = 0$.³³³

SOURCE: Yu Pin, HW5 problem G (p. 129).

S4.17. Prove the sequential criterion S4.15: f is uniformly continuous on E if and only if $d(f(p_n), f(q_n)) \rightarrow 0$ for every pair of sequences in E with $d(p_n, q_n) \rightarrow 0$. Then use it to show that $\sin(1/x)$ is not uniformly continuous on $(0, 1)$ and that x^2 is not uniformly continuous on $\mathbb{R}^1$.³³⁴

SOURCE: Tao I, Prop. 9.9.8; ECNU 例 10–11.

S4.18. Prove the gluing lemma S4.16, and use it with Theorem 4.19 to show that $\sqrt{x}$ is uniformly continuous on $[0, +\infty)$. Show likewise that $\cos \sqrt{x}$ is uniformly continuous there.³³⁵

SOURCE: ECNU 例 12 and 习题 4.2 #12, #20.

S4.19. Let f be continuous on $[a, +\infty)$ and suppose $\lim_{x \rightarrow +\infty} f(x)$ exists and is finite. Prove that f is bounded on $[a, +\infty)$ and uniformly continuous there. Must f attain a maximum? ³³⁶

SOURCE: ECNU 习题 4.2 #6, #16.

328 Four probes of the same two theorems. The first three are impossible — 4.15 makes the image compact, hence closed and bounded, and 4.22 makes it connected, hence an interval. The fourth is possible; construct one. A good check on whether you know what compactness and connectedness each contribute.

329 The average lies between $\min_i f(x_i)$ and $\max_i f(x_i)$, both of which are attained values of f ; now apply the intermediate value theorem between the two points where they are attained. This is the discrete ancestor of the mean value theorem for integrals (6.x) and of the notion of a barycentre.

330 Put $F(x) = f(x + 1/n) - f(x)$ on $[0, 1 - 1/n]$ and sum: $F(0) + F(1/n) + \cdots + F((n-1)/n) = f(1) - f(0) = 0$. A finite list of reals summing to zero either is all zeros or contains both signs, so F has a zero by 4.23. The telescoping sum is the whole idea, and it is why only reciprocal lengths work.

331 Existence and uniqueness from 4.23 plus strict increase of the left side on $(0, \infty)$. For the limit, note $x_n \in (\frac{1}{2}, 1)$ and that the sum telescopes: $x(1 - x^n)/(1 - x) = 1$. Show (x_n) is decreasing and bounded below, then pass to the limit in the closed form to get $\frac{1}{2}$. A clean synthesis of Chapters 3 and 4.

332 Uniqueness is immediate from the inequality. For existence, do not use 4.23 — use the iteration: $|x_{n+1} - x_n| \leq k^{n-1}|x_2 - x_1|$ makes (x_n) Cauchy by comparison with a geometric series (3.26), and 3.11 gives the limit, which is a fixed point by continuity. This is Rudin's 9.23 in one dimension, and the proof is identical — worth doing now so that 9.23 is recognition rather than novelty. Compare Rudin's Exercise 4.14, which gets a fixed point from the intermediate value theorem alone but no iteration.

333 Surprisingly hard, and the tool is Baire (Rudin's Exercise 3.22). Suppose the conclusion fails; build closed sets $F_N = \{\delta \geq 1 : |f(n\delta)| \leq \varepsilon \text{ for all } n \geq N\}$, whose union is $[1, \infty)$, so one has interior; then the dilates $\{n\delta : \delta \in I, n \geq N\}$ of an interval I eventually cover a half-line. The moral: a hypothesis quantified over every δ is a category hypothesis in disguise.

334 The forward direction is a substitution. For the converse, negate uniform continuity: some ε has, for each $\delta = 1/n$, a pair p_n, q_n within $1/n$ whose images are ε apart — the failing pair builds itself. For the examples take $1/(2n\pi)$ against $1/(2n\pi + \pi/2)$, and n against $n + 1/n$.

335 Split at 1: on $[0, 1]$ use compactness, on $[1, \infty)$ use $|\sqrt{x} - \sqrt{y}| = |x - y|/(\sqrt{x} + \sqrt{y}) \leq \frac{1}{2}|x - y|$. Note that $\sqrt{x}$ is uniformly continuous although its derivative is unbounded near 0 — so “bounded derivative” is sufficient for uniform continuity but far from necessary.

336 Both halves are the gluing pattern: the limit hypothesis makes f nearly constant past some M , and $[a, M]$ is compact. So a hypothesis at infinity substitutes for compactness of the domain. The last question is no: $f(x) = 1 - 1/x$ on $[1, \infty)$ has supremum 1, never attained.

337 Split the product:
 $|f(p)g(p) - f(q)g(q)| \leq |f(p)| |g(p) - g(q)| + |g(q)| |f(p) - f(q)|$
 — the identical decomposition as at 3.3(c). For the counterexample $f = g = x$ on $\mathbb{R}^1$. Sum, max and composition all survive with no extra hypotheses.

338 Uniform continuity sends Cauchy sequences to Cauchy sequences (Rudin's Exercise 4.11), completeness converts those to limits, and S4.1 promotes “every sequence converges” to “the limit exists.” Uniqueness is Rudin's Exercise 4.4. The example is merely continuous, not uniformly — by S4.15 with p_n, q_n as in exercise 17.

S4.20. Prove that if f and g are uniformly continuous and *bounded* on E , then fg is uniformly continuous on E ; and show by example that boundedness cannot be dropped. Which of $f+g$, $\max(f, g)$, $g \circ f$ are uniformly continuous without extra hypotheses?³³⁷

SOURCE: Tao II, Ex. 2.3.4–2.3.6.

S4.21. Prove S4.17: a uniformly continuous $f : E \rightarrow \mathbb{R}^k$ has a limit at every point of $\overline{E}$, and therefore extends uniquely to a uniformly continuous function on $\overline{E}$. Show that $\sin(1/x)$ on $(0, 1)$ has no such extension, and identify which hypothesis it violates.³³⁸

SOURCE: Tao I, Cor. 9.9.14; generalises Rudin's Exercise 4.13.

S4.22. Define the oscillation $\omega_f(p)$ as in S4.4. Prove that f is continuous at p if and only if $\omega_f(p) = 0$, that $\{x : \omega_f(x) \geq \varepsilon\}$ is closed for each $\varepsilon > 0$, and hence that the set of discontinuities of *any* f is a countable union of closed sets. Deduce, using Exercise 3.22, that no function is continuous at every rational and discontinuous at every irrational.³³⁹

SOURCE: Yu Pin, Lemmas 131–132; Tao I, Ex. 9.3.4.

S4.23. Let $A \subset \mathbb{R}^1$ be countable. Construct a monotonically increasing $f : \mathbb{R}^1 \rightarrow \mathbb{R}^1$ whose set of discontinuities is exactly A .³⁴⁰

SOURCE: Yu Pin, HW4 problem E (p. 110); cf. Tao I, Ex. 9.8.5.

S4.24. ★ (Cayley–Hamilton by continuity.) For A an $n \times n$ complex matrix let $P_A(X) = \det(XI - A)$. Prove: (a) if A is diagonalisable then $P_A(A) = 0$; (b) the map $A \mapsto P_A(A)$ is continuous on the space of matrices with the norm $\|\cdot\|$; (c) every A is a limit of diagonalisable matrices; (d) therefore $P_A(A) = 0$ for every A .³⁴¹

SOURCE: Yu Pin, winter problem set U (pp. 331–332).

S4.25. Prove that a path-connected set is connected, and that the closure of

$$S = \{(x, \sin(1/x)) : 0 < x \leq 1\}$$

is connected but not path-connected.³⁴²

SOURCE: Tao II, Ex. 2.4.7–2.4.9.

339 The last part is the converse question to Rudin's Exercise 4.18 (Thomae), and the answer is that the two cases are not symmetric. If such an f existed, $\mathbb{Q}$ would be a countable intersection of dense open sets, and intersecting further with the complements of the individual rationals would exhibit $\emptyset$ as such an intersection — contradicting Baire.

340 The exact converse of 4.30, and Rudin's Remark 4.31 sketches the construction: enumerate $A = \{a_1, a_2, \dots\}$ and put $f(x) = \sum_{a_n < x} 2^{-n}$. Convergence is comparison with a geometric series; the jump at a_n is exactly 2^{-n} ; continuity elsewhere follows because the tail beyond N contributes less than 2^{-N} . Together with 4.30 this characterises the possible discontinuity sets of monotone functions completely — a rare instance of a sharp answer.

341 The best advertisement in this chapter for Rudin's Exercise 4.4. Part (a) is one line in an eigenbasis; (b) holds because the entries of $P_A(A)$ are polynomials in the entries of A ; (c) perturbs the eigenvalues of an upper triangular form to make them distinct. Then a purely algebraic theorem falls out of “two continuous maps agreeing on a dense set are equal” — with the density supplied by linear algebra and the rigidity by Chapter 4.

342 The first part is 4.22 applied to the path. For the second, $\bar{S}$ adds the segment $\{0\} \times [-1, 1]$; connectedness holds because the closure of a connected set is connected, while a path from the segment into S would have to traverse infinitely many oscillations in finite time — make that precise by showing its first coordinate cannot leave 0. Rudin's favourite pathology at 4.27(d) turns out to separate two notions of connectedness he never distinguishes.